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Two Silhouettes of Strongly Correlated Topological Phases: Fractionalization and Crystallization

A DISSERTATION PRESENTED
BY
JUNKAI DONG
TO
THE DEPARTMENT OF PHYSICS

IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE DEGREE OF
DOCTOR OF PHILOSOPHY
IN THE SUBJECT OF
PHYSICS

HARVARD UNIVERSITY
CAMBRIDGE, MASSACHUSETTS
APRIL 2025

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Two Silhouettes of Strongly Correlated Topological Phases: Fractionalization and Crystallization

Abstract

Recent advances in two-dimensional materials have unveiled a plethora possibilities for exploring strongly correlated topological phases. This thesis focuses on predicting novel topological phases and examining their properties. Chapters 1 and 2 discuss phases characterized by fractionalization, where the system's excitations exhibit fractional electric charges. These phases, though arising without external magnetic fields, closely resemble fractional quantum Hall phases typically induced by strong magnetic fields. Specifically, Chapter 1 investigates fractional and integer Chern insulator phases within higher Chern bands that were proposed to exist in twisted graphene systems. Chapter 2 reports predictions of a composite Fermi liquid phase in twisted $MoTe_2$ through numerical simulations. Chapters 3 through 6 explore scenarios where electrons crystallization coexists with band topology. Contrary to the conventional belief that electronic crystals are topologically trivial, we propose that anomalous Hall crystals —- topologically nontrivial electronic crystals -— can exist. Chapter 3 demonstrates how the concept of anomalous Hall crystals can explain recent experimental findings in rhombohedral graphene moiré systems. Chapter 4 introduces a simplified theoretical model elucidating the origin of Chern numbers associated with these anomalous Hall crystals. Chapter 5 proposes a minimal model of the anomalous Hall crystal, bridging it to previous models of electronic crystallization. Finally, Chapter 6 investigates the elastic theory and phonon dynamics in electron crystals in the presence of Berry curvature.

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Chapter 1

Junkai Dong, Patrick J. Ledwith, Eslam Khalaf, Jong Yeon Lee, and Ashvin Vishwanath, Many-Body Ground States from Decomposition of Ideal Higher Chern Bands: Applications to Chirally Twisted Graphene Multilayers, Phys. Rev. Research 5, 023166 (2023).

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Chapter 6

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To my family,
To everyone hit by the storms.

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0

Introduction

The world is comprised of moving objects. Atoms. Humans. Stars. These objects and their potential movement patterns is the reason for the brimming diversity of the universe. There are two puzzling issues surrounding this diversity, however. All the objects that we described are made of the same three particles: electrons, protons, and neutrons. The magic power through which the wide-ranging phenomena is generated from these three ingredients is the concept of *emergence*. While the dimension of objects involved differ by many orders of magnitude, there are often patterns that occur over and over again in seemingly unrleated scenarios – this is the concept of *universality*. For example, ice and table salt are both crystalline solids, although the constituent atoms are completely different. The discovery of universal patterns, the classification of possible patterns, namely different *phases*, and the study of how these patterns can emerge in different systems, is a central theme of condensed matter physics.

One of the most useful tools to understand phases of matter is proposed by Landau, namely the Landau paradigm [1]. The Landau paradigm instructs us to look for symmetries of the theory that describes the system. Perhaps surprisingly, some systems spontaneously rearrange themselves such that the symmetries are broken: particles crystallize and break translation symmetry, magnets develop magnetism and break time reversal symmetry, and bosons condense into superfluids and break number conservation symmetry. From here, the Landau paradigm then provides a program to systematically investigate how the system goes between different phases, also known as phase transitions [2]. The Landau paradigm, now a cornerstone of condensed matter physics, has successfully revealed that many of these phases and phase transitions can be reinterpreted as one another: for example, the way that water turns into water vapor is identical to the way that some model of a magnet develops

magnetism.

Beyond these symmetry-broken phases are topological phases [3]. Mysteriously, the definition of "topological" phases is a negative one – topological phases are those that cannot be characterized by the Landau paradigm. In their purest form, they do not break any symmetries, and are usually characterized by long-range entanglement properties of its fundamental particles. The most prominent examples are the integer quantum Hall effect (IQHE) [4] and fractional quantum Hall effect (FQHE) [5], discovered in 1980 and 1982 respectively by applying huge magnetic fields to a two-dimensional electron gas. These phases are characterized by a non-local topological invariant called *Chern number*, which is physically related to its transverse electrical conductivity and can hence be measured experimentally [6]. They are found to be physically distinct from the value corresponding to a topologically trivial phase, which is zero. This distinction renders these two phases topological. Their nontrivial topology can manifest in different ways. In the IQHE, the electrons fill distinct Landau levels. These Landau levels can be interpreted as electronic bands with nontrivial band topology corresponding to the Chern number associated with each band. Within the band, a nonzero Chern number prohibits the electrons from becoming fully localized and thus described as classical particles: one cannot obtain a set of orthonormal wavefunctions of the band that simultaneously respects the underlying translation symmetry of the band and are exponentially localized [7]. If one increases the amount of magnetic field from the value needed to obtain IQHE, the system moves into the FQHE phases. In the FQHE, due to a combination of strong magnetic field and Coulomb interactions between the electrons, the electrons split up into particles with fractional charges of the electron – such as $e/3$ [8]. These exotic particles are called anyons, which are neither bosons nor fermions: they are characterized by a nontrivial braiding statistics angle. These anyons, particularly a variant called the non-abelian anyons, form the basis of topological quantum computation [9].

While these paradigmatic topological phases of matter have been experimentally realized, they do come at a price. These delicate phases require a triplet of stringent conditions to be met: sample quality, as the materials have to be essentially disorder-less; large magnetic field, typically on the order of $10 - 30\mathrm{T}$, ten times larger than the magnets used in power transformers; and low temperatures that are comparable to the temperatures of the outer space. Furthermore, there has been theoretical proposals that non-abelian anyons are most promisingly realized if superconductors are placed near quantum Hall phases. However, the strong magnetic field required to realize the quantum Hall phases can easily destroy the superconductivity. Given the potential of harnessing these topological phases of matter for engineering purposes such as quantum computing, it is necessary to find ways to

realize these phases in more accessible conditions. In particular, reducing the strength of magnetic fields required to realize these phases is a widely pursued research topic in recent years.

One of the most groundbreaking technical advances in the realm of two-dimensional materials is that of moiré engineering [10]. A moiré pattern is created when two lattice with slight mismatches are superimposed against each other – whether by a twist angle or by a difference in lattice constant. Such moiré patterns generate moiré lattices, a lattice that is orders of magnitude larger than the atomic lattices. Due to such a long lengthscale, the kinetic energy of itinerant electrons in the material is effectively suppressed and interactions become stronger. Strong interactions are important for the realization for the topological phases in three ways. First, if one aims to realize quantum Hall phases without magnetic fields, one has to break time reversal symmetry somehow. In conventional magnets, the electrons break time reversal symmetry by aligning their spins to lower the Coulomb energy. Similarly, in two-dimensional material, time reversal can be broken by valley-polarizing the electrons: low energy electrons in these moiré systems usually come from specific areas known as valleys in the Brillouin zone of the atomic lattice. If all the electrons come from the same region, the system will break time reversal symmetry. Once the time reversal symmetry is broken, the electron can reside in band structures, sometimes topological, that resemble those in magnetic fields magnitudes larger than the largest magnet on earth. Thus interactions can effectively generate a pseudo-magnetic field for the electrons. Second, the realization of the analogues of FQH phases require strong interactions to be present; without strong interactions, the electrons would never split into anyons. Third, it's often the interactions that set the energy scale for these topological phases: as long as the temperature is below the interaction scale, the system can still hold against thermal fluctuations that destroy the topological order. Therefore strong interactions also raises the temperature scales at which these phases will be seen.

Remarkably, some of these topological phases has been recently observed in moiré materials. Quantum anomalous Hall effect, where a Chern insulator is established without magnetic fields, was first seen in twisted bilayer graphene systems aligned with hexagonal boron nitride (hBN) [11, 12], and later in twisted multilayer graphene systems [13]. Quantum anomalous Hall effect is also seen in twisted transition metal dichalcogenide systems as well, and in both homobilayer [14, 15] and heterobilayer [16] varieties. Fractional Chern insulators were first seen at high magnetic fields in a graphene-hBN moiré system [17], and subsequently at lower magnetic fields in twisted bilayer graphene aligned with hBN with scanning probes [18]. Finally, in early 2023, fractional quantum anomalous Hall effect (FQAHE), fractional Chern insulators at zero magnetic field, was observed in twisted

$MoTe_2$, first optically [19] and subsequently through electrical transport [20]. In late 2023, FQAHE was observed in electrical transport in rhombohedral graphene moiré systems [21].

The scientific content of the thesis is either directly or indirectly inspired by these seminal experimental breakthroughs. Broadly speaking, the content can be divided into two parts:

In chapters 1 and 2 we present studies that explore fractional topological phases in moiré materials. They often view the electronic bands that host these fractional phases through the lens of "ideal quantum geometry" or "vortexability" [22, 23]. If a band is vortexable and has Chern number 1, its wavefunctions can be related to the wavefunctions of the lowest Landau level, which is known to host FQH phases. Inspired by this similarity, it has been proposed that fractional Chern insulators become exact ground states in vortexable bands. Chapter 1 concerns bands which are vortexable and has Chern number C larger than 1. We will show that a certain class vortexable bands with $C > 1$ can be decomposed, in a non-unitary way, into C Chern number 1 vortexable bands, and can thus be interpreted as C layers of lowest Landau levels, where each layer corresponds to a different translation-breaking pattern. This leads to an analytic construction of interesting many-body ground states such as generalized quantum Hall ferromagnets and multicomponent fractional Chern insulators. We apply this theory to the vortexable bands proposed to exist in twisted graphene systems in Ref. [24]. Chapter 2 is concerned with twisted $MoTe_2$, where fractional Chern insulators were observed first optically [19]. However, the optical probes are not sensitive to the gapless fractional phases, which include the composite Fermi liquids. We explain that the band in $MoTe_2$ that hosts these fractional topological phases is not only vortexable, but very similar to the lowest Landau level. We report numerical simulations involving both exact diagonalization and density matrix renormalization group methods that demonstrate the existence of a zero-field composite Fermi liquid phase.

The other central theme explored in this thesis is anomalous Hall crystals, which will be discussed in Chapters 3 through 6. Chapter 3 reports how the concept for anomalous Hall crystal was born: during a study of a Chern insulator in a rhombohedral graphene-hBN experiment, the Chern insulator was found to exist even in the absence of a moiré potential (that is, if the hBN layers are artificially removed), and thus must be due to the spontaneous breaking of continuous translation symmetry. This electronic crystalline phase that carries nonzero Chern number at zero magnetic field is named "anomalous Hall crystal". Chapter 4 discusses how quantum geometry of rhombohedral graphene gives rise to the Chern number of the anomalous Hall crystal. Chapter 5 proposes the λ-jellium model, a two-dimensional electron gas with Berry curvature, that serves as a minimal model for the anomalous Hall crystal. The λ-jellium model can be reduced to the jellium model, and is promising to be sim-

ulated using beyond mean-field techniques. Finally, chapter 6 discusses elasticity theory and phonon dynamics of electron crystals with Berry curvature, which includes anomalous Hall crystals. We report the existence of a kineo-elastic term in the elastic theory which renders the speed of sound assymetric under inversion.

1

Many-Body Ground States from Decomposition of Ideal Higher Chern Bands: Applications to Chirally Twisted Graphene Multilayers

1.1 Introduction

The discovery of correlated states in twisted bilayer graphene (TBG) has inspired interest in the study of topological flat bands [25–30]. The interplay of band topology and strong interactions in TBG makes it an ideal candidate to realize strongly correlated topological physics. A simplified "chiral" model[31], where intrasublattice moiré tunneling is ignored, yields exactly flat topological bands at the magic angle, thereby capturing the most essential features of the system. The chiral model hosts analytically-attainable wavefunctions [31] that are equivalent to those of the LLL in an inhomogeneous magnetic field [32]. Its mathematical properties and mysteries have turned the chiral model into a sub-field of study of its own right [31–43].

At the same time, it was shown [32] that the chiral model satisfies a quantum-geometric[22, 44–52] identity known as the "trace condition". It was recently shown that the trace condition enables intra-band vortex attachment[23, 24], which has strong implications for interacting physics. For example, it directly enables the construction of FQHE-like trial states which are *exact* ground states of the system under short-range repulsive interactions. For TBG, this led to the analytic prediction[32] of fractional Chern insulators (FCI)[53–62], states that host the fractional quantum Hall effect at fractional filling of a Chern band first observed[17] in Hofstadter

bands [63–67]. FCIs in TBG were simultaneously predicted in numerical works [68–70] and were recently experimentally observed [18] in a small magnetic field that effectively restored the ideal chiral limit [71]. We will refer to trace condition satisfying bands as *ideal* bands; they have also been referred to as "vortexable" [23]. The trace condition is also related to momentum space holomorphicity [22, 23, 45, 47, 48, 72], a powerful analytic tool.

Multilayer twisted graphene systems consisting of n chirally stacked[73–75] graphene layers, i.e. AB, ABC, etc., stacked with a single twist on top of m chirally stacked layers are a natural extension of TBG. Experimentally, twisted double bilayer graphene, $(n, m) = (2, 2)$ [76–78, 78–82], and twisted mono-bilayer graphene $(n, m) = (1, 2)$ [13, 83–88], have been fabricated and a nearly quantized anomalous Hall effect has repeatedly been observed with $C > 1$ [76, 81, 83, 84]. This has generated extensive theoretical interest in studying topological physics in these systems at different fillings [24, 89–101]. Furthermore, symmetry broken Chern insulators have been found experimentally at half filling [85]. A variety of these states was recently identified numerically together with signs of an approximate SU(2) symmetry [102], although its origin has not been understood. FCIs have also been studied in these systems [22, 95], and other systems with higher Chern bands [61, 64, 66, 103–108], though analytic progress has largely been limited to toy models. These advances motivate the need for a systematic understanding of correlation phenomena in higher Chern bands, including both symmetry-broken and fractional Chern insulators.

Chiral models have been constructed for the twisted chiral multilayers in Refs. [24, 96] through neglecting certain interlayer tunneling terms. Chiral twisted multilayer graphene yields flat and ideal higher Chern bands that are analytically tractable when tuned to the same magic angle as chiral TBG. By combining ideality, which allows the construction of trial wavefunctions for topologically ordered states, with higher Chern number in an experimentally feasible system, these models provide a novel and unexplored platform to study and realize interacting topological phases. Ref. [96] numerically found model FCI states with particle entanglement suggestive of particular "color-singlet" [61] Halperin states, though a general analytic understanding of the range of possible states in such a system remains lacking.

One approach to understanding correlated states in a $C > 1$ band is to employ the hybrid Wannier functions to decompose the band into C Chern 1 bands by enlarging the unit cell [109]. However, this approach does not respect the ideality condition: an ideal Chern $C > 1$ band generally decomposes into a set of $C = 1$ non-ideal bands such that analytic techniques cannot be applied to the decomposed basis. Furthermore, this decomposi-

tion assumes broken translation symmetry in a specific direction from the outset, making it difficult to understand translation-unbroken states like the FCIs observed in Ref. [96]. This naturally leads to the question: is it possible to decompose an ideal $C > 1$ band into ideal $C = 1$ bands and in a way that captures both translationally symmetric and translation-breaking states? A partial answer to this question was provided in Ref. [24] which showed that it is impossible to decompose a generic ideal $C > 1$ bands in terms of *orthogonal* and ideal $C = 1$ bands.

Here, we show that by lifting the orthogonality constraint, it is possible to decompose a generic ideal $C > 1$ into C ideal Chern 1 bands. From this decomposition, we reveal a hidden *non-unitary* $\mathrm{SU}(C)$ action among the decomposed bands combining real space and momentum space translations. We note that although this $\mathrm{SU}(C)$ action is generally non-unitary and is not a symmetry of the Hamiltonian, it *is* often a symmetry of the ground state manifold for short-range repulsive interactions, enabling us to make sharp predictions regarding the many-body ground state at partial filling. Furthermore, the $\mathrm{SU}(C)$ structure is reminiscent of the structure of multi-component Landau levels [61], allowing us to interpret these ground states in terms of more familiar correlated states in multi-component quantum Hall systems. It should be emphasized that the states we obtain are physically distinct and have non-trivial translation symmetry breaking patterns visible through real-space charge density, compared to multi-component Landau levels which have uniform charge density.

By employing this decomposition, we analytically construct ground states at a variety of fillings of ideal Chern $C > 1$ bands and make concrete predictions on their realization. First, we identify a ground state manifold of charge density waves (CDWs) at filling $1/C$ with emergent $\mathrm{SU}(C)$ symmetry. Such states can be interpreted as generalized quantum Hall ferromagnets in the decomposed basis. An immediate consequence of this identification is the existence of charged skyrmion textures which correspond to a characteristic winding pattern of the CDW order parameter (cf. Fig. 1.1). Such a winding pattern can be readily observed with local charge probes such as STM, which has already been employed in these systems [87, 88, 110–114]. Second, we characterize the structure of translation symmetric fractional Chern insulator states at fillings $1/(2Cs + 1)$, for each positive integer s, by establishing a direct analogy with flavor-singlet Halperin states. Finally, we discuss a manifold of translation-breaking Laughlin states that appear at low fillings $1/C(2s + 1)$ where fractionalization and topological order coexists with CDW order. These topological systems with broken translation symmetries are similar in nature to the Hall crystals in Ref. [115]. All our results are verified by numerical exact diagonalization (ED) on the chiral model for twisted mono-bilayer graphene. These results enable us to make experimental predictions for

graphene multilayers.

The paper is organized as follows. We begin with an overview of the central theoretical results as well as some consequences for experiment in Sec. 1.2. Next, we review the physical model for chiral graphene multilayers in Sec. 1.3 and quantum geometry techniques to understand the single particle physics in Sec. 1.4. We present the central technical result, the decomposition of an ideal higher Chern band, in Sec. 1.5 for $C = 2$. We study quantum ferromagnets, which manifest as topological CDWs, and their associated skyrmion description at half filling for a spinless $C = 2$ band in Sec. 1.6 and a spinful $C = 2$ band in Sec. 1.7. We discuss the possible fractional Chern insulators, both translation symmetric and translation broken, in Sec. 1.8. We generalize the physical implications to ideal bands with $C > 1$ in Sec. 1.9. We conclude with some future questions in Sec. 1.10.

1.2 Summary of Results

We begin by briefly summarizing our results and discussing their main implications. By now, a full understanding of ideal $C = 1$ bands has been achieved: they have been related to the LLL of a Dirac particle in a magnetic field [22, 32], which enables the construction of exact many body Laughlin-like ground states for short range interactions [23]. Our main technical achievement in this work is the generalization of such understanding for any ideal higher Chern band. In particular, we provide a general explicit construction to decompose any ideal higher Chern band into ideal but non-orthogonal Chern 1 bands with a non-unitary $\mathrm{SU}(C)$ action. Compared to the hybrid Wannier decomposition [109], our procedure has the distinct advantages of preserving the ideal band geometry and allowing access to translation broken states in all possible directions in the same basis. The preservation of ideal band geometry means we leverage the analytic knowledge of ideal $C = 1$ bands to analytically study interacting phases – in particular, FCIs – in $C > 1$ bands.

The decomposition provides analytic many-body states that are the *exact* ground states at certain fractional fillings for short range interactions. For example, for a spinless band with Chern number C these ground states include topological CDWs at $1/C$ filling. At lower fillings, we construct Halperin-like translationally symmetric FCIs at $\nu = 1/(2Cs + 1)$ and Laughlin-like translation broken FCIs at $\nu = 1/C(2s + 1)$ for any positive integer s.

The topological CDWs are best understood as a manifold of generalized quantum Hall ferromagnets (FM), where translations act as pseudospin rotations on the ferromagnets. The pseudospin ferromagnets are generically translation symmetric only in a $C \times C$ unit cell. Crystalline symmetries, both translations and discrete rotations,

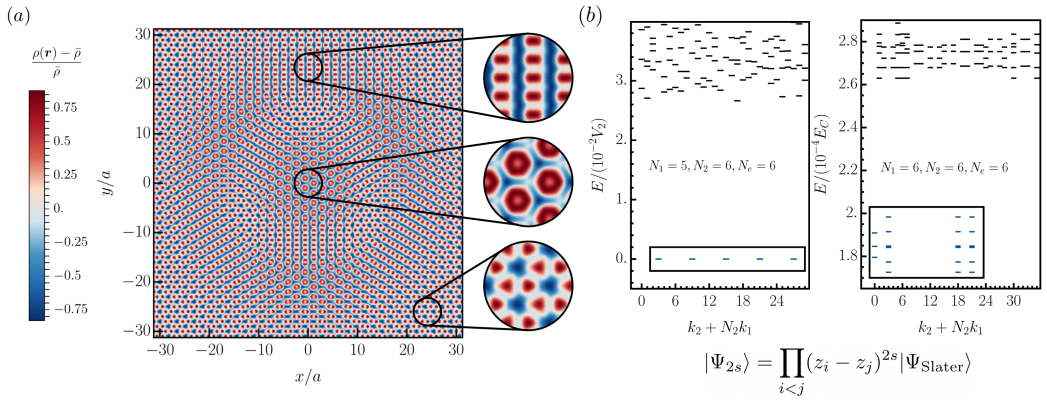
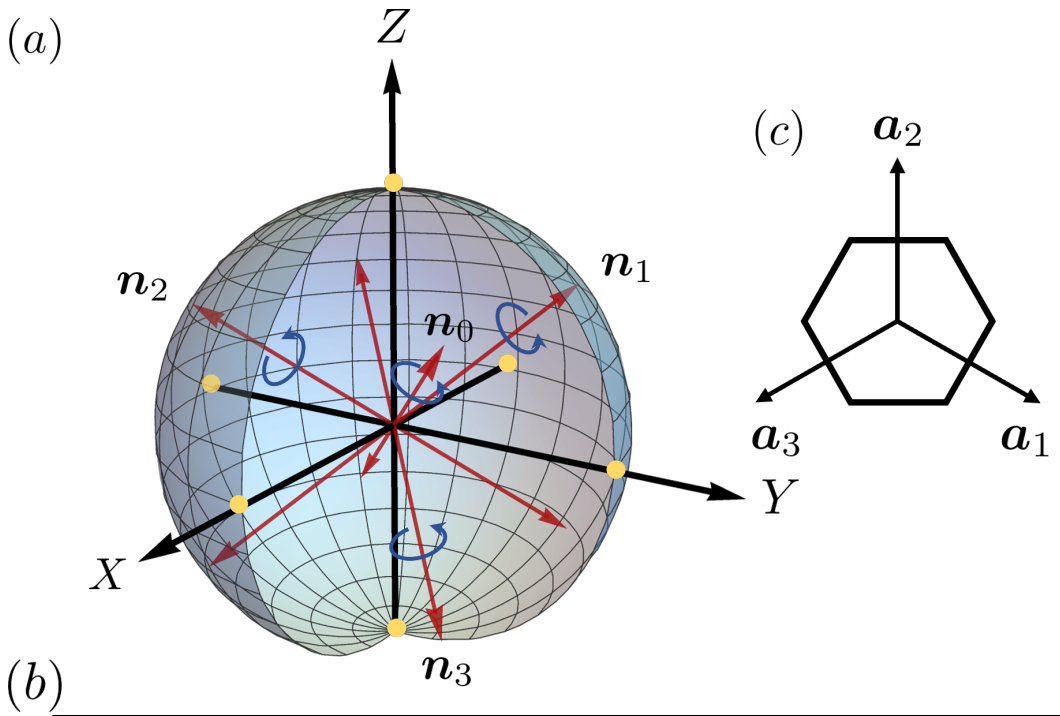


Figure 1.1: Main results of our paper. (a) Plots of the density profile of a CDW-skyrmion that arises out of a manifold of topological charge density waves from the ideal $C = 2$ band in chiral twisted monolayer-bilayer graphene. Windings of the order parameter corresponds to windings of the translation breaking pattern. We zoom into different regions to highlight different regions that correspond to particular topological charge density waves that repeat in a 2×2 unit cell. (b) shows many-body spectrum from ED at different fillings for the ideal $C = 2$ band in chiral twisted monolayer-bilayer graphene: boxed energy levels correspond to the ground state manifold of model FCIs. The first panel is the translation invariant (332) Halperin state at $\nu = 1/5$ with the interaction $V_1\delta''(\hat{\boldsymbol{r}}_i - \hat{\boldsymbol{r}}_j)$ and the expected ground state degeneracy of 5. The second panel is the translation broken Laughlin state at $\nu = 1/6$ under a screened Coulomb interaction with a ground state degeneracy of 21 that combines the Laughlin degeneracy of 3 with the generalized-ferromagnetic degeneracy of $N_e + 1 = 7$. Both of these collections of states originate from the depicted analytic vortex attachment construction summarized in Sec. 1.4.

emerge as rotations in the order parameter manifold. For $C = 2$, this manifold is a sphere and different topological CDW states may be understood as different pseudospin directions, pictorially represented in Fig. 1.2(a) and summarized in Tab. 1.2(b). The $\pm X, \pm Z, \pm Y$ axes of the pseudospin sphere correspond to states that preserve the translations along $\boldsymbol{a}_{1,2,3}$, where $\boldsymbol{a}_{1,2}$ are primitive lattice vectors and $\boldsymbol{a}_3 = -\boldsymbol{a}_1 - \boldsymbol{a}_2$. For rotational symmetry, we focus on the three-fold C_{3z} case which is relevant for graphene moirè systems, and other rotational symmetries may be understood similarly. We find that C_{3z} acts as a 120° rotation around a particular $\boldsymbol{n}_0$ axis of the pseudospin sphere. There are also rotations $C_{3z}^{(i)} = T_{\boldsymbol{a}_i} C_{3z} T_{-\boldsymbol{a}_i}$ around lattice vectors $\boldsymbol{a}_i$ and corresponding invariant axes $\boldsymbol{n}_i$; these are distinct operations because we have broken translations.

We construct exact model FCIs with the vortex attachment procedure [23, 24] reviewed in Sec. 1.4. By attaching vortices to the fully filled Chern C band, we arrive at an explicit first quantized wavefunction for the translationally-symmetric Halperin states. If we instead choose a $C = 1$ topological charge density wave as a parent state we obtain the translation broken Laughlin states. We numerically verify these predictions in exact diagonalization for the $C = 2$ band of chiral twisted monolayer-bilayer graphene, see Fig. 1.1(b).

Finally, based on our analysis in the ideal limit, we make concrete predictions for the ground states and excitations of twisted mono-bilayer graphene at several fillings as follows. Note, here we allow for valley and spin



(b)

Flavor Space	Real Space
$\pm X, \pm Y, \pm Z$ FMs	$T_{\boldsymbol{a}_i}$ preserving CDWs
$\pm \boldsymbol{n}_i$ FMs	C_{3z} symmetric CDWs
$R(\hat{x}/\hat{y}/\hat{z}, \pi)$	Translations $T_{\boldsymbol{a}_i}$
$R(\hat{n}_i, 2\pi/3)$	C_{3z} rotations around $\boldsymbol{a}_i$

Figure 1.2: Sphere of flavor ferromagnets (FMs) at half filling for a C_{3z} symmetric ideal band with $C = 2$. Each point on the sphere correspond to a topological CDW with unit cell $2\boldsymbol{a}_1 \times 2\boldsymbol{a}_2$. (a) We show the order parameter sphere of ferromagnets: red arrows correspond to C_{3z} symmetric CDWs (second and third insets of Fig. 1.1(a)) and yellow dots correspond to CDWs that preserve certain translation symmetries $T_{\boldsymbol{a}_i}$ (first inset of Fig. 1.1(a)). C_{3z} rotations in real space correspond to rotations around red arrows in order parameter space. Table (b) explains the representation of special CDWs as special ferromagnets on the sphere and crystalline symmetries as rotations. (c) We show the real space unit cell and the lattice vectors.

degeneracy, hence the filling ν ranges from $0 \leq \nu \leq 4$. We will focus on states doped on top of a $\nu = 3$ spin and valley polarized state such that there is a single empty $C = 2$ band left over.

1. At $\nu = 3 + 1/2$, we expect that the ground state will be a continuous manifold of spin polarized topological CDWs, or generalized quantum Hall ferromagnets. The observed topological CDWs in Ref. [85] are special points in a continuous manifold of generalized quantum Hall ferromagnets. Our main prediction is that there will be associated *charged* skyrmion textures with a particular winding of translation breaking patterns illustrated in Fig. 1.1(a). These textures can be directly probed by STM.

2. At $\nu = 3 + 1/5$, we expect that the ground state will be a translationally invariant Halperin (332) state: see first panel of Fig. 1.1(b) for numerical spectrum. In particular, the Hall conductivity (or the slope of the gapped feature on the Landau fan diagram) will be $2/5$.

3. At $\nu = 3 + 1/6$, we expect that the ground state will be translation breaking ferromagnetic Laughlin states: see second panel of Fig. 1.1(b) for numerical spectrum. These states will in general be translation breaking striped phases with $1/3$ Hall conductivity.

We end our summary of results with some cautionary comments on application to experiments. First, realistic systems will not be in the chiral limit, which means that the higher Chern band will not have exactly ideal quantum geometry, and may be dispersive. However, the band geometry and flatness could be improved by external perturbations such as out of plane electric and magnetic fields as believed to occur in magic angle graphene [71], bringing the system closer to the ideal limit. Second, in most of our discussions, we have assumed a spinless band for simplicity; exotic magnetic orders at half filling such as the tetrahedral antiferromagnetic (TAF) order exist in spinful bands [102, 116–121]. Remarkably, after adding spin we find that the TAF states lie in the same $\mathbb{CP}^3$ manifold as the previously mentioned charge density waves. Furthermore, a magnetic field may be used to favor the spin-polarized submanifold that hosts charge-density skyrmions. A final caveat is that typical twisted graphene systems contain twist angle disorder which leads to large inhomogeneous heterostrain. Strain greatly increases single particle dispersion and pushes the band away from ideality. We leave a detailed quantitative study of the realistic system that addresses the above issues to future work.

1.3 Chiral Twisted Graphene Multilayers

In this section, we review the chiral twisted graphene multilayers: they give rise to ideal higher Chern bands and closely describe experimentally relevant systems. We use them as our primary numerical example. Experimentally, these graphene multilayers have been fabricated both as twisted monolayer-bilayer (mono-bi) [83–86] and twisted bilayer-bilayer (bi-bi) [76–80] graphene systems. The following discussion will closely follow recent works which introduced these models [24, 96].

We consider $n + m$ graphene layers. The first n layers are untwisted and chirally stacked on each other: chiral stacking means that the successive layers have Bernal stacking AB or BA. These n chirally stacked layers are then twisted by the magic angle θ of chiral TBG, and put on top of another m layers of chirally stacked graphene. The mono-bilayer system corresponds to $n = 1, m = 2$ and the bilayer-bilayer system corresponds to $n = 2, m = 2$.

For concision, consider electrons of a specific valley and spin flavor. We make the approximation that the electron can only tunnel between neighboring layers, and that the interlayer tunneling is on-site (this neglects e.g. trigonal warping terms [122]). The Hamiltonian can be written as

$$H = \begin{pmatrix} h_{m,\sigma} & T_M \\ T_M^\dagger & h_{n,\sigma'} \end{pmatrix} \tag{1.1}$$

where $h_{n,\sigma}$ describes the n untwisted chirally stacked graphene layers with chirality $\sigma = \pm$ depending on whether the layers have AB $(+1)$ or BA (-1) stacking. T_M only tunnels between layers n and $n + 1$ and is identical to the moiré tunneling of chiral TBG, where intra-sublattice tunneling is switched off. Its explicit dimensionless form is given by

$$T_M = \alpha \begin{pmatrix} 0 & U(\boldsymbol{r}) \\ U^*(-\boldsymbol{r}) & 0 \end{pmatrix} \tag{1.2}$$

where $U_1(\boldsymbol{r}) = \sum_{n=0}^{2} e^{\frac{2\pi i n}{3} - i\boldsymbol{q}_n \cdot \boldsymbol{r}}$, $\boldsymbol{q}_n = C_3^n(0, -1)^T$, and C_3 is a 120° rotation. The parameter $\alpha = w_1/vk_\theta$ is a dimensionless tunneling strength, where $w_1 \approx 110$ meV and the angle-dependent kinetic energy scale is $vk_\theta = v\left|\boldsymbol{K}^{\text{top}} - \boldsymbol{K}^{\text{bot}}\right|$, where v is the Fermi velocity of the Dirac cone and $\boldsymbol{K}^{\text{top,bot}}$ are the graphene $\boldsymbol{K}$-points for the top and bottom stacks respectively. The chiral TBG magic angle corresponds to $\alpha \approx 0.586$ [31].

The form of $h_{n,\sigma}$ is tri-diagonal:

$$h_{n,\pm} = \begin{pmatrix} -i\sigma\cdot\nabla & T_\pm & 0 & \dots \\ T_\pm^\dagger & -i\sigma\cdot\nabla & T_\pm & \dots \\ 0 & T_\pm^\dagger & -i\sigma\cdot\nabla & \dots \\ \dots & \dots & \dots & \ddots \end{pmatrix} \tag{1.3}$$

The diagonal terms describe the Dirac cones of each layer, and the off diagonal terms describe the tunneling due to two kinds of Bernal stacking: $T_\pm = \beta\frac{\sigma_x \pm i\sigma_y}{2}$.

Most importantly, the model hosts zero energy flat bands at particular values of α identical to the magic angles for chiral TBG. These zero energy bands are closely related to the chiral symmetry of the model: $\{H, \sigma_z\} = 0$. The chiral symmetry arises because the model is purely off diagonal in the sublattice basis:

$$H = \begin{pmatrix} 0 & \mathcal{D}^\dagger \\ \mathcal{D} & 0 \end{pmatrix}_{\text{AB}}, \tag{1.4}$$

where the subscript AB denotes that the matrix is block diagonalized in its sublattice. Thus, we may choose zero energy eigenstates of H to be eigenstates of σ_z. At the magic angle $\alpha = 0.586$, two zero energy bands labeled by the sublattice index A and B emerge independent of the value of β [24, 96]. These bands carry Chern numbers, whose signs depend on the stacking chiralities (σ, σ') and whose values depend on (n,m).

For example, when both upper n layers and lower m layers are AB-stacked with $\sigma = \sigma' = 1$, the Chern number for the A-sublattice polarized flatband (zero-energy) is $C_A = n$, while for the B-sublattice polarized flatband $C_B = -m$. In particular, both mono-bilayer and bilayer-bilayer systems give rise to Chern bands with $|C| = 2$ for this stacking orientation. For an analytic derivation of the flat band wavefunctions $\psi_{\boldsymbol{k}}(\boldsymbol{r})$ for any stacking (σ, σ'), in terms of the $n = m = 1$ chiral TBG wavefunctions, see Ref. [24]. We numerically plot the band structure of mono-bi with the chiral model in Fig. 1.3; the higher Chern band is highlighted in red.

In the next section we will review ideal band geometry and show that the zero modes (flat bands) of magic-angle chiral graphene multilayers in Eq. (1.4) attain the ideal geometry. Crucially, the zero mode operator $\mathcal{D}$ only depend on coordinates x, y and the antiholomorphic derivative $\overline{\partial} = \frac{1}{2}(\partial_x + i\partial_y)$: $\mathcal{D}(\boldsymbol{r}, \partial_x, \partial_y) = \mathcal{D}(\boldsymbol{r}, \overline{\partial})$.

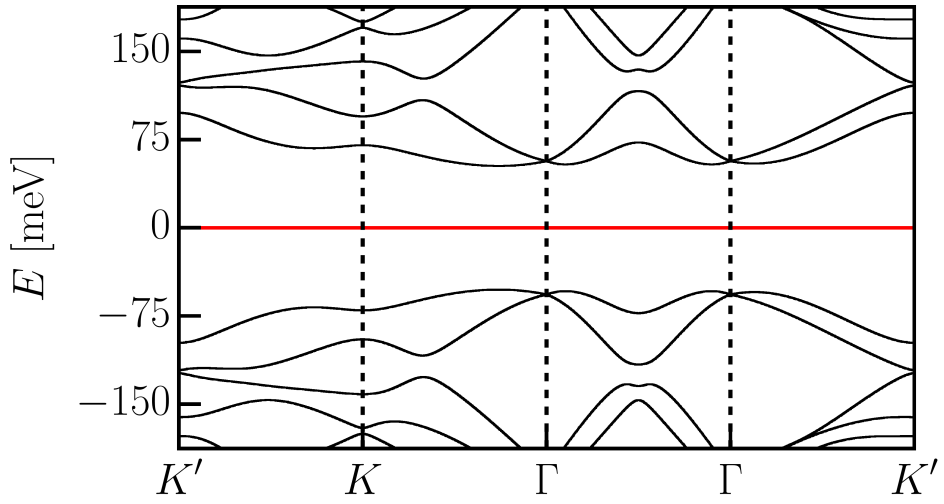


Figure 1.3: Band structure of twisted monolayer-bilayer graphene. The higher Chern band is highlighted in red. We use the chiral model (1.1) with $\alpha = 0.586$ and $\beta = 2$; the two Chern bands are degenerate and both of them have zero energy.

This means that

$$[\mathcal{D}, z] = 0, \quad \text{where } z = x + iy. \tag{1.5}$$

This property of $\mathcal{D}$ will be equivalent to the "vortex-attachment" description of ideal quantum geometry.

1.4 Quantum geometry of Ideal Chern bands

In this section we review our characterization of ideal band geometry via the equivalent perspectives of the "trace condition," momentum-space holomorphicity, and real-space "vortexability." We will review the utility of each of these perspectives. Finally, we will conclude by developing certain technical tools for working with ideal bands that we will use in the subsequent decomposition section.

Consider a set of bands with total Chern number $C \geq 0$ and periodic wavefunctions $u_{\boldsymbol{k}a}(\boldsymbol{r}) = e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}\psi_{\boldsymbol{k}a}(\boldsymbol{r})$, where a is a band index. The set of bands is *ideal* if they satisfy the trace condition defined as one of the following three equivalent conditions [23]:

(i) The inequality $\operatorname{tr} g(\boldsymbol{k}) \geq \Omega(\boldsymbol{k})$ is saturated, where $g_{\mu\nu}(\boldsymbol{k}) = \operatorname{Re}\eta_{\mu\nu}(\boldsymbol{k})$ is the Fubini Study metric and $\Omega(\boldsymbol{k}) = -\epsilon^{\mu\nu}\operatorname{Im}\eta_{\mu\nu}$ is the Berry curvature. [44, 47, 48, 54]. Here, η is the quantum metric defined by

$$\eta_{\mu\nu}(\boldsymbol{k}) = \sum_a \langle \partial_{k_\nu} u_{\boldsymbol{k}a} | \, Q(\boldsymbol{k}) \, | \partial_{k_\mu} u_{\boldsymbol{k}a} \rangle \tag{1.6}$$

and $Q(\boldsymbol{k}) = I - \sum_{\boldsymbol{k}a} |u_{\boldsymbol{k}a}\rangle \langle u_{\boldsymbol{k}a}|$.

(ii) The periodic wavefunctions $|u_{\boldsymbol{k}a}\rangle = e^{-i\boldsymbol{k}\cdot\boldsymbol{r}} |\psi_{\boldsymbol{k}a}\rangle$ may be chosen to be holomorphic functions of $k = k_x + ik_y$ in a suitable choice of gauge. Non-unitary gauge transformations are generically required to reach this gauge, and the wavefunctions $|u_{ka}\rangle$ are generically not orthonormal.

(iii) The Bloch wavefunctions ψ satisfy $z\psi = \mathcal{P}z\psi$ where $z = x + iy$ and $\mathcal{P} = \sum_{\boldsymbol{k}a} |\psi_{\boldsymbol{k}a}\rangle \langle \psi_{\boldsymbol{k}a}|$ is the projector onto the bands of interest, written here using an orthonormal basis $|\psi_{\boldsymbol{k}a}\rangle$.

See Ref. [48] for the equivalence of (i) and (ii) and detailed description of the associated Kähler geometry on the Brillouin Zone (BZ). The condition (iii) and its generalizations, together with its relationship to conditions (i) and (ii), is discussed in Ref. [23]. Pictorially, these relations are shown in Fig. 1.4. See also Ref. [123] which argued for a generalized version of condition (i); we will only use the traditional version here.

We note that the $z\psi = \mathcal{P}z\psi$ originates directly from the property (1.5) of chiral graphene multilayers [23]. If ψ is in the band of interest, the zero mode space of $\mathcal{D}$, so is $z\psi$ by (1.5); we calculate $\mathcal{D}(z\psi) = z\mathcal{D}\psi = 0$. Condition (ii) may also be understood from the fact that $\mathcal{D}(\partial_x, \partial_y) = \mathcal{D}(-2i\overline{\partial})$ [24, 32]. On the periodic wavefunctions $u_{\boldsymbol{k}}$ the relevant zero mode operator is $e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}\mathcal{D}(-2i\overline{\partial})e^{i\boldsymbol{k}\cdot\boldsymbol{r}} = \mathcal{D}(-2i\overline{\partial} + k_x + ik_y)$. Because the zero mode operator defining the periodic wavefunctions $u_{\boldsymbol{k}}$ only depends on $k_x + ik_y$, we may choose the wavefunctions $u_{\boldsymbol{k}}$ to only depend on $k_x + ik_y$. We therefore conclude that the flat bands of magic angle chiral multilayers host ideal quantum geometry.

Each of the above statements would be useful for a different purpose. Statement (i) provides a way to quantify the violation of ideality as $\mathrm{tr}(g) - \Omega$ [32, 44, 71]. Statement (iii) was recently interpreted [23, 24] as a "vortexability" condition: we may attach a vortex z to a wavefunction while remaining within the band of interest: $z\psi = \mathcal{P}z\psi$.

It was recently shown [23, 24] that the vortex attachment $\psi \to z\psi = \mathcal{P}z\psi$ enables us to generalize the Laughlin state construction to any ideal band with any Chern number $C \neq 0$. For example, starting with a Slater determinant state $|\Psi_{\text{Slater}}\rangle$ we may construct

$$|\Psi_{2s}\rangle = \prod_{i<j} (z_i - z_j)^{2s} |\Psi_{\text{Slater}}\rangle \tag{1.7}$$

for some s. We note that (1.7) should be understood on the infinite planar or expandable disk geometry, such

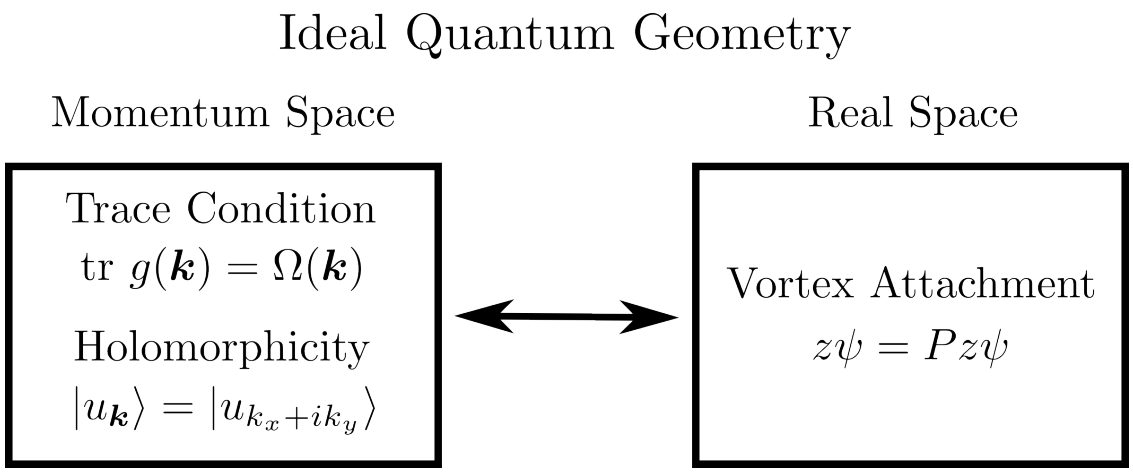


Figure 1.4: Equivalent descriptions of ideal quantum geometry in momentum space and real space corresponding to conditions (i)-(iii) in the main text. In momentum space we have (i) trace condition, useful for characterizing violations of ideal band geometry, and (ii) momentum holomorphicity, useful for analytic understanding of the single particle wavefunctions. In real space we have (iii) vortex attachment which enables us to directly construct many-body FCI states (1.7) that are the ground state in short-range interaction potentials.

that the electron density is allowed to change with the inclusion of the Jastrow factor $\prod_{i<j}(z_i - z_j)^{2s}$. A generalization of the wavefunction to torus geometry is shown in Ref. [124]. While ordinarily the Jastrow factor would introduce weight in remote bands, leading to increased kinetic energy, the condition (iii) ensures that this does not happen in ideal bands. The probability density $|\Psi_{2s}|^2$ has a zero of order $\geq 4s$ when two particles come together which minimizes the interaction energy and ensures that (1.7) is a ground state in the limit of short range repulsive interactions [22, 23, 125]. While this argument makes it clear that ideal FCI states exist in all ideal bands, a more detailed understanding of the wavefunctions is necessary to understand the character of the FCI states (1.7).

In this work we use statement (ii) to characterize the single particle wavefunctions and ensuing many-body states of single ideal bands with $C > 1$ which, among other results, enables us to directly characterize the FCIs generated by (1.7).

We consider a single ideal band with primitive lattice vectors as $\boldsymbol{a}_1$ and $\boldsymbol{a}_2$ and the corresponding reciprocal lattice vectors as $\boldsymbol{G}_1$ and $\boldsymbol{G}_2$ where $\boldsymbol{G}_i \cdot \boldsymbol{a}_j = 2\pi\delta_{ij}$. We shall always work with holomorphic, non-normalized, wavefunctions $u_{\boldsymbol{k}} = u_k = u_{k_x+ik_y}$. We furthermore ask that these wavefunctions are smooth throughout the

BZ [*]. The non-trivial topology of the band is encoded by the boundary conditions

$$\psi_{\boldsymbol{k}+\boldsymbol{G}} = \Xi_{\boldsymbol{G}}(k)\psi_{\boldsymbol{k}}, \qquad \Xi_{\boldsymbol{G}}(k) = \frac{u_{k+G}}{u_k} e^{i\boldsymbol{G}\cdot\boldsymbol{r}} \tag{1.8}$$

where the second equation guarantees that $\Xi(k)$ is holomorphic in k.

The Chern number of a single band is given by an integral of the Berry connection around the boundary of the BZ [22, 24, 126]:

$$\begin{aligned} 2\pi i C = &\log \Xi_{\boldsymbol{G}_1}(k+G_2) - \log \Xi_{\boldsymbol{G}_1}(k) \\ &+ \log \Xi_{\boldsymbol{G}_2}(k) - \log \Xi_{\boldsymbol{G}_2}(k+G_1). \end{aligned} \tag{1.9}$$

For a multiband system $\Xi_{\boldsymbol{G}}(k)$ is a matrix and the replacement $\Xi \to \det \Xi$ should be made in Eq. (1.9) for the total Chern number [23]. Importantly, this can be interpreted as the phase winding of the wavefunction around the BZ.

For a single band with Chern number $C > 0$ it is always possible to choose a gauge (see SI) such that

$$\begin{aligned} &\Xi_{\boldsymbol{G}_1}(k) = \xi_{C,k_0}(k) \equiv \exp(2\pi i C(k-k_0)/G_2), \\ &\Xi_{\boldsymbol{G}_2}(k) = 1 \end{aligned} \tag{1.10}$$

for some k_0 in the reduced first BZ: $k_0 \equiv k_0 + \boldsymbol{G}_i/C$. We call boundary conditions with this form *canonical*. The choice (1.10) leads to Chern number C in (1.9). We will make this choice for our ideal band of interest. The generality of (1.10) follows from the classification of holomorphic line bundles on the torus, as we explain in the SI.

The boundary conditions (1.10) are remarkably restrictive. Indeed, by fixing a gauge choice similar to (1.10), it can be shown that single particle wavefunctions for an ideal $C = 1$ band take the following form [22] (see SI for a review of this argument)

$$\psi_{\boldsymbol{k}\alpha}(\boldsymbol{r}) = \psi^{(\mathrm{LLL})}_{\boldsymbol{k}-\delta\boldsymbol{k}_0}(\boldsymbol{r})\mathcal{N}_\alpha(\boldsymbol{r}), \tag{1.11}$$

where $\delta\boldsymbol{k}_0 = \boldsymbol{k}_0 - \boldsymbol{k}_0^{\mathrm{LLL}}$. The equality (1.11) readily yields an interpretation of the FCIs (1.7) as generalized Laughlin states, modified from the usual Laughlin state through a periodic density modulation generated by $\sqrt{\sum_\alpha |\mathcal{N}_\alpha(\boldsymbol{r})|^2}$.

[*]We may always do this because we can remove a pole at k_0 through gauge transformations $u_k \to f(k-k_0)u_k$ where f is some quasi-periodic holomorphic function with $f(0) = 0$ (e.g. Jacobi theta functions of the first kind).

To understand the FCIs associated with higher Chern bands, we need to understand the wavefunctions in a manner analogous to (1.11). However, a key aspect of the classification (1.11) is the separation of the $\boldsymbol{k}$ dependence and the non-positional orbital α dependence. For higher Chern bands these are in general entangled and the resulting wavefunctions are much harder to work with as a result.

Instead, we will opt to *decompose* higher Chern bands into $C = 1$ bands of the form (1.11).

1.5 Decomposition of an ideal Chern number 2 band

In this section, we provide a framework of decomposing an ideal higher Chern band into athe basis of ideal Chern 1 bands where interesting interaction-driven physics becomes manifest. In particular, we illustrate a non-unitary $\mathrm{SU}(C)$ action on the wavefunctions, which will be promoted to an "emergent" symmetry on the ground state manifold of Hamiltonians with short-ranged interactions at particular fillings (see Sec. 1.6 and 1.8). As a concrete example, we will describe the decomposition of an ideal Chern band with $C = 2$.

1.5.1 Quantum Hall Bilayer

Before we move into the technical details of decomposing generic ideal higher Chern bands, it is useful to consider an easier example, quantum Hall multilayers, "in reverse". Quantum Hall multilayers can be considered to be a system with total Chern number $C > 1$, where each layer corresponds to a band with $C = 1$. Below, we will show how we can combine these layer bands to form a single $C > 1$ band. This decomposition will shed light on the general procedure to decompose a higher Chern band by reversing this process [61]. For simplicity, we start with the simplest case ofTo illustrate further, let us focus here on quantum Hall bilayer [127] with a constant magnetic field $B > 0$, which has a total Chern number 2. We define magnetic translation operators $T^{\mathrm{mag}}_{\boldsymbol{l}}$ that commute with the canonical momentum and satisfy the commutation relation

$$T^{\mathrm{mag}}_{\boldsymbol{l}_1} T^{\mathrm{mag}}_{\boldsymbol{l}_2} = e^{iB\hat{z}\cdot \boldsymbol{l}_1 \times \boldsymbol{l}_2} T^{\mathrm{mag}}_{\boldsymbol{l}_2} T^{\mathrm{mag}}_{\boldsymbol{l}_1} \tag{1.12}$$

If we use a unit cell spanned by lattice vectors $\boldsymbol{R}_1$ and $\boldsymbol{R}_2$ that encloses 2π magnetic flux such that the magnetic translations $T^{\mathrm{mag}}_{\boldsymbol{R}_1}$ and $T^{\mathrm{mag}}_{\boldsymbol{R}_2}$ commute, the system can be described as two $C = 1$ bands with explicit $\mathrm{SU}(2)$ action of Pauli matrices $\tau_{x,y,z}$ acting on layer degrees of freedom.

In order to "stitch" two bands into a single band of $C = 2$, let us instead consider magnetic translations $T^{\mathrm{mag}}_{\boldsymbol{a}_1}$

and $T^{\text{mag}}_{\boldsymbol{a}_2}$ that enclose π flux, $B\|\boldsymbol{a}_1 \times \boldsymbol{a}_2\| = \pi$ [61]. These translations anticommute, but if we combine them with layer operators

$$T_{\boldsymbol{a}_1} = \tau_x T^{\text{mag}}_{\boldsymbol{a}_1},\ T_{\boldsymbol{a}_2} = \tau_z T^{\text{mag}}_{\boldsymbol{a}_2},\ T_{\boldsymbol{a}_i}\psi_{\boldsymbol{k}}(\boldsymbol{r}) = e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\psi_{\boldsymbol{k}}(\boldsymbol{r}) \tag{1.13}$$

then the translations $T_{\boldsymbol{a}_i}$ commute and we may define Bloch states as above. Since the translations enclose half the usual area, the two $C = 1$ LLLs on each layer combine to form a single $C = 2$ band with wavefunctions $\psi_{\boldsymbol{k}}(\boldsymbol{r})$. In this language, $\nu = 1$ quantum Hall ferromagnets of the bilayer system become *charge density waves* in the newly defined unit cell since they break at least one of the translation symmetries $T_{\boldsymbol{a}_i}$. For example, polarizing the top layer, $\langle\tau_z\rangle \neq 0$, corresponds to breaking the translation symmetry $T_{\boldsymbol{a}_1} \propto \tau_x$ since it anticommutes with τ_z.

We remark that the layer operators τ_i can be expressed in terms of lattice translations $T_{\boldsymbol{a}_i}$ and magnetic translations:

$$\tau_z = (T^{\text{mag}}_{\boldsymbol{a}_2})^{-1}T_{\boldsymbol{a}_2},\ \tau_x = (T^{\text{mag}}_{\boldsymbol{a}_1})^{-1}T_{\boldsymbol{a}_1},\ \tau_y = (T^{\text{mag}}_{\boldsymbol{a}_3})^{-1}T_{\boldsymbol{a}_3} \tag{1.14}$$

where $\boldsymbol{a}_3 \equiv -(\boldsymbol{a}_1 + \boldsymbol{a}_2)$. This is the relation we want to use to construct $\mathrm{SU}(2)$ flavor actions when we decompose ideal higher Chern bands with $C = 2$. However, there is a problem: we do not have magnetic translation symmetries defined for an arbitrary higher Chern band. Therefore, in order to use the above relation to construct flavor actions, we have to define the magnetic translations in an arbitrary setting first. To do so, we first notice that the commutation relation in (1.12) implies that $T^{\text{mag}}_{\boldsymbol{l}}$ shifts the Bloch momenta associated to $T_{\boldsymbol{a}}$. We may thus identify magnetic translations as momentum translations:

$$T^{\text{mag}}_{\boldsymbol{l}}\psi^{\text{LLL}}_{\boldsymbol{k}}(\boldsymbol{r}) = c^{\text{LLL}}_{\boldsymbol{l}}(\boldsymbol{k})\psi^{\text{LLL}}_{\boldsymbol{k}-B\hat{\text{BZ}}\times\boldsymbol{l}}(\boldsymbol{r}). \tag{1.15}$$

Here $\psi^{\text{LLL}}_{\boldsymbol{k}}$ are $C = 1$ Bloch states of a single LLL with reciprocal lattice vectors $\boldsymbol{Q}_{1,2}$ corresponding to a 2π flux lattice: $B^{-1}\|\boldsymbol{Q}_1 \times \boldsymbol{Q}_2\| = 2\pi = 2B|\boldsymbol{a}_1 \times \boldsymbol{a}_2|$. The coefficient $c^{\text{LLL}}_{\boldsymbol{l}}(\boldsymbol{k})$ is not unique, depending on the momentum space gauge: under $\psi_{\boldsymbol{k}} \to \lambda_{\boldsymbol{k}}\psi_{\boldsymbol{k}}$ we have $c^{\text{LLL}}_{\boldsymbol{l}}(\boldsymbol{k}) \to \lambda_{\boldsymbol{k}}\lambda^{-1}_{\boldsymbol{k}-B\hat{\text{BZ}}\times\boldsymbol{l}}c^{\text{LLL}}_{\boldsymbol{l}}(\boldsymbol{k})$.

However, we can single out a unique decomposition and choice of $\mathrm{SU}(2)$ action through the use of a gauge-fixed $c_{\boldsymbol{l}}(\boldsymbol{k})$ due to our ability to pick a holomorphic gauge (1.10). Indeed, in the gauge (1.10), where the LLL has $\boldsymbol{k}^{\text{LLL}}_0 = -(\boldsymbol{Q}_1 + \boldsymbol{Q}_2)/2$, the magnetic translation operators of the lowest Landau level have the action (1.15)

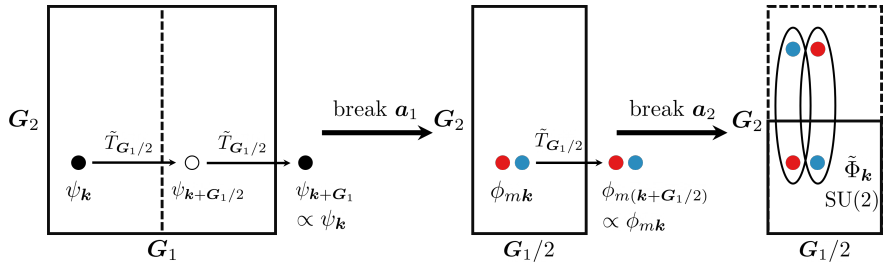


Figure 1.5: Schematic figure that shows the decomposition process in two steps. The first step corresponds to the diagonalization of boundary conditions from the original $C = 2$ band (shown in black and white) to form two ideal $C = 1$ bands $\phi_{m\boldsymbol{k}}$ (shown in red and blue), which breaks translation symmetry in $\boldsymbol{a}_1$. The second step identifies wavefunctions at different momenta as an SU(2) multiplet $\tilde{\phi}_{m\boldsymbol{k}}$; at each bloch momentum in quartered Brillouin Zone, there are two SU(2) multiplets that together give rise to a four component spinor $\tilde{\Phi}_{\boldsymbol{k}}$ (1.25).

with (see SI)

$$\begin{aligned} c_{\boldsymbol{l}}^{\mathrm{LLL}}(\boldsymbol{k}) &= e^{i\boldsymbol{k}_0^{\mathrm{LLL}}\cdot\boldsymbol{l}} c_{\boldsymbol{l}}^{(0)}(\boldsymbol{k}-\boldsymbol{k}_0^{\mathrm{LLL}}), \\ c_{\boldsymbol{l}}^{(0)}(\boldsymbol{k}) &= e^{i\boldsymbol{k}\cdot\boldsymbol{l}} e^{\frac{i}{2}(\boldsymbol{Q}_1+\boldsymbol{Q}_2)\cdot\boldsymbol{l}} e^{\frac{\pi Bl}{Q_2}-\frac{i}{2}\left(\frac{\overline{Q}_2}{Q_2}l+\bar{l}\right)\left(k-\frac{iBl}{2}\right)} \end{aligned} \tag{1.16}$$

We will soon use (1.15), with $c_{\boldsymbol{l}}(\boldsymbol{k})$ given in (1.16), to recover SU(2) operators (1.14) in an arbitrary ideal $C = 2$ band. Note that we have written (1.16) in a manner that will enable a straightforward generalization to bands with $\boldsymbol{k}_0 \neq \boldsymbol{k}_0^{\mathrm{LLL}}$. We will need the identities

$$\begin{aligned} e^{i\boldsymbol{k}_0\cdot\boldsymbol{R}_2} c_{\boldsymbol{R}_2}^{(0)}(\boldsymbol{k}-\boldsymbol{k}_0) &= -e^{i\boldsymbol{k}\cdot\boldsymbol{R}_2}\xi_{1,k_0}^{-1}(k), \\ e^{i\boldsymbol{k}_0\cdot\boldsymbol{R}_1/C} c_{\boldsymbol{R}_1/C}^{(0)}(\boldsymbol{k}-\boldsymbol{k}_0) &= e^{i\boldsymbol{k}\cdot\boldsymbol{R}_1/C} \end{aligned} \tag{1.17}$$

where $\boldsymbol{R}_i \cdot \boldsymbol{Q}_j = 2\pi\delta_{ij}$ are the associated lattice vectors and $C, \boldsymbol{k}_0$ are arbitrary. In deriving the first equation we used $\hat{\mathrm{BZ}} \cdot \boldsymbol{Q}_1 \times \boldsymbol{Q}_2 = 2\pi B$. Note that $\xi_{1k_0}(k) = e^{2\pi i C(k-k_0)/Q_2}$ is the k-space boundary condition defined in (1.10).

1.5.2 Decomposing a general ideal Chern band

With the understanding of the quantum Hall bilayer, we now decompose an arbitrary ideal Chern band with $C = 2$ into two ideal $C = 1$ bands that admit an SU(2) action. Note that, for a holomorphic Chern band of interest we do not generically have magnetic translation symmetry $T_{\boldsymbol{l}}^{\mathrm{mag}}$ defined at the UV scale unlike for the case of LLLs in quantum Hall problems. Nonetheless we will use (1.14) to recover an SU(2) action in any ideal $C = 2$ band through a more generalizable form of $T_{\boldsymbol{a}}^{\mathrm{mag}}$.

Consider an ideal $C = 2$ band defined on a lattice with lattice vectors $\boldsymbol{a}_{1,2}$ and reciprocal lattice vectors $\boldsymbol{G}_{1,2}$. As a naive attempt, one can decompose the band into two $C = 1$ bands by doubling the unit cell, for example, in the $\boldsymbol{a}_1$ direction; the new lattice vectors are $2\boldsymbol{a}_1$ and $\boldsymbol{a}_2$ and the corresponding reciprocal lattice vectors are $\boldsymbol{G}_1/2$ and $\boldsymbol{G}_2$. Accordingly, a single-particle state in the new BZ is expressed by a doublet $(\psi_{\boldsymbol{k}}, \psi_{\boldsymbol{k}+\boldsymbol{G}_1/2})^T$, and the boundary condition (1.8) takes the following form

$$\Xi_{\boldsymbol{G}_1/2}(k) = \begin{pmatrix} 0 & 1 \\ \xi_{2,k_0}(k) & 0 \end{pmatrix}, \quad \Xi_{\boldsymbol{G}_2}(k) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{1.18}$$

However, note that the $\boldsymbol{G}_1/2$ boundary condition is non-diagonal, implying that two bands are not really independent; for example, they may not be separately filled. In order to fully decompose the band, we seek holomorphic combinations of $\psi_{\boldsymbol{k}}, \psi_{\boldsymbol{k}+\boldsymbol{G}_1/2}$ that *diagonalize* the boundary conditions; furthermore, we require that the new, hybridized, bands have the canonical boundary conditions Eq. (1.10). To achieve this, we define a matrix U_k such that the transformed wavefunctions

$$\begin{pmatrix} \phi_{0\boldsymbol{k}} \\ \phi_{1\boldsymbol{k}} \end{pmatrix} = U_k \begin{pmatrix} \psi_{\boldsymbol{k}} \\ \psi_{\boldsymbol{k}+\boldsymbol{G}_1/2} \end{pmatrix} \tag{1.19}$$

satisfy our requirements. In doing so, we require U_k to be holomorphic so that ϕs respect the trace condition. To ensure the diagonal boundary condition are canonical for ϕ bands, U_k satisfies

$$\begin{aligned} U_{k+G_1/2}\Xi_{\boldsymbol{G}_1/2}(k)U_k^{-1} &= \begin{pmatrix} \xi_{1,g_0}(k) & 0 \\ 0 & \xi_{1,g_1}(k) \end{pmatrix}, \\ U_{k+G_2}\Xi_{\boldsymbol{G}_2}(k)U_k^{-1} &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned} \tag{1.20}$$

We need to solve for g_0 and g_1.

We find that there is a unique choice of U_k(See A.3):

$$U_k = \begin{pmatrix} 1 & \xi_{1,g_0}^{-1}(k) \\ 1 & \xi_{1,g_1}^{-1}(k) \end{pmatrix} \tag{1.21}$$

which yields $g_0 = k_0 + G_1/4$, $g_1 = g_0 - G_2/2$. Thus the boundary conditions for $\phi_{m\boldsymbol{k}}$ will be:

$$\phi_{m(\boldsymbol{k}+\boldsymbol{G}_1/2)} = \xi_{1,g_0-mG_2/2}(k)\phi_{m\boldsymbol{k}},\ \ \phi_{m(\boldsymbol{k}+\boldsymbol{G}_2)} = \phi_{m\boldsymbol{k}} \tag{1.22}$$

Note that the ϕ_m bands are exchanged under $T_{\boldsymbol{a}_1}$:

$$T_{\boldsymbol{a}_1}\phi_{m\boldsymbol{k}} = e^{i\boldsymbol{k}\cdot\boldsymbol{a}_1}\phi_{(m+1)\boldsymbol{k}} \tag{1.23}$$

where $m \equiv m+2$ is understood modulo two.

There is a subtlety in this "full" decomposition. The ϕ bands in general do not form an orthogonal basis. Orthogonality would only be obvious if $\psi_{\boldsymbol{k}}$ and $\psi_{\boldsymbol{k}+\boldsymbol{G}_1/2}$ had equal norm (which they do not since we have chosen a holomorphic basis) and if the matrix U_k was unitary. In principle these two sources of non-orthogonality could cancel out, but Ref. [24] showed that this is true if and only if the Berry curvature satisfies $\Omega(\boldsymbol{k}) = \Omega(\boldsymbol{k} + \boldsymbol{G}_1/2)$. However, this does not hold in general and is specifically untrue for the particular class of chiral graphene models we are considering. Nonetheless, we will see that this non-orthogonality does not hinder us from understanding interaction-driven physics based on decomposed basis.

Now that we have constructed two ideal bands, we want to hybridize them to construct an entire manifold of ideal bands. Soon we will see that the hybridization grants us the ability to access translation breaking states in all directions. Importantly, we want to hybridize two bands with the same boundary conditions in order to keep the boundary condition diagonal. Note that while $\phi_{0\boldsymbol{k}}$ and $\phi_{1\boldsymbol{k}}$ have different boundary conditions, $\phi_{0(\boldsymbol{k}+\boldsymbol{G}_2/2)}$ and $\phi_{1\boldsymbol{k}}$ have the same boundary conditions (see (1.22)).

We therefore define

$$\tilde{\phi}_{0\boldsymbol{k}} \equiv \phi_{0(\boldsymbol{k}+\boldsymbol{G}_2/2)}, \quad \tilde{\phi}_{1\boldsymbol{k}} \equiv \phi_{1\boldsymbol{k}}. \tag{1.24}$$

Since we are now hybridizing states with momenta shifted by $\boldsymbol{G}_2/2$ we should fold our Brillouin Zone accordingly. We therefore obtain *four* bands which may be written as

$$\tilde{\Phi}_{\boldsymbol{k}} \equiv \left(\tilde{\phi}_{0\boldsymbol{k}}, \tilde{\phi}_{1\boldsymbol{k}}, \tilde{\phi}_{0(\boldsymbol{k}+\boldsymbol{G}_2/2)}, \tilde{\phi}_{1(\boldsymbol{k}+\boldsymbol{G}_2/2)}\right)^T \tag{1.25}$$

which comprises of two pseudospin doublets from momentum points related by $\boldsymbol{G}_2/2$. While we now have four bands, corresponding to the fact that we have folded the Brillouin Zone in half twice, these should be understood

as two groups of two bands each – for example the first and third band taken together resemble the LLL with a 4π magnetic flux in our now-quadrupled unit cell spanned by $2\boldsymbol{a}_i$. These subbands cannot be filled independently without further hybridization; they are related by a shift by a reciprocal lattice vector as $\tilde{\Phi}_{1,\boldsymbol{k}+\boldsymbol{G}_2/2} = \tilde{\Phi}_{3,\boldsymbol{k}}$ and therefore cannot be individually defined on the first BZ. We will often work with $\tilde{\phi}_{m\boldsymbol{k}}$ as if they were two bands in a Brillouin Zone spanned by $\boldsymbol{G}_1/2$ and $\boldsymbol{G}_2$, but it is important to emphasize that this implicitly involves a momentum shift (1.24) and states built out of $\tilde{\phi}_{m\boldsymbol{k}}$ can break $T_{\boldsymbol{a}_2}$.

The real-space form $\tilde{\phi}_m$ is given by (see (1.11)):

$$\tilde{\phi}_{m\boldsymbol{k}} = \psi^{(\mathrm{LLL})}_{\boldsymbol{k}-\delta\boldsymbol{k}'_0}(\boldsymbol{r})\mathcal{N}^{(m)}(\boldsymbol{r}). \tag{1.26}$$

where $\boldsymbol{k}'_0 = \boldsymbol{g}_0 + \boldsymbol{G}_2/2$ and $\delta\boldsymbol{k}'_0 = \boldsymbol{k}'_0 - \boldsymbol{k}^{\mathrm{LLL}}_0$. Note that the momentum shift from ϕ to $\tilde{\phi}$, which made the boundary conditions m-independent, ensured that the origin of momentum space in the LLL part of $\tilde{\phi}$ is m-independent. On the other hand, the momentum shift implies $T_{\boldsymbol{a}_2}\tilde{\phi}_{m\boldsymbol{k}} = e^{i\boldsymbol{k}\cdot\boldsymbol{a}_2}(-1)^{m+1}\tilde{\phi}_{m\boldsymbol{k}}$, such that $\mathcal{N}^{(m)}(\boldsymbol{r})$ differ by a sign $(-)^m$ under translations by $\boldsymbol{a}_2$. We will see that using $\tilde{\phi}_{m\boldsymbol{k}}$ we can construct states that break $T_{\boldsymbol{a}_1}$, $T_{\boldsymbol{a}_2}$, or both. The actions of translation operators $T_{\boldsymbol{a}_i}$ on $\tilde{\Phi}_{\boldsymbol{k}}$ are

$$\begin{aligned} T_{\boldsymbol{a}_2} &= -e^{i\boldsymbol{k}\cdot\boldsymbol{a}_2}\mu_z \otimes \tau_z, \\ T_{\boldsymbol{a}_1} &= e^{i\boldsymbol{k}\cdot\boldsymbol{a}_1}\mu_x \otimes \tau_x \\ T_{\boldsymbol{a}_3} &= e^{i\boldsymbol{k}\cdot\boldsymbol{a}_3}\mu_y \otimes \tau_y \end{aligned} \tag{1.27}$$

where τ matrices act on the pseudospin index, the first and third versus the second and fourth components of $\tilde{\Phi}$, and μ matrices act on the $\boldsymbol{G}_2$ folding index, the first two components versus the second two components.

Inspired by the quantum Hall bilayer, we will combine the magnetic translations along with real space translations to derive an $\mathrm{SU}(2)$ action. However, since there is no physical magnetic field, there is no immediate notion of magnetic translation. Nevertheless, we can borrow the definition of the magnetic translation as a momentum shift (cf.(1.15),(1.16)):

$$T^{\mathrm{mag}}_{\boldsymbol{l}}\tilde{\phi}_{m\boldsymbol{k}}(\boldsymbol{r}) \equiv e^{i\boldsymbol{k}'_0\cdot\boldsymbol{l}}c^{(0)}_{\boldsymbol{l}}(\boldsymbol{k}-\boldsymbol{k}'_0)\tilde{\phi}_{m\boldsymbol{k}-B_{\mathrm{eff}}\hat{\mathrm{BZ}}\times\boldsymbol{l}}(\boldsymbol{r}) \tag{1.28}$$

where B_{eff} is such that $B_{\mathrm{eff}}\|2\boldsymbol{a}_1 \times \boldsymbol{a}_2\| = 2\pi$ and $c_{\boldsymbol{l}}(\boldsymbol{k})$ is given in (1.16). Note that (1.28) may be interpreted as a magnetic translation acting on the just the LLL part of $\tilde{\phi}$ in (1.26). Note that $T^{\mathrm{mag}}_{\boldsymbol{l}}$ is generically neither

unitary nor a symmetry.

In accordance with (1.14) we define

$$\mathsf{s}_z = (T^{\text{mag}}_{\boldsymbol{a}_2})^{-1} T_{\boldsymbol{a}_2}, \mathsf{s}_x = (T^{\text{mag}}_{\boldsymbol{a}_1})^{-1} T_{\boldsymbol{a}_1}, \mathsf{s}_y = (T^{\text{mag}}_{\boldsymbol{a}_3})^{-1} T_{\boldsymbol{a}_3} \tag{1.29}$$

with $\boldsymbol{a}_3 = -(\boldsymbol{a}_1 + \boldsymbol{a}_2)$ (see e.g. Fig. 1.2, though the above discussion does not require C_3 symmetry). The magnetic translations then correspond to translations by half reciprocal lattice vectors; we have $-B_{\text{eff}}\hat{\text{BZ}} \times \boldsymbol{a}_2 = \boldsymbol{Q}_1 = \boldsymbol{G}_1/2$ and $-B_{\text{eff}}\hat{\text{BZ}} \times \boldsymbol{a}_1 = -\boldsymbol{Q}_2/2 = -\boldsymbol{G}_2/2$. Accordingly, the action of Eq. (1.29) on the spinor $\tilde{\Phi}$ yields

$$\mathsf{s}_z = \mu_0 \otimes \tau_z, \;\; \mathsf{s}_x = \mu_0 \otimes \tau_x, \;\; \mathsf{s}_y = \mu_0 \otimes \tau_y. \tag{1.30}$$

In deriving (1.30) we used the action of translations on the spinor (1.27), the boundary conditions $\tilde{\phi}_{m\boldsymbol{k}+\boldsymbol{G}_1/2} = \xi_{1,k_0'}(k)\tilde{\phi}_{m\boldsymbol{k}}$ and $\tilde{\phi}_{m\boldsymbol{k}+\boldsymbol{G}_2} = \tilde{\phi}_{m\boldsymbol{k}}$, and the identities (1.17).

We give a final note on translation breaking patterns of different states. Given the representation of translation (1.27), states with nonzero s_z polarization break $T_{\boldsymbol{a}_1}$ and states with nonzero s_x polarization break $T_{\boldsymbol{a}_2}$. Therefore, we can access all sorts of translation breaking patterns within the $\tilde{\Phi}$ basis. The procedure through which we decompose the $C = 2$ band is shown pictorially in Fig. 1.5.

Although in this basis the operators s_i look Hermitian, they are not. Indeed, the basis is written with respect to the $\tilde{\phi}_{m\boldsymbol{k}}$ wavefunctions, and these wavefunctions are not orthogonal to each other. However, this does not prevent us from building interesting many-body quantum states.

1.6 Manifold of Charge Density Waves and Skyrmion textures

We now move onto the construction of many-body states. In this section, we construct a continuous manifold of topological CDW states at half filling of an ideal $C = 2$ band; this manifold will be identified with a sphere of generalized quantum Hall ferromagnets. These states arise out of the $\text{SU}(2)$ action on the two decomposed $C = 1$ bands. We will see that the $\text{SU}(2)$ action on the band becomes an emergent symmetry of the ground state manifold. This manifold admits skyrmion textures of CDWs that carry electric charge.

We consider a repulsive density-density interaction in the short-range interaction limit. For this section, this

amounts to choosing a contact interaction

$$H = \frac{V_0}{2} \sum_{i<j} \delta(\hat{\boldsymbol{r}}_i - \hat{\boldsymbol{r}}_j) \tag{1.31}$$

with $V_0 > 0$. We have written the interaction in first quantization and $\hat{\boldsymbol{r}}_i$ is the position operator associated to particle i.

We now construct the manifold of charge density waves that emerges as the zero-energy ground state manifold of (1.31). Consider fully filling a single band formed by a generic linear combination $\sum_m \alpha_m \tilde{\phi}_{m,\boldsymbol{k}}$ of the $\tilde{\phi}_m$ bands constructed in the previous section. The band formed by the linear combination will have wavefunctions

$$\begin{aligned} \tilde{\varphi}_{\theta,\phi,\boldsymbol{k}}(\boldsymbol{r}) &\equiv \cos(\theta/2)\tilde{\phi}_{0\boldsymbol{k}}(\boldsymbol{r}) + \sin(\theta/2)e^{i\phi}\tilde{\phi}_{1\boldsymbol{k}}(\boldsymbol{r}) \\ &= \psi^{(\mathrm{LLL})}_{\boldsymbol{k}-\boldsymbol{k}'_0}(\boldsymbol{r})\mathcal{N}^{(\theta,\phi)}(\boldsymbol{r}) \end{aligned} \tag{1.32}$$

for some (θ, ϕ) determined by α_m and $\mathcal{N}^{(\theta,\phi)} = \cos(\theta/2)\mathcal{N}^{(0)} + \sin(\theta/2)e^{i\phi}\mathcal{N}^{(1)}$. Thus, for every point (θ, ϕ) on the Bloch sphere we obtain an ideal $C = 1$ band. Fully filling the band (1.32) will generate a Slater determinant state

$$|\theta, \phi\rangle \equiv \mathcal{A} \bigotimes_{\boldsymbol{k}} |\tilde{\varphi}_{\theta,\phi,\boldsymbol{k}}\rangle \tag{1.33}$$

where $\mathcal{A}$ antisymmetrizes the particles and $\boldsymbol{k}$ runs through the BZ with reciprocal lattice vectors $\boldsymbol{G}_1/2$ and $\boldsymbol{G}_2$ (the states (1.33) still generically break $T_{\boldsymbol{a}_2}$ translation symmetry: see the discussion below (1.25)). The Slater determinant forces the momentum dependence to be antisymmetric, which, together with the form (1.11), forces the $\boldsymbol{r}$-coordinate dependence to be antisymmetric. Thus, $|\theta, \phi\rangle$ is annihilated by the contact interaction.

We have constructed a ferromagnetic ground state $|\theta, \phi\rangle$ associated with each point (θ, ϕ) on the pseudospin-sphere. Each ferromagnet is a topological CDW that is translationally symmetric with a $2\boldsymbol{a}_1 \times 2\boldsymbol{a}_2$ unit cell. At special points — on the X, Y, Z axes — the state further preserves one of the translation symmetries $T_{\boldsymbol{a}_i}$ for $i = 1, 3, 2$ (see (1.27)). The order parameter sphere is graphically shown in Fig. 1.2(a).

The manifold of charge density waves appears in finite-size numerics through an extensive ground state degeneracy. The degeneracy is given by the dimension of the $\mathrm{SU}(2)$ irreducible representation that ferromagnetic states belong to. Such an irrep is given by the symmetric tensor product of N_e spin-$1/2$ systems. This is the spin

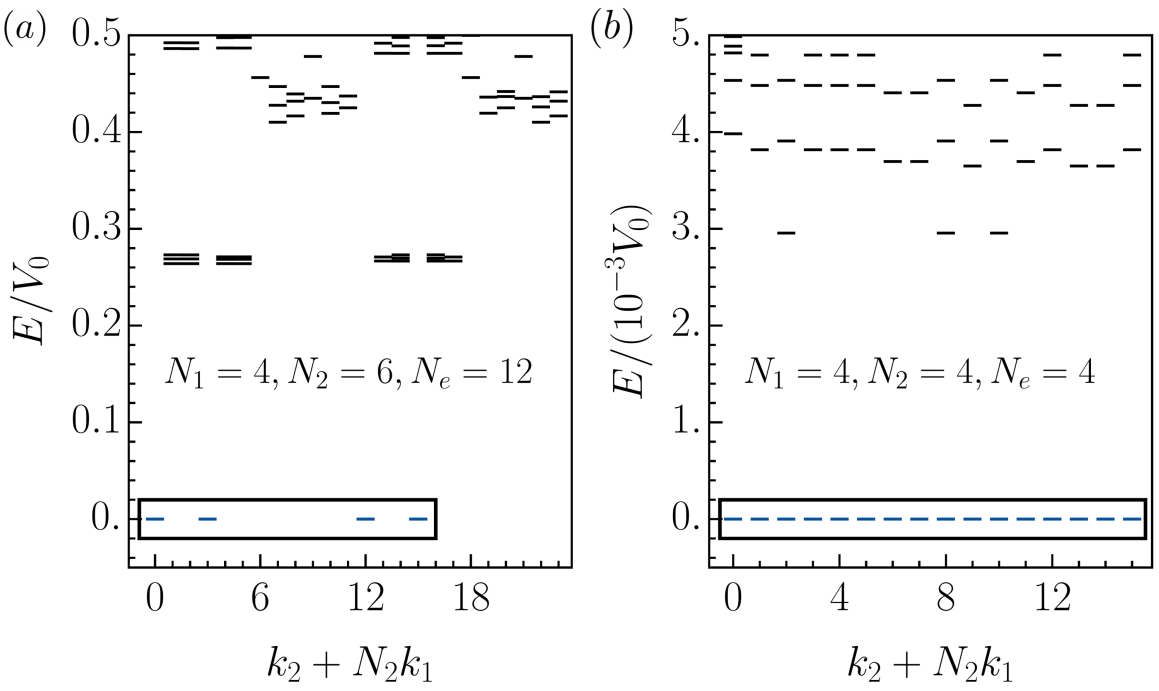


Figure 1.6: Momentum resolved spectrum for chiral twisted graphene multilayers with contact interaction from ED. Ground states are exact zero modes surrounded by black boxes (there are multiple zero modes within one momentum sector): they are identified with the analytically constructed charge density waves explored in this section. In panel (a): $C = 2$, $\nu = 1/2$, and GSD = 13. (b): $C = 4$, $\nu = 1/4$ and GSD = 35. The ground state degeneracies match exactly with Eq. (1.34) for $C = 2$ and for higher Chern numbers Eq. (A.48).

$N_e/2$ respresentation of SU(2) whose dimension is

$$\text{GSD} = N_e + 1. \tag{1.34}$$

In our numerical calculations, we use the ideal Chern bands of chiral twisted graphene multilayers [24] with contact interaction. The ground state degeneracies are confirmed via ED, see Fig. (1.6), for systems that have an even number of $\boldsymbol{a}_1 \times \boldsymbol{a}_2$ unit cells in each direction. For $m\boldsymbol{a}_1 \times n\boldsymbol{a}_2$, where either m or n is odd, the SU(2) operators cannot be defined and only the special ferromagnets that preserve $T_{\boldsymbol{a}_1}$ or $T_{\boldsymbol{a}_2}$ for m or n odd respectively will survive, leading to a GSD of two. We indeed see that this is the case.

We now investigate the role of rotational crystalline symmetries. They further relate ferromagnets to each other, so they are identified as rotations on the order parameter sphere. The representation of the states as points on the sphere and crystalline symmetries as rotations is summarized in Tab. 1.2(b). In particular, we focus on the three-fold rotational symmetry C_{3z} relevant to graphene moiré systems [85, 102]. Since C_{3z} permutes the

lattice vectors $\boldsymbol{a}_1 \to \boldsymbol{a}_2 \to \boldsymbol{a}_3$ it also permutes the states that preserve the translation symmetries $T_{\boldsymbol{a}_{1,2,3}}$ which correspond to the X, Z, Y axes of the order parameter sphere respectively. We therefore conclude that C_{3z} is a rotation around the $\boldsymbol{n}_0 = (111)^T$ axis of the order-parameter sphere. There are other rotationally symmetric axes $\boldsymbol{n}_i$ related by translations $T_{\boldsymbol{a}_i}$; they correspond to the rotation symmetry $C_{3z}^{(i)} = T_{\boldsymbol{a}_i} C_{3z} T_{-\boldsymbol{a}_i}$ that rotates about the point $(x, y) = \boldsymbol{a}_i$. It is convenient to identify $C_{3z}^{(0)} = C_{3z}$ and $\boldsymbol{a}_0 = 0$.

The crystalline symmetries C_{3z} and $T_{\boldsymbol{a}_i}$ together form a subgroup of the emergent $\mathrm{SO}(3)$ symmetry of the order parameter sphere. We identify this subgroup as the alternating group on four elements A_4. The four elements being permuted may be taken to be the rotationally symmetric axes $\boldsymbol{n}_i$, for $i = 0, 1, 2, 3$. The rotations $C_3^{(i)}$ are the four different three-cycles, and the translations $T_{\boldsymbol{a}_i}$ are products of two transpositions.

We now discuss topological textures of the order parameter sphere. Because $\pi_2(S^2) = \mathbb{Z}$, there are non-trivial solitonic textures of the order parameter known as skyrmions. These skyrmions carry electric charge $Ce = e$ where $C = 1$ is the Chern number of each ferromagnet [128, 129]. Note that these charge e skyrmions have arisen out of a $C = 2$ band, where one may a priori expect charge $2e$ skyrmions. We numerically plot the density profile of such a skyrmion in Fig. 1.1(a). We assume a skyrmion texture whose core points at $\boldsymbol{n}_0$ and at infinity points at $-\boldsymbol{n}_0$. The azimuthal angle and the polar angle from $\boldsymbol{n}_0$ is assumed to be $\theta(\boldsymbol{r}) = \pi(1 - \exp(-r^2/2\lambda^2))$, $\phi(\boldsymbol{r}) = -\phi$.

We also note that the ferromagnets still have very isotropic energy even for longer ranged interactions and realistic band geometry. This suggests skyrmion may well be the lowest energy charged excitations in this system.

1.7 Adding real electron spin: $\mathrm{SU}(4)$ ferromagnets and skyrmion crystals

Inclusion of an additional degree of freedom, such as the real electron spin, greatly enriches the manifold of ferromagnets. With spin, we now have four ideal $C = 1$ bands $\tilde{\phi}_{m\sigma}$, where $\sigma = \uparrow, \downarrow$ labels the physical electron spin. We obtain a zero energy ground state under a contact interaction at $\nu = 1/2$ of the $\boldsymbol{a}_1 \times \boldsymbol{a}_2$ unit cell through filling any linear combination of these four Chern bands: this space of ground states has a natural $\mathrm{U}(4)$ action that rotates the filled Chern band into the empty ones and can be identified as the coset manifold $\mathrm{U}(4)/\mathrm{U}(3) \times U(1) = \mathbb{CP}^3$. Here $\mathrm{U}(4)$ rotates the four ideal Chern bands into each other and $\mathrm{U}(3)$, $\mathrm{U}(1)$ are redundancies that correspond to changes of bases of the empty and filled bands respectively. The charge density waves of the previous section are encompassed here as states that are both pseudospin polarized and spin polarized. These form an $S^2 \times S^2$ submanifold of $\mathbb{CP}^3$.

More interesting states from the ferromagnetic manifold emerge when the pseudospin and real spin are entangled, such that both polarizations are individually zero. For example, consider filling the band described by the wavefunctions

$$\tilde{\phi}_{\boldsymbol{k}} = \frac{1}{\sqrt{2}}(\tilde{\phi}_{0,\uparrow,\boldsymbol{k}} - \tilde{\phi}_{1,\downarrow,\boldsymbol{k}}). \tag{1.35}$$

The state (1.35) breaks both spin rotations as well as translations $T_{\boldsymbol{a}_i}$. However, the combinations

$$\tilde{T}_{\boldsymbol{a}_1} = T_{\boldsymbol{a}_1}\sigma_x, \qquad \tilde{T}_{\boldsymbol{a}_2} = T_{\boldsymbol{a}_2}\sigma_z \tag{1.36}$$

are preserved such that the charge density of (1.35) is symmetric in the original 1×1 unit cell. We identify these states with the tetrahedral antiferromagnets (TAFs) in [102].

We may apply spin rotations to (1.35) to obtain all the states that have both zero spin and zero pseudospin polarization. Since there is no fixed axis in the spin texture of the TAF state, the spin rotation action is faithful. Thus, the manifold of all TAF states is equivalent to the group manifold $\mathrm{SO}(3)$. Due to its special spin texture throughout the unit cell, it can also be understood as a skyrmion crystal. In Fig. 1.7 we plot the charge density and spin texture of an analytically constructed skyrmion crystal state in the 2×2 unit cell. We choose a state where the spin is pointing in the $+z$ direction at the center of the unit cell for aesthetic purposes. We note that the charge density is visibly periodic with respect to the 1×1 unit cell.

1.8 Fractional Chern Insulator

Having established the quantum Hall ferromagnets in the ideal $C = 2$ band, we now investigate the various FCIs that arise in this system. We classify the FCIs into two categories: translation symmetric FCIs and translation broken FCIs. The first category corresponds to flavor singlets after the decomposition, whereas the second category corresponds to a flavor ferromagnet.

1.8.1 Translation Symmetric FCIs

Vortex attachment in ideal Chern bands ensure that FCIs exist through (1.7). However, what is the topological order and filling of the state given by (1.7)? At $C = 1$, the expression gives a Laughlin topological order at filling $1/(2s + 1)$ due to the wavefunction classification (1.11) [32, 34]. What about bands with $C > 1$?

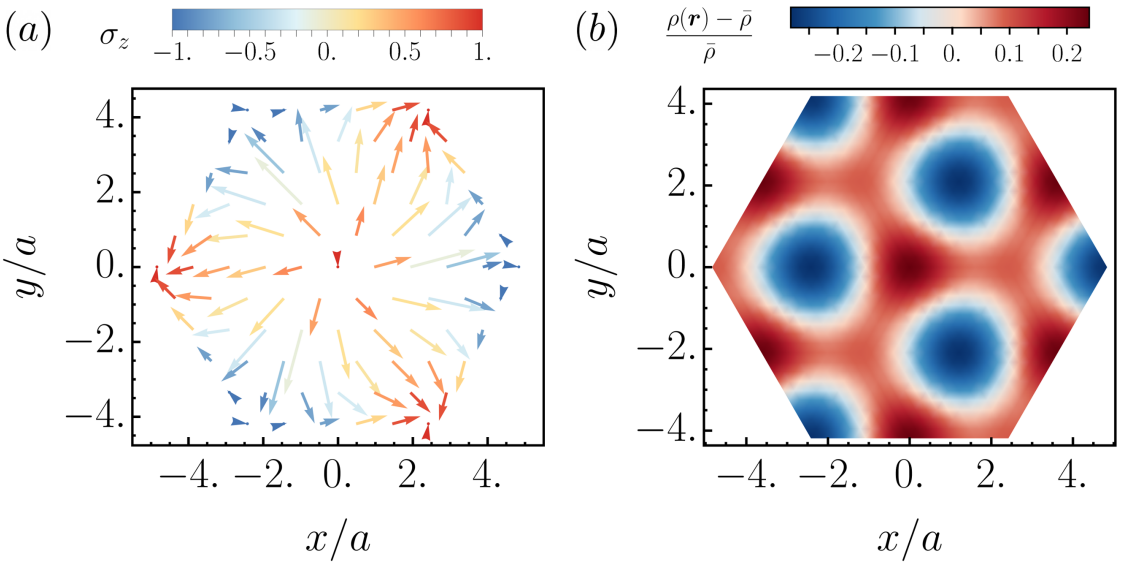


Figure 1.7: (a) Spin texture and (b) density profile of a TAF state: it is invariant under combinations of single translations and magnetic rotations. Thus the spin texture is not periodic in the quadrupled unit cell but the density profile is periodic. In the quadrupled unit cell, the spin texture winds around like a skyrmion, which is why this state is also identified with a skyrmion crystal. For presentation, we rotated the spin texture such that the spin is pointing in the $+z$ direction at the center of the unit cell.

Before discussing FCIs in higher Chern bands, it is instructive to review possible FQH states in multi-component Landau levels. For example, consider a system consisting of two LLLs, either labeled by layer or flavor index. In such a system, it is well-known that the so-called Halperin (mmn) wavefunction can be written as the following [130]:

$$\begin{aligned}&\Psi^{(mmn)}_{l_1,\ldots,l_{2N}}(\boldsymbol{r}_1,\ldots,\boldsymbol{r}_{2N})\\&=\mathcal{A}\Big(\prod_{i<j}^{N}(z_i-z_j)^m(w_i-w_j)^m\prod_{i,j=1}^{N}(z_i-w_j)^n\\&\times\prod_{i=1}^{N}e^{-\frac{|\boldsymbol{r}_i|^2}{4\ell_B^2}}e^{-\frac{|\boldsymbol{r}_{i+N}|^2}{4\ell_B^2}}\delta_{1,l_i}\delta_{2,l_{i+N}}\Big).\end{aligned}\tag{1.37}$$

Here we have taken the first N particles to be in the first layer and the second N particles to be in the second later and defined associated complex coordinates $z_i = x_i + iy_i$ and $w_i = x_{i+N} + iy_{i+N}$. The antisymmetrization operator $\mathcal{A}$ enforces antisymmetry between the coordinates $i < N$ and $i > N$; within each class antisymmetry is guaranteed by restricting m to be an odd integer. We have used the symmetric electromagnetic gauge for convenience. The state describes an electronic state at filling $\nu = 2/(m+n)$ with respect to a unit cell of volume $2\pi\ell_B^2$, which can be computed in the standard manner by counting the maximal power of z. Accordingly,

$\Psi^{(110)}$ corresponds to the fully filled state with $\nu = 2$.

By direct computation, one can show that the states $\Psi^{(2s+1,2s+1,2s)}$ arise from the vortex attachment (1.7) applied to the fully filled Slater determinant state $\Psi^{(110)}$. Note that the states $\Psi^{(2s+1,2s+1,2s)}$ are singlets since neither $\Psi^{(110)}$ nor the Jastrow factor in (1.7) breaks the $\mathrm{SU}(2)$ flavor symmetry.

We now apply (1.7) to a generic ideal $C = 2$ band, where the Slater state is the fully filled state. Since an ideal $C = 2$ band can be decomposed into two non-orthogonal $C = 1$ bands, the natural expectation is that vortex attachment (1.7) yields a state analagous to $\Psi^{(2s+1,2s+1,2s)}$. We now show precisely how this works.

The fully filled state $|\Psi^{(C=2)}_{\nu=2}\rangle$ is obtained by filling each band labeled by m successively, regardless of orthogonality. Indeed under a non-unitary gauge transformation $c_\alpha \to d_\alpha = S_{\alpha\beta} c_\beta$ we have $\varepsilon^{\alpha_1,\dots\alpha_N} c_{\alpha_1} \dots c_{\alpha_N} = \det S^{-1} \varepsilon^{\alpha_1,\dots\alpha_N} d_{\alpha_1} \dots d_{\alpha_N}$ such that the fully filled state takes the same form in any basis regardless of orthonormality. Using the relationship (1.11), we obtain

$$\begin{aligned}
&\Psi^{(C=2)}_{\alpha_1,\dots\alpha_N}(\boldsymbol{r}_1, \dots, \boldsymbol{r}_{2N}) \\
&= \mathcal{A}\Bigg(\prod_{i<j}^{N} (z_i - z_j)(w_i - w_j) \\
&\times \prod_{i=1}^{N} e^{-\frac{|\boldsymbol{r}_i|^2}{4\ell^2_{B_{\text{eff}}}}} e^{-\frac{|\boldsymbol{r}_{i+N}|^2}{4\ell^2_{B_{\text{eff}}}}} \mathcal{N}^{(0)}_{\alpha_i}(\boldsymbol{r}_i) \mathcal{N}^{(1)}_{\alpha_{i+N}}(\boldsymbol{r}_{i+N}) \Bigg)
\end{aligned} \tag{1.38}$$

where $\mathcal{N}^{(m)}$ are defined in (1.26). As before, the antisymmetry operator $\mathcal{A}$ antisymmetrizes the particles $i < N$ with $i > N$, and we have used z and w to distinguish the particle positions on the top and bottom layer respectively. As in previous sections, $CB_{\text{eff}}|\boldsymbol{a}_1 \times \boldsymbol{a}_2| = 2\pi$ such that the original unit cell of the $C = 2$ band has π effective flux.

Then, if we act with the vortex attachment factor $\prod_{i<j}(z_i - z_j)^{2s}$, we obtain a state directly analogous to the Halperin state at the same filling

$$\begin{aligned}
&\Psi^{(C=2,2s)}_{\alpha_1,\dots\alpha_N}(\boldsymbol{r}_1, \dots, \boldsymbol{r}_{2N}) \\
&= \mathcal{A}\Bigg(\prod_{i<j}^{N} (z_i - z_j)^{2s+1} (w_i - w_j)^{2s+1} \prod_{i,j=1}^{N} (z_i - w_j)^{2s} \\
&\times \prod_{i=1}^{N} e^{-\frac{|\boldsymbol{r}_i|^2}{4\ell^2_{B_{\text{eff}}}}} e^{-\frac{|\boldsymbol{r}_{i+N}|^2}{4\ell^2_{B_{\text{eff}}}}} \mathcal{N}^{(0)}_{\alpha_i}(\boldsymbol{r}_i) \mathcal{N}^{(1)}_{\alpha_{i+N}}(\boldsymbol{r}_{i+N}) \Bigg)
\end{aligned} \tag{1.39}$$

The only modification is the extra data associated with the non-positional orbital indices α, and a periodic charge modulation generated by the $\boldsymbol{r}$-dependence of $\mathcal{N}_\alpha^{(m)}(\boldsymbol{r})$.

The filling and topological order is independent of the functions $\mathcal{N}_\alpha^{(m)}(\boldsymbol{r})$: these functions simply modify the orbital and positional textures at the unit cell scale, while the filling is set by the competition between the high power of z_i and the exponential decay of $e^{-\frac{|\boldsymbol{r}_i|^2}{4\ell_{B_{\text{eff}}}^2}}$ at long distances. We obtain a filling of $2/(4s+1)$ electrons per $2\pi\ell_{B_{\text{eff}}}^2$ area, or $1/(4s+1)$ electrons in the $C=2$ unit cell. The states (1.37) have a ground state degeneracy $m^2 - n^2 = 4s+1$ in the flavor singlet case. We expect this to be true for (1.39) as well, and indeed find that it is the case in exact diagonalization. We remark that the constructed state $\Psi^{(C=2,2s)}$ is a flavor singlet, which implies that this family of FCI states is translation symmetric. As reported in Ref. [96], the authors have also verified an FCI in the $C=2$ band, at filling factor $\nu = 1/5$ (see Fig. 1.1(b)) with ground state degeneracy 5. It is compatible with the flavor singlet Halperin (332) state (1.39) which is a translation symmetric zero-mode of the interaction $V_1\delta''(\hat{\boldsymbol{r}}_i - \hat{\boldsymbol{r}}_j)$. A straightforward extension of the above construction to C decomposed flavors (see Sec. 1.9 and SI) yields flavor-singlet Halperin states at fillings $1/(2Cs+1)$ of the Chern C band.

Note that ground state degeneracy and particle-entanglement-spectrum counting consistent with the Halperin state was obtained previously by Ref. [96]. Our analytic understanding complements this result.

We may straightforwardly generalize our construction above to construct analogs of all Halperin states (1.37) entirely within the band of interest. While for $m \neq n+1$ the states are not SU(2) flavor singlets or zero modes of density-density pseudopotentials, they can still arise as translationally symmetric ground states. Let us focus on the (112) state which is familiar in the context of spin-degenerate LLLs at LLL filling-factor $2/3$[131–136] and TBG [69]. In our $C=2$ bands, the (112) state appears at filling $1/3$ in the $C=2$ unit cell and has a topological [†] ground state degeneracy of 3. While it is not a zero mode of the short-range interaction $V_1\delta''(\hat{\boldsymbol{r}}_i - \hat{\boldsymbol{r}}_j)$, by analogy with spinful LLLs we are optimistic that it appears as a ground state. Indeed, a previously-unidentified FCI in Ref. [95] at filling factor $1/3$ in the $C=2$ bands of TDBG is consistent with a (112) state. In particular, the PES counting of 3250[95] with a 5×6 system size and 3 particles in a subregion *exactly* matches the PES counting expected[137] for a (112) state[‡].

[†]The total ground state degeneracy is likely enhanced on large system sizes with C_3 symmetry, e.g. 6×6 or higher.
[‡]Ahmed Abouelkomsan, private communication

1.8.2 Translation Broken FCIs

After studying translation symmetric FCIs, we now study translation broken FCIs. Since the translation symmetric FCIs are flavor singlets, we can break translation symmetry by polarizing one of the flavors, generating ferromagnetic FCIs. These states can be obtained by applying the vortex attachment procedure (1.7) to a quantum Hall ferromagnet (1.33). Consider attaching $2s$ vortices to the ferromagnet:

$$|\Psi_{2s,\theta,\phi}\rangle = \prod_{i<j}(z_i - z_j)^{2s} |\theta, \phi\rangle \,. \tag{1.40}$$

Given the similarity of the ferromagnet to a fully filled LLL, the filling fraction would be $1/(2s+1)$ for the ferromagnet. In the original $C = 2$ unit cell, the filling fraction would be $1/(4s+2)$. These states correspond to ferromagnetic Laughlin states.

We consider the simplest case where $s = 1$, which corresponds to $\nu = 1/6$. It's interesting to study whether these ferromagnetic Laughlin states will be ground states of the Coulomb interaction projected to the $C = 2$ band. We consider the dual-gate screened Coulomb interaction

$$V_C(\boldsymbol{q}) = \frac{E_C}{|\boldsymbol{q}|a} \tanh(|\boldsymbol{q}|d) = \sum_{n\geq 0} V_n |\boldsymbol{q}|^{2n} d^{2n+1} \tag{1.41}$$

where d is the distance from the system to the gates. Since the $V_{0,1,2}$ terms annihilate the Laughlin state, the lowest term in the Hamiltonian that contributes to energy of the Laughlin states is $|\boldsymbol{q}|^6$. Thus we would expect a ground state energy dependence of d^7. On the other hand, any other state that is not the Laughlin state at this filling would have a lower order zero in the wavefunction, and in general its energy will gain contribution from the $|\boldsymbol{q}|^4$ term, resulting in a power law of d^5 or lower. We perform ED to compute the power law dependence of the ground state energies on d and the energy of the first excited state in Fig. (1.8). The numerical spectrum clearly shows that the ground states have the same power law scaling in d, which resembles the scaling of Laughlin states, whereas the first excited states have a different scaling behavior, which we expect to be $\sim d^5$ for sufficiently small d. We thus conclude that the ground states are ferromagnetic Laughlin states.

The numerical studies further confirm our result on the emergent $\mathrm{SU}(2)$ action on the ferromagnetic states. For N_e electrons, there are $N_e + 1$ generalized ferromagnetic states (1.34). Each ferromagnetic state will then

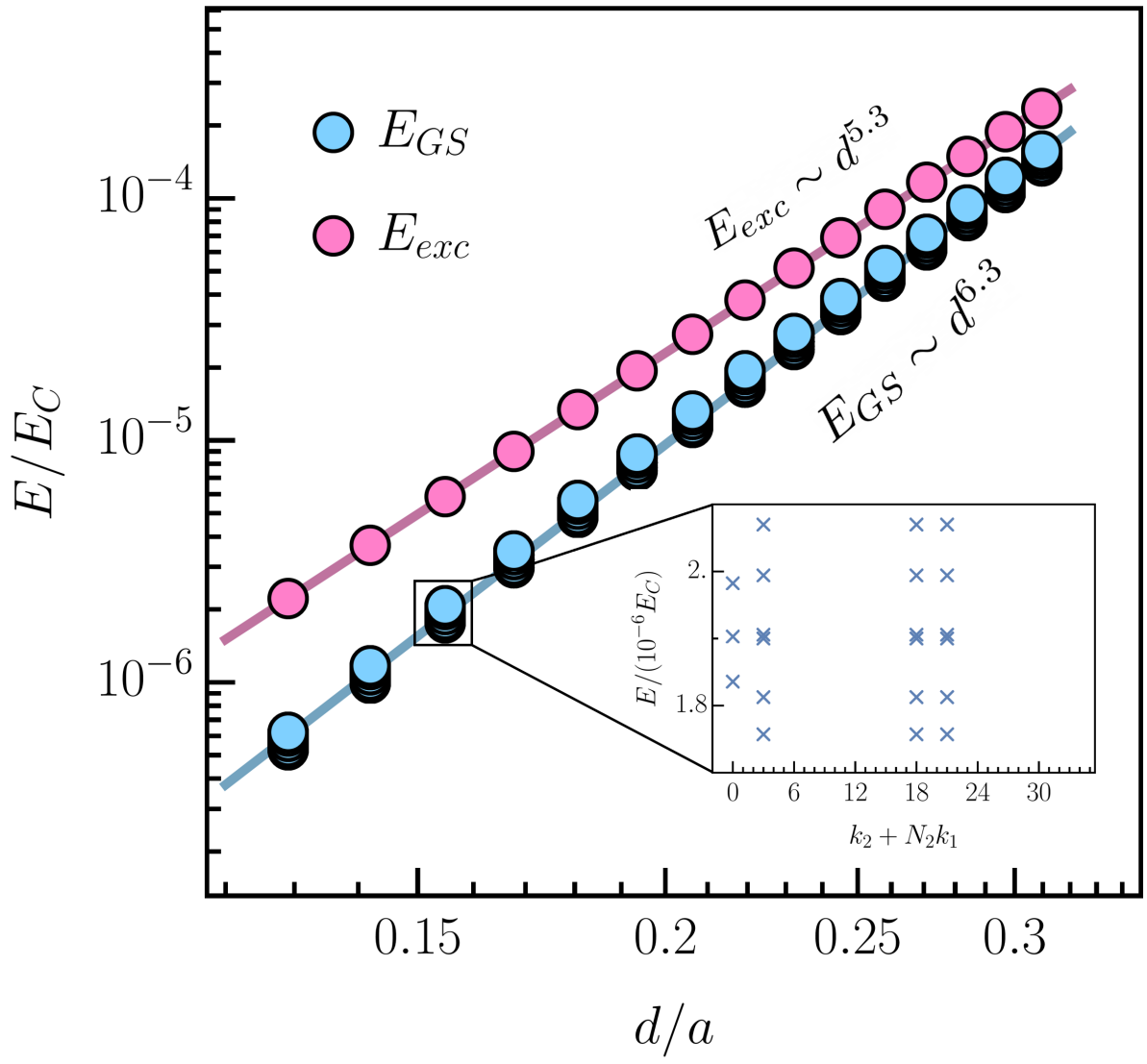


Figure 1.8: Scaling form of ground state energies E_{GS} and the first excited state energy E_{exc} for a $C = 2$ band at $\nu = 1/6$ with $N_1 = N_2 = N_e = 6$ under screened Coulomb interaction at different gate distances. The dots correspond to the numerically-obtained exact diagonalization spectrum and the lines are log-linear fits to the spectrum. The extracted power laws $E \sim d^{\eta}$ are comparable to the theoretical expectation $\eta_{\mathrm{GS}} = 7$ and $\eta_{\mathrm{exc}} = 5$, with deviations being attributed to the finite range of d, in particular since very small d is hard to access numerically. The inset shows the spectrum of $21 = 3(N_e + 1)$ Laughlin states, which follows from the SU(2) representation theory (1.34), (1.42).

generate $2s+1$ topologically-degenerate Laughlin states, so in total

$$\text{GSD} \approx (2s+1)(N_e+1) \tag{1.42}$$

which matches with the numerics in Fig. 1.1c and Fig. 1.8 for $N_e = 6$ and $s = 1$. For a Chern $C > 2$ band we similarly decompose into C Chern 1 bands (see Sec. 1.9 and SI) and make a Laughlin state in one; we obtain a filling factor $\nu = 1/C(2s+1)$ and a ground state degeneracy of $(2s+1)$ multiplied by the associated ferromagnetic degeneracy (A.48).

1.9 Ideal $C > 2$ Bands

After studying the ideal $C = 2$ bands in detail, we briefly summarize our results on ideal bands with $C > 2$. Detailed derivations can be found in SI.

Ideal bands with arbitrary Chern number C can also be decomposed into ideal subbands with lower Chern number by breaking translation symmetry (i.e. choosing a larger unit cell). If the unit cell is expanded by a factor of r, where r divides C, then we show that there are r subbands which may be hybridized such that each band is ideal and carries Chern number C/r.

We will focus on the case $r = C$, where we obtain C ideal Chern 1 bands ϕ_m by breaking translation symmetry by C times in one direction, say $\boldsymbol{a}_1$. These bands will have different boundary conditions at the same momentum point. However, similar to the $C = 2$ case, we are able to construct a hidden $\mathrm{SU}(C)$ action if we break translation symmetry in the other direction too, such that the final unit cell is expanded by a factor of C in both directions. The bands $\tilde{\phi}_{m\boldsymbol{k}} = \phi_{m(\boldsymbol{k}+\boldsymbol{G}_2/2-m\boldsymbol{G}_2/C)}$ will again have the same boundary conditions. We may now form linear superpositions

$$\phi_{v\boldsymbol{k}}(\boldsymbol{r}) = \sum_m v_m \tilde{\phi}_{m\boldsymbol{k}}(\boldsymbol{r}) \tag{1.43}$$

where v is a vector in $\mathbb{C}^C$. The states (1.43) are generically translationally symmetric symmetric only with respect to a $C\boldsymbol{a}_1 \times C\boldsymbol{a}_2$ unit cell. The mixed translations $(T^{\text{mag}}_{\boldsymbol{a}_i})^{-1} T_{\boldsymbol{a}_i}$ generate an $\mathrm{SU}(C)$ action similar to the $\mathsf{s}_{x,y,z}$ operators for $C = 2$ (see SI).

We now discuss the ground states of an ideal Chern band at $\nu = 1/C$ for the contact interaction (1.31). We

are now also able to construct a manifold of topological CDWs for this system

$$|v\rangle = \mathcal{A} \bigotimes_{\boldsymbol{k}} |\phi_{v\boldsymbol{k}}\rangle \,. \tag{1.44}$$

Due to the redundancy $v_m \to \lambda v_m$ we identify this manifold with $\mathbb{CP}^{C-1}$. This manifold of CDWs appear in finite size numerics too, with appropriate degeneracy counting (A.48), see Fig. 1.6(b).

FCI states can also be constructed for general C via the vortex attachment procedure. As with $C = 2$, attaching $2s$ vortices to the fully filled Chern C band constructs a flavor symmetric Halperin state at filling $\nu = 1/(2Cs+1)$. We may also attach $2s$ vortices to a flavor polarized (translationally broken) ferromagnet (1.44) to obtain a ferromagnetic Laughlin state at $\nu = 1/C(2s+1)$.

1.10 Conclusion

In this work we have explored the correlation physics in higher Chern bands, using an analytically tractable starting point, the ideal Chern bands. This approach allows us to make several rigorous statements regarding the energetics of ground states, which continue to hold even when we relax the conditions for strictly ideal bands. They provide a framework to understanding correlated states in realistic higher Chern band systems such as chiral graphene multilayers, that are already the subject of intense numerical and experimental study.

Our approach involves decomposing ideal flat Chern bands with arbitrary Chern number C into C ideal Chern bands with Chern number 1. We constructed combinations of real space and momentum space translations to form a non-unitary $\mathrm{SU}(C)$ action between the bands. We then analytically wrote down ideal many-body ground states such as generalized quantum Hall ferromagnets, flavor-singlet Halperin states, and Laughlin ferromagnets that are exact ground states in the limit of short-range interaction potentials. We found that the $\mathrm{SU}(C)$ action often becomes an emergent symmetry in the ground state manifold of short range interactions. For $C = 2$, we confirm our analytic predictions with numerical evidence that these states are indeed ground states of interactions projected to the ideal flat Chern bands of chiral twisted graphene multilayers and make concrete experimental predictions for twisted graphene multilayers at various fillings. Our work opens a route to an analytic understanding of correlated phenomena in higher Chern bands.

Many interesting questions arise naturally after this work. Since the decomposition in general gives non-orthogonal bands, when are the bands orthogonal? The $C = 2$ case is solved in Ref. [24] but remains to be

extended for general C. Beyond ideality, how can we understand the manifold of charge density waves? The manifold of states survives outside the ideal limit and long range interactions with suitable anisotropies, but thus far it can only be accessed through exact diagonalization[102]. While we have shown FCIs exist in the ideal limit, a phase diagram with realistic parameters, including band dispersion enhancement from integrating out remote bands [71] remains to be constructed. While there are caveats, exciting experiments have already seen $C = 1$ correlated insulator phases at $\nu = 3 + 1/2$, where realistic models predict that the ground state is in the same manifold as our pseudospin ferromagnets. What external parameters (displacement field, out of plane magnetic field) are optimal for their realization? Are there any FCI states in this system that is not generated through the typical vortex attachment procedure, for example non-Abelian states? We believe that future works will be inspired by these questions.

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Note added: During the preparation of the draft we became aware of parallel work by Jie Wang *et al.* [138] which overlaps with part of our results.

2

Composite Fermi Liquid at Zero Magnetic Field in Twisted $MoTe_2$

2.1 Introduction

Strong interactions can lead to exotic phases of matter such as non-Fermi liquids. A remarkable example is the composite Fermi liquid (CFL) that occurs in a half or quarter filled lowest Landau level (LLL). The CFL is a non-Fermi liquid with an emergent Fermi sea composed of charge neutral "composite fermions" [139–142] and has anomalous responses to a wide variety of experimental probes [143–148]. The gapless CFL state has provided an elegant interpretation for various Abelian [139–142] and non-Abelian gapped topological phases [149].

This work proposes an alternative route to realize CFLs. Our proposal is based on twisted 2D transition metal dichalcogenides (TMD), a family of platforms that have realized a wealth of interesting phenomena [14–16, 19, 150–161], and generated much theoretical interest for their topological properties [162–175]. A recent experiment [19] provided strong evidence for zero field fractional Chern insulators (FCIs) [53–55, 176] at fillings $\nu = -2/3$ and $-3/5$ in twisted bilayer $MoTe_2$ ($tMoTe_2$). The $\nu = -2/3$ FCI was separately found by Ref. [161]. These experiments were preceded by theoretical models of Chern bands in $tMoTe_2$ [163], as well as numerical works that found FCIs at partial fillings in $MoTe_2$ [177] and in WSe_2 [178, 179]. More recently, theoretical studies combining *ab-initio* lattice relaxation and exact diagonalization on $tMoTe_2$ [180, 181] have also obtained FCIs.

FCIs were previously reported at high magnetic fields [17] by partially filling Hofstadter bands [182] of a substrate-induced moiré potential in graphene. Shortly thereafter, with the discovery of correlated phenom-

ena [10, 25] and spontaneous Chern insulators [11, 12, 79] in twisted bilayer graphene (TBG), FCIs in zero field were theoretically anticipated in magic-angle TBG [32, 69, 183]. Experimental observations of FCIs in this setting soon appeared [18], albeit in a small magnetic field that theory [71] found was needed to improve the bandwidth and quantum geometry. These barriers are strikingly absent in $tMoTe_2$, motivating us to go beyond zero field FCIs to an exotic gapless state — the CFL.

We will focus on the gapless CFL phase, which presents challenges [184–187] relative to the well-understood spectral and entanglement signatures present in gapped FCI phases [57, 188–192]. Combining large scale exact diagonalization (ED) with density matrix renormalization group (DMRG) numerics, we find a broad CFL phase at experimentally realistic parameters of $tMoTe_2$. Furthermore, we present an explicit trial wavefunction that captures the essential features of the zero field CFL and its low energy spectrum. Finally, we discuss experimental signatures that distinguish the CFL from Fermi liquids, enabling experimental exploration.

2.2 Continuum model

We consider a model [163] for the valence bands of a twisted TMD with gate-screened [193] Coulomb interactions

$$\hat{H} = -\hat{h} + \frac{1}{2A}\sum_{\boldsymbol{q}} V_{\boldsymbol{q}} :\hat{\rho}_{\boldsymbol{q}}\hat{\rho}_{-\boldsymbol{q}}:, \quad V_q = \frac{2\pi \tanh(qd)}{\epsilon_r \epsilon_0 q}, \tag{2.1}$$

where $\hat{\rho}_{\boldsymbol{q}}$ is the density operator, A is the sample area, normal ordering is relative to filling $\nu = 0$, d is gate distance, and $\epsilon_r \approx 8 - 40$ is the dielectric constant [180]. Due to spin-valley locking [163], the low energy holes of the K (K') valley are locked to spin up (down). The total kinetic term is $h = h_K + h_{K'}$ with [163]

$$h_K = \begin{bmatrix} h^b(\boldsymbol{r}) + V/2 & T(\boldsymbol{r}) \\ T^\dagger(\boldsymbol{r}) & h^t(\boldsymbol{r}) - V/2 \end{bmatrix}, \tag{2.2}$$

where $h^\ell(\boldsymbol{r}) = -(\boldsymbol{p} - \hbar v_F \boldsymbol{K}^\ell)^2/2m^* + \Delta^\ell(\boldsymbol{r})$ and $h_{K'}$ is determined by time-reversal. Here the layer-diagonal terms include the quadratic monolayer TMD dispersion centered at rotated monolayer K-points $\boldsymbol{K}^{t/b}$, shifted by the displacement field V, and the moiré potentials $\Delta^{b/t}(\boldsymbol{r}) = 2v\sum_{j=1,3,5}\cos(\boldsymbol{b}_j \cdot \boldsymbol{r} \pm \psi)$. The off-diagonal terms are interlayer tunnelings $T(\boldsymbol{r}) = \omega\left(1 + e^{i\boldsymbol{b}_2\cdot\boldsymbol{r}} + e^{i\boldsymbol{b}_3\cdot\boldsymbol{r}}\right)$, where $\boldsymbol{b}_j$ are the reciprocal vectors obtained by counterclockwise $(j-1)\pi/3$ rotations of $\boldsymbol{b}_1 = (4\pi 3^{-1/2}\theta/a_0, 0)$. We focus on $tMoTe_2$, where recent first-principles calculations [180] (see also [163, 181]) found $(a_0, m^*, V, \psi, \omega) =$

$(3.52\,, 0.6m_e, 20.8\,\text{meV}, -107.7°, -23.8\,\text{meV})$. We take $\theta = 3.7°$ throughout.

2.3 Flat Almost-Ideal Chern Band

Fig. 2.1(a) shows the bandstructure for electrons h_K. The top moiré band has Chern number $C = 1$, due to the skyrmionic character of the layer spinor [163].

Recent experiments [19, 161] demonstrate that the many-body ground state is ferromagnetic (valley-polarized) in at least the range $-1.2 \lesssim \nu \lesssim -0.4$. The "parent state" for this regime is the correlated insulating state at $\nu = -1$. Fig. 2.1(b) shows its bandstructure within self-consistent Hartree-Fock (SCHF), which is strongly renormalized by interactions. Strikingly, the renormalized $C = 1$ band (red) becomes almost exactly flat, with bandwidth 1.6 meV at $\theta = 3.7°$. This *reduction* [194] in bandwidth from interaction effects is highly unusual [195].

The many-body physics of such flat bands is determined by the Bloch wavefunctions, often through their "quantum geometry". Recent theories [22, 23, 31, 32, 44, 45, 47–49, 52, 54, 96, 138, 196] emphasize the role of Kähler geometry in FCI stability. We say that a band has "ideal quantum geometry" if the trace inequality $T = \int d^2\boldsymbol{k}\,(\text{Tr}\, g_{\text{FS}}(\boldsymbol{k}) - \Omega(\boldsymbol{k})) \geq 0$ is saturated [22, 31, 32, 51]; here g_{FS} is the Fubini-Study metric and Ω is the Berry curvature. Ideal bands are "vortexable" in the sense that $\hat{z}P = P\hat{z}P$ where P is the projector onto the band and $\hat{z} = \hat{x} + i\hat{y}$ [23, 24]. Vortexability enables the direct construction of Laughlin-like FQHE trial states that are exact many-body ground states for ideal bands with short-range interactions [23, 24, 125]. Fig. 2.1(c) shows T, the deviation from ideality, and $\sigma[\Omega]$, the standard deviation of Berry curvature. Both are small in $tMoTe_2$ for $3° \leq \theta \leq 4°$. The top valence band thus has the rare combination of excellent quantum geometry and negligible bandwidth that favors lattice realizations of exotic quantum Hall states at zero magnetic field.

The interacting physics of the flat band is modelled by projecting Eq. (2.1) via $-\hat{h} \to \sum_{\boldsymbol{k}} \epsilon(\boldsymbol{k}) \hat{c}^\dagger_{\boldsymbol{k}} \hat{c}_{\boldsymbol{k}}$ and $\hat{\rho}_{\boldsymbol{q}} \to \overline{\rho}_{\boldsymbol{q}} = \sum_{\boldsymbol{k}} \hat{c}^\dagger_{\boldsymbol{k}} \langle u_{\boldsymbol{k}} | u_{\boldsymbol{k}+\boldsymbol{q}} \rangle \hat{c}_{\boldsymbol{k}+\boldsymbol{q}}$ where $\epsilon(\boldsymbol{k})$ and $u_{\boldsymbol{k}}$ are the dispersion and periodic part of Bloch wavefunction. Fig. 2.1(d) shows the bare ($\nu = 0$) and renormalized ($\nu = -1$) bandwidths versus twist angle, minimized near 3° and 3.6°, respectively. Fig. 2.1(e) confirms that FCIs are stabilized at $\nu = -1/3, -2/3$ — in accordance with previous results [177, 180, 181]. The mild angular dependence should make FCIs relatively robust to twist angle disorder. Notably the gap at $\nu = -2/3$ is largest where the bandwidth at $\nu = -1$ is smallest [197]. We therefore expect ∼3.6° to be optimal for FQH physics at half filling.

2.4 Composite Fermi liquid at $\nu = -1/2$

We now go beyond gapped FCIs and examine the more exotic gapless CFL state [139, 149]. We focus on $\nu = -1/2$ but our conclusions also apply to $\nu = -3/4$ (data in SM [198]).

(i) Many body spectrum: Fig. 2.1(f, g) compares the spectra of twisted $MoTe_2$ and the lowest Landau level (LLL) at half filling with 14 electrons, showing a one-to-one correspondence at low energy. The LLL spectrum uses the same geometry as $tMoTe_2$ with Coulomb interactions. This one-to-one similarity holds at all system sizes $N_e = 8 - 14$. We thus conclude that the ground state of $\hat{H}$ at $\nu = -1/2$ is the same phase as the half-filled LLL with Coulomb interactions — the CFL. The ground state and low-energy excitations are at precisely the momenta expected for compact composite Fermi sea (CFS) configurations [199]. See SM for other system sizes, and detailed matching of degeneracies, momenta, and excitations to CFL expectations.

(ii) Absence of electron Fermi surface: A finite quasiparticle weight $Z > 0$ gives the jump in electron occupations $n(\boldsymbol{k})$ at the Fermi surface in a regular Fermi liquid (FL). As a non-Fermi liquid, composite fermions have vanishing Z, leading to the absence of Fermi surface occupation discontinuities.

To characterize the CFL, we employ large-scale iDMRG [200, 201] calculations with the TenPy library [202]. We use an infinite cylinder geometry with circumference $L_y = 5 - 10$, corresponding to L_y evenly spaced horizontal wires through the Brillouin zone (Fig. 2.2(c) inset). We take a computational basis of hybrid Wannier orbitals [59, 203, 204], and use "MPO compression" [205, 206] to accurately capture gate-screened Coulomb interactions in the flat band. Under weak interactions ($\epsilon_r = 100$), we find the FL expected from band theory at $\nu = -1/2$, with an almost-circular Fermi surface centered at Γ (Fig. 2.2(a), left) with radius $k_F = (A_{\mathrm{BZ}}/2\pi)^{1/2}$. The SM shows electrons, holes, and particle-hole pairs are likely gapless [207], confirming the Fermi liquid.

Under realistic interactions ($\epsilon_r = 15$) with the same parameters, the ground state has quasi-uniform occupations $|n(\boldsymbol{k}) - \frac{1}{2}| < 0.17$ (Fig. 2.2(a), right). Because charge $Q_E = 1$ correlations are short-ranged, the state is inconsistent with an electronic Fermi liquid. However, the state has high entanglement and significant electrically-neutral correlations, consistent with the gapless density fluctuations expected from an emergent CFS. To reveal the "hidden" CFS, we turn to the structure factor.

(iii) Scattering across the composite Fermi sea: Fig. 2.2(b) contrasts the connected structure factor $S(\boldsymbol{q}) = \langle \hat{\rho}_{\boldsymbol{q}} \hat{\rho}_{-\boldsymbol{q}} \rangle - \langle \hat{\rho}_{\boldsymbol{q}} \rangle \langle \hat{\rho}_{-\boldsymbol{q}} \rangle$ between the FL and the CFL. Both nearly vanish when $|\boldsymbol{q}| > 2k_F$, strongly implying

that there is a Fermi surface in the CFL phase whose constituent fermions aren't electrons. We then match the features of $S(\boldsymbol{q})$ to scattering events with different momentum transfers across the putative CFS in Fig. 2.2(c), e.g. $\hat{c}^\dagger_{\boldsymbol{k}=G}\hat{c}_{\boldsymbol{k}=B}$ scattering with $q_x \approx 1.94k_F$. The *tour-de-force* work of Geraedts *et al* [184] showed such features are emblematic of the CFS arising from the half-filled LLL. As every feature in $S(\boldsymbol{q})$ corresponds to such a scattering (quantitative matching in SM), we conclude the state has an almost-circular [208] CFS composed of non-Landau quasiparticles. These two independent numerical methods establish a CFL state at $\nu = -1/2$ (see SM for $\nu = -3/4$).

2.5 Zero Field CFL Wavefunction

Standard theories of composite fermions apply at $B > 0$, where emergent gauge flux cancels external magnetic flux. These cannot apply directly here at zero magnetic field. We therefore construct an explicit zero-field CFL wavefunction. To start, we approximate the geometry of the top $tMoTe_2$ band as ideal. Such bands have the general "LLL-like" wavefunction [22, 32],

$$\psi_l(\boldsymbol{r}) = \phi(\boldsymbol{r})\zeta_l(\boldsymbol{r}) = f(z)e^{-K(\boldsymbol{r})}\zeta_l(\boldsymbol{r}), \tag{2.3}$$

where $f(z)$ is holomorphic and $\zeta_l(\boldsymbol{r})$ is an orbital-space spinor where $\sum_l |\zeta_l(\boldsymbol{r})|^2 = 1$. Here $\phi(\boldsymbol{r})$ is the wavefunction of a Dirac particle in an inhomogeneous, periodic, magnetic field $B(\boldsymbol{r}) = \nabla^2 \mathrm{Re}K(\boldsymbol{r})$ with one flux per unit cell [32, 209, 210]. While ψ is symmetric under ordinary translations, $\phi(\boldsymbol{r})$ and $\zeta_l(\boldsymbol{r})$ are symmetric under *magnetic* translations, with opposite magnetic twists [34], giving a gauge redundancy $\phi(\boldsymbol{r}) \to e^{+i\lambda(\boldsymbol{r})}\phi(\boldsymbol{r}), \zeta_l(\boldsymbol{r}) \to e^{-i\lambda(\boldsymbol{r})}\zeta_l(\boldsymbol{r})$. The form Eq.(2.3) implies that *all* many-body wavefunctions within the band of interest have the form $\Psi = \Psi_\phi \prod_i \zeta_{l_i}(\boldsymbol{r}_i)$, where Ψ_ϕ is a wavefunction of flux-feeling particles; in the SM we interpret this fractionalization in terms of a new type of Chern band parton theory [211, 212]; see also [213–218]. For example, we may use Read & Rezayi's LLL ansatz for the CFL [219] to obtain:

$$\Psi(\{\boldsymbol{r}_i\}) = \mathcal{P} \det_{ij} \psi^{\mathrm{CF}}_{\boldsymbol{k}_i}(\boldsymbol{r}_j) \prod_{i<j}(z_i - z_j)^2 \prod_i e^{-K(\boldsymbol{r}_i)}\zeta_{l_i}(\boldsymbol{r}_i). \tag{2.4}$$

Here $\mathcal{P} = \prod_i P_i$ is the many-body projector to the top band, and $\psi^{\mathrm{CF}}_{\boldsymbol{k}_i}$ fill a Fermi sea [220].

2.6 Experimental Signatures

We conclude with experimental implications of the quantum geometry and CFL phase. Fig. 2.3(a,b) show phase diagrams of $tMoTe_2$. At $\nu = -1$, SCHF finds the $|C| = 1$ phase transitions to an valley and layer polarized phase at large V. At $\nu = -1/2$, we find a broad CFL phase centered around $3.8°$ that competes with layer polarized phases and $C = 1$ Fermi liquids at larger V. The layer polarized region is estimated from SCHF at $\nu = -1$, where an interaction-driven layer-polarized state is more favorable. The phase diagram at $\nu = -3/4$ is similar (see SM), except the CFL is more sensitive to displacement field.

The almost ideal quantum geometry manifests optically. If a band with projector P is vortexable, then $\hat{z}P = P\hat{z}P$ implies the velocity operator $\hat{v}^{\pm} = -i[\hat{x} \pm i\hat{y}, \hat{H}]$ must obey $(I-P)\hat{v}^{+}P = 0$, i.e. left-circularly polarized transitions are "extinguished". This gives perfect circular dichroism:

$$\frac{\sigma^{+} - \sigma^{-}}{\sigma^{+} + \sigma^{-}} = 1; \ \sigma^{\pm}(\omega) = \frac{ie^2}{\hbar} \sum_{\boldsymbol{k}, a \neq b = 0} \frac{f_{ab}}{\epsilon_{ab}} \frac{|\langle \psi_{\boldsymbol{k}a} | \hat{v}^{\pm} | \psi_{\boldsymbol{k}b} \rangle|^2}{\omega - \epsilon_{ab}}. \tag{2.5}$$

Here $\epsilon_{ab} = \epsilon_a - \epsilon_b$ are energy differences and $f_{ab} = f(\epsilon_a) - f(\epsilon_b)$ are Fermi factors. Fig. 2.3(c) shows $\sigma^{\pm}$ for $tMoTe_2$ at $\nu = -1$. As the $C = +1$ band is nearly vortexable, transitions from the second and third valence bands to the empty top valence band nearly vanish, giving nearly-perfect circular dichroism > 0.9 at resonance. The inset shows a control experiment: the Haldane model has Chern bands $C = \pm 1$ but not ideal geometry; σ^{-} is not extinguished there.

Finally, we discuss direct experimental probes of the zero-field CFL. While the CFL and the FL are both compressible and metallic, they differ in that the CFL's excitations have vanishing overlap with the electron in the limit of low energies, and CFs themselves are best thought of as (doubled) *vortices* in the electronic fluid [141, 142, 221–223]. This observation leads to a number of striking physical responses that differ strongly from Fermi liquids. These include (i) a "pseudogap" in the tunneling density of states $A(\omega) \propto e^{-\omega_0/\omega}$ [224] as a function of bias ω, which has been observed between two CFLs with a tunnel barrier [143]; (ii) distinct DC conductivity in the clean limit: $\sigma_{xx} \to 0$ in a CFL in the absence of disorder $k_F l \to \infty$, whereas in the FL, even in a Chern band, σ_{xx} diverges [225]; (iii) strong violation of the Wiedemann-Franz law [222, 223] which compares heat and charge transport; (iv) quantum oscillations with doping, that CFs feel a magnetic field $\propto (\nu - 1/2)$ and can fill Landau levels, leading to Jain-like [139] FCIs when fully developed, which can further be probed using geomet-

ric resonance with a one-dimensional periodic grating [141, 144, 145]; (v) vanishing thermoelectric conductance $\alpha_{xx} = j_x/(-\partial_x T)$ due to approximate emergent particle-hole symmetry [226, 227]; (vi) surface acoustic wave attenuation, a contactless probe that measures $\sigma_{xx}(\boldsymbol{q}) \propto |\boldsymbol{q}|$ in the CFL [141], as opposed to $\sigma_{xx} \propto |\boldsymbol{q}|^{-1}$ in a clean FL [146].

Finally we highlight properties of zero field CFLs that transcend LLL physics. First, the Chern bands of $MoTe_2$ have one effective magnetic flux quantum per moiré unit cell, translating to 160 T at 3.7°. This vastly exceeds laboratory magnetic fields, leading to enhanced energy scales. The lack of *real* quantizing magnetic fields, however, opens up the possibility of employing zero field experimental probes such as high resolution angle-resolved photoemission spectroscopy (ARPES). Furthermore, the exponentially suppressed tunneling density of states of the CFL could be probed through tunneling from a proximate Fermi liquid state, or via spatial variation of the twist angle, which can be used to create a CFL-FL interface within the same sample. Our work does not rule out the possibility of a continuous quantum phase transition, driven by displacement field, between the CFL and FL [217], which could be studied experimentally. Since the effective magnetic field of the TMD originates from spontaneous breaking of time reversal symmetry through valley polarization, rather than external magnetic field, domains between opposite valley polarizations and hence between time-reversal-related CFLs are expected. Transport properties across such a domain wall would interrogate composite fermions in an entirely new regime, and potentially shed light on their proposed Dirac character [142, 221, 223]. Finally, we note that moiré phonons occur on the same scale as the effective magnetic length in this system; their interplay with CFL physics is unclear at present and worthy of future study.

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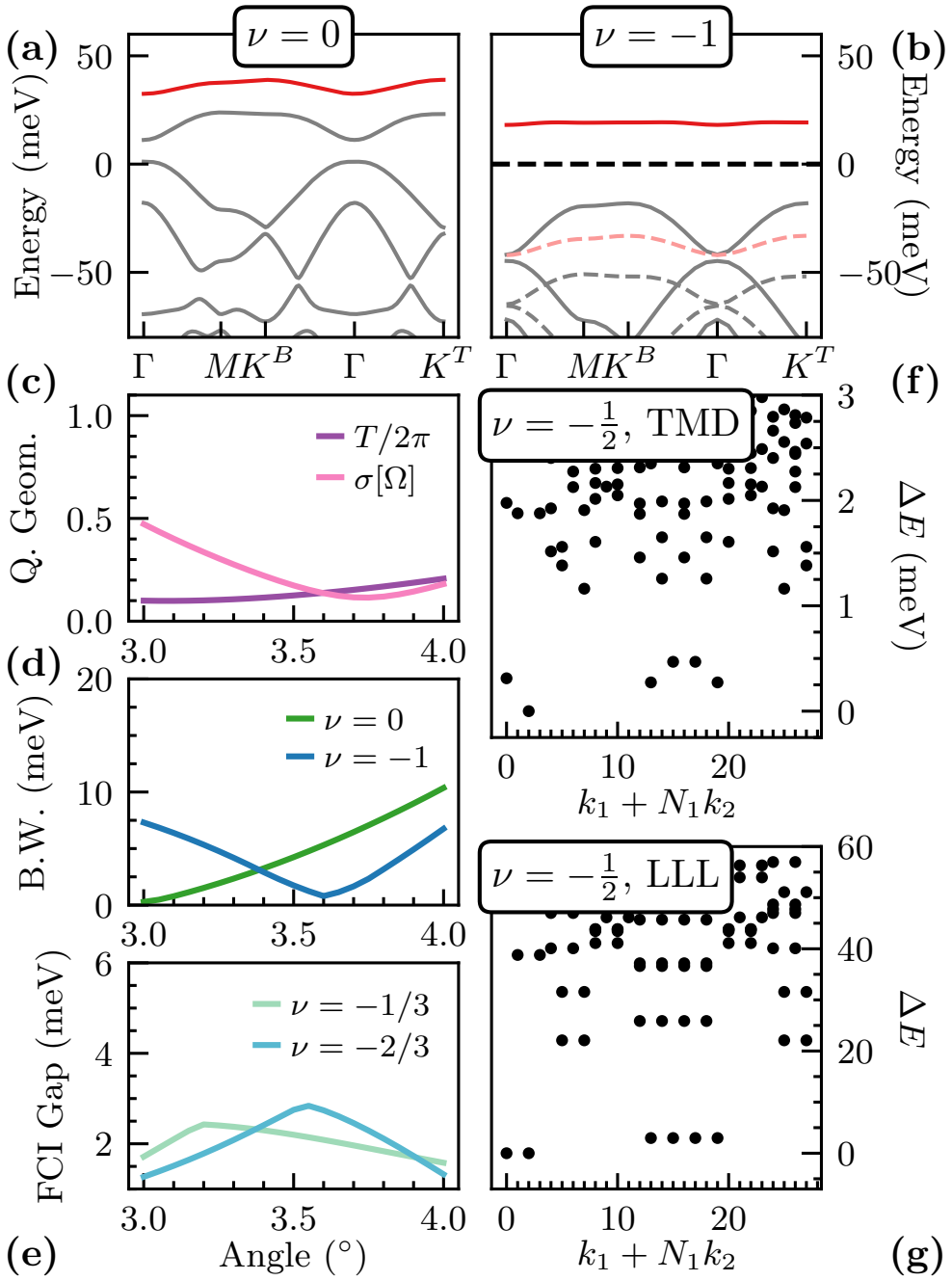


Figure 2.1: The top valence band has favorable conditions for fractionalized topological phases. Bandstructure as seen from (a) charge neutrality and (b) from $\nu = -1$ computed from self-consistent Hartree-Fock. (c) Quantum geometry in terms of trace condition T and Berry curvature deviation $\sigma[\Omega]$. (d) Bare and SCHF bandwidths and (e) the many-body gap of FCIs at $\nu = -1/3$ and $\nu = -2/3$ as a function of twist angle. The FCI gaps are obtained from ED with $N_e = 8$ and 16 respectively. (f) and (g): ED spectrum for 14 particles at half filling for Coulomb interaction in lowest Landau level and screened Coulomb interaction in twisted $MoTe_2$, respectively. Parameters: $(\theta, \epsilon_r, d) = (3.7°, 15, 300\,)$ unless otherwise noted.

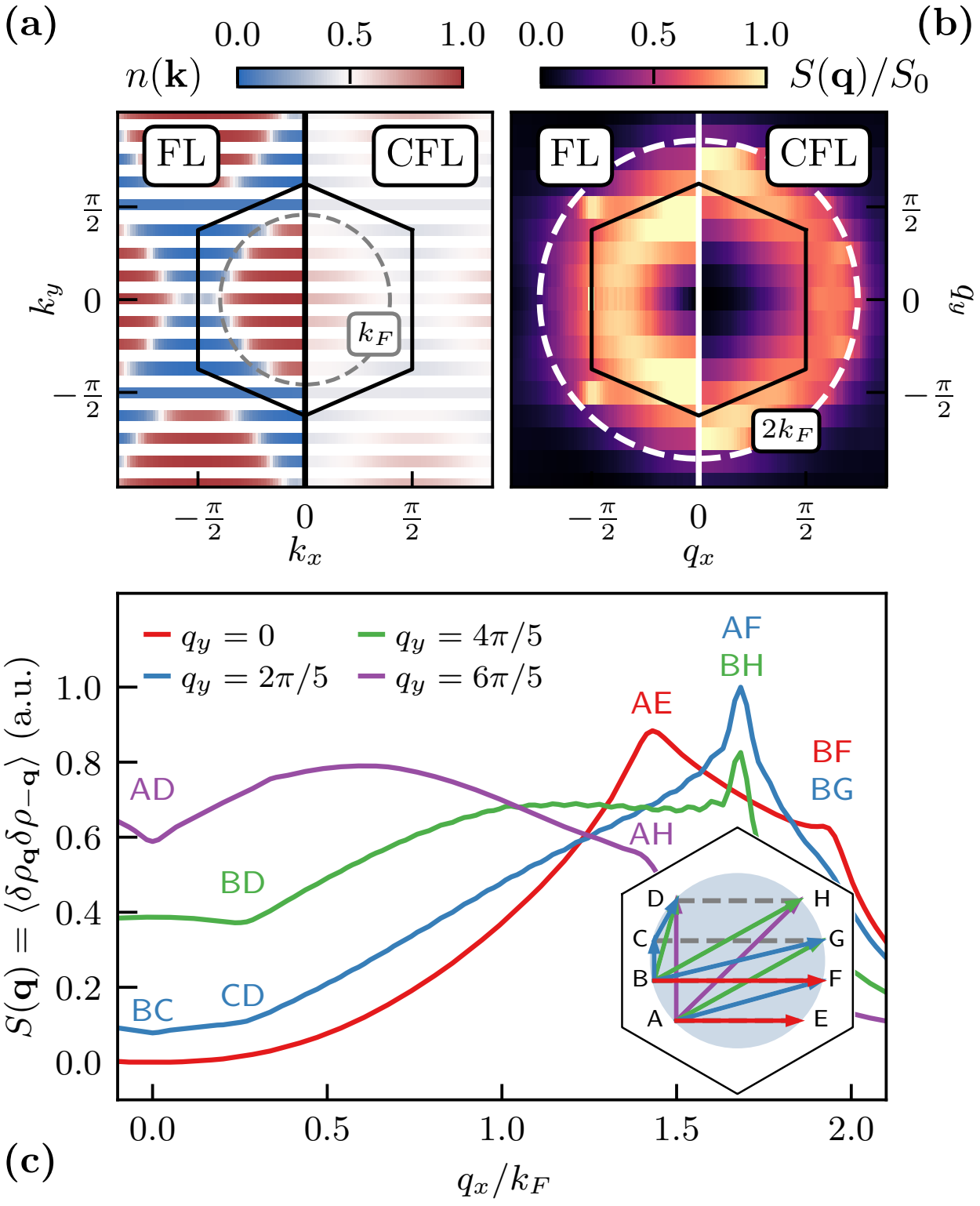


Figure 2.2: Numerical identification of the composite Fermi liquid (CFL) from iDMRG. (a) Occupations $n(\boldsymbol{k})$ in the Brillouin zone at $L_y = 8$ for the Fermi liquid (FL, left side) versus the CFL (right side). (b) Connected structure factor $S(\boldsymbol{q}) = \langle \hat{\rho}_{\boldsymbol{q}} \hat{\rho}_{-\boldsymbol{q}} \rangle - \langle \hat{\rho}_{\boldsymbol{q}} \rangle \langle \hat{\rho}_{-\boldsymbol{q}} \rangle$ at $L_y = 8$. Characteristic features of a Fermi surface are visible for both the FL and CFL: near-vanishing weight outside $|\boldsymbol{q}| \approx 2k_F$, and peaks corresponding to momentum transfers inside that radius. (c) Cuts of $S(\boldsymbol{q})$ at constant q_y for $L_y = 5$ for the CFL. Each peak or inflection in $S(\boldsymbol{q})$ quantitatively matches scattering events across the almost-circular composite Fermi surface (Inset). Parameters match Fig. 2.1 with $\epsilon_r = 15$ (100) for the CFL (FL).

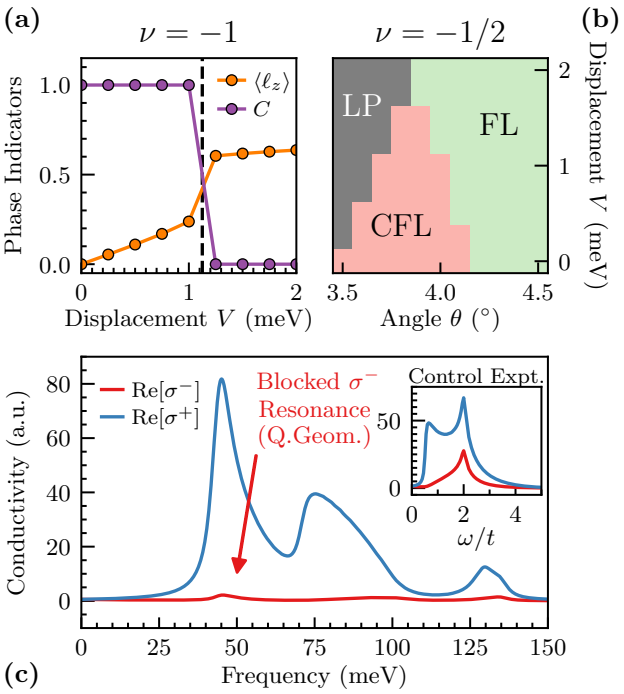


Figure 2.3: Many-body phase diagrams and optical responses. (a) Phase diagram at $\nu = -1$ with $\theta = 3.7°$ showing a transition from $C = 1$ layer-unpolarized state to a $C = 0$ layer polarized state. (b) Phase diagram of the topological regime at $\nu = -1/2$: The CFL phase is shown in red, whereas the green region corresponds to the FL phase. Here 'LP' indicates a layer polarization instability determined from $\nu = -1$ SCHF. (c) Direct optical probe of almost-ideal quantum geometry via an "extinguished" valence-valence optical responses in σ^-, Inset: the Haldane model at $(t, t_2) = (1, 0.05)$ has non-ideal geometry. Parameters match Fig. 2.1.

3

Anomalous Hall Crystals in Rhombohedral Multilayer Graphene I: Interaction-Driven Chern Bands and Fractional Quantum Hall States at Zero Magnetic Field

3.1 Introduction

Tunable experimental platforms that host correlated topological states are fertile grounds for exotic phenomena. In recent years, a menagerie of candidates have arisen, including twisted graphene multilayers [10, 13, 18, 25, 26, 26, 28, 29, 76–78, 83, 85, 231–248], twisted transition metal dichalcogenide (TMD) bilayers [14–16, 19, 20, 158, 159, 161, 163–168, 171, 173–175, 177–181, 228, 229, 249–256], and rhombohedral multilayer graphene (RMG) [17, 21, 79, 257–274]. Topological phenomena have been reported in RMG systems with $N_L = 2 - 5$ layers, both with [258, 259, 262] and without [257, 260, 261, 271] moiré potentials due to an hBN substrate. A number of theoretical studies have examined this hBN moiré potential [275–279] and the resulting topological bands [79, 272, 273, 273, 274, 280]. A recent experiment [21] reported that rhombohedral pentalayer graphene aligned to an hBN substrate hosts a $|C| = 1$ quantized anomalous Hall (QAH) insulator, and a whole series of fractional quantum anomalous Hall (FQAH) states that spontaneously break time-reversal symmetry.

Finding Chern bands that are favorable to realize such fractional states is an ongoing challenge [53, 55, 57, 176, 189, 192]. In initial realizations of fractional Chern insulators (FCI), magnetic field was necessary to induce [17], or improve the Chern band [18, 71] in order to favor fractional states. A zero-field fractional quantum Hall effect was only recently realized in $MoTe_2$ [19, 20, 161, 249]. For all of the aforementioned systems, the Chern band has been understood at a single-particle level, with interactions providing isospin polarization and thence the fractional states [89, 163, 193, 228, 281, 282]. In contrast, the origin and the character of the Chern band in rhombohedral pentalayer graphene [21] has so far been a mystery.

In this work, we shed light on this mystery by numerically solving the interacting many-body problem in $N_L = 4 - 6$ layer RMG with an hBN potential. Using self-consistent Hartree-Fock (SCHF) calculations, we find that interactions open a gap at filling $\nu = 1$, stabilizing an isolated $|C| = 1$ Chern band. Using exact diagonalization (ED) and density-matrix renormalization group (DMRG) calculations, we confirm that this Chern band hosts an FQAH state.

The Chern band is remarkably robust to perturbations. Crucially, it arises when the electrons are polarized *away* from the aligned hBN substrate, which we refer to as the "moiré-distant" case (Fig. 3.1(a)). It follows that the Chern band is insensitive to many details; similar Chern bands arise for different moiré potentials, different hopping parameters, and for $N_L = 4 - 6$ rhombohedral layers (Fig. 3.1(c)). In fact, the Chern band persists even as the moiré potential is turned off within our SCHF calculations (Fig. 3.1(b), 3.3(a)(d)). This is reminiscent of the "Hall crystal" phase proposed in [115, 283], where electrons in a magnetic field crystallize into a state with nonzero Hall conductance. Thus, we propose that the Chern insulator at $\nu = 1$ is an anomalous Hall crystal (AHC) state that spontaneously breaks both continuous translation symmetry and time reversal symmetry (with the moiré potential providing a pinning field). This establishes an entirely new mechanism for Chern band formation that arises from the interplay of translation breaking and topology.

3.2 Microscopic Model

We consider a model [284, 285] of RMG with a moiré potential [275] due to an hBN substrate at twist angle θ (Fig. 3.1(a)):

$$\hat{H} = \hat{h}_{\text{kin}} + \sum_{\boldsymbol{q}} \frac{U_{|\boldsymbol{q}|}}{2A} :\hat{\rho}_{\boldsymbol{q}}\hat{\rho}_{-\boldsymbol{q}}:, \quad U_q = \frac{2\pi \tanh(qd)}{\epsilon_r \epsilon_0 q}, \tag{3.1}$$

where $\hat{\rho}_{\boldsymbol{q}}$ is the density operator at momentum $\boldsymbol{q}$, A is the sample area, normal ordering is relative to the charge neutrality gap of RMG in the presence of displacement field, and the screened Coulomb potential $U_{|\boldsymbol{q}|}$ has gate distance $d = 250$ and dielectric constant $\epsilon_r = 5$.

The kinetic term in the K-valley takes the form

$$h_{\text{kin}} = h_{RG}^{(N_L)} + h_D + V_{\text{hBN}}, \tag{3.2}$$

where $h_{RG}^{(N_L)}$ is a standard model [259] for N_L-layer rhombohedral graphene with in-layer and interlayer hoppings t_{0-4}. The displacement term $[h_D]_{\ell\ell'} = u_D(\ell - 1 - (N_L - 1)/2)\delta_{\ell\ell'}$ creates a potential difference of u_D per layer with the zero point in the middle. Physically, $u_D > 0$ polarizes electrons in the conduction band *away* from the hBN moiré potential. The potential acts on the bottom layer as $V_{\text{hBN}}(\boldsymbol{r}) = V_0\sigma^0 + V_1\boldsymbol{f}(\boldsymbol{r}, \psi)\cdot\boldsymbol{\sigma}$, which is a sum of first harmonics with strength V_1 and phase ψ, plus a layer potential shift V_0 (σ^μ are sublattice Pauli matrices). While a variety of models have been proposed for V_{hBN} [275–278], our focus is on the moiré-distant side $u_D > 0$, which we find is largely V_{hBN}-independent. As the displacement field and moiré potential break C_{2x} and M_x symmetries of standalone RMG, the moiré Hamiltonian $\hat{H}$ retains only C_{3z} rotation, time-reversal, and moiré translation symmetries. See App. I [286] for complete model details.

3.3 Interaction-Induced Chern Band

To understand the origin of the Chern insulator, we performed self-consistent Hartree-Fock (SCHF) calculations on up to 48×48 moiré unit cells. We project the $N_L = 5$ graphene layers to the lowest 7 conduction bands, which is well-controlled whenever $|u_D|$ is sizable. The relevant Chern insulator recently observed in rhombohedral pentalayer graphene was at electron filling $\nu = 1$, which suggests a flavor polarization into a single species. Therefore we assume polarization to a single valley and spin flavor.

We first consider $\theta = 0.77°$ and $u_D = 50\,\text{meV}$, estimated to match the displacement field $D/\epsilon_0 = 0.81\,\text{V/nm}$ that stabilizes the QAH insulator experimentally *. The single-particle band structure of moiré-distant electrons (Fig. 3.2(b)) *does not* have a direct gap above the first band. Within SCHF at $\nu = 1$ (Fig. 3.2(c)), we find interactions open a gap, where the occupied band (red) has strictly positive Berry curvature (inset) and

*We estimate the conversion between D and u_D by comparing the reported HF phase diagram with the experiment in rhombohedral trilayer graphene [259, 287].

Chern number $|C| = 1$. We refer to this as the AHC phase, for reasons justified below. Its narrow 6 meV bandwidth and favorable quantum geometry, discussed below, make it eminently suitable for fractional topological states.

3.4 Phase Diagram at $\nu = 1$

Fig. 3.2(a,e) shows the phase diagram of moiré rhombohedral pentalayer graphene at filling $\nu = 1$. On the moiré-distant side $u_D > 0$, the AHC phase is stabilized over a broad range of twist angles and displacement fields near the experimentally relevant point (yellow star). Decreasing the displacement field drives a transition into a metallic phase, consistent with experiment [21]. Reducing the twist angle θ instead drives a transition to a $C = 0$ insulator that we call a "Wigner-like insulator" (WLI), based on its charge distribution $\rho(r)$. The band structures of a representative WLI state is shown in Fig. 3.2(g). The charge density of the state (Fig. 3.2(h)) shows that $\rho(r)$ varies strongly in the WLI, with peaks on a triangular lattice, reminiscent of a Wigner crystal. Its single-particle band structure in Fig. 3.2(f) reveals the origin of this charge density variation: at the γ point, the back-folded bands decrease in energy as θ decreases, bringing many within 30 meV of the minimum. These bands may then be strongly mixed by interactions, selecting the configuration that minimizes Hartree energy, i.e. a triangular lattice. Conversely, the AHC has a more uniform charge distribution without distinct peaks (Fig. 3.2(d)).

We comment briefly on the moiré-proximate side $u_D < 0$, where the dominant phase is a $C = 0$ insulator. Though consistent with experiment [21], we caution that this phase governed by details of V_{hBN}, and Fig. 3.3(b) shows that varying the moiré potential strength and phase can drive a transition to a metal or even a Chern insulator. Models incorporating lattice relaxation [278], other *ab initio* modelling [277, 288], and comparison with experiment [279] may be required here. [†]

3.5 Robustness of the Anomalous Hall Crystal

The AHC state is robust to a wide range of perturbations. In Fig. 3.3(c), we show the Chern number as a function of u_D and dielectric constant ϵ_r. The AHC state survives up to $\epsilon_r \sim 15$ for a range of u_D, showing that relatively weak interactions are sufficient to stabilize it. Fig. 3.3(a, d) shows the moiré-distant charge gap as a function of ϵ_r and moiré potential strength V_1 and phase ψ. The charge gap increases with interaction strength,

[†]Near $u_D \approx 0$ valence bands, which we ignore, also become relevant; our model is most reliable for the moiré-distant regime $u_D > 0$.

but is essentially independent of the moiré potential. The charge gap is peaked near $u_D \approx 55\,\mathrm{meV}$ at $\theta \approx 0.77°$, despite the effective moiré potential decreasing monotonically with u_D (see App. II [286]). Conversely, tuning V_1 and ψ changes the ground state on the moiré-proximate side (Fig 3.3(b)).

Remarkably, the AHC phase does not require a particular number of rhombohedral layers. Fig. 3.1(c) shows SCHF ground state at $\nu = 1$ for $N_L = 4 - 6$ layers, with an $|C| = 1$ AHC phase appearing in all cases at large u_D. Moreover, the critical u_D^* decreases significantly as the number of layers increases, easing the experimental requirement of a large displacement field $D \propto u_D$. Finally, the phase diagram is robust to modelling differences in the hopping parameters of rhombohedral graphene. For instance, we show in the Supplemental Material that the AHC remains when all further neighbor hoppings $t_{i\geq 2}$ are set to zero.

3.6 Spontaneous translation symmetry breaking

The resilience of the $|C| = 1$ phase to the change in moiré potential raises a natural question: Is the moiré potential necessary at all for stabilizing the state? We now argue that within SCHF the ground state is an anomalous Hall crystal [115, 283] that spontaneously breaks translation symmetry.

We consider the Hamiltonian with *zero* moiré potential. Under this condition, SCHF calculations at $n \approx 0.93 \times 10^{12}\mathrm{cm}^{-2}$ find a ground state with *spontaneously* broken graphene translation symmetry, which has much lower energy than (metallic) graphene-translation symmetric states when $u_D > 25\,\mathrm{meV}$ (Fig. 3.1(b)). We note that the AHC state is most stable compared to the metallic phase around $u_D = 55\,\mathrm{meV}$, which is consistent with our estimate for the experimental displacement field and where the charge gap reaches maximum.

The AHC state is a compressible state in the clean limit, since the lattice constant can change continuously. However, a small spatially-varying potential or disorder can pin its periodicity [115], whereupon its nontrivial Chern number can be revealed through the Streda formula and Hall conductance. This is likely the role played by moiré potential in experiment [21]: both the topology and the charge gap of the ground state are determined by the interaction effects that are already present in the moiré-less limit, but the moiré pinning field stabilizes the crystal at $\nu = 1$.

3.7 Fractional quantum anomalous Hall (FQAH) effect

One of the main features of the experimental discovery [21] was a full series of FQAH states found at fractional fillings in the absence of magnetic field. Employing exact diagonalization (ED) and density matrix renormalization group (DMRG) calculations, here we numerically confirm that partially filling the Chern band generated by the AHC gives us FQAH states.

Recent progress on the study of FCI phases has emphasized the role of quantum geometry [22, 23, 31, 32, 44, 45, 47–49, 52, 54, 96, 138, 175, 196, 210, 290], in addition to flat dispersion [57, 188–192]. Small trace condition violation $T = \int d^2\boldsymbol{k}(\mathrm{Tr}(g_{\mathrm{FS}}(\boldsymbol{k})) - \Omega(\boldsymbol{k}))$ gives a band suitable for finding FCIs. Here, $g_{\mathrm{FS}}(\boldsymbol{k})$ is the Fubini-Study metric, and $\Omega(\boldsymbol{k})$ is the Berry curvature. The AHC band has both flat bandwidth and relatively small trace condition violation (Fig. 3.4(a)), making it likely to host FQAH phases. Indeed we find that a Laughlin state appears at $\nu = 2/3$ of the AHC band, with clear signatures from both ED (Fig 3.4(b)) and DMRG calculations (Fig 3.4(d)). These calculations [286] are done in the many-body Hilbert space of the lowest SCHF band at $\nu = 1$, assuming spin and valley polarization, which numerically biases the formation of fractional topological states. The FQAH phase yields to a competing metallic phase at small and large u_D (Fig 3.4(c)). The FQAH state is likely a ground state in the spin- and valley-polarized sector. Similar to the case in the lowest Landau level [131, 291, 292], this state is likely in competition with a spin-unpolarized state, which merits future investigation.

Examining the stability of the FQAH phases down to the zero moiré potential limit is an excellent topic for future studies, where we expect competition between an FQAH phase with the same unit cell size, versus Wigner or anomalous Hall crystals with larger unit cells. Experimentally, the existence of the $\nu = 2/3$ plateau points to the realization of the former scenario, at least in the weak moiré potential limit, while a profusion of insulating states at smaller densities [21] might be due to the latter scenario.

3.8 Discussion

We have demonstrated the existence of the AHC phase within SCHF calculations. While such a phase can be stabilized *in principle*, SCHF is known to overestimate the tendency to crystallize, for example in Wigner crystals [293]. Pinpointing the location of the phase boundary between the AHC and the metal with more sophisticated techniques (just as variational Monte Carlo is used in the jellium model [294–296]) is left to future work.

We note that the AHC phase at the optimal u_D survives even after decreasing the interaction strength by a factor of three (Fig. 3.3(c,d)), hinting at the robustness of the phase beyond mean-field theory.

The interpretation of the Chern state as an AHC clarifies an experimental mystery. Experimental phase diagrams in [21] are strongly asymmetric between positive and negative displacement fields D, suggesting that the hBN is only aligned on one side (Fig. 3.1(a)). Comparing experimental data at $\nu = 4$ [21] to SCHF calculations of the charge gap [286], we independently infer that Chern insulator appears when the electrons are polarized away from the hBN potential (moiré-distant) instead of towards the hBN potential (moiré-proximate). We also predict that AHC in samples without moiré would have a characteristic density pattern and $\sigma^{xy} = e^2/h$ over extended density ranges.

The ubiquity of the AHC phase in our phase diagram hints at a universal physical mechanism stabilizing the formation of a Chern band, while the relatively uniform charge density of the AHC state points to a translation-breaking mechanism different from that of the Wigner crystal. A deeper analytic understanding of the origin of the AHC phase is clearly needed, which will be addressed in a forthcoming work [297].

Note added

Towards the completion of the project, four interesting related papers [298–301] were posted, which agree with our finding in the areas of overlap.

Acknowledgements

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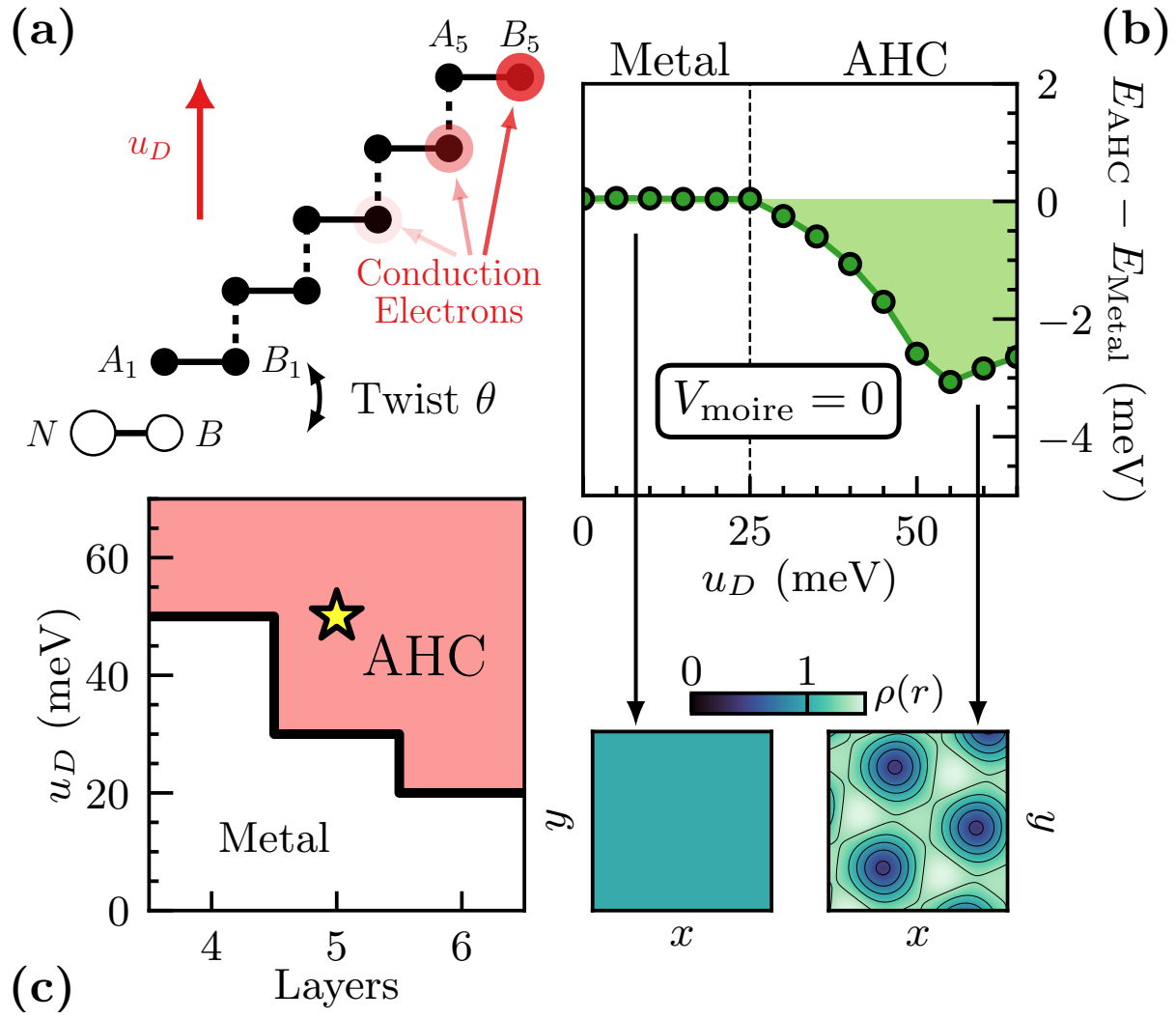


Figure 3.1: (a) Side view of rhombohedral pentalayer graphene aligned to an hBN substrate on the bottom with a twist angle θ. The potential $u_D > 0$ polarizes electrons away from the hBN, making them "moiré-distant". (b) The energy difference per electron between a graphene translation symmetric metal and an anomalous Hall crystal (AHC) within SCHF as a function of u_D in rhombohedral pentalayer graphene at *zero* moiré potential $V_1 = 0$. The AHC is the ground state beyond $u_D = 25$ meV. Typical charge densities $\rho(r)$ (lower right) show spontaneous translation symmetry breaking in the AHC. (c) Phase diagram of rhombohedral multilayer graphene for $N_L = 4 - 6$ layers with phase boundary from SCHF. The critical displacement field u_D^* decreases with increasing N_L. Yellow star: estimated experimental parameters [21].

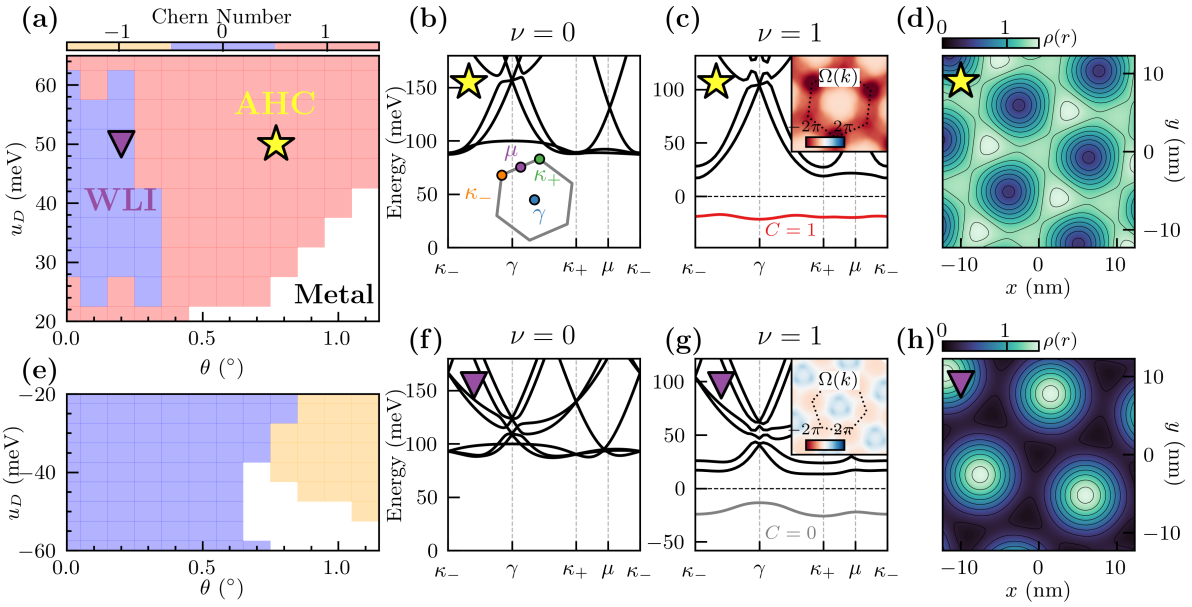


Figure 3.2: (a,e) The phase diagram of moiré rhombohedral pentalayer graphene as a function of the interlayer potential difference u_D, and the twist angle θ. The colors label the Chern number of the SCHF ground states, with gapless regions shown in white. On the moiré-distant side ($u_D > 0$), the anomalous Hall crystal (AHC) phase has $C = 1$, whereas the Wigner-like insulator (WLI) has $C = 0$. (b,f) Non-interacting band structures at the yellow star $(u_D, \theta) = (50\,\text{meV}, 0.77°)$ and purple triangle $(u_D, \theta) = (50\,\text{meV}, 0.2°)$. Inset in (b): plot of the moiré Brillouin zone. (c,g) SCHF bandstructure at $\nu = 1$, finding an AHC and WLI at the yellow star and purple triangle respectively. Insets: Berry curvature $\Omega(\boldsymbol{k})$ of the occupied band. (d,h) Charge densities $\rho(\boldsymbol{r})$ of the AHC (d) and WLI (h). The AHC charge density resembles a honeycomb lattice, whereas the WLI charge density resembles a triangular lattice.

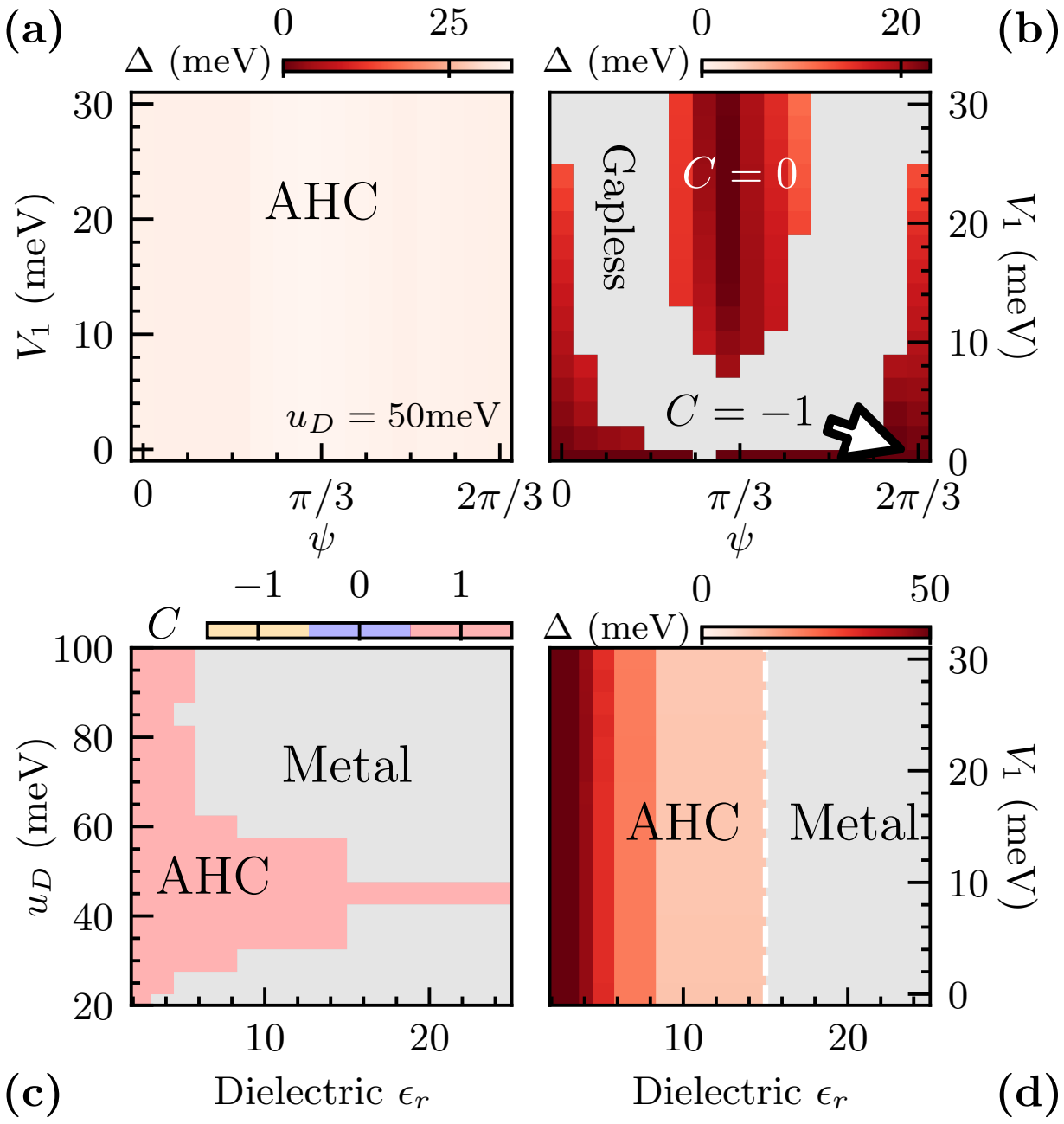


Figure 3.3: (a) The featureless charge gap of rhombohedral pentalayer graphene at $\nu = 1$ on the moiré-distant side ($u_D = 50$ meV) as the moiré strength V_1 and phase ψ vary. (b) Same on the moiré-proximate side ($u_D = -50$ meV), with a trivial insulator separated from a small Chern insulating phase by a metallic region. (c) The Chern number with respect to u_D and dielectric constant ϵ_r on the moiré-distant side, showing the AHC survives to relatively weak interaction strengths near $u_D \sim 50$ meV. (d) The charge gap at $u_D = 50$ meV versus moiré strength V_1 and dielectric ϵ_r. The AHC phase reaches $V_1 = 0$, demonstrating interaction-driven spontaneous translation symmetry breaking. All data from SCHF at $\theta = 0.77°$.

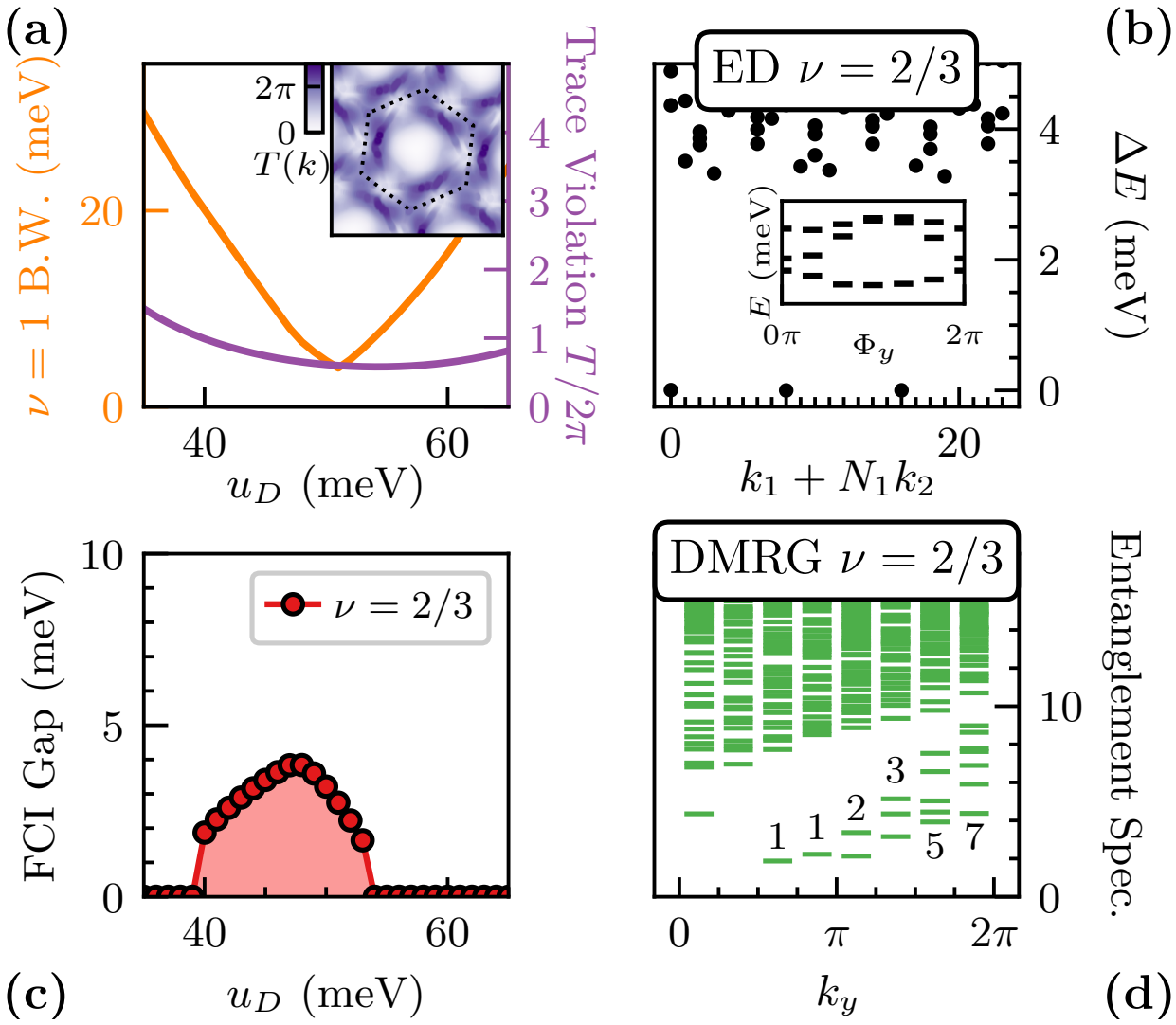


Figure 3.4: (a) Band conditions of the AHC phase as a function of u_D. Both the bandwidth and the violation of trace condition are minimized near $u_D = 50$ meV. (b) Many-body spectrum from ED on a 4×6 system at $\nu = 2/3$. Inset: spectral flow of the three degenerate ground states, a signature of the FCI ground state [57, 189, 192]. (c) The FCI gap at $\nu = 2/3$ as a function of u_D obtained in ED. The FCI gap is nonzero around 40 to 55 meV, and gives way to metals beyond this regime due to large bandwidths. (d) Entanglement spectrum obtained in DMRG at $\nu = 2/3$ with $L_y = 8$. The partition number counting of states below the entanglement gap is consistent with the chiral edge modes associated with the Laughlin states, confirming the state is an FCI [289]. Parameters: $(u_D, \theta) = (50\,\text{meV}, 0.77°)$ unless specified.

4

Anomalous Hall Crystals in Rhombohedral Multilayer Graphene II: General Mechanism and a Minimal Model

4.1 Introduction

Interacting electrons in the absence of external periodic potential can spontaneously break continuous translation symmetry, a phenomenon first predicted by Wigner in the context of the jellium model [302]. Solid state systems, where translation symmetry is explicitly broken by the atomic potential, seem incompatible with realizing such spontaneous translation breaking. However, continuous translation symmetry can be (approximately) restored at very low electron densities: experiments both with strong magnetic field [303–307] and without [308–315] have reported Wigner crystals. The profusion of two-dimensional materials with controllable electron densities therefore presents an excellent opportunity to explore the physics of electron crystallization, yet little has been realized, or even predicted, beyond Wigner crystals.

While Wigner crystals are essentially classical crystals of localized electrons, it is in principle possible to generate a topologically nontrivial phase under spontaneous translation symmetry breaking. This was first borne out in *Hall crystals*, a topologically nontrivial state formed by electrons in a Landau level spontaneously breaking translation symmetry [115, 283, 316, 317]. Surprisingly, recent experiments on rhombohedral multilayer graphene (RMG) [21] suggested an avenue for realizing such a topological Wigner crystal at zero magnetic field.

There, pentalayer RMG, with a moiré potential induced by an aligned hBN crystal on one side, shows both Chern insulating (CI) and fractional Chern insulating (FCI) states at zero magnetic field, also called fractional quantum anomalous Hall (FQAH) states [53, 55, 188, 191, 318, 319]. These states appear under a sizable perpendicular displacement field that polarizes the electrons to one side — and combined experimental [21, 258] and theoretical evidence [298–301, 320] indicate it is the side *away* from the hBN moiré, strongly suppressing the moiré potential. This creates a puzzle: how can such a weak moiré potential stabilize a Chern insulator?

Remarkably, self-consistent Hartree-Fock (SCHF) calculations in models of RMG revealed that the Chern insulator is stable even in the absence of moiré potential [299, 320]. In this limit the state is interpreted as spontaneously breaking translation symmetry. To emphasize that it appears at *zero* magnetic field, the state was named an *anomalous* Hall crystal. In mean-field treatments of models without the moiré potential, the state is surprisingly stable to various perturbations, such as changes in the microscopic hopping parameters. The role of the moiré potential in experiment is currently under debate [301]. However, the ubiquity of the AHC phase in theoretical models without a moiré potential is in strong contrast to the case of strong moiré limit.

The appearance of AHC ground states in these models poses a fundamental conceptual question: when and why do these models give rise to a topological state? Due to its nontrivial topology, the AHC state does not admit a simple classical explanation for its origin á la Wigner crystals. In this paper, we shed light on a simple yet fully quantum-mechanical mechanism underlying the origin of the AHC ground states.

Beyond the fundamental interest in its stability, the AHC provides a mechanism to generate nontrivial band topology different from previous examples such as nonzero magnetic field, band inversion [321, 321–323] and background skyrmionic textures [163, 324–329]. Here, it is the spontaneously broken translation symmetry that predominantly generates the requisite Berry curvature. The source of this Berry curvature is the interference of plane waves: after translation breaking, the state at each crystalline momentum is a superposition of several plane waves whose phase winding generates Berry curvature [330]. In fact, the Chern number is highly constrained by the interference pattern between a small subset of plane waves near the high-symmetry points.

To study the stability and origin of topology, we construct a minimal phenomenological model, the "three-patch model," wherein we keep a minimal number of plane waves to represent translation-broken states with different Chern numbers. This amounts to keeping only three electrons and seven plane wave states.

Despite its simplicity, the three-patch model both illustrates when the AHC state is energetically favorable, and matches realistic phase diagrams closely. Particularly, we show that the stability of the AHC phase depends

on the scattering phases between plane waves, which are in turn dictated by the quantum geometric phases of the plane wave states. Furthermore, the three-patch model reproduces the full Hartree-Fock prediction for the phase boundary between the AHC and a trivial crystal with striking accuracy, serving as *a posteriori* confirmation of its generality and applicability. The three-patch model thus gives a tractable way to understand the anomalous Hall crystal mechanism of generating quantized anomalous Hall states.

The remainder of this work is organized as follows. In Section 4.2 we introduce the "three-patch model", an analytical-solvable model with a stable anomalous Hall crystal phase. Based on this simple picture, we make a number of specific predictions about microscopic models of RMG. By studying the model with RMG spinors, we predict that both the Hartree and Fock interactions cooperate to stabilize the state in a large part of the parameter space. We further predict a quantum phase transition between the $C = 0$ state and $C = 1$ state, driven by a change in quantum geometry.

We then provide an intuitive picture to understand the charge densities of the three-patch states. This also allows us to graphically examine the energetics of the states, which reveals that the Hartree interaction always favors the AHC state. In Section 6.3.2, we perform SCHF calculation of a microscopic Hamiltonian of RMG. We confirm that ground states found in SCHF match the three-patch characteristics, and verify that the SCHF phase diagram is closely mirrored by the prediction by the three-patch model. We conclude with some discussion about future directions in Section 6.4.

4.2 A Three-patch model of an anomalous Hall crystal

We discuss a mechanism for generating Chern bands in models of interacting electrons where both crystallinity, i.e. spontaneous breaking of the continuous translation, *and* band topology are spontaneously generated. Some of the ingredients, which are made more precise below, are (i) a relatively flat energy dispersion in the momentum range of interest, and a large dispersion outside it, (ii) quantum geometry encoded in a multicomponent wavefunction that varies over the Brillouin zone, which seeds the Chern character of the final state and (iii) C_3 symmetry, retained partly for analytical convenience. This general mechanism is encapsulated in a minimal "three-patch model". Solving the minimal model gives a simple criterion based on quantum geometry for obtaining the AHC as the ground state, which is satisfied in a wide range of parameters.

4.2.1 Construction of the Three-Patch Model

We now construct the three-patch model, highlighting the hypotheses it requires.

Consider an electronic band of a multi-orbital continuum quantum system with continuous translation symmetry, C_3 symmetry, and broken time-reversal symmetry (for example, the K valley of rhombohedral multilayer graphene). The Hilbert space is spanned by plane wave states $|\boldsymbol{q}, a\rangle$, where $\boldsymbol{q}$ is the wavevector and a denotes the component (i.e. spin, layer, sublattice, etc.). We choose a basis for components $|a\rangle$ that are irreps of C_3, so that the action of C_3 on the basis states is

$$\hat{C}_3 |\boldsymbol{q}, a\rangle = \exp\left[\frac{2\pi i \ell_a}{3}\right] |C_3 \boldsymbol{q}, a\rangle , \quad \ell_a \in \mathbb{Z}/3\mathbb{Z}. \tag{4.1}$$

Eigenstates $\hat{c}^\dagger_{\boldsymbol{q}} |0\rangle = |\phi_{\boldsymbol{q}}\rangle$ in the band of interest are indexed by unrestricted momentum $\boldsymbol{q}$ such that

$$\hat{h} |\phi_{\boldsymbol{q}}\rangle = E_{\boldsymbol{q}} |\phi_{\boldsymbol{q}}\rangle , \quad \hat{T}_{\boldsymbol{r}} |\phi_{\boldsymbol{q}}\rangle = e^{i\boldsymbol{q}\cdot\boldsymbol{r}} |\phi_{\boldsymbol{q}}\rangle , \tag{4.2}$$

where $\hat{h}$ is the single-particle Hamiltonian of the system, and $\hat{T}_{\boldsymbol{r}'}\phi(\boldsymbol{r}) = \phi(\boldsymbol{r} + \boldsymbol{r}')$. These wavefunctions therefore take the unnormalized plane wave form

$$\phi^a_{\boldsymbol{q}}(\boldsymbol{r}) = \langle \boldsymbol{r}, a | \phi_{\boldsymbol{q}} \rangle = \chi^a(\boldsymbol{q}) e^{i\boldsymbol{q}\cdot\boldsymbol{r}}. \tag{4.3}$$

We also define $\chi(\boldsymbol{q}) = \sum_a \chi^a(\boldsymbol{q}) |a\rangle$ to be the orbital-space spinor.

We add density-density interactions in the usual way:

$$\begin{aligned} \hat{H}_0 &= \hat{h} + \hat{H}_{\text{int}} \\ &= \hat{h} + \frac{1}{2}\int d^2\boldsymbol{r}\, d^2\boldsymbol{r}'\, V(\boldsymbol{r} - \boldsymbol{r}') : \hat{\rho}(\boldsymbol{r})\hat{\rho}(\boldsymbol{r}') :, \end{aligned} \tag{4.4}$$

where $\hat{\rho}(\boldsymbol{r})$ is the sum of real space densities over all components.

We restrict our attention to a fixed electron density n, where we will study the energetic competition between crystalline insulating ground states. In anticipation of spontaneous translation symmetry breaking, we consider a hexagonal "superlattice" generated by $\boldsymbol{a}_1, \boldsymbol{a}_2$, shown in Fig. 4.1(a). The superlattice size is chosen such that there is one electron per unit cell i.e. filling $\nu = 1$. While this is *not* due to an explicitly translation-breaking

superlattice potential (e.g. moiré potential), we adopt notation from that context. The band structure is folded (but not yet electronically hybridized) onto a hexagonal mini Brillouin zone (mBZ), Fig. 4.1(b), with reciprocal lattice vectors $\boldsymbol{g}_1, \boldsymbol{g}_2$ and three high-symmetry points γ, κ, and κ'. Fig 4.1(b) shows $\boldsymbol{\kappa}_1$, $\boldsymbol{\kappa}_3$, and $\boldsymbol{\kappa}_5$ get folded to κ, while $\boldsymbol{\kappa}_2$, $\boldsymbol{\kappa}_4$ and $\boldsymbol{\kappa}_6$ are folded to κ'. We choose $\boldsymbol{\gamma}$ at a C_3 high symmetry point before translation breaking. The folded single-particle band structure is three-fold degenerate on the edge of the bottom band at κ and κ', which we call mini-valleys. The C_3 symmetry enforce degeneracy of the kinetic energy $E_{\boldsymbol{\kappa}_j} = E_{\boldsymbol{\kappa}_{j+2}}$, so the folded band structure is three-fold degenerate.

To make analytic progress, we impose three physical conditions on the ground states:

1. C_3-preserving Crystallization: spontaneous symmetry breaking of continuous translations while preserving C_3. This gives a hexagonal "superlattice" with three C_3 invariant momenta γ, κ, and κ'.

2. Weak coupling mean field: the ground state is a Slater determinant state that only hybridizes energetically proximate bands.

3. Three-patch assumption: the single-particle orbitals that make up the ground state wavefunction is essentially homogeneous in large patches around the three C_3 invariant momenta. This is in spirit similar to previous "patch models" or "hotspot models" in the superconductivity community [331–345].

We will argue in App. D.4 that these three conditions can be physically motivated if the band dispersion has a flat bottom and a dispersive edge. Moreover, we verify these three conditions are consistent with SCHF ground states of microscopic model of RMG in Sec. 6.3.2.

Under these assumptions, the ground states of $\hat{H}_0$ can be physically modeled by C_3 symmetric Slater determinants with one electron at each of the high symmetry points. This procedure is detailed in App. D.4. Briefly, this follows by taking a single representative $\boldsymbol{k}$-point for each high-symmetry patch, which we conveniently take to be the high-symmetry points. Since the state is weakly coupled, only low-energy bands are allowed to hybridize, giving three states at κ, κ' due to symmetry-enforced band crossings, but generically only one at γ, as shown in

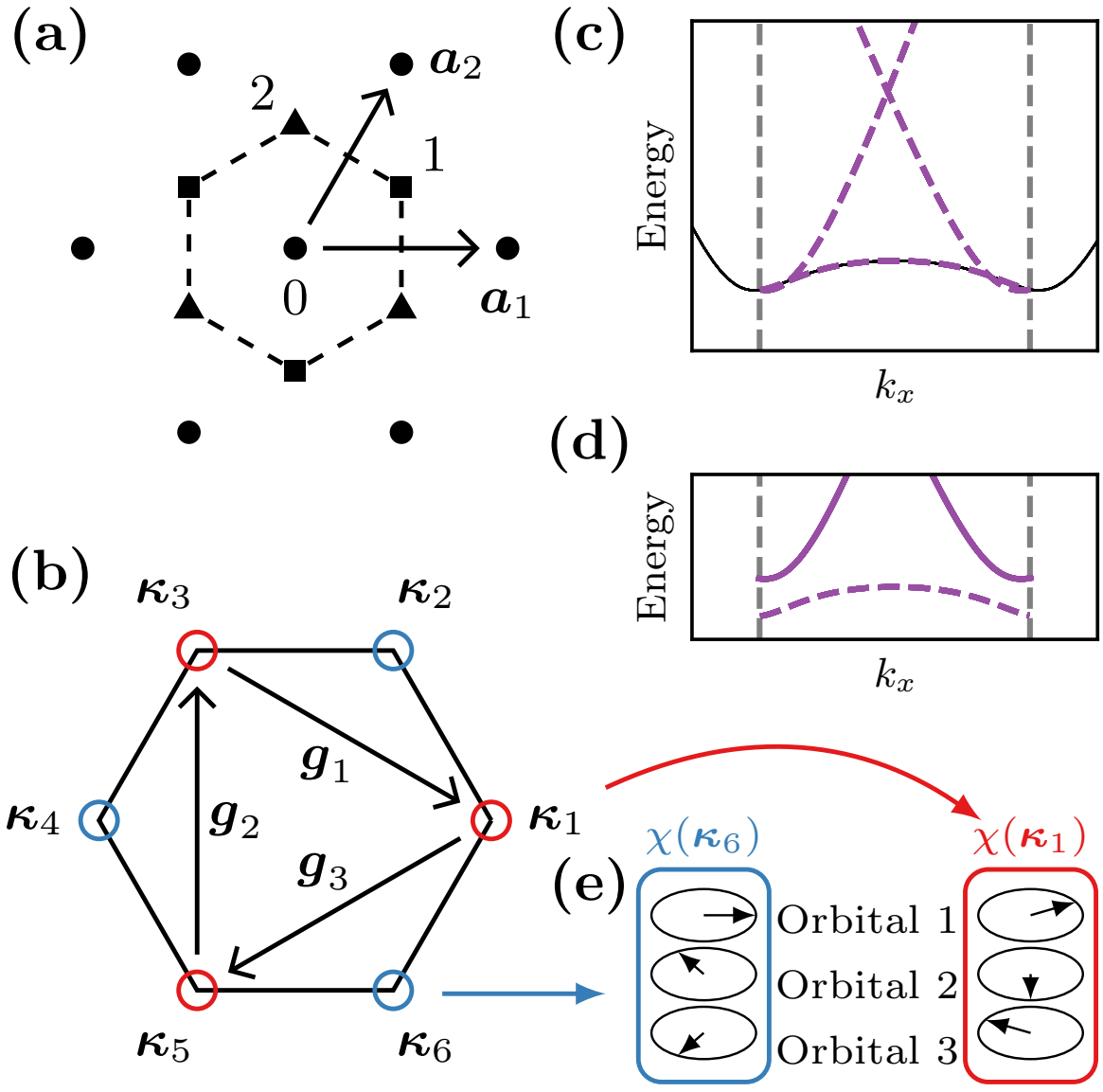


Figure 4.1: Setup of the three-patch model. (a) Real space geometry of the superlattice. The three inequivalent Wyckoff positions are labeled by circles, squares, and triangles. (b) Mini Brillouin zone (mBZ) and associated high symmetry points. The points $\boldsymbol{\kappa}_1$, $\boldsymbol{\kappa}_3$, $\boldsymbol{\kappa}_5$ span the κ point, while $\boldsymbol{\kappa}_2$, $\boldsymbol{\kappa}_4$, and $\boldsymbol{\kappa}_6$ span the κ' point. (c) Schematic folded band structure of the model in the mBZ. The κ and κ' points have three-fold degeneracies. (d) Band gaps are opened up along mBZ boundaries by interactions, which crucially depend on (e) the nontrivial quantum geometry of multi-component wavefunctions.

Fig. 4.1(c). The only possible single-particle wavefunctions are

$$\begin{aligned}\psi_{\gamma,\ell_\gamma=0}(\boldsymbol{r}) &= \chi(\gamma) \\ \psi_{\kappa,\ell_\kappa}(\boldsymbol{r}) &= \frac{1}{\sqrt{3}}\sum_{j=1,3,5}\exp\left[\frac{2\pi i\ell_\kappa}{3}\frac{j}{2}\right]e^{i\boldsymbol{\kappa}_j\cdot\boldsymbol{r}}\chi(\boldsymbol{\kappa}_j), \\ \psi_{\kappa',\ell_{\kappa'}}(\boldsymbol{r}) &= \frac{1}{\sqrt{3}}\sum_{j=2,4,6}\exp\left[\frac{2\pi i\ell_{\kappa'}}{3}\frac{j}{2}\right]e^{i\boldsymbol{\kappa}_j\cdot\boldsymbol{r}}\chi(\boldsymbol{\kappa}_j),\end{aligned} \tag{4.5}$$

where we fix the gauge $\chi(\boldsymbol{\kappa}_{j+2}) = \hat{C}_3\chi(\boldsymbol{\kappa}_j)$, and the states are indexed by C_3 angular momenta ℓ_ξ defined by

$$C_3\left|\psi_{\xi,\ell_\xi}\right\rangle = \exp\left[\frac{2\pi i\ell_\xi}{3}\right]\left|\psi_{\xi,\ell_\xi}\right\rangle. \tag{4.6}$$

In the above, we fixed the gauge such that $\ell_\gamma = 0$.

Given the discrete choices at each momentum point, there are exactly 9 Slater determinant states $|\Psi_{\ell_\kappa,\ell_{\kappa'}}\rangle$ at filling one electron per k-point, which are fully characterized by ℓ_κ and $\ell_{\kappa'}$. In first quantization,

$$\Psi_{\ell_\kappa,\ell_{\kappa'}}(\boldsymbol{r}_1,\boldsymbol{r}_2,\boldsymbol{r}_3) = \mathcal{A}[\psi_\gamma(\boldsymbol{r}_1),\psi_{\kappa,\ell_\kappa}(\boldsymbol{r}_2),\psi_{\kappa',\ell_{\kappa'}}(\boldsymbol{r}_3)], \tag{4.7}$$

where $\mathcal{A}$ is the antisymmetrization operator. Each $|\Psi_{\ell_\kappa,\ell_{\kappa'}}\rangle$ captures the character of a full 2D state whose Chern number is given by

$$C \equiv \ell_\kappa + \ell_{\kappa'} \mod 3. \tag{4.8}$$

This follows from the theory of symmetry indicators [322, 346–351], which posits that the sum of angular momentum at C_3 symmetric momenta determines the Chern number mod 3 i.e. $C \equiv \ell_\gamma + \ell_\kappa + \ell_{\kappa'}$. This can be thought of as a generalization of the famous Fu-Kane formula for inversion symmetric topological insulators [322], which was later generalized to Chern bands with discrete rotation symmetries [346, 347]. We choose the C_3 origin such that $\ell_\gamma = 0$ above, implying Eq. (4.8). We will refer to three possible values of C as simply $C = 1, 0, -1$, although in principle the Chern number can be larger.

Due to C_3, these Slater determinant states have degenerate kinetic energy, so their energetic competition is determined solely through interactions:

$$E(\ell_\kappa,\ell_{\kappa'}) = \langle\Psi_{\ell_\kappa,\ell_{\kappa'}}|\hat{H}_{\text{int}}|\Psi_{\ell_\kappa,\ell_{\kappa'}}\rangle. \tag{4.9}$$

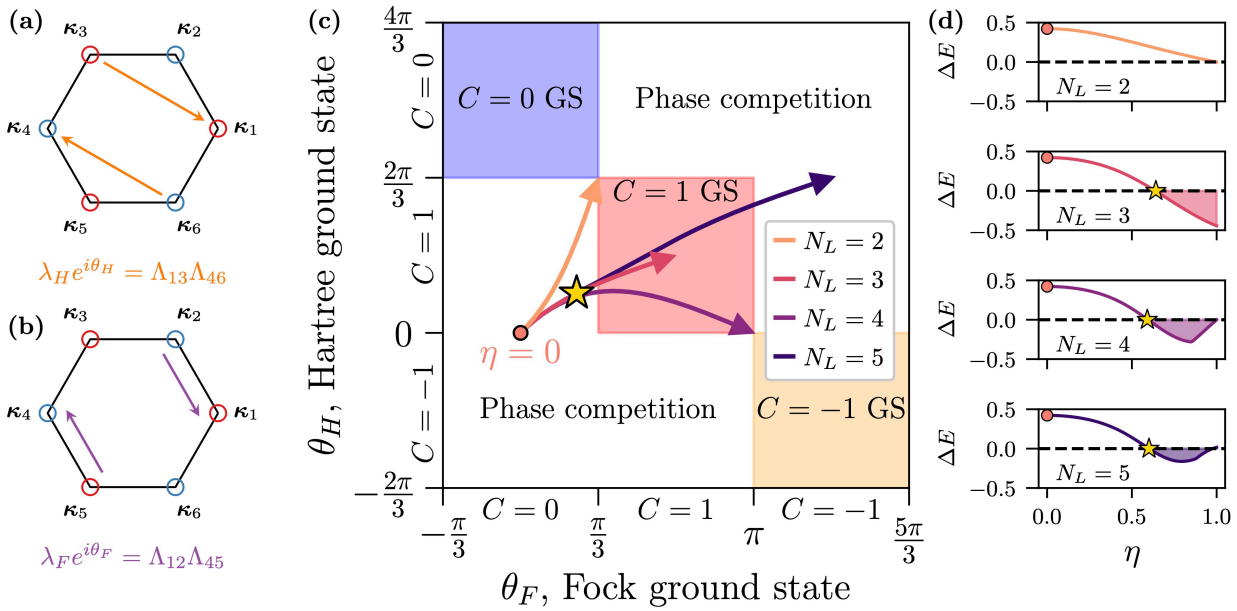


Figure 4.2: Predicted phase diagram by the three-patch model. (a) and (b) show the scattering processes contributing to the Hartree and Fock terms $E_H(C)$ and $E_F(C)$, and their associated form factors. (c) The angles of the form factors, θ_H and θ_F, determine the Chern number favored by the Hartree and Fock terms respectively. Colored regions are where the Hartree and Fock ground states agree. Curves correspond to the evolution of θ_H and θ_F as a function of $0 \leq \eta \leq 1$, which tunes the RMG-inspired spinors $\chi^{(N_L)}$ for different number of components N_L,. The transition points where the $C = 1$ state becomes the ground state in the three-patch model for $N_L = 3, 4, 5$ are very close to each other, and are marked by the yellow star in the phase diagram. The orange dot marks the $\eta = 0$ point. (d) The energy difference between the $C = 1$ state and the lowest-energy $C \neq 1$ state in the three-patch model for different number of components N_L as a function of η. Transition points to the $C = 1$ states are marked by yellow stars.

Below we will evaluate the energies $E(\ell_\kappa, \ell_{\kappa'})$ analytically to determine when a $C = 1$ state is stabilized.

4.2.2 Analytical solution of the three-patch model

We will now perform a full analytical evaluation of the three-patch energy in Eq. (4.9). As it is the energy of a Slater determinant state, we can perform Wick contraction to separate Hartree and Fock contributions:

$$E(\ell_\kappa, \ell_{\kappa'}) = \frac{1}{2} \sum_{\substack{\xi,\xi' \in \\ \{\gamma,\kappa,\kappa'\}}} E_H(\xi, \xi') + E_F(\xi, \xi'). \tag{4.10}$$

Here, the Hartree and Fock (exchange) energies E_H, E_F between single particle states are defined as

$$E_H(\xi,\xi') = \int d^2\boldsymbol{r} d^2\boldsymbol{r}' \overline{\psi^a_\xi(\boldsymbol{r})}\psi^a_\xi(\boldsymbol{r}) V(\boldsymbol{r}-\boldsymbol{r}') \overline{\psi^b_{\xi'}(\boldsymbol{r})}\psi^b_{\xi'}(\boldsymbol{r}), \quad E_F(\xi,\xi') = -\int d^2\boldsymbol{r} d^2\boldsymbol{r}' \overline{\psi^a_{\xi'}(\boldsymbol{r})}\psi^a_\xi(\boldsymbol{r}) V(\boldsymbol{r}-\boldsymbol{r}') \overline{\psi^b_\xi(\boldsymbol{r}')}\psi^b_{\xi'}(\boldsymbol{r}' \tag{4.11}$$

where again $\psi^a_\xi(\boldsymbol{r})$ is the component a of the single-particle wavefunctions at high symmetry point ξ, and the

repeated indiced are summed over.

Naively, the interaction energy Eq. (4.10) comes with three pairs of terms. However, translation by $\tilde{\boldsymbol{a}} = (\boldsymbol{a}_1 + \boldsymbol{a}_2)/3$ shifts angular momenta $\ell_\kappa \to \ell_\kappa + 1$, and $\ell_{\kappa'} \to \ell_{\kappa'} - 1$ as we show in App. D.3. Thus, the energy can only depend on $\ell_\kappa + \ell_{\kappa'} \equiv C \mod 3$. Since the $(\xi, \xi') = (\gamma, \kappa)$ term does not depend on $\ell_{\kappa'}$, it cannot have dependence on angular momenta; the same goes for the (γ, κ') term. The only term with dependence on C is the (κ, κ') term.

By explicit computation in Fourier space, we obtain the energy of the (κ, κ') interaction term as a function of $\ell_\kappa, \ell_{\kappa'}$:

$$E_H(\kappa, \kappa') = \frac{1}{9} \sum_{m,n \in \{1,3,5\}} \sum_{o,p \in \{2,4,6\}} e^{2\pi i[(n-m)\ell_\kappa + (p-o)\ell_{\kappa'}]/6} \Lambda_{m,n} \Lambda_{o,p} V_{\boldsymbol{\kappa}_n - \boldsymbol{\kappa}_m} \delta_{\boldsymbol{\kappa_n} - \boldsymbol{\kappa}_m, \boldsymbol{\kappa}_o - \boldsymbol{\kappa}_p}, \tag{4.12}$$

$$E_F(\kappa, \kappa') = -\frac{1}{9} \sum_{m,n \in \{1,3,5\}} \sum_{o,p \in \{2,4,6\}} e^{2\pi i[(n-m)\ell_\kappa + (p-o)\ell_{\kappa'}]/6} \Lambda_{m,p} \Lambda_{o,n} V_{\boldsymbol{\kappa}_p - \boldsymbol{\kappa}_m} \delta_{\boldsymbol{\kappa_p} - \boldsymbol{\kappa}_m, \boldsymbol{\kappa}_o - \boldsymbol{\kappa}_n}, \tag{4.13}$$

where we defined form factors $\Lambda_{m,n} = \chi^\dagger(\boldsymbol{\kappa}_m)\chi(\boldsymbol{\kappa}_n)$, and $V_{\boldsymbol{q}}$ is the Fourier transform of interaction $V(\boldsymbol{r})$ at momentum $\boldsymbol{q}$. The physical meaning of each term in this sum is clear: they correspond to different scattering processes with momentum transfer $\boldsymbol{\kappa}_m - \boldsymbol{\kappa}_n$.

Only the scattering processes depicted in Fig. 4.2(a, b) and their C_3 rotated counterparts have $\ell_\kappa, \ell_{\kappa'}$ dependence. Collecting those together, and using the C_3 action $\Lambda_{n,m} = \Lambda_{n+2,m+2}$ to simplify the expression, we get

$$E(C) = V_{\boldsymbol{g}_1} \epsilon_H(C) + V_{\boldsymbol{\kappa}_1} \epsilon_F(C) + E_0, \tag{4.14}$$

$$\begin{aligned} \epsilon_H(C) &= \frac{2}{3} \mathrm{Re}\left[e^{2\pi i C/3} \Lambda_{1,3}\Lambda_{4,6}\right] \\ &= \frac{2}{3}\lambda_H \cos\left(\theta_H + \frac{2\pi C}{3}\right), \end{aligned} \tag{4.15}$$

$$\begin{aligned} \epsilon_F(C) &= -\frac{2}{3} \mathrm{Re}\left[e^{-2\pi i C/3} \Lambda_{1,2}\Lambda_{4,5}\right] \\ &= -\frac{2}{3}\lambda_F \cos\left(\theta_F - \frac{2\pi C}{3}\right). \end{aligned} \tag{4.16}$$

Here E_0 is a term that does not depend on the Chern number C. We parameterized the form factors in terms of two angles

$$\Lambda_{1,3}\Lambda_{4,6} = \lambda_H e^{i\theta_H}, \quad \Lambda_{1,2}\Lambda_{4,5} = \lambda_F e^{i\theta_F}. \tag{4.17}$$

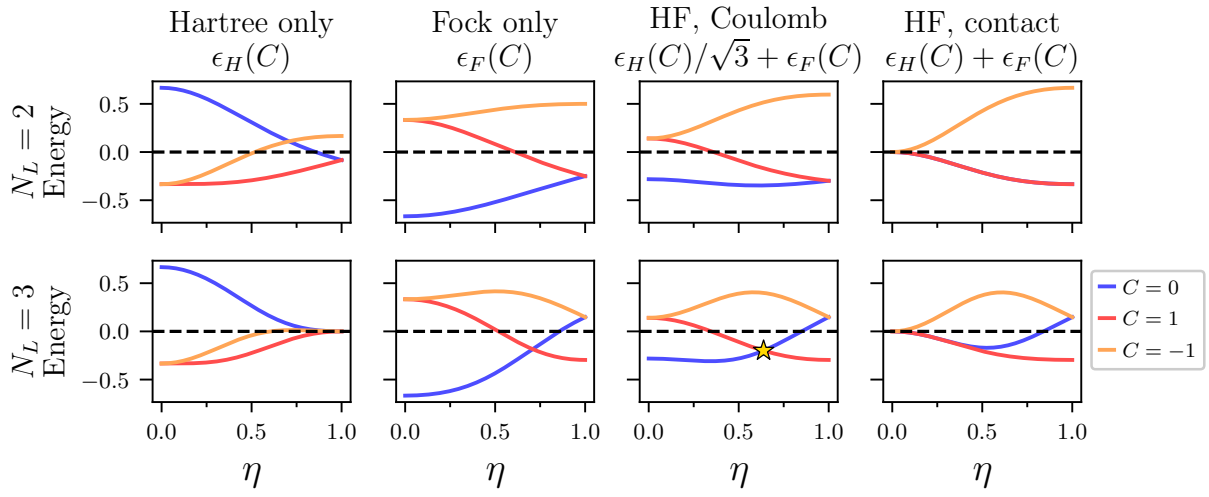


Figure 4.3: Hartree and Fock energies of different C states. The rows correspond to different number of components. The first two columns correspond to the Hartree and Fock terms $E_H(C)$ and $E_F(C)$, plotted with $V_{\boldsymbol{g}} = V_{\boldsymbol{\kappa}} = 1$. "HF, contact" is the sum of Hartree and Fock energies with $V_{\boldsymbol{g}} = V_{\boldsymbol{\kappa}} = 1$, while "HF, Coulomb" is computed with $V_{\boldsymbol{\kappa}} = 1, V_{\boldsymbol{g}} = 1/\sqrt{3}$. The yellow star marks the phase transition point between $C = 0$ and $C = 1$ using Coulomb interaction.

The ground state of $\epsilon_H(\epsilon_F)$ is determined by $\theta_H(\theta_F)$ alone. For certain values of θ_H and θ_F, the ground states of the Hartree and the Fock terms agree, uniquely determining the ground state. In Fig. 4.2(c), we show a 2D phase diagram of the three-patch model as a function of θ_H and θ_F. The "frustration free" regions where the Hartree ground state and the Fock ground state agree are shaded with color.

We note these angles are connected to the Pancharatnam overlap

$$P = \prod_{i=1}^{6} \Lambda_{i-1,i} = \lambda_F^3 e^{3i\theta_F}, \tag{4.18}$$

where χ_0 is defined to be χ_6. The angle $\arg P$ records the accumulated geometric phase by a path along the mBZ boundaries. In the case where spinors vary slowly along the path, the phase $\arg P = 3\theta_F$ can be approximated by the total Berry curvature within the Brillouin zone. The Fock term starts favoring a $C = 1$ state when $\theta_F > \pi/3$, which approximately translates to $\int_{\mathrm{mBZ}} \Omega(\boldsymbol{k}) \gtrsim \pi$, i.e. half the Berry curvature necessary for the $C = 1$ state. The *Hartree* term, however, favors the $C = 1$ state at arbitrarily small η, whose competition with the Fock term determines the ultimate ground state.

4.2.3 Topological Transition from RMG spinors

The phases θ_H and θ_F can depend on the detail of the spinor structure, which in turn determine whether the $C = 1$ is stable. We now consider a concrete example of the spinor to examine the behavior of $\epsilon_H(C)$ and $\epsilon_F(C)$. We take the following spinor motivated by the physics of RMG systems:

$$\chi^{(N_L)}(\boldsymbol{\kappa}_{n+1}) = \mathcal{N}(1, \eta e^{2\pi i n/6}, \ldots, \eta^{N_L-1} e^{2\pi i (N_L-1)n/6}), \tag{4.19}$$

where η is a positive number less than 1, $\mathcal{N} = 1/\sqrt{\sum_{a=0}^{N_L-1} \eta^{2a}}$ is the normalization factor. The components have C_3 action given by $\hat{C}_3 = \mathrm{diag}(1, e^{2\pi i/3}, \ldots, e^{2\pi i (N_L-1)/3})$.

In Fig. 4.2(c), we show parametric curves $(\theta_F(\eta), \theta_H(\eta))$ for $0 \leq \eta \leq 1$ for various values of $N_L = 2 - 5$. We see that at $N_L = 2$, the curve never enters the region where both Hartree and Fock terms favor the $C = 1$ ground state (labeled as "$C = 1$ GS" in the figure), while for other number of layers, a significant portion of the curve is inside the "$C = 1$ GS" region. We conclude that the AHC state forms at a large range of η for $N_L = 3 - 5$, regardless of the values of $V_{\boldsymbol{g}}$ and $V_{\boldsymbol{\kappa}}$.

To see this more quantitatively, we evaluate the Hartree and Fock energies explicitly. They can be organized in powers of η (See App. D.1):

$$\epsilon_H(C) = \frac{2}{3} \sum_{a,b=0}^{N_L-1} \mathcal{N}^4 \eta^{a+b} \cos\left(\frac{2\pi(a+b+C)}{3}\right) \tag{4.20}$$

$$\epsilon_F(C) = -\frac{2}{3} \sum_{a,b=0}^{N_L-1} \mathcal{N}^4 \eta^{a+b} \cos\left(\frac{2\pi(a+b-2C)}{6}\right). \tag{4.21}$$

We consider two types of interactions: short-range interaction, i.e. contact interaction $V(\boldsymbol{r}) \propto \delta(\boldsymbol{r}), V_{\boldsymbol{q}} =$ const. ($V_{\boldsymbol{\kappa}_1} = V_{\boldsymbol{g}_1}$), and long-range interaction, i.e. Coulomb interaction $V(\boldsymbol{r}) \propto 1/|\boldsymbol{r}|, V_{\boldsymbol{q}} \propto 1/|\boldsymbol{q}|$ ($V_{\boldsymbol{\kappa}_1} = \sqrt{3} V_{\boldsymbol{g}_1}$).

Let us define ΔE to be the energy difference between the Coulomb energy of the $C = 1$ state and the lowest energy state:

$$\Delta E \coloneqq \frac{\epsilon_H(1)}{\sqrt{3}} + \epsilon_F(1) - \min_{C=0,-1} \left[\frac{\epsilon_H(C)}{\sqrt{3}} + \epsilon_F(C)\right]. \tag{4.22}$$

Values $\Delta E < 0$ signal that $C = 1$ is the ground state with the Coulomb interaction. We plot this for $N_L = 2 - 5$ in Fig. 4.2(d). We see that for $N_L \geq 3$, $C = 1$ becomes the ground state at $\eta \sim 0.6$, marked by a yellow star, while the $C = 1$ never becomes the unique ground state for $N_L = 2$. The corresponding critical point is marked in Fig. 4.2(c) as well, where the single yellow star covers all the critical points. As the Hartree term always favors the $C = 1$ state, the transition point is slightly outside of the "$C = 1$ GS" region. We conclude that the AHC state is favored for large enough η for $N_L \geq 3$.

Having understood the case of Coulomb interaction for $N_L = 2 - 5$, we now look at Hartree and Fock contributions separately. Since many of the qualitative features are similar for all $N_L \geq 3$, we focus on $N_L = 2, 3$. We plot the Hartree and Fock energies $(\epsilon_H(C), \epsilon_F(C))$, as well as contact and Coulomb energies $(\epsilon_H(C) + \epsilon_F(C), \epsilon_H(C)/\sqrt{3} + \epsilon_F(C))$ in Fig. 4.3. We note the following features of these plots, which can also be analytically confirmed: 1) Hartree always favors the $C = 1$ state, 2) Fock favors the $C = 0$ for all values of η for $N_L = 2$, but favors $C = 1$ above $\eta \approx 1/\sqrt{2}$ for $N_L \geq 3$, 3) Coulomb interaction energy behaves qualitatively similarly to the Fock interaction energy, and 4) The contact interaction makes $C = 0$ and $C = 1$ exactly degenerate at $N_L = 2$, which is a general feature of $N_L = 2$ spinors (which is true in general for $N_L = 2$, see App. D.2).

We stress again that the energetics of the phase transition for this choice of spinors is mostly governed by the Fock exchange energy, manifesting in the similarity between the Fock energy curve and the Coulomb energy curve. While the Hartree energy always favors the $C = 1$ state, it only shifts the phase transition point slightly.

4.2.4 Three-patch states in real space

In this section, we inspect the charge density of the three-patch states with RMG spinors, discussed in Sec. 4.2.3. These charge densities are not only characteristic features of the different three-patch states, but also let us peek into their energetics in different regimes.

Let us consider the single particle state at κ with angular momentum ℓ_κ. The C_3 symmetry of the spinors $\hat{C}_3\chi(\boldsymbol{\kappa}_i) = \chi(\boldsymbol{\kappa}_{i+2})$, fixes the i dependence of $\chi^a(\boldsymbol{\kappa}_i \in \kappa)$ to the form $\chi^a(\boldsymbol{\kappa}_j) = \exp[2\pi i \ell_a (j-1)/6]\chi^a(\boldsymbol{\kappa}_1)$, where ℓ_a is the angular momentum of spinor component a. Using this fact, we can rewrite each component of $\psi_{\kappa,\ell_\kappa}$ (Eq. (4.5) as follows:

$$\psi^a_{\kappa,\ell_\kappa}(\boldsymbol{r}) = \varphi_{\kappa,\ell_\kappa+\ell_a}(\boldsymbol{r})\chi^a(\boldsymbol{\kappa}_1). \tag{4.23}$$

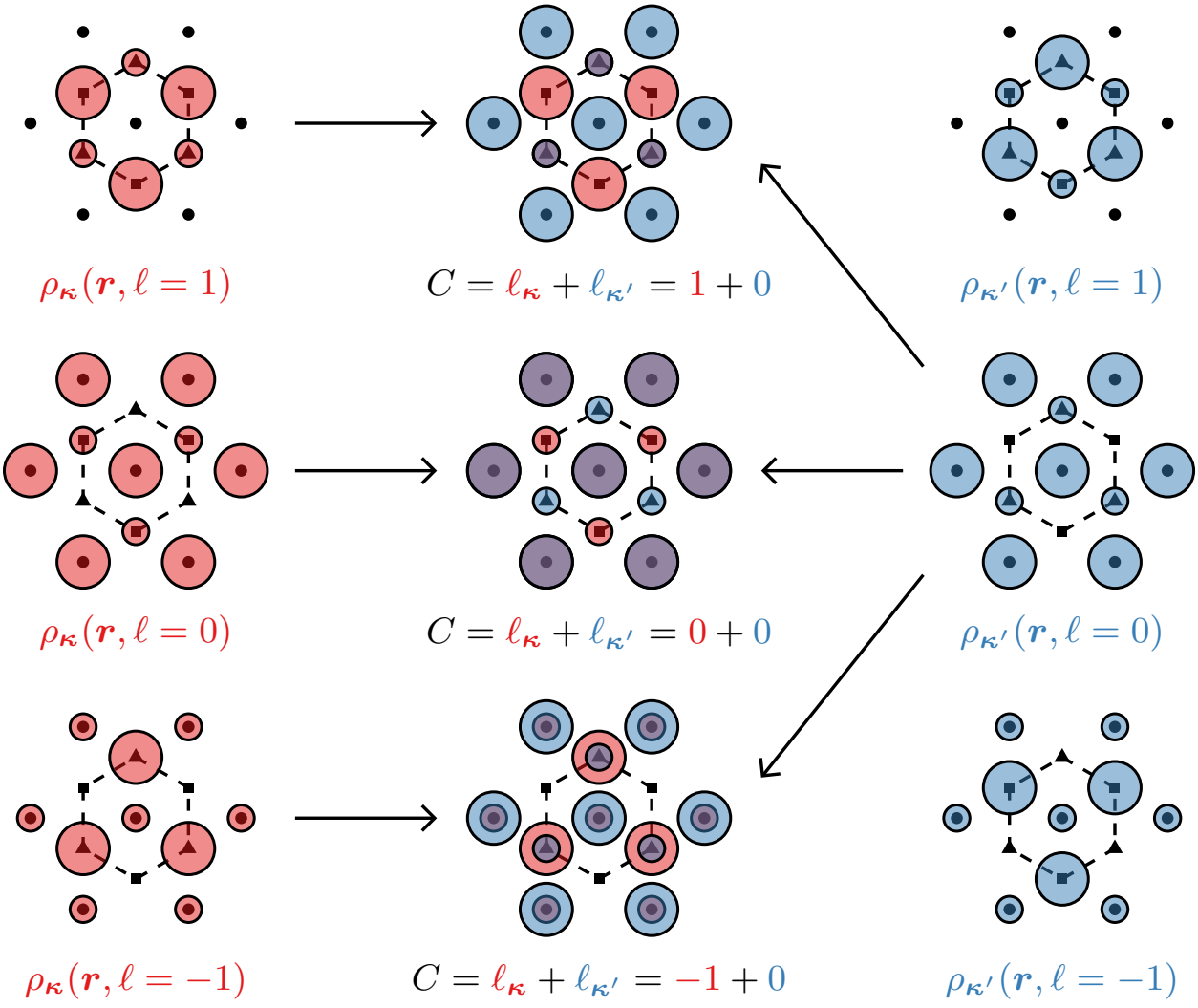


Figure 4.4: Charge densities of competing candidate states of the three patch model. (left): the charge densities at the κ point. The size of the circles corresponds to different amplitudes of charge densities, which come from different components of the spinor. Different angular momentum states labeled by ℓ are related to each other by translation. (right): the charge densities in the κ' point. (center): the total charge densities of states with different Chern numbers.

The spatial wavefunction is given by

$$\varphi_{\kappa,\ell_{\rm tot}}(\boldsymbol{r}) \coloneqq \frac{1}{\sqrt{3}} \sum_{j=1,3,5} e^{2\pi i \ell_{\rm tot} j/6} e^{i\boldsymbol{\kappa}_j \cdot \boldsymbol{r}}, \tag{4.24}$$

in which $\ell_{\rm tot}$ is the total angular momentum of the component. At κ point, it can be written as $\ell_{\rm tot} = \ell_\kappa + \ell_a$. The functions $\varphi_{\kappa'}$ for κ' can be obtained by complex conjugation. Some of the key properties of these wavefunctions are reviewed in App. D.3.

Let us now consider the RMG-inspired spinor, Eq. (4.19), with $N_L = 2$:

$$\chi^{(2)}(\boldsymbol{\kappa}_n) \propto (1, \eta e^{2\pi i (n-1)/6}). \tag{4.25}$$

Since the components have different angular momenta ℓ_a, their charge densities $\rho_{\kappa,\ell_{\rm tot}}(\boldsymbol{r}) = \sum_a |\psi^a_{\kappa,\ell_\kappa}(\boldsymbol{r})|^2$ appear offset from each other, resulting in the charge density pattern shown in the left column of Fig. 4.4. We likewise show the charge densities in κ' in the right column.

The total charge densities are simply the sum of these charge densities. We note that for each Chern number, there are three possible charge densities corresponding to different C_3 angular momenta. In Sec. 4.3.4, we show that these schematic charge densities agree well with actual SCHF charge densities.

These spatial density patterns can be used to understand some aspects of the energetics of the three-patch states. In App. D.5, we show that the three-patch charge densities can be used to understand 1) Hartree energy competition between $C = 1$ and $C = 0$ state, 2) Fock energy competition between $C = 0$ and $C = 1$ at small η, and 3) the stability of the $C = 0$ state under honeycomb potential.

We caution the readers that even though the charge densities provide a simple graphical picture to understand some aspects of the energy competition, it also fails to predict many important features. In particular, it does not capture the strength of the Fock interaction at large η, thus failing to capture the stability of the AHC state.

4.3 Application of the three-patch model to Rhombohedral Multilayer Graphene

We now introduce a streamlined microscopic model of rhombohedral multilayer graphene (RMG) [284]. We note that our model neglects detailed effects such as longer-range hoppings within the rhombohedral graphene, which was shown to not affect the phase diagram strongly [320], lattice relaxation, and the layer-dependence

of the Coulomb interactions. We also neglect the hBN moiré potential [79, 276, 288, 352], but allow for interaction generated spontaneous translation breaking. We show this simplified model hosts an AHC state within SCHF, and furthermore provides a concrete justification for the three-patch model and its approximations.

4.3.1 Hamiltonian

Consider N_L layers of graphene with rhombohedral stacking with creation operators $\hat{c}^\dagger_{\sigma,\ell}$ where $\sigma \in \{A, B\}$ labels sublattice and $\ell \in \{1, \ldots, N_L\}$ labels layer. From the side (Fig. 4.5(d)), RMG forms a staircase with different hopping strengths within and between layers — akin to the SSH chain. The Hamiltonian is

$$\begin{aligned}
\hat{h}_{\mathrm{RMG}} &= \sum_{\boldsymbol{k}} \hat{h}_0(\boldsymbol{k}) + \hat{h}_D(\boldsymbol{k}) \\
\hat{h}_0(\boldsymbol{k}) &= -t_0 S(\boldsymbol{k}) \sum_{\ell=1}^{N_L} c^\dagger_{B,\ell}(\boldsymbol{k}) c_{A,\ell}(\boldsymbol{k}) \\
&\quad + t_1 \sum_{\ell=2}^{N_L} c^\dagger_{B,\ell-1}(\boldsymbol{k}) c_{A,\ell}(\boldsymbol{k}) + \mathrm{h.c.} \\
\hat{h}_D(\boldsymbol{k}) &= -\sum_{\ell=1}^{N_L} u_D(\ell-1)\hat{n}_\ell(\boldsymbol{k})
\end{aligned} \tag{4.26}$$

where $t_0 = 3100\,\mathrm{meV}$ is the intralayer hopping, $t_1 \approx 380\,\mathrm{meV}$ is the interlayer hopping, $S(\boldsymbol{k}) = \sum_{n=0}^{2} e^{i\boldsymbol{k}\cdot\boldsymbol{\delta}_n}$ in terms of graphene interatomic vectors $\boldsymbol{\delta}_n = C_3^{n-1}(0, a/\sqrt{3})$, $a = 0.246\,\mathrm{nm}$ is the graphene lattice constant, and $\hat{n}_\ell(\boldsymbol{k}) = c^\dagger_{A,\ell}(\boldsymbol{k})c_{A,\ell}(\boldsymbol{k}) + c^\dagger_{B,\ell}(\boldsymbol{k})c_{B,\ell}(\boldsymbol{k})$ is the electron density on layer ℓ. Finally, $u_D > 0$ is a displacement field that polarizes electrons in the conduction bands towards the top layer. We have neglected further-neighbor hoppings $t_{i\geq 2}$. While they change the topology of the Fermi surface at small electron density from an annulus to three distinct pockets, a previous SCHF study has shown that the AHC state is unaffected by these further-neighbor hoppings [320]. The model enjoys C_3 rotation symmetry, two-fold rotation C_{2x}, mirror M_x, time-reversal, and standard discrete translations.

We can define a natural length scale of this Hamiltonian by the condition that the first and second term of $\hat{h}_0(\boldsymbol{k})$ in Eq. (4.26) have equal magnitude. Approximating the intra-layer dispersion linearly as $t_0 S(K + \Delta\boldsymbol{k}) = \hbar v_F \Delta\boldsymbol{k} \cdot \sigma$ with $v_F = 1.004 \times 10^6 \mathrm{m\,s^{-1}}$, we find the length scale to be $L_0 = hv_F/t_1 \approx 11\mathrm{nm}$. This natural length scale should be contrasted against the length scale of translation symmetry breaking, coming from either explicit moiré potential or spontaneous symmetry breaking. We expect qualitatively new physics to happen when

these two length scales match. We note that this is precisely what happens for the magic angle of twisted bilayer graphene, whose moiré length scale is also around $10\mathrm{nm}$.

We consider an interacting Hamiltonian

$$\hat{H}_{\mathrm{RMG}} = \hat{h}_{\mathrm{RMG}} + \frac{1}{2A}\sum_{\boldsymbol{q}} V_{\boldsymbol{q}} : \hat{\rho}_{\boldsymbol{q}}\hat{\rho}_{-\boldsymbol{q}} :, \; V_{\boldsymbol{q}} = \frac{2\pi \tanh |\boldsymbol{q}| d}{\epsilon_r \epsilon_0 |\boldsymbol{q}|} \tag{4.27}$$

where $\hat{\rho}_{\boldsymbol{q}}$ is the charge density operator at wavevector $\boldsymbol{q}$, A is the sample area, $V_{\boldsymbol{q}}$ corresponds to the double-gated Coulomb interactions with gate distance $d = 25\,\mathrm{nm}$ and dielectric $\epsilon_r = 5$ (unless otherwise specified). Finally, normal ordering is with respect to the fermionic vacuum at charge neutrality for simplicity. We note this vacuum is strongly renormalized in realistic models in the limit $u_D \to 0$, necessitating "Hartree-Fock subtraction" or other techniques to property renormalize the kinetic term in order to model the realistic system [301]. In the simplified model used here we take only the lowest conduction band to be dynamical.

To perform Hartree-Fock calculations, we focus on the low-energy part of the single-particle spectrum and only keep the states near the K point of graphene. We use graphene Bravais lattice vectors

$$\boldsymbol{R}_1 = a(1,0), \quad \boldsymbol{R}_2 = a(1/2, \sqrt{3}/2), \tag{4.28}$$

with corresponding reciprocal lattice vectors

$$\boldsymbol{G}_1 = \frac{2\pi}{a}(1, 1/\sqrt{3}), \quad \boldsymbol{G}_2 = \frac{2\pi}{a}(0, 2/\sqrt{3}), \tag{4.29}$$

and $\boldsymbol{K} = 2\pi/a(2/3, 0)$. The vicinity of the K point is shown in Fig. 4.5(a).

To account for the possibility of translation symmetry breaking, we consider a "ghost" superlattice generated by the Bravais vectors

$$\boldsymbol{r}_1 = L_s(1,0), \quad \boldsymbol{r}_2 = L_s(1/2, \sqrt{3}/2), \tag{4.30}$$

where L_s is the superlattice scale defined below. The corresponding reciprocal lattice vectors are given by

$$\boldsymbol{g}_1 = \frac{2\pi}{L_s}(1, 1/\sqrt{3}), \quad \boldsymbol{g}_2 = \frac{2\pi}{L_s}(0, 2/\sqrt{3}). \tag{4.31}$$

These generate the mini-Brillouin zone (mBZ) shown in Fig. 4.5(a). Here, we have chosen the mBZ to be cen-

tered at the graphene K point and oriented the same way as the graphene Brillouin zone, with high-symmetry points γ, κ, and κ'.

The superlattice scale L_s is fixed by the electron density n. As we are interested in the behavior of the system at the filling of one electron per superlattice unit cell i.e. $\nu = 1$, we choose L_s to enforce this filling. Since the area of the superlattice unit cell is $\sqrt{3}L_s^2/2$, we have

$$L_s^2 = \frac{2}{\sqrt{3}n}. \tag{4.32}$$

While we do not consider the effect of a moiré potential in this work, we provide a conversion between moiré parameters and superlattice parameters for ease of comparison to experiments with moiré patterns from proximate hBN layers. In the presence of moiré potential, the superlattice constant L_s should be taken to be the same as moiré lattice constant. In the case of the moiré pattern generated by hBN/graphene lattice mismatch, the lattice constant goes as

$$L_s^{\text{moiré}} = a\frac{1+\epsilon}{\sqrt{\epsilon^2 + 2(1+\epsilon)(1-\cos\theta)}}, \tag{4.33}$$

where $\epsilon \approx 0.018$ is the lattice mismatch, and θ is the relative rotation angle between hBN and graphene. The experimental value of $\theta = 0.77^\circ$ [21] corresponds to $L_s \approx 11.5$nm, putting it close to the natural length scale L_0. Plots of this relation are shown in Appendix D.7.

With the above choice of geometry, the single particle states in our mean-field calculations take the form

$$\psi_{\boldsymbol{k}}(\boldsymbol{r}) = \sum_{m,n} c_{\boldsymbol{k}}^{m,n} \phi_{\boldsymbol{k}+m\boldsymbol{g}_1+n\boldsymbol{g}_2}(\boldsymbol{r}), \tag{4.34}$$

where $\phi_{\boldsymbol{k}}$ are eigenstates of the single particle Hamiltonian, and $c_{\boldsymbol{k}}^{m,n}$ are complex coefficients.

4.3.2 Spinor structure of RMG

We now investigate the spinor structure of RMG. As has been shown in Sec. 4.2, a fundamental input to the three-patch model are the spinors at the corners of the mBZ, i.e. the $\boldsymbol{\kappa}_i$ points.

We assume the time-reversal symmetry has been broken by spontaneous valley polarization, and focus on near the graphene K point. We can simplify Eq. (4.26) by approximating $-t_0 S(\boldsymbol{K}+\boldsymbol{k}) \approx v_F \bar{k}$, where $v_F \approx$

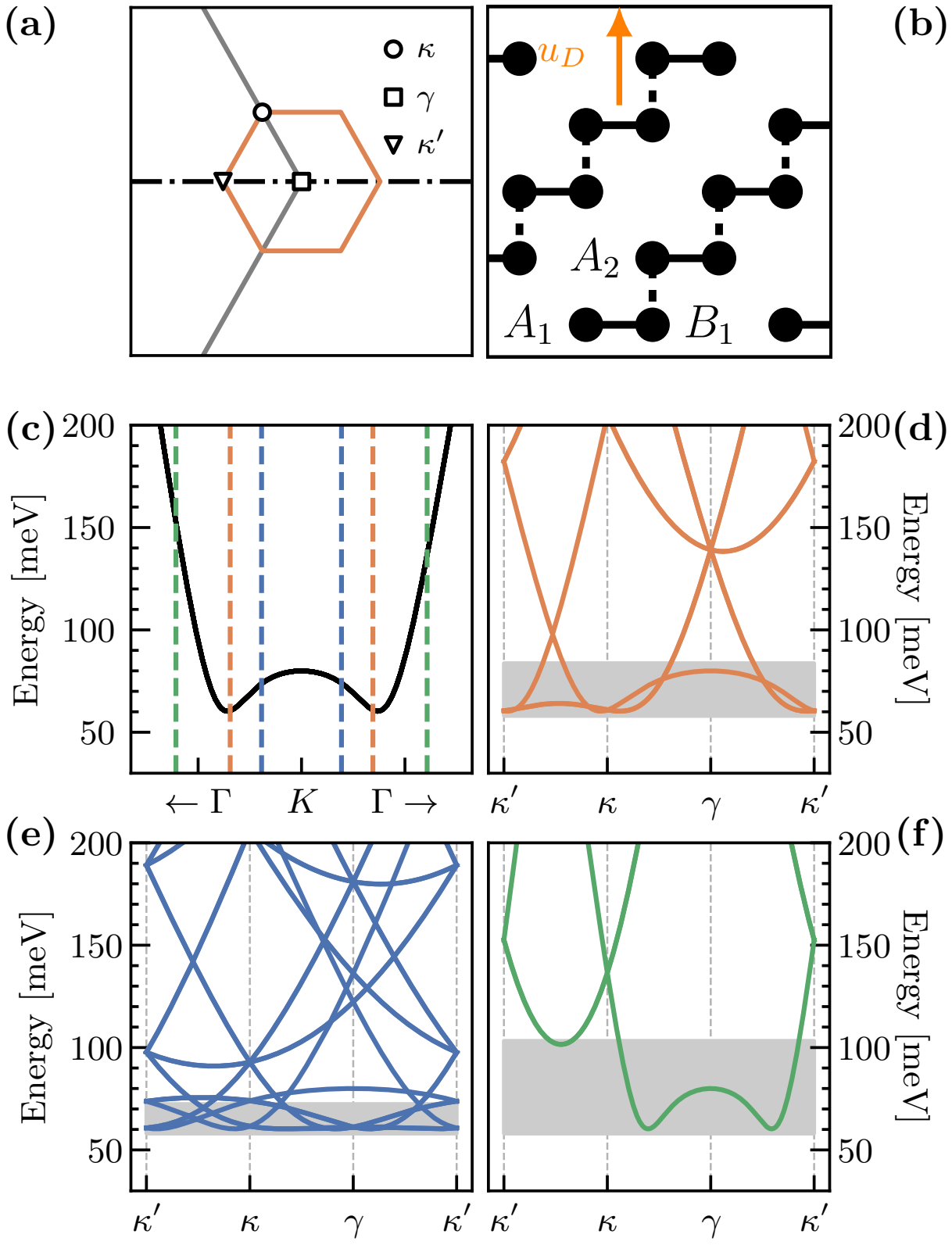


Figure 4.5: (a) The choice of mBZ for folding. The γ point is chosen to coincide with the K point of RMG. (b) Schematic side view of the structure of RMG, showing the staircase-like structure. We simplify the RMG model such that only the hoppings $t_{0,1}$ along the staircase are taken into account. (c) The band structure of the simplified Hamiltonian of RMG at $N_L = 5$ (Eq. (4.26)) along the k_x axis at $u_D = 40\,\mathrm{meV}$. Vertical lines correspond to the position of $\boldsymbol{\kappa}$ point after folding the band structure at densities $(29, 9.2, 2.9) \times 10^{11}\mathrm{cm}^{-2}$ from outside to inside. (d), (e), (f) The RMG band structure after folding according to (c). Electronic densities are $(9.2, 2.9, 29) \times 10^{11}\mathrm{cm}^{-2}$ respectively. Gray shading corresponds to Coulomb interaction strengths at the interparticle distance corresponding to these densities. Only the electronic density in (d) is suitable for the three-patch model. (e) is below the optimal range of density for the three-patch

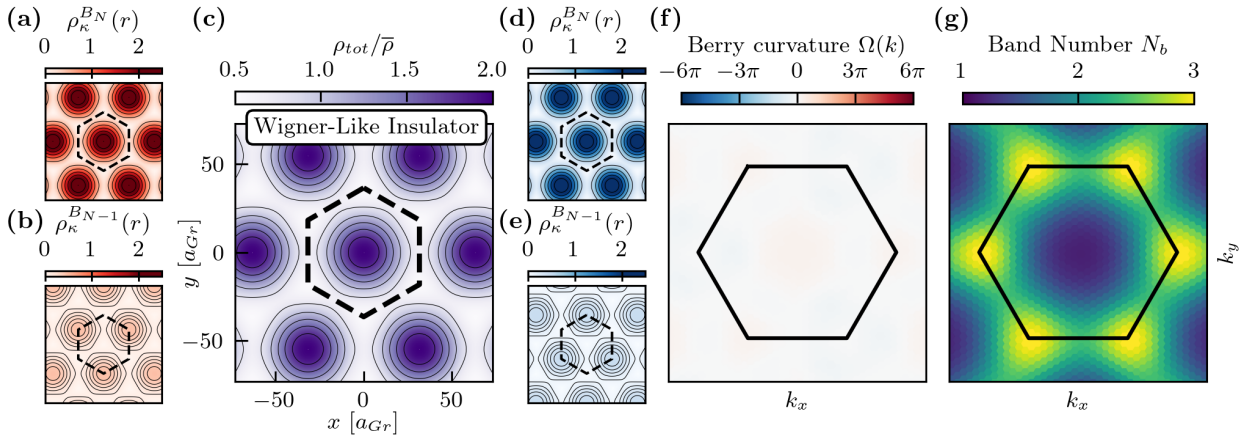


Figure 4.6: Properties of a $C = 0$ state (Wigner-like insulator) found in SCHF at $N_L = 5$ with parameters $(t_0, t_1) = (3100, 380)$meV, $u_D = 15$meV, $n = 4.8 \times 10^{11}\text{cm}^{-2}$, analogous to that in Fig. 4.7. We use 24×24 k-points and project onto the lowest 7 bands. (a-e) Charge densities of the $C = 0$ state. The left panels (a,b) correspond to orbitals B_{N_L} and B_{N_L-1} of the wavefunction at κ, while the right panels (d,e) correspond to charge densities at κ' of the same orbitals. The central panel (c) is the total charge density, including all the momenta points. (f) The Berry curvature, normalized such that the average of $\Omega(\boldsymbol{k})$ being 2π corresponds to $C = 1$ over the BZ.(g) The band number $N_\text{b}(\boldsymbol{k})$ in the mBZ. We can clearly see the existence of three patches.

10^6m/s is the graphene Fermi velocity, and $\bar{k} = k_x - ik_y$. When $v_F|\boldsymbol{k}| < t_1$, the spectrum corresponds to the topological phase of the SSH chain [353, 354]. It hosts two localized boundary modes, polarized on A and B sublattices. The A mode has (unnormalized) wavefunction

$$\tilde{\chi}_B(\boldsymbol{K}+\boldsymbol{k}) = (1, 0, -\eta(\boldsymbol{k}), 0, \eta(\boldsymbol{k})^2 \dots) \text{ where } \eta(\boldsymbol{k}) = \frac{v_F|\boldsymbol{k}|}{t_1} \tag{4.35}$$

in the basis $B_{N_L}, A_{N_L}, B_{N_L-1}, A_{N_L-1}, \dots$. Its normalized counterpart is $\chi_B = \tilde{\chi}_B/|\tilde{\chi}_B|$. The A boundary mode is $\chi_A = \sigma_x P \chi_B^*$ where σ_x flips sublattice, and P reverses the order of the layers.

If $u_D, v_F|\boldsymbol{k}| < t_1$, then the layer edge modes give the that give the highest energy valence band and lowest energy conduction band of RMG. We can thus project the single-particle Hamiltonian $H(\boldsymbol{k})$ into a two-dimensional Hilbert space spanned by $\chi_A(\boldsymbol{k})$ and $\chi_B(\boldsymbol{k})$:

$$H^{\text{eff}}(\boldsymbol{K}+\boldsymbol{k}) = \begin{pmatrix} -u_D\delta(\boldsymbol{k}) & t_1\eta(\boldsymbol{k})^{N_L} \\ t_1\eta(\boldsymbol{k})^{*N_L} & u_D(-(N_L-1)+\delta(\boldsymbol{k})) \end{pmatrix} \tag{4.36}$$

where $\delta(\boldsymbol{k}) = \frac{1}{u_D}\langle\chi_A(\boldsymbol{k})|H_D|\chi_A(\boldsymbol{k})\rangle \approx |\eta(\boldsymbol{k})|^2/(1-|\eta(\boldsymbol{k})|^2)$. At small enough $\boldsymbol{k}$, the diagonal part dominates, and we can approximate the energy eigenstate by $\tilde{\chi}_B$.

Dropping the A sublattices, and performing a layer-dependent gauge transformation, we find that the spinor

at the $\boldsymbol{k}$ are given by

$$\chi_B(\boldsymbol{K}+\boldsymbol{k}) \propto (1, \eta(\boldsymbol{k}), \eta(\boldsymbol{k})^2, \dots). \tag{4.37}$$

Thus, we recover the spinor structure used in Sec. 4.2.3, allowing us to compare the phenomenology of the section to that of the simplified RMG Hamiltonian.

We now consider the mBZ at density n to determine the value of η. The mBZ corners, κ_j, measured from the graphene K point, are given by

$$\boldsymbol{\kappa}_j(n) = \frac{2\pi}{L_s(n)}(C_6)^{j-1}(2/3, 0), \tag{4.38}$$

where C_6 is the six fold counterclockwise rotation, and L_s is determined by n via Eq. (4.32). Correspondingly, the spinors are given by

$$\chi_B(\boldsymbol{K}+\boldsymbol{\kappa}_{n+1}) \propto (1, \eta(\boldsymbol{\kappa}_i(n))e^{2\pi i n/6}, \eta(\boldsymbol{\kappa}_i(n))^2 e^{4\pi i n/6}, \dots) \tag{4.39}$$

where

$$\eta(\boldsymbol{\kappa}_i(n)) \coloneqq \frac{v_F|\boldsymbol{\kappa}_i(n)|}{t_1} = \frac{L_0|\boldsymbol{\kappa}_i(n)|}{2\pi} = \frac{L_0}{2\pi^2}\left(\frac{n}{6\sqrt{3}}\right)^{\frac{1}{2}}. \tag{4.40}$$

The critical $\eta \sim 0.6$ in the three-patch model, where the $C = 0$ state transitions to a $C = 1$ state, corresponds to an electronic density of $7.9 \times 10^{11}\ \mathrm{cm}^{-2}$, with a corresponding superlattice length scale of approximately 12 nm, slightly larger than the length scale $2\pi v_F/t_1 = 11$ nm. In terms of the angle between hBN and graphene, this corresponds to roughly $0.6°$, close to the experimental value $0.77°$ [21].

4.3.3 Density range of validity

The band structure of RMG under displacement field, Fig. 4.5(c), has a relatively flat band bottom, suggesting it could host three-patch states. We now carefully demarcate the density regime where expect the three-patch phenomenology.

We then fold the band structure onto the mini-Brillouin zone given according to the reciprocal lattice vectors in Eq. (4.31). Note that since there is no superlattice-scale potential, there is no eigenvalue repulsion.

There is a range of densities in which the minimal model is applicable. The dashed lines in Fig. 4.5(c) show the location mBZ boundaries at $k - K = \pm 2\pi/L_s(2/3, 0)$ corresponding to three different densities. Corre-

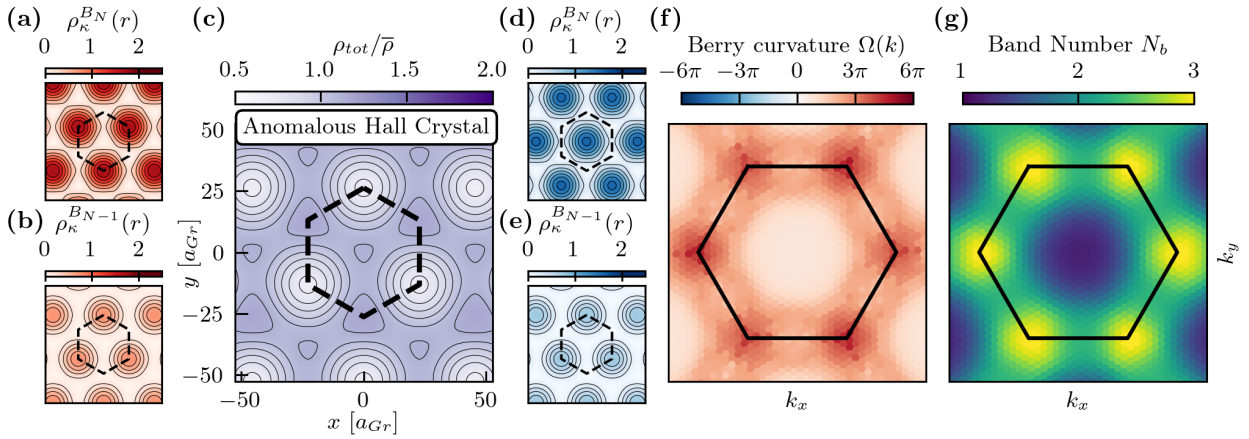


Figure 4.7: Properties of a $C = 1$ state (anomalous Hall crystal) found in SCHF at parameters $N_L = 5$, $(t_0, t_1) = (3100, 380)$meV, $u_D = 40$meV, $n = 9.2 \times 10^{11}\text{cm}^{-2}$. Panels are arranged similarly to Fig. 4.6.

spondingly, the xaxis labels (κ', κ, and γ) on Fig. 4.5(d-f) refer to different points for different panels. Fig. 4.5(d) shows the band structure at optimal density (orange line), whose lowest band is relatively flat compared to the interaction scale (gray shading), promoting symmetry breaking. Moreover, the large gap to the second band exceeds the interaction scale significantly at γ, disfavoring mixing. Thus, the ground state is likely to be a weakly coupled symmetry-broken state.

Outside this range of densities, the three-patch assumptions can break down. If the density is too low [Fig. 4.5(e)], then the other bands will mix strongly at γ, violating the assumed form of the wavefunction in Eq. (4.5). This will potentially alter the angular momentum ℓ_γ. Alternatively, too large a density [Fig. 4.5(f)] makes the band too dispersive at the mBZ boundaries, potentially favoring a Fermi liquid over a crystal. Henceforth we focus on the "Goldilocks zone" of densities corresponding to folding the bands near the edge of the flat band bottom.

4.3.4 Three-patch states as SCHF ground states

In Sec. 4.2, we showed that there is a close energetic competition between $C = 0$ and $C = 1$ states in the three-patch model using RMG spinors. We now show that the microscopic model Eq. (4.27) not only realizes this energetic competition, but recapitulates the same real space charge density patterns.

We briefly comment on our SCHF numerics. We explicitly allow translation symmetry breaking by using the folded band structure in the mBZ defined in Eq. (4.31). We assume valley and spin polarization, resulting from the flavor Stoner ferromagnetism due to the large density of states [259]. We choose the superlattice scale L_s

such that the filling is one electron per superlattice unit cell at each density considered. We project to the lowest 7 conduction bands for simplicity, although we note that it may become uncontrolled due to the reduced Hilbert space. See Ref. [320] for further numerical details.

At low densities, corresponding to low η, the ground state is a topologically trivial insulator. Its real-space charge density is shown in Fig. 4.6(c), which has an overall triangular lattice of a Wigner-like insulator — just as in Fig. 4.4. This charge density is essentially determined by the sum of charge densities from κ and κ'. In the dominant B_{N_L} spinor components, shown in (a) and (d), the charge is centered at the same Wyckoff positions, with the subdominant B_{N_L-1} components "filling in the gaps" to produce the same structure shown in Fig. D.2. The Berry curvature, shown in Fig. 4.6(f) is small everywhere, giving $C = 0$.

At higher densities, i.e. higher η, there is a transition to a $C = 1$ state. Fig. 4.7(c) shows its charge density, which is fairly uniform with peaks on a honeycomb lattice. Now the dominant B_{N_L} components are centered on *different* Wyckoff positions at κ versus κ', but the subdominant components are on top of each other. Referring to Fig. 4.4, we can see this is precisely the same structure as the $C = 1$ state there, whose dominant and subdominant components are represented by large and small circles, respectively. Here the Berry curvature, Fig. 4.7(f), is strongly peaked at κ and κ', with a slight C_2 breaking.

The success of the three-patch model, which only contains three single-particle wavefunctions, to reproduce the SCHF charge density is consistent with three-patch assumptions. We now explicitly verify that the ground states are weak-coupling states that mix only a few low-lying bands in the folded band structure: this can be revealed by studying the number of single-particle bands of the folded band structure mixed into the occupied band of the SCHF ground state. To quantify this, we note that SCHF eigenstates have the form

$$\psi_{\boldsymbol{k}}(\boldsymbol{r}) = \sum_{\boldsymbol{g}} \chi_{\boldsymbol{k}}(\boldsymbol{g}) e^{i(\boldsymbol{k}+\boldsymbol{g})\cdot\boldsymbol{r}}, \tag{4.41}$$

where $\boldsymbol{g}$ runs over superlattice reciprocal vectors, i.e. different (unhybridized) mini-bands. We define a mini-band distribution function

$$p_{\boldsymbol{k}}(\boldsymbol{g}) \coloneqq |\chi_{\boldsymbol{k}}(\boldsymbol{g})|^2. \tag{4.42}$$

When the eigenstate is an equal superposition of N plane waves (i.e. N mini-bands after folding), the von Neumann entropy of the band distribution function $S_{\text{v.N.}}(\boldsymbol{k}) = -\sum_{\boldsymbol{g}} p_{\boldsymbol{k}}(\boldsymbol{g}) \log p_{\boldsymbol{k}}(\boldsymbol{g})$ equals $\log N$. Therefore,

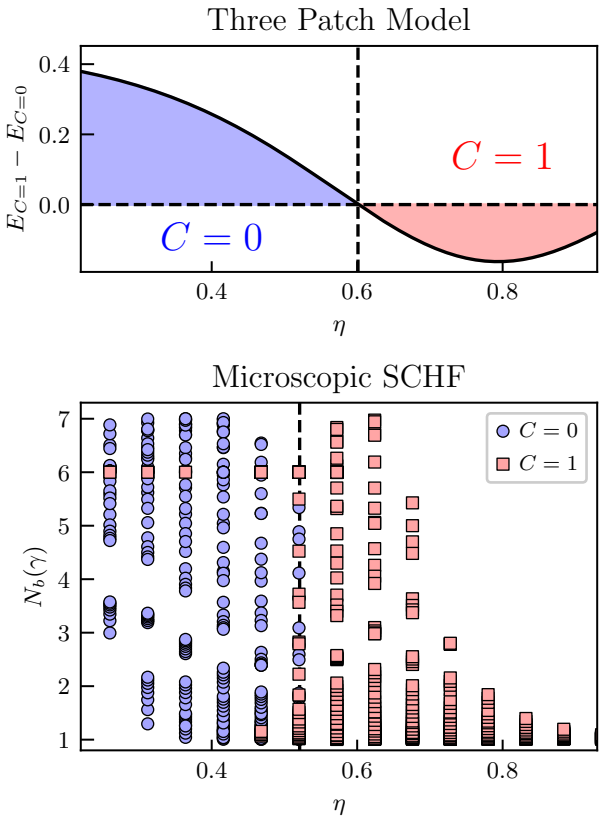


Figure 4.8: (a) Energy differences between $C = 1$ and $C = 0$ ground states in the three-patch model. We take spinors of the form Eq. (4.19) with $N_L = 5$, finding a transition at $\eta \approx 0.6$. (b) Dimensionally-reduced phase diagram of insulating states in Eq. (4.27), simplified microscopic RMG. Here η is proportional to the density, while $N_b(\gamma)$ quantifies how much remote bands are populated in the γ point wavefunction of the SCHF ground state (details in text). The line at $\eta = 0.52$ approximately separates the $C = 0$ and $C = 1$ phases, in good agreement with the phenomenological three-patch model.

we define the band number

$$N_b(\boldsymbol{k}) := e^{S_{\text{v.N.}}(\boldsymbol{k})} = \exp\left(-\sum_{\boldsymbol{g}} p_{\boldsymbol{k}}(\boldsymbol{g}) \log p_{\boldsymbol{k}}(\boldsymbol{g})\right) \tag{4.43}$$

to quantify the number of bands that mix at each k-point.

The band number $N_b(\boldsymbol{k})$ is shown in Fig. 4.6,4.7 (g). It is approximately 1 near γ, but is close to 3 in at κ, κ'. This confirms that only minibands close to the band bottom appear in the SCHF wavefunction, explicitly confirming the "weak coupling" assumption. In fact, we can see the appearance of large patch-like features near γ, κ and κ'. The assumptions of the three-patch model are therefore borne out in microscopic SCHF, so we should expect the phenomenology and predictions of the three-patch model to hold.

4.3.5 Microscopic verification of the three-patch phase diagram

The previous sections demonstrated translation breaking, weak band mixing and clear patches in SCHF ground states of RMG, putting the model into the regime of the three-patch model. We therefore expect the predictions and phenomenology of the three-patch model to appear in the microscopic model. We now show a striking manifestation of this: the competition between $C = 0$ and $C = 1$ is set by a single parameter η.

Fig. 4.8(a) shows the energy difference between $C = 0$ state and the $C = 1$ state in the three patch model with Coulomb interaction using spinors χ_B from Eq. (4.35) parametrized by η with $N_L = 5$. There is a phase transition around $\eta = 0.6$. We recall that in RMG, the value of η is given by

$$\eta(n) = \frac{L_0}{2\pi^2}\left(\frac{n}{6\sqrt{3}}\right)^{\frac{1}{2}} \tag{4.44}$$

where $L_0 \approx 11nm$ is the natural length scale of RMG. We note that effect of the ignored zero mode χ_A is negligible at large u_D, where its weight is suppressed. However, at small u_D, its weight can become as large as $O(10)\%$. This gives rise to a large "gray area" where the three-patch model is inconclusive about the ground state (App. D.6), and effects neglected in the three-patch model can determine the ground state. For simplicity, we apply the result of the three-patch model with χ_B in our analysis below.

If η is the primary driver of the phase transition between $C = 0$ and $C = 1$ — as predicted by the three-patch model — there should be a clean phase boundary in terms of η alone. To verify this, we first compute the microscopic ground state over a high dimensional parameter space, then dimensionally reduce the results in terms of just two variables. Explicitly, we select η and $N_{\mathrm{b}}(\gamma)$, which quantifies the number of remote bands that mix at the γ point, and must be close to 1 for the three patch model to be applicable. Concretely, we take $N_L = 5$, and vary $u_D \in [5, 70]\,\mathrm{meV}$, $\epsilon_r \in [3, 10]$, and $n \in [1.4, 17] \times 10^{11}\mathrm{cm}^{-2}$. We identify metallic data points either by a large occupation difference in the mBZ or by a small indirect gap less than $2\,\mathrm{meV}$. We discard such metallic data points as they are outside the scope of the three-patch model.

The dimensionally-reduced SCHF phase diagram is shown in Fig. 4.8(b). We see that all of the points above $\eta = 0.52$ (dashed line) belong to the $C = 1$ phase, while most of the points below $\eta = 0.52$ are in the $C = 0$ phase. We note that the $C = 0$ and $C = 1$ phase boundary in SCHF at $\eta = 0.52$ goes far beyond the weak coupling $N_b(\gamma) = 1$ point, potentially hinting that the three-patch model works beyond the weak-coupling assumption. We may thus conclude that η indeed controls the topological phase transition, and that the three-

patch model accurately predicts the critical value.

It is instructive to fix a realistic value of ϵ_r to see a density v. u_D phase diagram(Fig. 4.9(a)). We see that the AHC state is favored in the "Goldilocks" region at intermediate density and a narrow range of displacement field u_D. The qualitative features of this phase digram remains the same even when we include further-neighbor hoppings(Fig. 4.9(b)). As predicted by the three-patch model, the AHC phase gives way to the WC phase at low density. In fact, the critical value can be shifted to lower density/η by changing the gate distance of the interaction, exactly as predicted by the three-patch model (App. D.7). We will comment further on the striking success of the three-patch model in the discussion.

4.3.6 Three-patch states from analytical mean-field equation

In this brief section, we use a mean field gap equation to analyze the interaction-driven gap. Surprisingly, we find gap opening induces nearly constant wavefunctions in the κ, κ' patches.

Let us focus on a small momentum patch of radius S at κ. At each point in the patch, there are three low-lying states in the Hilbert space of the folded Brillouin zone corresponding to plane waves $|\phi_{\boldsymbol{\kappa}_1+\boldsymbol{k}}\rangle$, $|\phi_{\boldsymbol{\kappa}_3+\boldsymbol{k}}\rangle$, and $|\phi_{\boldsymbol{\kappa}_5+\boldsymbol{k}}\rangle$, defined in Eq. (4.2). In this three-dimensional Hilbert space, the linearized dispersion of each plane-wave state (before folding) is given by

$$H_0(\boldsymbol{k}) = \begin{pmatrix} v_F^{\text{eff}} \boldsymbol{k} \cdot \hat{\boldsymbol{e}}_1 & 0 & 0 \\ 0 & v_F^{\text{eff}} \boldsymbol{k} \cdot \hat{\boldsymbol{e}}_3 & 0 \\ 0 & 0 & v_F^{\text{eff}} \boldsymbol{k} \cdot \hat{\boldsymbol{e}}_5 \end{pmatrix}, \tag{4.45}$$

where $\hat{\boldsymbol{e}}_i$ is the unit vector parallel to $\boldsymbol{\kappa}_i$, and v_F^{eff} is the effective Fermi velocity at the κ patch. We assume the dispersion v_F^{eff} is small, since the κ patch is near a local minimum of dispersion in experimental relevant regimes [see Fig. 4.5(c)].

We consider a mean-field Hamiltonian $H_{\text{mf}}(\boldsymbol{k})$ by projecting the Hatree and Fock terms into this subspace. By C_3 symmetry, it takes the form

$$H_{\text{mf}}(\boldsymbol{k}) = \begin{pmatrix} 0 & \Delta(\boldsymbol{k}) & \Delta^*(\boldsymbol{k}) \\ \Delta^*(\boldsymbol{k}) & 0 & \Delta(\boldsymbol{k}) \\ \Delta(\boldsymbol{k}) & \Delta^*(\boldsymbol{k}) & 0 \end{pmatrix}. \tag{4.46}$$

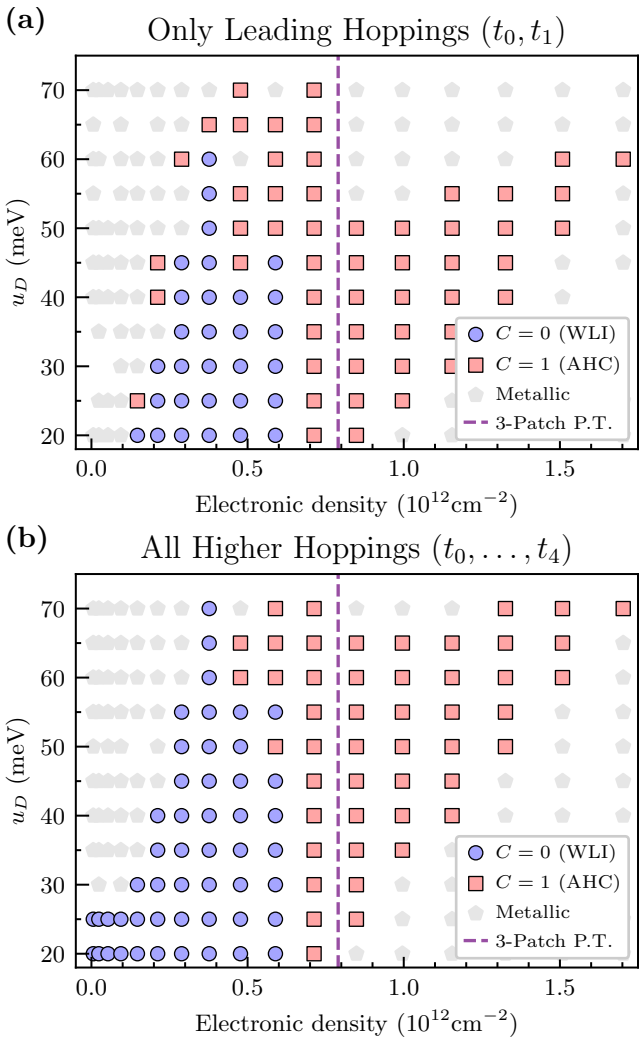


Figure 4.9: Phase diagram of RMG without a moiré potential in self-consistent Hartree-Fock. Red square indicate anomalous Hall crystals, blue circles indicate Wigner-crystal-like insulating states. Gray pentagons are states that are likely metallic (diagnosed by $n(\boldsymbol{k})$ varying by $> 1\%$). The dashed purple line shows the phase transition predicted by the three-patch model — close to the true mean field transition. (a) Phase diagram in simplified model Eq. (4.36). (b) Phase diagram including higher hoppings $(t_2, t_3, t_4) = (-21\,\mathrm{meV}, 290\,\mathrm{meV}, 141\,\mathrm{meV})$ in the expanded model from Ref. [320]. We note that the addition of higher hoppings does not significantly alter the phase diagram, suggesting these phase are stable to a variety of perturbations. Parameters: 24×24 unit cells, $\epsilon_r = 5$, $d_{\mathrm{gate}} = 10\,\mathrm{nm}$.

where the mean field $\Delta(\boldsymbol{k})$ can be written as

$$\Delta(\boldsymbol{k}) = V_{\boldsymbol{g}_1}\Lambda_{\boldsymbol{\kappa}_1+\boldsymbol{k},\boldsymbol{\kappa}_3+\boldsymbol{k}}\int d^2\boldsymbol{k}'\Lambda_{\boldsymbol{\kappa}_3+\boldsymbol{k}',\boldsymbol{\kappa}_1+\boldsymbol{k}'}\left\langle\hat{c}^\dagger_{\boldsymbol{\kappa}_3+\boldsymbol{k}'}\hat{c}_{\boldsymbol{\kappa}_1+\boldsymbol{k}'}\right\rangle \tag{4.47}$$

$$-\int d^2\boldsymbol{k}' V_{\boldsymbol{k}-\boldsymbol{k}'}\Lambda_{\boldsymbol{\kappa}_3+\boldsymbol{k}',\boldsymbol{\kappa}_3+\boldsymbol{k}}\Lambda_{\boldsymbol{\kappa}_1+\boldsymbol{k},\boldsymbol{\kappa}_1+\boldsymbol{k}'}\left\langle\hat{c}^\dagger_{\boldsymbol{\kappa}_3+\boldsymbol{k}'}\hat{c}_{\boldsymbol{\kappa}_1+\boldsymbol{k}'}\right\rangle, \tag{4.48}$$

where $\Lambda_{\boldsymbol{k},\boldsymbol{k}'} = \chi^\dagger(\boldsymbol{k})\chi(\boldsymbol{k}')$ is the form factor of plane wave states. The first term is the Hartree contribution, whereas the second term is the Fock contribution.

Since we are only interested in the vicinity of κ, we assume that all the spinors around the $\boldsymbol{\kappa}_i$ points are roughly unchanging:

$$\chi(\boldsymbol{\kappa}_i+\boldsymbol{k}) \approx \chi_i. \tag{4.49}$$

We also approximate $V_{\boldsymbol{k}-\boldsymbol{k}'} = V_0$. With these approximations, Eq. (4.47) can be drastically simplified:

$$\Delta(\boldsymbol{k}) = (V_{\boldsymbol{g}_1}|\Lambda_{1,3}|^2 - V_0)\int_{|\boldsymbol{k}'|<S}\langle\hat{c}^\dagger_{\boldsymbol{\kappa}_3+\boldsymbol{k}'}\hat{c}_{\boldsymbol{\kappa}_1+\boldsymbol{k}'}\rangle \equiv \Delta. \tag{4.50}$$

Note that now the mean-field potential Δ is independent of $\boldsymbol{k}$, and thus so is the mean-field Hamiltonian Eq. (4.47).

To solve the self-consistent equation Eq. (4.50), we consider the single-particle Hamiltonian $H(\boldsymbol{k}) = H_0(\boldsymbol{k}) + H_{\text{mf}}(\boldsymbol{k})$, and solve for the lowest energy single-particle state of $H(\boldsymbol{k})$. In the absence of the kinetic part $H_0(\boldsymbol{k})$, the lowest energy eigenstate takes the form

$$|\boldsymbol{k}^{(0)}\rangle = \frac{1}{\sqrt{3}}(|\phi_{\boldsymbol{\kappa}_1+\boldsymbol{k}}\rangle + |\phi_{\boldsymbol{\kappa}_3+\boldsymbol{k}}\rangle + |\phi_{\boldsymbol{\kappa}_5+\boldsymbol{k}}\rangle), \tag{4.51}$$

if we take Δ to be real and negative. We note that the eigenstate at $\boldsymbol{k} = 0$ has angular momentum $\ell_\kappa = 0$. Now we treat $H_0(\boldsymbol{k})$ perturbatively, assuming $\varepsilon = |v_F^{\text{eff}}\boldsymbol{k}/\Delta|$ to be small. The lowest energy eigenstate of the full Hamiltonian $H(\boldsymbol{k})$ is given by

$$|\boldsymbol{k}\rangle = \frac{1}{\sqrt{3}}\sum_{i=1,3,5}\left(1 - \frac{(\hat{\boldsymbol{k}}\cdot\hat{\boldsymbol{e}}_i)}{3|\Delta|}\varepsilon\right)|\phi_{\boldsymbol{\kappa}_i+\boldsymbol{k}}\rangle + O\left(\varepsilon^2\right). \tag{4.52}$$

Evaluating $\langle \hat{c}^\dagger_{\boldsymbol{\kappa}_3+\boldsymbol{k}'} \hat{c}_{\boldsymbol{\kappa}_1+\boldsymbol{k}'} \rangle$ with $|\boldsymbol{k}\rangle$, we find

$$\Delta = \left(V_{\mathbf{0}} - V_{\boldsymbol{g}_1}|\Lambda_{1,3}|^2\right) \int d^2\boldsymbol{k}' \frac{\sqrt{2}\Delta}{\sqrt{|v_F^{\text{eff}}\boldsymbol{k}'|^2 + 18|\Delta|^2}}. \tag{4.53}$$

Cancelling Δ from each side, it is easy to see the equation admits a nonzero solution only if the prefactor is positive.

The positivity condition on the prefactor has a quantum geometric interpretation. It can be written as

$$V_{\mathbf{0}} - V_{\boldsymbol{g}_1}|\Lambda_{1,3}|^2 = V_{\mathbf{0}} - V_{\boldsymbol{g}_1} + V_{\boldsymbol{g}_1}|d_{1,3}|^2 > 0, \tag{4.54}$$

where $|d_{1,3}|^2 = 1 - |\Lambda_{1,3}|^2$ is the quantum distance [330]. This inequality is satisfied for physical interactions $V_{\boldsymbol{g}_1} < V_{\mathbf{0}}$, revealing a mean field tendency to open a gap. In fact, in the limit of the contact interaction $V_{\boldsymbol{g}_1} = V_{\mathbf{0}}$, the left-hand side of the inequality reduces to $V_{\boldsymbol{g}_1}|d_{1,3}|^2$, revealing that the quantum distance between spinors is a major driving force behind the symmetry breaking tendency. We also note that the positivity of the prefactor is a necessary, but not sufficient, condition for opening a finite gap. Unlike in BCS theory, the k-integral does not diverge at $\Delta = 0$, and thus the equation does not hold for an arbitrarily small prefactor. It is amusing to note that the magnitude of the form factors controls the gap, while their phase controls the topology of a resulting insulator.

From Eq. (4.52), we see that the wavefunctions $|\boldsymbol{k}\rangle$ are all perturbatively close to $|\boldsymbol{k}^{(0)}\rangle$, controlled by the small ratio $|v_F^{\text{eff}}\boldsymbol{k}/\Delta|$, giving rise to a patch-like behavior. The physical interpretation of this is straightforward: the Fock energy favors a large overlap between wavefunctions, similar to how it favors ferromagnetic configuration between spins with large, negative Fock energy. The states around κ are therefore perturbatively close to each other, controlled by the dispersion v_F^{eff} that is small around the flat band bottom.

4.4 Conclusion and outlook

In this work, we constructed a physically-motivated minimal "three-patch model" for the AHC state composed of seven spinor-valued plane waves at the high symmetry points κ, κ', and γ. In the three-patch model, these high symmetry point wavefunctions completely determine both the energetics and the topology of the states, through which we predict the phase diagram of the RMG. The success of the phenomenological three-

patch model is striking. Despite its extreme simplicity — with analytic solvability, no kinetic energy, and a vastly reduced Hilbert space — it qualitatively and quantitatively captures the results of microscopic SCHF calculations. This has a fascinating implication: the competition between $C = 0$ and $C = 1$ is mostly or fully determined by interactions.

Given the example of Wigner crystals, one might expect the topologically trivial electronic crystal to win the energetic competition. But here a crucial extra ingredient is present: the spinor structure. The spinors enter via two geometrical phases, θ_H and θ_F, that determine the ground state favored by the Hartree and Fock terms respectively. These can compete, but there is a broad "frustration free" regime where Hartree and Fock terms collaborate to favor $C = 1$. In fact, the spinors of RMG fall in this "frustration free" regime in the broad range of the phase diagram.

When comparing careful self-consistent Hartree-Fock results on the simplified RMG Hamiltonian and Coulomb interaction with $N_L = 5$, we found that the three-patch model successfully described many features of RMG, at least in the weak-coupling regime. In particular, we found that the three-patch model qualitatively reproduced the phase diagram as shown in Fig. 4.8. The phase boundary between the topological and trivial state was mostly determined by a single parameter η of the spinor. Rather surprisingly, the three-patch model also captured the precise charge density pattern obtained from numerics, which in part justifies the effectiveness of our three-patch model.

It is fruitful to consider the unusual mechanism that generated the Berry curvature in the AHC state. Explicitly, the Berry curvature comes from phase-ful superpositions of plane waves produced from gap opening — *not* just the intrinsic Berry curvature present in the single particle bands before folding. Even if there is only an infinitesimal amount of intrinsic Berry curvature, such as for the $N_L = 3$ spinor at $\eta \to 0$ with contact interactions (Fig. 4.3), the AHC state can still be favored. The AHC thus represents a novel scenario where interaction-driven spontaneous translation breaking is the *source* of topology. The high density of states regimes without much Berry curvature, which are often realized near the bottom of a band, may thus permit AHC states. We suggest this mechanism opens an unexplored regime where quantized anomalous Hall effects might occur.

In fact, the three-patch model provides a simple algorithm to look for AHC candidate materials: 1) Perform *ab initio* calculations to look for compound with the band structure with a flat band bottom, 2) Compute the spinor structure near the band bottom, and 3) Compute the quantum geometric phases θ_H and θ_F. If these phases lie in the "frustration free" region, the compound is a potential AHC candidate.

While we formulated the three-patch model for continous symmetry breaking, a similar treatment based on symmetry indicators and the patch model might be applicable in more generality. For example, the appearance of QAH-crystal states in moiré systems [355] can be subject to three-patch style analysis. It is also interesting to note that the combination of a weak translation-breaking potential and commensurate filling might stabilize AHC with other Bravais lattice geometries.

We note a simple experimental prediction of the three-patch model. Since the critical value of η is proportional to $\sqrt{n}/t_1$, increasing the interlayer hopping t_1 by applying pressure would increase the density necessary to stabilize the AHC state. Pressure therefore might drive a quantum phase transition between the AHC state and the Wigner-like insulator state.

An essential future direction is to go beyond mean-field theory. Although the three-patch model here has given a simple physical picture of the AHC phase, we expect that — just as for the Wigner crystal [356–358] — its energetic competition should change significantly in beyond mean-field treatments. This would shed light on the energy competition with metallic states, as well as correlated insulators e.g. an analogue of composite fermion crystals in the lowest Landau level [359–362]. Moreover, there is a fascinating possibility that the AHC state can compete with a *fractional anomalous Hall crystal*, where a larger unit cell is fractionally occupied, resulting in a translation broken topologically ordered state. Understanding these questions requires analytical or numerical techniques that can treat many strongly-correlated bands, and will be a topic for future work. We hope that the three-patch model will provide valuable insight and phenomenological guidance for such future studies.

Note added—After the completion of this work, Refs. [363–365] appeared. Refs. [363, 364] proposed different systems to realize the AHC. Ref. [365] discussed a model similar to ours and the results agree in overlapping areas.

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5

A Jellium Model for the Anomalous Hall Crystal

5.1 Introduction

In the classic problem considered by Wigner [302], electron crystallization driven by strong Coulomb interactions was examined within the jellium model, in which an electron fluid embedded in a uniform neutralizing background spontaneously breaks translation symmetry upon decreasing the density. The two-dimensional incarnation of the jellium model, the two-dimensional electron gas (2DEG), has since been studied extensively using advanced numerical techniques [293–296, 366–373], which revealed the importance of beyond-mean-field effects for the energy competition of the Wigner crystal (WC) phase with the Fermi liquid.

Recent experiments [21, 374–377] in rhombohedral multilayer graphene (RMG), where new phases including integer and fractional quantum Hall phases emerge at low electronic densities and zero magnetic field, have inspired an explosion of theoretical interest in topological crystalline phases of matter. We, along with our collaborators and other simultaneous works, proposed [298, 299, 320] the possibility of an interaction-driven Chern insulator that spontaneously breaks continuous translation symmetry, which we dubbed the anomalous Hall crystal (AHC).

The complexity of the microscopic RMG Hamiltonian, consisting of ten orbitals and a morass of single particle hopping parameters, makes it difficult to distill simple physics, preventing a unified understanding despite intensive efforts [297, 301, 365, 378–382]. This has led to a flurry of interest in simplified models for AHCs [363, 364, 383]. However, these models are either phenomenological mean-field models, or contain a large number of spinor components, thus precluding the possibility of beyond mean-field numerics. Indeed, determining the fate of the AHC in the presence of beyond-mean-field quantum fluctuations is an urgent chal-

lenge.

To begin tackling this challenge, we propose a simple model: λ-jellium. This two-band model endows the quadratic band of the jellium model with a nontrivial skyrmionic texture in momentum space, which is responsible for its non-zero Berry curvature. The model contains two tuning parameters: r_s and λ. The familiar r_s is the ratio between the quadratic dispersion and Coulomb interaction terms, whereas $1/\lambda$ controls the extent of the skyrmionic texture. In the $\lambda \to 0$ limit, the model reduces to the standard two-dimensional jellium model, thus allowing us to systematically understand the effect of Berry curvature. Our model, given its simplicity, limited spinor dimensions, and connection to the jellium model, can be studied with beyond mean-field techniques employed such as variational and diffusion Monte-Carlo methods developed over many decades [294–296, 367–373].

To pave the way for these advanced methods, we perform extensive self-consistent Hartree-Fock calculations of the phase diagram of the λ-jellium model. We find that the AHC phase occupies a significant region of the phase diagram. In addition, we also find two distinct Wigner crystal phases, a conventional Wigner crystal and a halo Wigner crystal with distinct symmetry properties. Two Fermi liquids, whose Fermi surfaces are circular and annular, also appear in the phase diagram. We find that the halo Wigner crystal and the annular Fermi liquid arises due to the interplay between interactions and non-trivial quantum geometry. We further find a quantum critical point between the AHC phase and the halo Wigner crystal, which is described by an emergent Dirac cone.

Finally, we find that in some parts of the AHC phase, the lattice geometry can be non-triangular. This subtlety in the choice of lattice geometry already exists in the case of Wigner crystals, where the large r_s classical electrostatic limit selects the triangular lattice. However, deep in the quantum regime, the preferred lattice geometry is not *a priori* clear. For instance, with spinful electrons at small interaction strengths, a square antiferromagnet wins over the triangular lattice within HF [293]. Similarly, in λ-jellium we find that non-triangular AHC states can be energetically preferred over triangular AHC states. This echoes earlier works on RMG, which found unit cells with multiple electrons [376, 380] or non-triangular unit cells [384] can be stabilized.

The rest of the paper is organized as follows. Section 5.2 introduces the λ-jellium model. Sec. 5.3, then analyzes its mean-field phase diagram in detail. In Sec. 5.4, we examine the second-order phase transition between the AHC and WC. Sec. 5.5 examines the close competition within the AHC phase between triangular and square unit cells. Finally, we close with a discussion in Sec. 6.4.

5.2 The λ-jellium model

We now introduce the Hamiltonian for λ-jellium, a two-band generalization of jellium with a skyrmionic texture in momentum space that provides a tunable amount of Berry curvature.

Recall the Hamiltonian for the (spinless) 2DEG is [294]

$$\hat{H}_{\text{2DEG}} = \frac{1}{r_s^2}\sum_{i=1}^{N}\nabla_i^2 + \frac{2}{r_s}\sum_{i<j}^{N}\frac{1}{|\boldsymbol{r}_i - \boldsymbol{r}_j|}, \tag{5.1}$$

where length is measured in units of the typical interparticle distance a (defined below), and energy has units of Rydbergs Ry *. These are related to microscopic units by [294]

$$r_s = \frac{a}{a_0}, \quad a = \frac{1}{\sqrt{\pi\rho}}, \quad a_0 = \frac{\hbar^2}{me^2}, \quad \text{Ry} = \frac{me^4}{2\hbar^2}. \tag{5.2}$$

Here ρ is the number density of electrons, a circle of radius a encloses one particle on average, and r_s is the dimensionless potential/kinetic ratio. Eq. (6.47) sets the Fermi momentum to be $\frac{2}{a}$, which means $k_F = 2$ in the unit system used in Eq. (5.1).

We now present a *local* two-band generalization of this model whose lower band has identical dispersion and interaction to Eq. (5.1), but includes a tunable amount of Berry curvature concentrated at the band minimum. To maintain locality, we use a kinetic energy that is a simple polynomial in the derivative operators:

$$\hat{h} = \Delta\begin{bmatrix} -\lambda^2\nabla^2 & i\lambda\partial \\ i\lambda\bar{\partial} & 1 \end{bmatrix}, \tag{5.3}$$

where $\partial = \partial_x - i\partial_y$, $\nabla^2 = \partial_x^2 + \partial_y^2$ and Δ is a large and positive constant.

One can think of this as a Dirac equation with a momentum-dependent mass. The lower band of $\hat{h}$ is exactly flat: $\epsilon_{\boldsymbol{q}} = 0$. Its wavefunction is $\phi(\boldsymbol{r}) = \frac{1}{\sqrt{A}}\int d^2\boldsymbol{q}\ \phi_{\boldsymbol{q}}\ e^{i\boldsymbol{q}\cdot\boldsymbol{r}}$ with spinor

$$\phi_{\boldsymbol{q}} = \frac{1}{\sqrt{1+\lambda^2 q^2}}\begin{bmatrix} 1 \\ \lambda(q_x + iq_y) \end{bmatrix}, \tag{5.4}$$

*Note that 1 Ry differs from the other standard measure of energy, a Hartree, by a factor of two.

where $q = |\boldsymbol{q}|$. The upper band is separated by a gap with minimum size Δ. Henceforth we take $\Delta \to \infty$ and focus on the lower band.

We combine Eqs. (5.1,5.3) to form the full model, which we call λ-jellium:

$$\hat{H} = \Delta \sum_{i=1}^{N} \begin{bmatrix} -\lambda^2 \nabla_i^2 & i\lambda\partial_i \\ i\lambda\bar{\partial}_i & 1 \end{bmatrix} - \sum_{i=1}^{N} \hat{I}_2 \frac{\nabla_i^2}{r_s^2} + \frac{2}{r_s} \sum_{i<j}^{N} \frac{1}{|\boldsymbol{r}_i - \boldsymbol{r}_j|}, \tag{5.5}$$

where $\hat{I}_2$ is the identity in spinor space and the interaction is spinor isotropic. We fix the number density to be ρ; Eq.(6.47) applied unchanged. This Hamiltonian is controlled by two dimensionless parameters: the familiar potential/kinetic ratio r_s and a new parameter λ that controls the concentration of Berry curvature in the lower band. We note that the single-particle part of this model can be obtained as a particular limit of the two-band models discussed in Refs. [321, 364].

Let us set out the single-particle properties of λ-jellium. In addition to continuous translation symmetry, the Hamiltonian has a continuous rotation symmetry

$$\hat{R}_\theta \phi(\boldsymbol{r}) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix} \phi(R_\theta \boldsymbol{r}), \tag{5.6}$$

where the 2 by 2 matrix acts on the spinor components. Its single-particle energy is purely kinetic, with dispersion (Fig. 5.1a) $\epsilon_{\boldsymbol{q}} = \frac{q^2}{r_s^2}$ in the lower band, matching the 2DEG. The non-trivial spinor $\phi_{\boldsymbol{q}}$ gives the lower band a skyrmionic texture in momentum space, with a spin up "skyrmion core" near $\boldsymbol{q} = 0$ and spin down at infinity. It wraps the Bloch sphere exactly once as $\boldsymbol{q}$ varies, producing Berry curvature (Fig. 5.1b)

$$\Omega(\boldsymbol{q}) = 2 \left(\frac{\lambda}{\lambda^2 |\boldsymbol{q}|^2 + 1} \right)^2. \tag{5.7}$$

Within a disk of radius κ, the enclosed Berry curvature is a Lorentzian

$$I(\kappa) = 2\pi \frac{\lambda^2 \kappa^2}{1 + \lambda^2 \kappa^2}, \tag{5.8}$$

with full-width half-max $\frac{1}{\lambda}$. As λ grows, the total Berry curvature is always 2π, but becomes concentrated at $\boldsymbol{q} = 0$, reaching $I(k_F = 2) = \pi$ at $\lambda = \frac{1}{2}$. The Berry curvature vanishes when $\lambda = 0$, whereupon the lower

band of $\hat{H}$ reduces to the spinless 2DEG. λ-jellium therefore provides a minimal modification to the standard 2DEG whose interaction strength and band topology can be tuned continuously and independently.

We note that due to the holomorphicity of the unnormalized spinors $\sqrt{1+\lambda^2 q^2}\phi_{\boldsymbol{q}}$ with respect to $\boldsymbol{k}$, the lower band of our continuum λ-jellium model has "ideal quantum geometry" [22, 23]. Another model for the AHC, the parent band model [364], also possesses ideal quantum geometry due to its (anti-)holomorphic spinors. However, the parent band model is usually studied in the limit where the dimension of the spinors becomes large and the Berry curvature becomes uniform. In contrast, the the λ-jellium model always has two-component spinors with concentrated Berry curvature around $\boldsymbol{k} = 0$.

5.3 Mean-Field Phase Diagram

We now study the phase diagram of λ-jellium where we find a Berry-curvature induced anomalous Hall crystal competing with other crystalline phases and liquids. We work at mean-field level, employing self-consistent Hartree Fock (SCHF) with three possible translation patterns imposed: (I) continuous translation symmetry (CTS), (II) a discrete translation symmetry (DTS) with a triangular lattice with one electron per unit cell, (III) discrete translation symmetry (DTS) with a *square* lattice with on electron per unit cell. The calculation with CTS is performed in the thermodynamic limit, while the DTS calculation is done on $N_k \times N_k$ unit cells, with N_k going up to 36. See Appendices for further numerical details. The resulting phase diagram is shown in Fig. 5.1 (c).

The long-ranged Coulomb interaction makes energy convergence with the system size difficult. We incorporate the standard correction from the Madelung energy, which effectively reduces the energy error scaling from N_k^{-1} to N_k^{-3} [385]. Details of this finite-size scaling can be found in Apps. E.2,E.3,E.4.

5.3.1 Circular and Annular Fermi Liquids

The phase diagram (Fig. 5.1 (c)) supports two regions with Fermi liquid ground states. These two regions correspond to physically distinct Fermi liquids, whose energy competition is driven by the presence of Berry curvature.

To understand the competition between the Fermi liquids, we analyze the phase diagram while imposing continuous translation symmetry. The mean field states are entirely characterized by their momentum space occu-

pations $n(\boldsymbol{k}) = \langle c^\dagger_{\boldsymbol{k}} c_{\boldsymbol{k}} \rangle$ with *unbounded* momentum $\boldsymbol{k}$. At small λ, we find the textbook circular Fermi liquid, with $n(\boldsymbol{k}) = \theta(|\boldsymbol{k}| - k_F)$. At moderate interaction strengths, λ drives a transition to an annular Fermi liquid. This phase has two concentric circular Fermi surfaces, so that $n(\boldsymbol{k}) = 1$ when $k_m \leq |\boldsymbol{k}| \leq k_M$ and otherwise vanishes. The competition between the circular and annular Fermi liquids, which we call FL(C) and FL(A) respectively, is shown in Fig. 5.2(a).

Interactions drive the formation of the annular Fermi liquid via a mechanism that can be understood qualitatively from the skyrmionic spinor texture. To understand this, let us write down the energy functional for mean-field Fermi liquid states in terms of $n(k)$. The kinetic energy is

$$E_{\text{kin}} = \sum_{\boldsymbol{k}} n(\boldsymbol{k}) \frac{|\boldsymbol{k}|^2}{r_s^2}. \tag{5.9}$$

As detailed in App. E.1 the interaction projected into the lower band is

$$H_{\text{int}} = \frac{1}{2A} \sum_{\boldsymbol{q} \neq 0} \sum_{\boldsymbol{k}, \boldsymbol{k}'} V(\boldsymbol{q}) \Lambda_{\boldsymbol{q}}(\boldsymbol{k}) \Lambda_{-\boldsymbol{q}}(\boldsymbol{k}') : c^\dagger_{\boldsymbol{k}} c_{\boldsymbol{k}+\boldsymbol{q}} c^\dagger_{\boldsymbol{k}'} c_{\boldsymbol{k}'-\boldsymbol{q}} :, \tag{5.10}$$

where $\Lambda_{\boldsymbol{q}}(\boldsymbol{k}) = \phi^\dagger_{\boldsymbol{k}} \phi_{\boldsymbol{k}+\boldsymbol{q}}$ is the form factor and $V(\boldsymbol{q}) = 4\pi / r_s |\boldsymbol{q}|$ is the Fourier transform of the Coulomb interaction. Using Wick's theorem with $n(\boldsymbol{k})$

$$E_{\text{int}} = -\frac{1}{2A} \sum_{\boldsymbol{k}, \boldsymbol{k}'} \tilde{V}(\boldsymbol{k}, \boldsymbol{k}') n(\boldsymbol{k}) n(\boldsymbol{k}'), \tag{5.11}$$

where $\tilde{V}(\boldsymbol{k}, \boldsymbol{k}') = V(\boldsymbol{k} - \boldsymbol{k}') |\Lambda_{\boldsymbol{k} - \boldsymbol{k}'}(\boldsymbol{k}')|^2$. The Hartree-Fock bandstructure for $n(\boldsymbol{k})$ is therefore

$$E_{\text{HF}}(\boldsymbol{k}) = \frac{|\boldsymbol{k}|^2}{r_s^2} - \frac{1}{A} \sum_{\boldsymbol{k}'} \tilde{V}(\boldsymbol{k}, \boldsymbol{k}') n(\boldsymbol{k}'). \tag{5.12}$$

We can now qualitatively understand how the spinors control the Hartree-Fock band structure. A nontrivial form factor $|\Lambda| < 1$ will reduce the exchange energy gain from the Fock term. This is consistent with the fact that exchange interaction favors ferromagnetic spinor configurations. How does this work with the skyrmionic spinor texture Eq. (5.4)? At small λ the "skyrmion core" region, where spins point up, is large compared to k_F. The form factors are therefore close to unity, giving a large negative exchange energy. Conversely, large λ spinors have a small "skyrmion core" compared to k_F, with most states inside the non-interacting Fermi surface are

pointing down. Occupying the mis-aligned spinors near the core incurs a significant energy penalty. This drives the state to deplete the skyrmion core, resulting in an annular Fermi surface, Fig. 5.2(a). The depletion of $\boldsymbol{k} = 0$ will be a recurring feature throughout the phase diagram.

5.3.2 Halo Wigner Crystals

Wigner crystal phases are characterized by spontaneous breaking of continuous symmetry, as well as zero Chern number. The 2DEG is known to host triangular Wigner crystalline states at large r_s [386–388]. Mean-field calculations underestimate the value of critical r_s between ferromagnetic FL and WC to be around 2 [293]. The λ-jellium model phase diagram, which is continuously connected to 2DEG in the $\lambda \to 0$ limit, also hosts Wigner crystalline phases, shown in Fig. 5.1(c).

At large r_s and λ, there is a distinct Wigner crystal phase — also with a triangular lattice — which we call a halo Wigner crystal. A similar crystalline phase was recently studied in Bernal bilayer graphene [389]. To see this, we plot the momentum space occupation number $n(\boldsymbol{k}) = \langle c^\dagger_{\boldsymbol{k}} c_{\boldsymbol{k}} \rangle$ for a normal Wigner crystal and a halo Wigner crystal in Fig. 5.3(a) and (b). We see that $n(\boldsymbol{k} = 0)$ takes the maximal value for the normal Wigner crystal, while $n(\boldsymbol{k} = 0) = 0$ for the halo Wigner crystal.

The vanishing of $n(\boldsymbol{k} = 0)$ is in fact enforced by symmetry. We find these Wigner crystals have a C_3 symmetry compatible with the triangular unit cell. Thus, viewing the Wigner crystal as a band insulator, the orbitals occupied at the high symmetry points must be eigenstates of the spatial symmetries, characterized by their angular momenta. Concretely, the C_3 angular momentum is

$$\hat{C}_3 \psi^{\mathrm{WC}}_\Gamma = \psi^{\mathrm{WC}}_\Gamma, \quad \hat{C}_3 \psi^{\mathrm{haloWC}}_\Gamma = e^{i2\pi/3} \psi^{\mathrm{haloWC}}_\Gamma, \tag{5.13}$$

where Ψ_Γ is the single-particle orbital at the Γ point. Since the angular momentum at Γ does not depend on the choice of C_3 centers, the halo Wigner crystal and the normal Wigner crystal are distinct crystalline insulators with different symmetry properties [350].

We can understand the change in angular momentum from a simple semiclassical analysis for the single particle problem [389]. Each individual semiclassical electron can be treated as a particle with dispersion and Berry curvature specified by Eq. (5.5). On top of that, it is subjected to a harmonic potential generated by the Coulomb interactions due to all other electrons present. The electron must follow the single-particle Schrodinger's equa-

tion in momentum space [389, 390]:

$$\left(\epsilon_{\boldsymbol{k}} + \frac{\alpha}{2}|i\boldsymbol{\nabla}_{\boldsymbol{k}} - \boldsymbol{A}_{\boldsymbol{k}}|^2 + \alpha\frac{\mathrm{Tr}\, g_{\mu\nu}(\boldsymbol{k})}{2}\right)\psi_\ell(\boldsymbol{k}) = E_\ell\psi_\ell(\boldsymbol{k}). \tag{5.14}$$

Here, we approximated the confining potential by $V(\boldsymbol{r}) = \alpha|\boldsymbol{r}|^2/2$, $\boldsymbol{A}_{\boldsymbol{k}}$ is the Berry connection of the band, and $g_{\mu\nu}(\boldsymbol{k})$ is the quantum metric of the band. When $\lambda = 0$, it reduces to the standard harmonic oscillator, whose ground state has angular momentum $\ell = 0$. When $\lambda \to \infty$, the Berry curvature is $\Omega(\boldsymbol{k}) = 2\pi\delta(\boldsymbol{k})$, corresponding to threading 2π flux at the origin. Furthermore, $\mathrm{Tr}\, g_{\mu\nu}(\boldsymbol{k})$ becomes a delta function at the origin. These two effects collaborate to suppress the occupation of the $\boldsymbol{k} = 0$ region and change the angular momentum of the ground state to $\ell = 1$. reproducing the numerical observation [†].

5.3.3 Anomalous Hall Crystal

Due to the coexistence of single-particle Berry curvature and a trend towards crystallization at large interactions, we expect the anomalous Hall crystal (AHC) — a crystalline phase that becomes a Chern insulator when pinned — to appear. Previous works have emphasized the importance of nontrivial quantum geometry and Berry curvature in stabilizing the AHC phase [297–299, 320, 363–365]. In particular, Ref. [365] argued that enclosing more than π Berry curvature in the unit cell will tend to drive a transition from a WC to an AHC. For the λ−jellium, this Berry curvature can be approximated by the enclosed Berry curvature within the non-interacting Fermi surface, which is given by Eq. (5.8). The enclosed Berry curvature reaches $I(k_F) = \pi$ at $\lambda = 1/2$, which is indeed roughly the location of the phase transition between WC and AHC at moderate r_s. At large r_s, the system is no longer weakly-coupled, and AHC is overtaken by the halo WC.

At even larger $\lambda \gtrsim 2.5$, the AHC undergoes a transition to the annular Fermi liquid. We note, however, that the large λ regime suffers from significant finite-size effects due to the tiny skyrmion core in momentum space. Within the AHC phase, there is a subtle competition between different unit cell geometries, which we examine in Section 5.5.

[†] We sincerely thank Ophelia Evelyn Sommer for pointing out the quantum metric contribution to us.

5.4 Second-order topological phase transition between AHC and WC

We now examine the crystal-to-crystal phase transitions in λ-jellium, shown in Fig. 5.1(c). We diagnose the order of the phase transition from the direct gap of the Hartree-Fock band structure near the phase boundary. The transition from a normal WC to a halo WC is first order, consistent with its origin from level-crossing in the single-particle spectrum. The AHC to WC transition near $\lambda = 1/2$ is also first order, similar to the first-order WC-AHC transition observed for RMG [297].

The transition between halo WC and AHC, on the other hand, appears to be continuous. Fig. 5.4(b) shows that as r_s approaches the transition from above, the mean-field charge gap closes in an unambiguously continuous manner. Approached from below, the charge gap closes but eventually jumps near the transition. We attribute this jump to finite-size effects, which become increasingly severe as one approaches the transition, requiring extremely fine grids and a large number of bands.

The origin of this continuous transition can be understood from the Hartree-Fock band structure in Fig. 5.4(a,b). There the bands are colored according to their spinor polarization $z_a(\boldsymbol{k}) = \langle \psi_{a\boldsymbol{k}} | \sigma_z | \psi_{a\boldsymbol{k}} \rangle$, where $|\psi_{a\boldsymbol{k}}\rangle$ corresponds to the Bloch state at momentum $\boldsymbol{k}$ of band $a = 1, 2$, counting from below. This thus reveals the band inversion nature of the transition: before the transition, $z_1(\Gamma)$ is large whereas $z_2(\Gamma)$ is small, but after the transition $z_2(\Gamma)$ is large whereas $z_1(\Gamma)$ is small. The vicinity of Γ is therefore well-described by a massive Dirac model. Tuning the mass through zero not only gives the observed band inversion, but also produces the angular momentum change required for the halo WC phase. The transition from the AHC to the halo WC is therefore captured by a standard Dirac mass closing.

We note that a similar transition has been analyzed in the context electrons in a magnetic field. Ref. [115] analyzed the phase transition between the "Hall crystal" phase [115, 283, 316, 317] with a nonzero Chern number and the "Wigner crystal" phase with zero Chern number. Within the Hartree approximation, the transition was found to be precisely through the Dirac cone, suggesting the universality of this mechanism.

5.5 AHC Lattice Geometry

In the *classical* Wigner crystal $r_s \to \infty$, the competition between different geometries is set purely through electrostatics, and the triangular lattice is preferred over the square lattice by about 0.5% [386]. This preference survives in the presence of quantum corrections, with the triangular Wigner crystal remaining favored all the

way down to the Fermi liquid transition at $r_s \approx 30$ [373]. However, there is a small region near $r_s \approx 2$ in the spinful Wigner crystal, where a square lattice antiferromagnetic crystal is preferred at mean field level [293]. The preference for a triangular lattice therefore depends on the specific details of the Wigner crystal phase.

Does the AHC prefer a triangular lattice? To understand this, we turn to numerics and examine the energetic competition within the AHC phase between the triangular lattice and the square lattice. Our result, shown in Fig. 5.1(c), is that while the triangular lattice is preferred over the square in most of the AHC phase diagram, the reverse is true near the phase boundary between normal WC and AHC. The energetic competition remains close, with the square lattice favored by $< 0.5\%$ at most at $\lambda = 2/3$. We note that finite-size energy corrections are significant, and methods detailed in Apps. E.2,E.3 are required to compare the energies reliably. In contrast, the triangular lattice is more stable at larger λ and r_s. We caution that another unit cell shape may have yet-lower energy in this region; future work will examine the full landscape of possible unit cells [391].

The preference for the triangular lattice at large λ and moderate r_s can be understood from the single-particle wavefunctions. When $\lambda \to \infty$, the Berry curvature distribution becomes concentrated at Γ, which is similar to concentrated Berrry curvature models of [392]. Electrons can thus be split into two groups: those in the small skyrmion core with Bloch momentum less than $O(1/\lambda)$ — which carry all the topology — and topologically trivial ones away from the core. The trivial electrons dominate the phase space and therefore control the leading order energetics. Thus, one expects that they will crystallize in a fashion similar to electrons in the jellium model, forming a triangular lattice. This argument is fully consistent with our empirical observation: the competition between the triangular and square lattices within the AHC phase is most relevant when λ is smallest.

5.6 Discussion

In this paper, we proposed the λ-jellium model where a new parameter λ adds Berry curvature to the two-dimensional electron gas without modifying its quadratic dispersion. We investigated its phase diagram as a function of both λ and the standard potential/kinetic ratio r_s with self-consistent Hartree-Fock methods, finding a rich phase diagram, Fig 5.1(c). We find a large anomalous Hall crystal (AHC) phase, an electronic crystal that possesses a Chern number. Beyond that, increasing λ drives two transitions out of the usual Fermi liquid and Wigner crystal phases of the 2DEG. This is because λ produces a skyrmionic texture in the single-particle spinors, creating a "core" of mis-aligned spinors at large λ, incurring a significant exchange energy penalty. In the small r_s Fermi liquid, this makes occupying states close to zero momentum expensive, driving a circular-to-annular

transition. At large r_s, the Wigner crystal transitions to a distinct "halo" Wigner crystal phase that depletes the $\boldsymbol{k} = 0$ region as well. We also observed a continuous transition between crystals of different Chern numbers driven by r_s, similar to a phase transition analyzed in Ref. [115]. A careful understanding of this transition and its experimental implications is left for a future work.

The presence of nontrivial Berry curvature in λ-jellium not only gives rise to topological states, but also changes the geometry of the crystalline ground states. While the usual Wigner crystal favors the triangular lattice, we found that in the λ-jellium model, AHC can favor the square lattice over the triangular lattice in some parts of the phase diagram.

A crucial next step is to move beyond mean-field techniques. As mentioned above, the two-component nature of our model together with the fact that only first and second derivatives appear make it well-suited for a variety of many-body numerical techniques, including variational monte carlo and neural network wavefunction methods. λ-jellium may therefore be a good model to establish crystallization with non-trivial Chern number at the many-body level.

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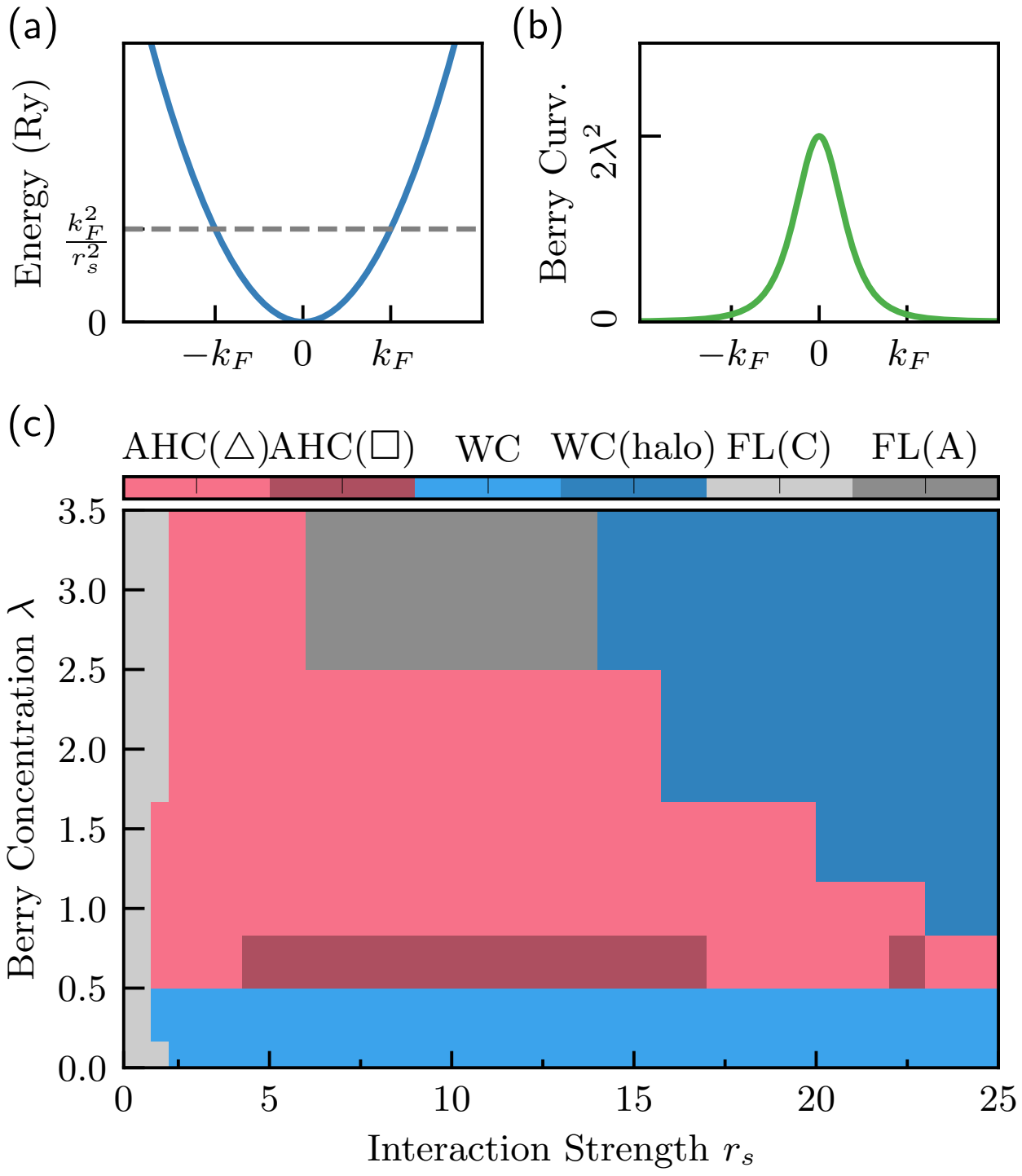


Figure 5.1: (a) Quadratic dispersion and (b) Berry curvature of the lower single-particle band of the topological electron gas model, Eq. (5.5). (c) Mean-field phase diagram. The limit $\lambda = 0$ is identical to the spinless 2DEG. Two significant Fermi liquid (FL) regions are present: one at low interaction strengths, and one at large interaction strength and Berry curvature concentration. Under strong interactions, a Wigner crystal (WC) appears that undergoes a first-order transition to an anomalous Hall crystal (AHC) at $\lambda \approx \frac{1}{2}$. A putative second order transition back to a WC appears at larger λ. A significant region of AHC is present, with the competition between triangular and square AHCs shown by light pink (AHC(△)) when the triangular AHC has lower energy than the square, and dark pink (AHC(□)) when the reverse is true. Within the AHC phase, a square unit cell is preferred to a triangular unit cell in a region near $\lambda = 2/3$.

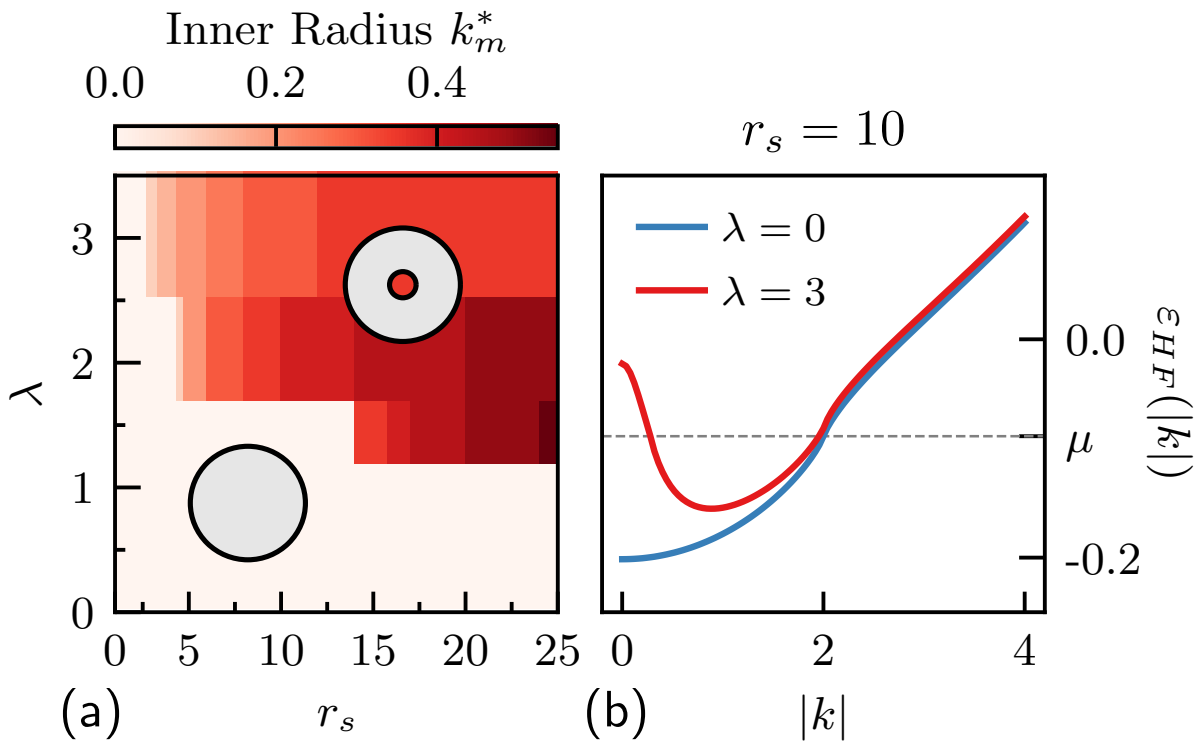


Figure 5.2: Fermi liquid phenomenology of the model. (a) Self-consistent inner radius k_m^* of the Fermi surface when continuous translation symmetry is imposed. (b) The Fermi liquid develops a strong peak at $\boldsymbol{k} = 0$ at large λ, r_s due to exchange (Fock) interactions, leading to annular Fermi surfaces.

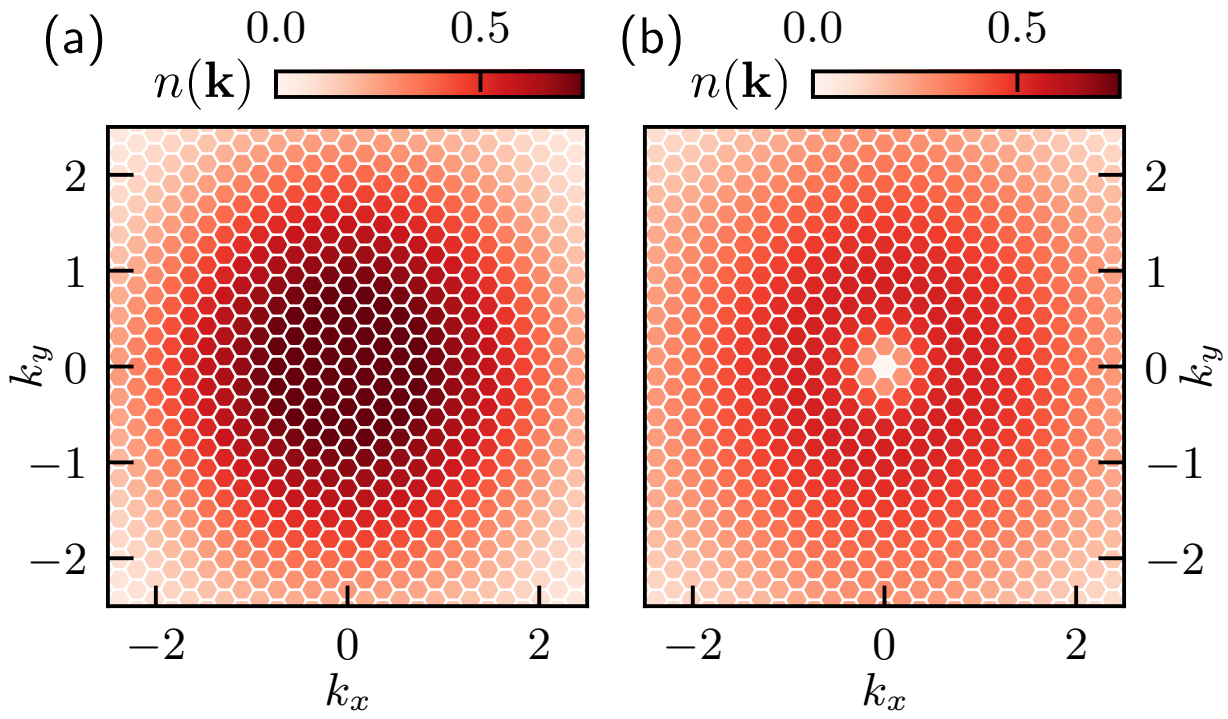


Figure 5.3: Occupations of plane wave states in the lower band for different Wigner crystals. (a) Normal Wigner crystal at $(r_s, \lambda) = (20, 0)$. (b) halo Wigner crystal at $(r_s, \lambda) = (20, 2.5)$. The $\boldsymbol{k} = 0$ region is depleted due to the Fock interactions at large λ.

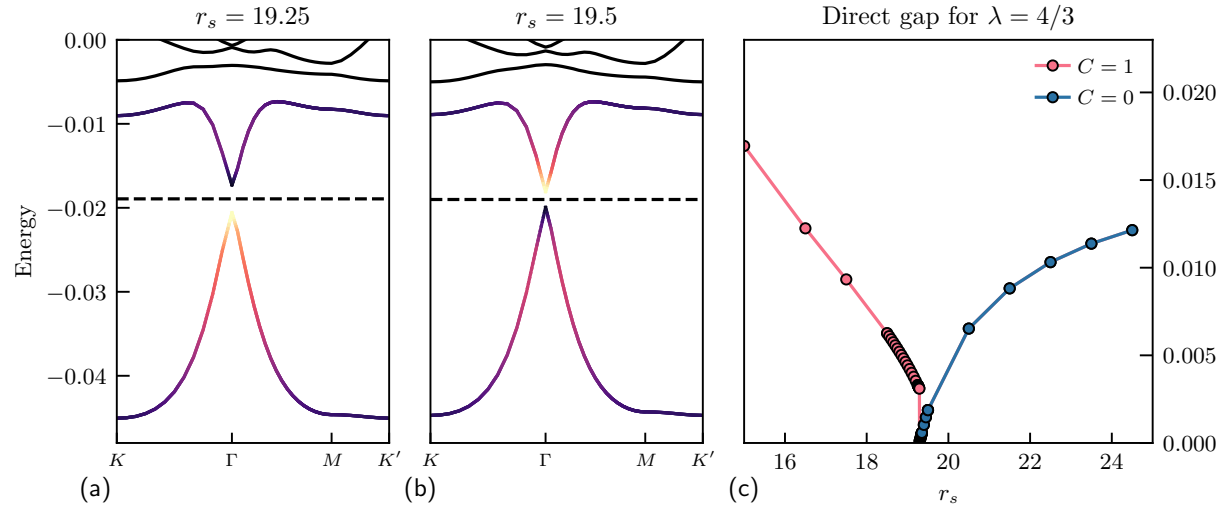


Figure 5.4: Continuous phase transition driven by r_s at $\lambda = 4/3$. (a,b): self-consistent Hartree-Fock band structures before and after the transitions. Coloring corresponds to $z_a(\boldsymbol{k})$, the polarization on the first versus second spinor components. Clearly, the r_s tuned transition corresponds to a band inversion transition. (c): Direct charge gap near the phase boundary between AHC and the halo WC.

6

Phonons in Electronic Crystals with Berry Curvature

6.1 Introduction

The topic of electron crystallization has a rich history, originating with Wigner's analysis [302]. Since then, research has expanded to include the crystallization of electrons in strong magnetic fields and on the surface of liquid helium [303–305, 308]. Driven by significant advances in the synthesis and tunable doping of two-dimensional (2D) materials there has been a surge of renewed interest in this problem [306, 307, 309, 311–315, 393]. Of particular interest are several cases involving valley degrees of freedom like graphene and TMD heterostructures, where spontaneous valley polarization leads to broken time reversal symmetry and parent bands with Berry curvature [394, 395]. A low density of electrons doped into these parent bands sets the stage for electron crystallization in the presence of inherent Berry curvature [298, 299, 320]. An intriguing possibility is the emergence of a Chern insulator upon spontaneous breaking of translation symmetry, dubbed the anomalous Hall crystal (AHC) [299, 320], which may be thought of as a zero-field counterpart of the Hall crystal proposed in Refs. [115, 283, 316, 317]. The intriguing observation of quantum anomalous Hall effects in rhombohedral 4-5-6-7 layer graphene [21, 374–377, 396] under a strong displacement field and a weak moiré potential strongly motivates further study of the AHC including its stability [297, 301, 363–365, 378–383, 397].

Despite the growing interest in electronic crystals, the formulation of an effective theory of low energy phonons in the most general setting remains an open challenge. In this work, we derive the general form of the low-energy phonon action for electronic crystals, and outline a calculational scheme to obtain the parameters of this theory

from a microscopic model. Our focus will be on crystals that are insulating once they are pinned, encompassing both anomalous Hall and traditional Wigner crystals. We work within a general framework with minimal symmetry constraints, allowing for the possible absence of time-reversal and rotational symmetries, in addition to breaking Galilean invariance. Surprisingly, we find that certain terms in the effective theory, which initially seem permissible based on symmetry considerations, are actually forbidden. Conversely, we identify a kineo-elastic term that intriguingly couples strain with the time derivative of the displacement vector, which to our knowledge had not been pointed out in previous studies.

We derive a general formalism that enables us to immediately address a fundamental question: What is the low-energy dispersion of phonons in the anomalous Hall crystal (AHC)? It is well known that the phonon dispersions of a regular Wigner crystal differ considerably depending on whether a magnetic field is present [386, 398]. While at first glance the AHC seems to share many properties in common with the Wigner crystal in a magnetic field, including broken time reversal symmetry, we find surprisingly that the low energy phonon dispersion actually follows that of the zero field Wigner crystal. This conclusion rests purely on the fact that translations in orthogonal directions commute and mirrors arguments for the counting of Goldstone modes [399–408]. This highlights the delicate interplay between time reversal symmetry breaking, translation symmetry and magnetic fields.

We extract the phonon action coefficients by matching the phonon response with the long wavelength response of the microscopic electronic action. Due to the sensitivity of the ground state to the boundary conditions, the response can be calculated by considering a translational gauge twist, .i.e. by modifying the periodic boundary conditions by a translation. It is worth briefly describing the key steps in this approach. A useful analogy is with *superfluid* effective actions, where U(1) charge symmetry is spontaneously broken and the relevant parameters governing the effective theory are superfluid stiffness and charge compressibility. In crystals, where translations are broken the analogous quantities we need to calculate are the elastic moduli and the effective mass. We bridge the gap to microscopic models by providing a method for determining the parameters of the effective theory in terms of the intrinsic properties of microscopic electronic models, which can be calculated with any method that provides an estimation on ground-state energies.

In this work, we will work with the self-consistent Hartree-Fock (SCHF) approximation. For the elastic moduli, we consider the variation of the SCHF energy upon deforming the unit cell of the crystal. For the effective mass, we extract the 'momentum compressibility' evaluating the change in the ground state energy in the pres-

ence of a boost transformation, which applies a chemical potential for the broken symmetry charge, i.e. momentum. In the absence of Galilean invariance this is no longer just the electron mass, and is a key characteristic of the crystalline state. The mixed kineo-elastic term, is obtained by combining these procedures.

We then apply our procedure to λ-jellium, a model with tunable Berry curvature that encompasses both traditional Wigner crystals and anomalous Hall crystals, as well as models of rhombohedral multilayer graphene. A useful check on our calculations is provided by the time dependent Hartree Fock (TDHF) calculations that we present, which agree well at low energies with our phonon effective theory with no fitting parameters.

Let us briefly summarize the results of the numerical investigation. In the λ-jellium model, we find that the addition of Berry curvature tends to make the crystals softer and heavier, particularly when their Berry curvature is not too concentrated. In some parts of the phase diagram, the crystal becomes soft enough to destabilize the triangular lattice. We compute the full energy landscape at those points to find square or rhombic crystals as global minima.

The subtlety of the crystal lattice shape carries over to the rhombohedral multilayer graphene, so we focus on a parameter regime where the triangular lattice is stabilized. By applying our procedure, we find that the kineo-elastic term is non-zero and large: the velocity in the $+x$ and $-x$ directions differ by as much as 20%. Furthermore the transition temperature can be determined from the stiffness using the KTHNY theory of melting [409–413].

We now review the relationship of this work to previous works. Wigner crystallization in a band with Berry curvature was previous studied in Ref. [389]. In deriving the phonon effective action, our approach is similar to obtaining the effective action by gauging the broken symmetry, which has been considered widely for internal symmetries, see [414, 415] and references therein, and in the context of spacetime symmetries in [416]. In the context of unidirectional ordering in superconductors, a similar approach, taking only the pairing field into account appeared in [417].

The instability of triangular crystals have been studied previously. In the context of Wigner crystals, Ref. [293] found a square lattice antiferromagnetic crystal in Hartree-Fock theory. Meanwhile, in the context of AHC, its instability to lattice deformation has been explored using in time dependent Hartree Fock [301] and elasticity theory [384]. An instability to expanding unit cell [380, 418] has also been explored. Moreover, the purely kinetic part of the effective action and some additional terms were extracted by [397].

In contrast, our analysis considers *all* symmetry allowed terms, including previously-overlooked kineo-elastic

term. We thus present the first full low-energy elastic theory of electron crystals. Furthermore, we provide simple methods for computing *all* coefficients in the phonon effective action from microscopic models, validated by a rigorous comparison with TDHF numerics. This provides a simple and easily computed recipe to construct a quantitatively accurate low energy phonon theory of any electronic crystal.

The rest of the paper is organized as follows. In Sec. 6.2, we review the elastic theory for phonons. We then develop a formalism for computing the effective action from "gauge twists", which changes the boundary condition of the system. This gives rise to a straightforward numerical algorithm for computing all coefficients of the effective theory for phonons. In Sec. 6.3, we apply this method to the λ-jellium model and rhombohedral multilayer graphene. We benchmark our results against TDHF calculations and find excellent agreement of the low frequency spectrum. Armed with these tools, we take a detailed tour of the phase diagram of λ-jellium model, and compute its full energy landscape versus the crystal lattice shapes. We apply a similar method to rhombohedral pentalayer graphene, finding a locally stable anomalous Hall crystal with large kineo-elastic coupling. We close with Sec. 6.4, where we discuss future directions.

6.2 Effective Field theory of Phonon Modes in Quantum Crystals

In this section, we derive the method for computing the coefficients of the low-energy effective field theory. Our main technique is to couple the system to background *translation* gauge twist.

6.2.1 The elastic theory for phonons

To set the stage, let us review the elastic theory of phonons, while taking care not to drop any symmetry-allowed terms. We consider an electronic crystal whose equilibrium electron locations are given by $\boldsymbol{R}_i \in \Lambda$, where Λ is the crystalline lattice. We assume that all elctrons crystallize so in the presence of weak pinning we have an insulator. We take the average density of electrons to be n. Defining the *displacement field* $\boldsymbol{u}(\boldsymbol{R}_i)$ such that $\boldsymbol{u}(\boldsymbol{R}_i) + \boldsymbol{R}_i$ is the location of the electron originally at $\boldsymbol{R}_i$. By taking the long-wavelength limit, we convert $\boldsymbol{u}$ to a continuous field and analyze its action.

In the presence of Coulomb interaction, is it convenient to split the action to the local part and the Coulomb part:

$$S^{\text{phonon}}[\boldsymbol{u}] = S^{\text{local}}[\boldsymbol{u}] + S^{\text{Coulomb}}[\boldsymbol{u}] \tag{6.1}$$

We will impose translation symmetry, implemented by $\boldsymbol{u} \to \boldsymbol{u} + \boldsymbol{\epsilon}$ for arbitrary constant $\boldsymbol{\epsilon}$. Crucially, we do not require either time-reversal symmetry or other lattice symmetries such as inversion. The local action is obtained by keeping all symmetry allowed terms up to two orders in derivatives and quadratic in the displacement fields:

$$\begin{aligned} S^{\text{local}}[\boldsymbol{u}] = \rho \int dt d\boldsymbol{r} \, \Big[& \frac{1}{2} m_{ab} \dot{u}_a \dot{u}_b - \frac{1}{2} \lambda_{abcd} u_{ab} u_{cd} \\ & + \beta \epsilon_{ab} u_a \dot{u}_b - \ell_{abc} \dot{u}_a \partial_b u_c \Big] , \end{aligned} \tag{6.2}$$

Here, we introduced $u_{ab} = \frac{1}{2}[\partial_a u_b + \partial_b u_a]$ for convenience and ϵ_{ab} is the two dimensional antisymmetric tensor. The equation of motion derived from the phonon action is invariant under constant shift in $\boldsymbol{u}$, as required by translation symmetry. Let us now go through through the coefficients one by one: 1) m_{ab} is the effective mass that controls the kinetic energy of sliding crystal. 2) λ_{abcd} is the stiffness tensor, which measures the energy cost associated with deforming the crystal. 3) β is the Berry phase term, which generates Lorentz force on the center of mass motion of the crystal. 4) ℓ_{abc} is a *kineo-elastic* term, which couples the strain on the crystal with its velocity.

We note that the Berry phase term is at first sight not invariant under translation. However, upon shift by Δu it generates a term of the form $\beta \epsilon_{ab} \Delta u_a \dot{u}_b$. This is a total derivative and does not affect the equation of motion. Time-reversal symmetry forbids this term. We consider Hamiltonians that break time-reversal, however, so this term is *a priori* allowed in the action.

Coulomb term, on the other hand, is given by

$$S^{\text{Coulomb}}[\boldsymbol{u}] = \rho^2 e^2 \int dt d\boldsymbol{r} d\boldsymbol{r}' \frac{1}{\epsilon |\boldsymbol{r} - \boldsymbol{r}'|} u_{aa}(\boldsymbol{r}) u_{bb}(\boldsymbol{r}') \tag{6.3}$$

We will set the electron charge to be $e = 1$ subsequently.

As an example, we will now impose C_3 symmetry, which is respected by all models we consider in this paper. The C_3 symmetry will constrain the parameters in the effective field theory. The kinetic term becomes isotropic: $m_{ab} = m\delta_{ab}$. The elastic term also simplifies greatly to be parameterized by two parameters:

$$\frac{1}{2}\lambda_{abcd}u_{ab}u_{cd} = \frac{K}{2}(u_{xx}+u_{yy})^2 + \frac{\mu}{2}[(u_{xx}-u_{yy})^2 + (2u_{xy})^2] \tag{6.4}$$

in which K characterizes a response to the expansion of volume, and μ characterizes the response to volume-preserving deformations, such as strains. This is also why we will refer to this term as "shear stiffness" below. Surprisingly, all of these terms behave as if the system is isotropic: they are all invariant under continuous rotations.

The kineo-elastic term deserves more attention. The presence of this term changes the canonical momentum in the presence of strain to

$$\pi_a = \frac{\partial L}{\partial \dot{u}_a} = m\dot{u}_a - \ell_{abc}\partial_b u_c. \tag{6.5}$$

For a fixed value of $\dot{\boldsymbol{u}}$ and $\partial_a \boldsymbol{u}$, however, the Hamiltonian is fully determined by the effective mass and stiffness, and does not depend on the kineo-elastic term.

To further analyze this term, we note that ℓ_{abc} carries three vector indices. Thus, it will carry angular momentum either 3 or 1; the C_3 symmetry will constrain such that only the angular momentum 3 terms remain. Those can be parameterized by two real numbers:

$$\begin{aligned}\ell_{abc}u_a\partial_b u_c = \ell(\dot{u}_x(u_{xx}-u_{yy}) - \dot{u}_y(2u_{xy})) \\ + \ell'(\dot{u}_y(u_{xx}-u_{yy}) + \dot{u}_x(2u_{xy})).\end{aligned} \tag{6.6}$$

We can always choose to rotate the system such that $\ell' = 0$. Since this leaves the rest of the Lagrangian invariant, we will only consider the case where $\ell \neq 0$. We will later prove $\beta = 0$ for crystals of interest. Assuming this, we can solve for the poles of the Green's function for the phonons. We then find that the leading order dispersion in the $\boldsymbol{q} \to 0$ limit of the longitudinal and transverse phonons are:

$$\begin{aligned}\omega_l(\boldsymbol{q}) &= \sqrt{\frac{2\pi\rho|\boldsymbol{q}|}{\epsilon m}}, \\ \omega_t(\boldsymbol{q}) &= v_t(\hat{\boldsymbol{q}})|\boldsymbol{q}|, \\ v_t(\hat{\boldsymbol{q}}) &= \frac{\cos(3\theta_{\boldsymbol{q}})\ell + \sqrt{4m\mu + \cos^2(3\theta_{\boldsymbol{q}})\ell^2}}{2m}\end{aligned} \tag{6.7}$$

in which $\theta_{\boldsymbol{q}}$ is the angle between the x axis and $\boldsymbol{q}$. We note that while the longitudinal phonon has a dispersion $\propto \sqrt{q}$, the transverse phonons have a finite velocity that varies in direction. Notably, as a result of the kineo-elastic term, the forward and backward speed of phonons is in general different and an angle dependent quantity,

where $v_t(\hat{\boldsymbol{q}}) - v_t(-\hat{\boldsymbol{q}}) = \cos(3\theta_{\boldsymbol{q}})\ell/m$. We remark that the presence of the kineo-elastic term is purely a consequence of the low symmetry of the system, and is generically present for any crystal.

We find that K does not appear in the leading order contributions to the phonons; its contributions are one order higher in $|\boldsymbol{q}|$ compared to the Coulomb term $S^{\text{Coulomb}}[\boldsymbol{u}]$. Thus, we drop the consideration for K whenever we discuss C_3 symmetric crystals.

6.2.2 Low-energy action from the translational gauge twist

Recall, in deriving an effective low energy theory for superfluids it is helpful to consider the system with periodic boundary conditions, that we then twist by phase rotation. The superfluid density can be extracted from the energy cost of the twist, at least when the only low energy modes are Nambu-Goldstone modes. As an analogous procedure here, we will twist boundary conditions by the spontaneously broken symmetry i.e. translations, and extract the energy cost to obtain the stiffness. This procedure is similar to obtaining the effective goldstone theory by gauging the symmetry, and appealing to the Anderson-Higgs mechanism, see for example [415] and references therein, with the notable advantage that we don't need to consider the effect of the curvature of the gauge field, which is a significant practical simplification.

A translational gauge twist is a time and spatially dependent translation $\boldsymbol{r} \to \boldsymbol{r}' = \boldsymbol{r} - \boldsymbol{\theta}(\boldsymbol{r}', t)$. We will only need gauge twists corresponding to constant gradients. In analogy with the superfluid, we consider a system on a torus with a fixed periodic boundary condition. This gauge twist may be obtained by acting with a unitary that takes the form $U_{\boldsymbol{\theta}} = \exp(-i \int d\boldsymbol{r}\, \boldsymbol{\pi}(\boldsymbol{r}) \cdot \boldsymbol{\theta})$, where $\boldsymbol{\pi}$ is the momentum density, which is the charge density of the translation symmetry. Let H be the Hamiltonian obtained in the absence of gauge twist. By considering the time evolution of a state $U_{\boldsymbol{\theta}} |\psi\rangle$ as being governed by H, the time evolution of $|\psi\rangle$ is governed by the twisted Hamiltonian

$$H[\boldsymbol{\theta}] = U_{\boldsymbol{\theta}}^\dagger H U_{\boldsymbol{\theta}} - U_{\boldsymbol{\theta}}^\dagger \dot{U}_{\boldsymbol{\theta}}, \tag{6.8}$$

which we take to act of a Hilbert space with fixed $\boldsymbol{\theta}$ independent boundary conditions. Its precise form depends on the choice of the microscopic Hamiltonian, and we provide the form for fermionic Hamiltonians in Sec. 6.2.5.

While this is a unitary transformation in the infinite plane, it relates sectors with different choices for periodic boundary conditions. Therefore, with a fixed boundary condition, the spectrum changes before and after the

transformation. With this choice of Hamiltonian Eq. 6.8, we can write the effective action as

$$\epsilon^{\Gamma[\boldsymbol{\theta}]} = \mathrm{tr}\left[T\epsilon^{-\int_{t_i}^{t_f} H[\boldsymbol{\theta}]dt}\right] \tag{6.9}$$

where T is the time ordering symbol, and by trace we really mean inserting appropriate boundary conditions at the initial and final time corresponding to the ground state satisfying $\boldsymbol{\theta}$ independent boundary conditions.

$\Gamma[\boldsymbol{\theta}]$ contains all dynamical information coming from low-energy modes of the system that couple to $\boldsymbol{\theta}$. By understanding long-time long-wavelength behavior of $\Gamma[\boldsymbol{\theta}]$, we can extract the effective theory of the system.

6.2.3 Matching with the phonon theory

Let us now suppose there exists a phonon field $\boldsymbol{u}$ whose action characterizes the low energy response. Let us also assume that $\boldsymbol{u}$ transforms under the translational gauge twist as $\boldsymbol{u} \to \boldsymbol{u}-\boldsymbol{\theta}$. Therefore the action is modified by the gauge twist, becoming:

$$S^{\mathrm{phonon}}[\boldsymbol{u}] \to S^{\mathrm{phonon}}[\boldsymbol{u}-\boldsymbol{\theta}]. \tag{6.10}$$

The effective action for $\boldsymbol{\theta}$ arises from integrating phonon modes

$$\epsilon^{\Gamma[\boldsymbol{\theta}]} = \int \mathcal{D}\boldsymbol{u}\, \epsilon^{S^{\mathrm{phonon}}[\boldsymbol{u}-\boldsymbol{\theta}]}, \tag{6.11}$$

where the integration $\mathcal{D}\boldsymbol{u}$ is only over boundary-condition-respecting $\boldsymbol{u}$. Now consider gauge twisting by $\boldsymbol{\theta}$, that obeys different boundary conditions

$$\boldsymbol{\theta}_i(\boldsymbol{r}) = \boldsymbol{\theta}(\boldsymbol{r}, t_i), \quad \boldsymbol{\theta}_f(\boldsymbol{r}) = \boldsymbol{\theta}(\boldsymbol{r}, t_f). \tag{6.12}$$

If $\boldsymbol{\theta}$ is chosen to satisfy the equations of motions in the bulk, then we may consider expanding the action to quadratic order in $\boldsymbol{u}$, suppressing space-time indices for brevity

$$S^{\mathrm{phonon}}[\boldsymbol{u}-\boldsymbol{\theta}] = S^{\mathrm{phonon}}[-\boldsymbol{\theta}] + \int \frac{\delta S[-\boldsymbol{\theta}]}{\delta\theta^a} u^a \tag{6.13}$$

$$+ \frac{1}{2}\int \frac{\delta^2 S[-\boldsymbol{\theta}]}{\delta\theta^a \delta\theta^b} u^a u^b \tag{6.14}$$

$$= S^{\mathrm{phonon}}[-\boldsymbol{\theta}] + S^{\mathrm{phonon}}[\boldsymbol{u}] \tag{6.15}$$

The linear term vanishes since $\boldsymbol{\theta}$ obeys the bulk equation of motion and the quadratic $\boldsymbol{u}$ term must be precisely the original action evaluated at $\boldsymbol{u}$ since the action is *quadratic*. Since the phonon no longer couples to $\boldsymbol{\theta}$, we can trivially perform the path integral, and noting that $S[-\boldsymbol{\theta}] = S[\boldsymbol{\theta}]$ we see that up to an irrelevant additive constant

$$\Gamma[\boldsymbol{\theta}] = S^{\text{phonon}}[\boldsymbol{\theta}]. \tag{6.16}$$

Therefore, the effective action in terms of an on-shell configuration of $\boldsymbol{\theta}$ contains all information about the effective action $S^{\text{phonon}}[\boldsymbol{u}]$.

6.2.4 Extraction of the coefficients

Let us first consider the Berry phase term β in the phonon action. Here we show that the coefficient of this Berry phase term is given by:

$$\beta = -\frac{}{2N} \langle [\hat{p}_x, \hat{p}_y] \rangle \,. \tag{6.17}$$

where $\hat{\boldsymbol{p}} = \sum_i \hat{\boldsymbol{p}}_i$ is the total momentum operator for all the particles labeled by i, and N is the number of particles. This formula echoes the argument for counting of Nambu-Goldstone bosons in nonrelativistic systems [399–408].

In the models we consider the momentum operators $\hat{p}_x,\ \hat{p}_y$ *commute*, regardless of whether we have a regular Wigner crystal or an AHC. Therefore we find that $\beta = 0$ and this term is *absent* in these models. Conversely, systems with non-commuting translations, such as charged particles in a magnetic field, can pick up nonzero Berry phase term β. It is well known that the dispersion relations for Wigner crystal phonons are strongly modified by a magnetic field, due to the onset of this term [386, 398]. Indeed we see that the translation keeps the path integral invariant only up to an overall phase, which indicates magnetic translation symmetry.

Let us now derive Eq. (6.17). We extract β by considering the boost response in a system that has been displaced a constant amount Δx in an orthogonal direction. From representation of the effective action in terms of phonons, this response depends only on β:

$$\frac{\partial}{\partial \dot{\theta}_y} \Gamma[\theta_x = \Delta x, \theta_y = \dot{\theta}_y (t - t_i)] = \int dt N \beta \Delta x. \tag{6.18}$$

By comparison, we may also evaluate the response from the Hamiltonian $H[\boldsymbol{\theta}] = U_{\boldsymbol{\theta}}^\dagger H U_{\boldsymbol{\theta}} - U_{\boldsymbol{\theta}}^\dagger \dot{U}_{\boldsymbol{\theta}}$. Since the

Hamiltonian is invariant under uniform translations, the only change in the action must arise from the change in the Maurer Cartan form. A time dependent uniform translation is generally implemented by the unitary $U_{\boldsymbol{\theta}} = e^{-\boldsymbol{\theta}\cdot\hat{\boldsymbol{p}}}$. As such, the Maurer-Cartan form to quadratic order in $\boldsymbol{\theta}$ is

$$-U_{\boldsymbol{\theta}}^{\dagger}\dot{U}_{\boldsymbol{\theta}} = -\dot{\theta}_y\hat{p}_y + \frac{}{2}\Delta x\dot{\theta}_y[\hat{p}_x,\hat{p}_y] + \dots \tag{6.19}$$

Thus we can also see that

$$\frac{\partial}{\partial\dot{\theta}_y}\Gamma = -\frac{\partial}{\partial\dot{\theta}_y}\int dt\,\langle H[\boldsymbol{\theta}]\rangle \tag{6.20}$$

$$= \int dt\,\langle\hat{p}_y\rangle - \frac{}{2}\langle[\hat{p}_x,\hat{p}_y]\rangle\,\Delta x \tag{6.21}$$

where expectation values are take in the translated but stationary ground state. Clearly $\langle\hat{p}_y\rangle = 0$, from which equation (6.17) immediately follows.

We now move on to extract other coefficients of the effective action. When the momenta commute so that $\beta = 0$, the effective action only depends on gradients of $\boldsymbol{\theta}$. Then $\boldsymbol{\theta}$ with arbitrary constant spatial and temporal gradients are guaranteed to satisfy the equations of motion. In such cases, the Mauer-Cartan form $-iU_{\theta}^{\dagger}\dot{U}_{\theta} = -\dot{\theta}_x\hat{p}_x - \dot{\theta}_y\hat{p}_y$ is also time independent. Therefore, $H[\boldsymbol{\theta}]$ is time-independent, and we can write the action as:

$$\epsilon^{\Gamma[\boldsymbol{\theta}]} = \epsilon^{-i(t_f-t_i)\langle H[\boldsymbol{\theta}]\rangle}, \tag{6.22}$$

where $\langle H[\boldsymbol{\theta}]\rangle$ is the ground state energy of $H[\boldsymbol{\theta}]$ in the original, fixed boundary condition. Therefore we get

$$\Gamma[\boldsymbol{\theta}] = -(t_f - t_i)\,\langle H[\boldsymbol{\theta}]\rangle \tag{6.23}$$

Thus expanding the ground state energy to quadratic order and taking derivatives, we get

$$m_{ab} = -\frac{1}{N}\,\frac{\partial^2}{\partial\dot{\theta}_a\partial\dot{\theta}_b}\,\langle H[\boldsymbol{\theta}]\rangle\Big|_{\boldsymbol{\theta}=0} \tag{6.24}$$

$$\ell_{abc} = \frac{1}{N}\,\frac{\partial^2}{\partial(\partial_b\theta_c)\partial\dot{\theta}_a}\,\langle H[\boldsymbol{\theta}]\rangle\Big|_{\boldsymbol{\theta}=\mathbf{0}} \tag{6.25}$$

$$\lambda_{abcd} = \frac{1}{N}\,\frac{\partial^2}{\partial\theta_{ab}\partial\theta_{cd}}\,\langle H[\boldsymbol{\theta}]\rangle\Big|_{\boldsymbol{\theta}=\mathbf{0}} \tag{6.26}$$

where N is the total number of particles, and we introduced $\theta_{ab} = \frac{1}{2}(\partial_a\theta_b + \partial_b\theta_a)$. We emphasize again that these expectation values are taken with respect to the ground state of $H[\boldsymbol{\theta}]$, subject to boundary conditions that are independent of $\boldsymbol{\theta}$. The details of the extraction procedure, including convenient choices for $\boldsymbol{\theta}$ are given in Sec. 6.2.6.

Although we will compute these expectation values within Hartree-Fock theory, it is important to note that the formalism developed here is applicable to any other numerical method for computing the ground state energy.

6.2.5 Application of the general procedure to electronic Hamiltonians

We now consider how to implement the previously described procedure for electronic system. Consider a two-dimensional translationally invariant electron gas with first quantized Hamiltonian of N particles

$$H = \sum_{i=1}^{N} h(\hat{\boldsymbol{p}}_i) + \frac{1}{2}\sum_{i\neq j} V(\hat{\boldsymbol{r}}_i - \hat{\boldsymbol{r}}_j) \tag{6.27}$$

where h is the bare Hamiltonian of the bands, which generally has internal state structure, and V is the (screened) Coulomb interaction. We consider a system on a finite torus geometry with periods $\boldsymbol{L}_{1,2}$. The gauge twist labeled by $\boldsymbol{\theta}$ physically represents deforming and boosting this torus. Clearly due to the long range nature of the Coulomb interaction, a deformation that does not preserve the area of the system, will result in a super-extensive change in the total energy. The corresponding response is universal, and given in the phonon theory by equation (6.3). As such, we may focus on unimodular deformations, and boosts so that for any single-particle wavefunction $\psi(\boldsymbol{r})$ the deformation takes the form $U_{\boldsymbol{\theta}}\psi(\boldsymbol{r}) = \psi(\boldsymbol{r} - \boldsymbol{\theta})$. For a uniform boost $\boldsymbol{v}$ and a deformation M_{ab}, that is

$$\theta_a(\boldsymbol{r}, t) = v_a t + M^{-1}_{ab} r_b, \tag{6.28}$$

the operator factorizes

$$U_{\boldsymbol{\theta}} = \epsilon^{-\boldsymbol{v}t\cdot\boldsymbol{p}}\epsilon^{-\frac{1}{2}M^{-1}_{ab}[\hat{p}_a\hat{r}_b + \hat{r}_b\hat{p}_a]} = U^{\text{boost}}_{\boldsymbol{\theta}} U^{\text{unimod}}_{\boldsymbol{\theta}} \tag{6.29}$$

The corresponding Maurer-Cartan form is

$$-U^{\dagger}_{\boldsymbol{\theta}}\dot{U}_{\boldsymbol{\theta}} = -\left(U^{\text{unimod}}_{\boldsymbol{\theta}}\right)^{\dagger} \boldsymbol{v}\cdot\hat{\boldsymbol{p}}\, U^{\text{unimod}}_{\boldsymbol{\theta}} \tag{6.30}$$

Thus the gauged Hamiltonian takes the form

$$H[\boldsymbol{\theta}] = \left(U_{\boldsymbol{\theta}}^{\text{unimod}}\right)^{\dagger} (H - \boldsymbol{v}\cdot\hat{\boldsymbol{p}})\, U_{\boldsymbol{\theta}}^{\text{unimod}} \tag{6.31}$$

This Hamiltonian should be understood to act on states with fixed boundary conditions, which for a single particle wavefunction take the form $\psi(\boldsymbol{r} + \boldsymbol{L}_{1,2}) = \psi(\boldsymbol{r})$. On the other hand, it is straightforward to see that for any wavefunction $\psi(\boldsymbol{r})$ satisfying the original boundary conditions, the wavefunction $\tilde{\psi} = U_{\boldsymbol{\theta}}^{\text{unimod}}\psi$ satisfies boundary conditions $\tilde{\psi}(\boldsymbol{r} + (1+M)\boldsymbol{L}) = \tilde{\psi}(\boldsymbol{r})$. The state $\tilde{\psi}$ now transforms under the Hamiltonian

$$\tilde{H}[\boldsymbol{\theta}] = H - \boldsymbol{v}\cdot\hat{\boldsymbol{p}} \tag{6.32}$$

Numerically we find it most convenient to work with this boosted Hamiltonian acting on states obeying the unimodularly changed boundary conditions.

Note, using Eqs. (6.28),(6.32) and the Güttinger-Hellmann-Feynman theorem [419–421], Eqs. (6.24),(6.25) can equivalently be cast as:

$$m_{ab} = \frac{1}{N}\frac{\partial}{\partial\dot{\theta}_b}\langle\hat{p}_a\rangle \tag{6.33}$$

$$\ell_{abc} = -\frac{1}{N}\frac{\partial}{\partial(\partial_b\theta_c)}\langle\hat{p}_a\rangle \tag{6.34}$$

We find that these formulas are much more stable numerically. We will now briefly discuss how we numerically implement these derivatives.

6.2.6 Numerical implementation

In the discussion above, we have established that the coefficients of the phonon effective action can be extracted (1) by choosing different periods $\boldsymbol{L}_{1,2}$ for the torus geometry and (2) by boosting the system with velocity $\boldsymbol{v}$. We now discuss how each of these are numerically implemented.

Different Torus Geometries

To be concrete, let's consider implementing a non-zero shear strain on the system. This is equivalent to taking a pair of new lattice vectors

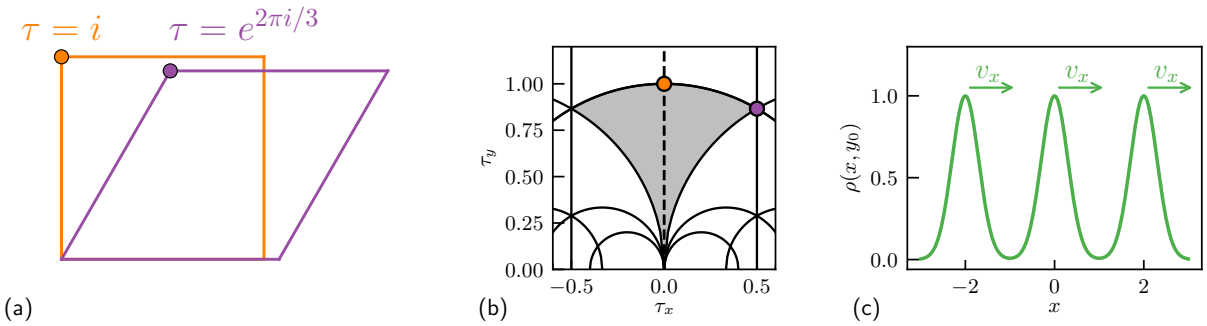


Figure 6.1: The numerical procedure by which the parameters are extracted. (a) Different unit cells of the same area can be chosen by specifying the orientation ϕ and modular parameter τ, which describes the relative relation between the lattice vectors $\boldsymbol{L}_2$ and $\boldsymbol{L}_1$ that define the torus geometry. The square lattice (orange) corresponds to $\tau = i$ whereas the triangular lattice (purple) corresponds to $\tau = e^{2\pi i/3}$. (b) All inequivalent choices of τ can be chosen by sampling a fundamental domain of the modular group (one choice is shaded in gray). Solid lines represent the boundaries of the fundamental domain, which correspond to rhombic lattices. The dashed line corresponds to rectangular lattices. (c) Boosting the system gives the crystal a center-of-mass velocity. Here we are showing a line cut along the x direction of the density profile of a crystal, which is sliding with velocity v_x in the x direction.

$$\boldsymbol{L}'_a = (I + M(\theta_{xy}))\boldsymbol{L}_a, \quad M(\theta_{xy}) = \begin{bmatrix} 0 & 2\theta_{xy} \\ 0 & 0 \end{bmatrix} \tag{6.35}$$

Thus, if we can extract the energy landscape of all possible torus geometries, we can find the shear stiffness μ easily. To do that, we need a good parametrization of the space of all torus geometries. Since all of our finite size calculations are performed at constant electronic densities and electron numbers, the lattice vectors are constrained that the area $|\boldsymbol{L}_1 \times \boldsymbol{L}_2|$ remains unchanged. All such lattices are thus parametrized by ϕ the orientation of lattice vector $\boldsymbol{L}_1$ and a modular parameter τ in the upper half complex plane:

$$\tau = \frac{L_{2x} + iL_{2y}}{L_{1x} + iL_{1y}} \tag{6.36}$$

Given (ϕ, τ) and the area of the torus A, there is an unique pair of lattice vectors that correspond to these values:

$$\boldsymbol{L}_1 = \sqrt{\frac{A}{\tau_y}} R(\phi) \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \boldsymbol{L}_2 = \sqrt{\frac{A}{\tau_y}} R(\phi) \begin{bmatrix} \tau_x \\ \tau_y \end{bmatrix} \tag{6.37}$$

in which $R(\phi)$ corresponds to a rotation by ϕ.

The triangular lattice corresponds to $\tau = e^{2\pi i/3}$, and the square lattice corresponds to $\tau = i$, as shown in

Fig. 6.1(a). Different choices of the unit cell for the same torus correspond to a modular transformation on τ:

$$\tau \to \frac{a\tau + b}{c\tau + d}, \tag{6.38}$$

where $a, b, c, d \in \mathbb{Z}, ad - bc = 1$. All the inequivalent values for τ, corresponding to difference choices of lattices, form the fundamental domain of the modular group, one of which is shown in Fig. 6.1(b). When the Hamiltonian possesses continuous rotational symmetry, the energy of the ground state cannot depend on ϕ. Thus, the modular parameter τ is the only variable that determines the energy. In that case, we can choose $\phi = 0$ without loss of generality. The strain θ_{xy} thus takes $\tau_x \to \tau_x + 2\theta_{xy}\tau_y, \tau_y \to \tau_y$. The stiffness corresponding to the rest of volume-preserving deformations can be extracted in a similar way.

Boost transformation and Galilean invariance

Once the unimodular transformation is implemented, we boost the Hamiltonian according to Eq. (6.32), which we reproduce here for convenience:

$$\hat{H}'(\boldsymbol{v}) = \hat{H} - \boldsymbol{v} \cdot \hat{\boldsymbol{p}}, \tag{6.39}$$

where $\hat{\boldsymbol{p}} = \sum_i \hat{\boldsymbol{p}}_i$ is the center of mass momentum. We note the same boosting procedure was performed in Ref. [397] for models of AHC and RMG.

With a concrete procedure for the boost operator, it is now easy to see why Galilean invariance fixes the effective mass. When the system is Galilean invariant, the many-body Hamiltonian of the system factorizes:

$$\hat{H} = \frac{\hat{\boldsymbol{p}}^2}{2Nm_b} + \hat{H}_{rel} \tag{6.40}$$

in which $\hat{H}_{rel}$ describes the relative motion of the electrons, whereas the first term describes the center of mass motion. m_b is the bare mass of the electron. The boost only couples to the center of mass motion, Since the two terms decouple, the minimization of $\hat{H}_{rel}$ can be performed independently; its ground state energy will be denoted by E_0. Now we shall consider minimizing the Hamiltonian that describes the center of mass motion $\hat{\boldsymbol{p}}^2/2Nm_b - \boldsymbol{v} \cdot \hat{\boldsymbol{p}}$: it is minimized when the state is an eigenstate of $\hat{\boldsymbol{p}}$ with eigenvalue $Nm_b\boldsymbol{v}$, leading to $m = m_b$ by Eq. (6.33).

Full extraction procedure for a C_3 symmetric crystal

For a C_3 symmetric crystal, the elastic Lagrangian simplifies greatly. The effective mass tensor becomes isotropic: $m_{ab} = m\delta_{ab}$. From Eq. (6.4), the elastic tensor simplifies greatly to be only described by two parameters, K and μ. However, given that K describes compression modes, its effects are dominated by the Coulomb interactions. From Eq. (6.6), the kineo-elastic couplings also simplify to be described by two terms ℓ and ℓ'. Thus, there are four numbers to be extracted in total: m, μ, ℓ, ℓ'. For convenience, we give explicit formulas for their extraction below.

$$\begin{aligned} m &= \frac{1}{N}\frac{\partial \langle \hat{p}_x \rangle}{\partial v_x}, \\ \mu &= \frac{1}{4N}\frac{\partial^2 \langle H[\theta_{xy}] \rangle}{\partial \theta_{xy}^2}, \\ \ell &= \frac{1}{2N}\frac{\partial \langle \hat{p}_y \rangle}{\partial \theta_{xy}}, \\ \ell' &= -\frac{1}{2N}\frac{\partial \langle \hat{p}_x \rangle}{\partial \theta_{xy}}. \end{aligned} \tag{6.41}$$

6.2.7 Electromagnetic response of anomalous Hall crystals

The feature distinguishing an (anomalous) Hall crystal from an ordinary Wigner crystal is the electromagnetic response. At energies below the gap to electrons, the electromagnetic response is governed by bound charges and currents and a topological Chern-Simons term.

$$\Gamma^{E<E_{\text{pinning}}}[A] = \int dt d\boldsymbol{r}\ [MB + \boldsymbol{P}\cdot\boldsymbol{E}] \tag{6.42}$$

$$+ \frac{C}{4\pi}\int AdA + \dots \tag{6.43}$$

corresponding to a Hall conductivity $\sigma_{xy} = C\frac{e^2}{h}$, magnetization density M and polarization density $\boldsymbol{P}$. In general there will also be higher order derivative terms. However, phonons also enter the electromagnetic response by modifying the polarization and magnetization. The leading coupling can be understood by considering a uniform translation by $\boldsymbol{\theta}$ which leaves the average M unchanged, but shifts the polarization density $\boldsymbol{P} \to \boldsymbol{P} + e\rho\boldsymbol{\theta}$

where ρ is the average density, leading to the coupling:

$$S^{\text{phonon}-\text{EM}}[\boldsymbol{u}, A] = \int dt d\boldsymbol{r}\, e\rho\, \boldsymbol{u} \cdot \boldsymbol{E} + \ldots \tag{6.44}$$

which can be directly verified by including electromagnetic potentials in the microscopic theory at the outset.

We can obtain the dominant response to external fields by first setting $\boldsymbol{k} = 0$ and then integrating out the phonons leading to:

$$\sigma_{ab}(\mathbf{0}, \omega \to 0) = \frac{e^2\rho}{\omega + i0^+} [m]^{-1}_{ab} \tag{6.45}$$

at which point the Drude weight may be read off $\mathcal{D}_{ab} = e^2\pi\rho\,[m]^{-1}_{ab}$.

To obtain subleading terms in the conductivity one needs higher derivative terms in the phonon effective action which we do not retain here, but see Ref. [397]. We expect that in general the transverse conductivity is not quantized in the absence of pinning, since anomalous phonon velocity contributions destroy the quantization of the transverse conductivity. However, if we introduce pinning, in the form of a mass term $\boldsymbol{u}^2$ to the phonon action, then at low frequencies the quantized Hall conductance obtains $\sigma_{ab} = \epsilon_{ab}\frac{C}{2\pi}$.

6.3 Numerical Results

We now use the general formalism developed above to study the low-energy excitations of three increasingly-complex electronic crystals. First, we study the Wigner crystal phase of jellium, where we show our linear response theory reproduces standard results as a sanity check. Second, we showcase how our method works in the λ-jellium model with non-trivial Berry curvature, accurately reproducing the phonon velocity from far-more-expensive time-dependent Hartree-Fock calculations. The Berry curvature drives a transition to an anomalous Hall crystal, which we find to be *heavier* and *softer* than the jellium Wigner crystal — to the extent that the softness destabilizes the hexagonal lattice in one regime. Performing a global stability analysis reveals its fate: it remains within the AHC phase but prefers a rhombic or square lattice. Third and finally we consider a microscopic model of rhombohedral pentalayer graphene. This model has neither time-reversal nor inversion symmetry, thus allowing the kineo-elastic term. At a point where a triangular AHC phase is stablized, we find a giant sound velocity and a non-zero kineo-elastic term that gives rise to strong direction anisotropies in the speed of sound.

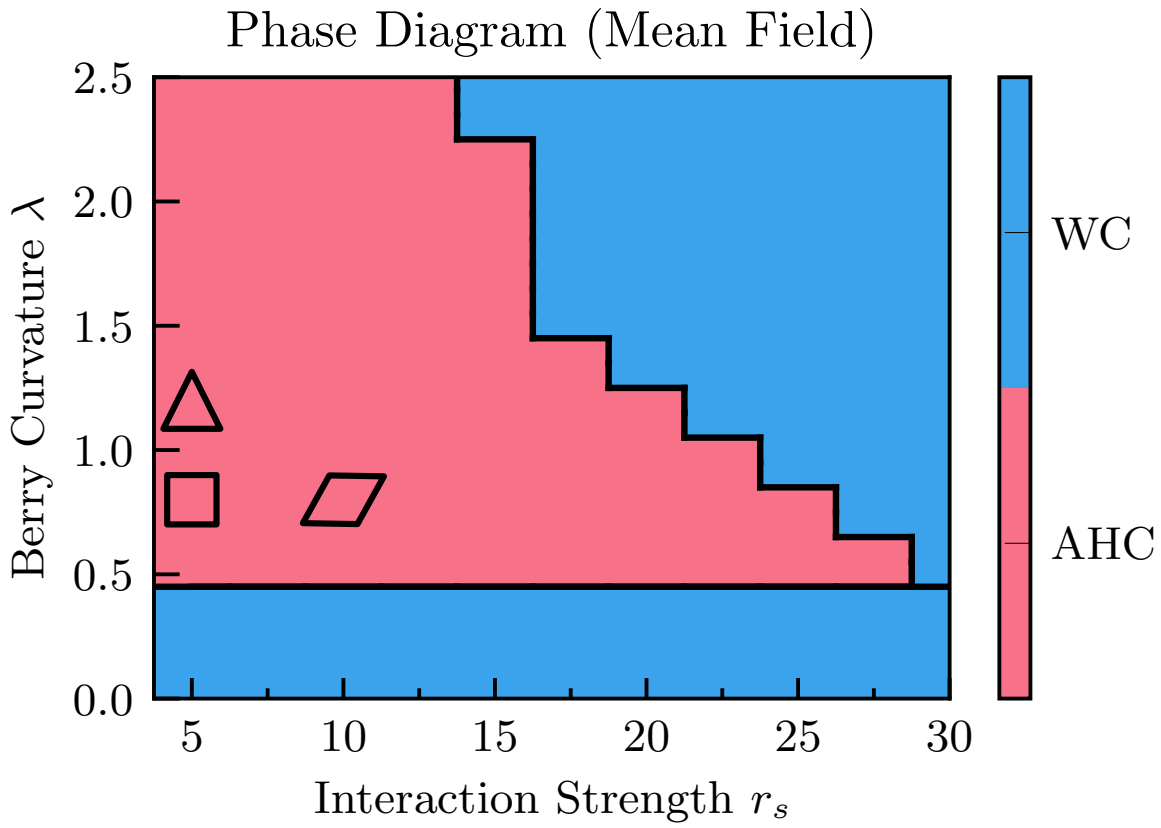


Figure 6.2: Hartree-Fock phase diagram of the λ-jellium model, which shows the competition between crystalline phases. Fermi liquid phases in the phase diagram are out of the parameter regimes shown here. Colors correspond to Wigner Crystals and Anomalous Hall Crystal. The parameter combinations $(\lambda, r_s) = (0.8, 5), (0.8, 10), (1.2, 5)$ will be studied in more careful detail in a later part of the manuscript; the anomalous Hall crystals take different shapes at those parameter points. All data are computed from SCHF (18×18) with 13 bands taken, where a triangular unit cell was assumed.

Hamiltonian and Mean-Field Phase Diagram

We now apply our linear response procedure to study the elastic theory of λ-jellium. λ-Jellium is a model, recently proposed by some of us, that extends the spinless jellium model with a parameter λ that adds Berry curvature without modifying Coulomb interactions or the quadratic dispersion. When $\lambda = 0$, the model reduces to normal jellium, whose Wigner crystal we study first as a sanity check.

The Hamiltonian of λ-Jellium is

$$\hat{H} = \Delta \sum_{i=1}^{N} \begin{bmatrix} -\lambda^2 \nabla_i^2 & i\lambda\partial_i \\ i\lambda\bar{\partial}_i & 1 \end{bmatrix} - \sum_{i=1}^{N} \hat{I}_2 \frac{\nabla_i^2}{r_s^2} + \frac{2}{r_s} \sum_{i<j}^{N} \frac{1}{|\boldsymbol{r}_i - \boldsymbol{r}_j|}, \tag{6.46}$$

where Δ is taken to be large.

Following conventions of Ref. [294], length is measured in units of the interparticle distance a and energy is measured in Rydbergs (Ry), with potential/kinetic ratio r_s and density ρ from

$$r_s = \frac{a}{a_0}, \quad a = \frac{1}{\sqrt{\pi\rho}}, \quad a_0 = \frac{\hbar^2}{me^2}, \quad \mathrm{Ry} = \frac{me^4}{2\hbar^2}. \tag{6.47}$$

The lower single-particle band of λ-jellium has quadratic dispersion $\epsilon_{\boldsymbol{q}} = q^2/r_s^2$ where $q = |\boldsymbol{q}|$ as usual. The wavefunction of the bottom band is

$$\phi_{\boldsymbol{q}} = \frac{1}{\sqrt{1 + \lambda^2 |\boldsymbol{q}|^2}} \begin{pmatrix} 1 \\ \lambda(q_x + iq_y) \end{pmatrix}, \tag{6.48}$$

which has a skyrmionic texture in momentum space: the spinor points up at $\boldsymbol{q} = 0$, and winds the Bloch sphere once and points down as $\boldsymbol{q} \to \infty$.

The texture carries Berry curvature $\Omega(\boldsymbol{q}) = 2\lambda^2[\lambda^2 |q|^2 + 1]^{-2}$ where as usual the second band is pushed up above energy $\Delta \to \infty$, and is irrelevant. This gives a quadratic band independently adjustable potential/kinetic ratio r_s and Berry curvature concentration λ, as claimed. Fig. 6.2 shows the mean-field phase diagram of λ-jellium away from small r_s. The anomalous Hall crystal phase occupies a large region of the phase diagram, where it competes Wigner crystals at large r_s. For a detailed tour of the mean-field phase diagram, we refer the readers to Ref. [422].

We note that Coulomb interaction induces significant finite-size correction to energy. We mitigate this by the standard technique of computing the Madelung energy, as detailed in Ref. [422].

Benchmarking Assessment: Case of Wigner Crystals

As a sanity check on our linear response methods, we focus first on $\lambda = 0$. There Eq. (6.46) reduces to the standard jellium model. Above $r_s = 3$ the mean field ground state is a triangular lattice Wigner crystal whose

elastic theory is well-understood [386]. Galilean invariance of the Hamiltonian fixes the effective mass to be $m = r_s^2/2$, while the shear stiffness can be computed in classical electrostatics as $\mu \approx 0.276/r_s + O(r_s^{-2})$. The kineo-elastic term vanishes by time-reversal symmetry. Fig. 6.3 shows the stiffness and mass computed via our linear response formalism, Eq. (6.41), from the mean-field ground states. The elastic coefficients match the above analytical expectations with great accuracy.

Furthermore, our linear-response elastic theory accurately describes the low-energy excitations of the Wigner crystal. Wigner crystals have two gapless collective modes: the transverse and longitudinal phonons whose dispersions are

$$\begin{aligned} \omega_t(\boldsymbol{q}) &= \sqrt{\frac{\mu}{m}}|\boldsymbol{q}| + \dots \\ \omega_l(\boldsymbol{q}) &= \sqrt{\frac{V(\boldsymbol{q})|\boldsymbol{q}|^2\rho}{m}} + \dots, \end{aligned} \tag{6.49}$$

respectively. For long-ranged Coulomb interactions $V(\boldsymbol{q}) = 4\pi/r_s|\boldsymbol{q}|$, leading to a standard plasmon dispersion $\omega_\ell \propto \sqrt{q}$, just as in Fermi liquids. Fig. 6.3(c) shows the dispersions $\omega_{t,l}$ predicted by the phonon effective action using the linear response coefficients. To verify them, we performed time-dependent Hartree-Fock (TDHF) computations of the entire spectrum of neutral excitations over the SCHF ground state, finding excellent quantitative agreement Fig. 6.3(c).

We comment that extracting the elastic coefficients from linear response theory requires care, particularly close to first-order phase transitions where numerical derivatives can be unstable. Altogether our results on Wigner crystals validate the linear response formulas, which we now apply to more complex models.

Soft and Heavy Crystals with Berry Curvature

We now add Berry curvature to the mix, studying λ-Jellium at $\lambda > 0$. There are two other crystalline states in that regime: a large anomalous Hall crystal phase that eventually undergoes a *continuous* transition into a second Wigner crystal phase at large (r_s, λ). In fact this second Wigner crystal has different angular momentum than the usual Wigner crystal, and its momentum occupations have a "halo" pattern around $\boldsymbol{q} = 0$; we call it the "halo" Wigner crystal (see also Ref. [389]). This seems counterintuitive — how can increasing λ drive a transition into a trivial phase? The reason lies in the single-particle bandstructure: at large λ the Berry curvature is sharply concentrated at $\boldsymbol{q} = 0$, approaching a δ-function [392]. When λ becomes large, the electrons near $\boldsymbol{q} = 0$ have a large exchange energy penalty. Heuristically, it is then preferable to not to occupy those states, which shifts

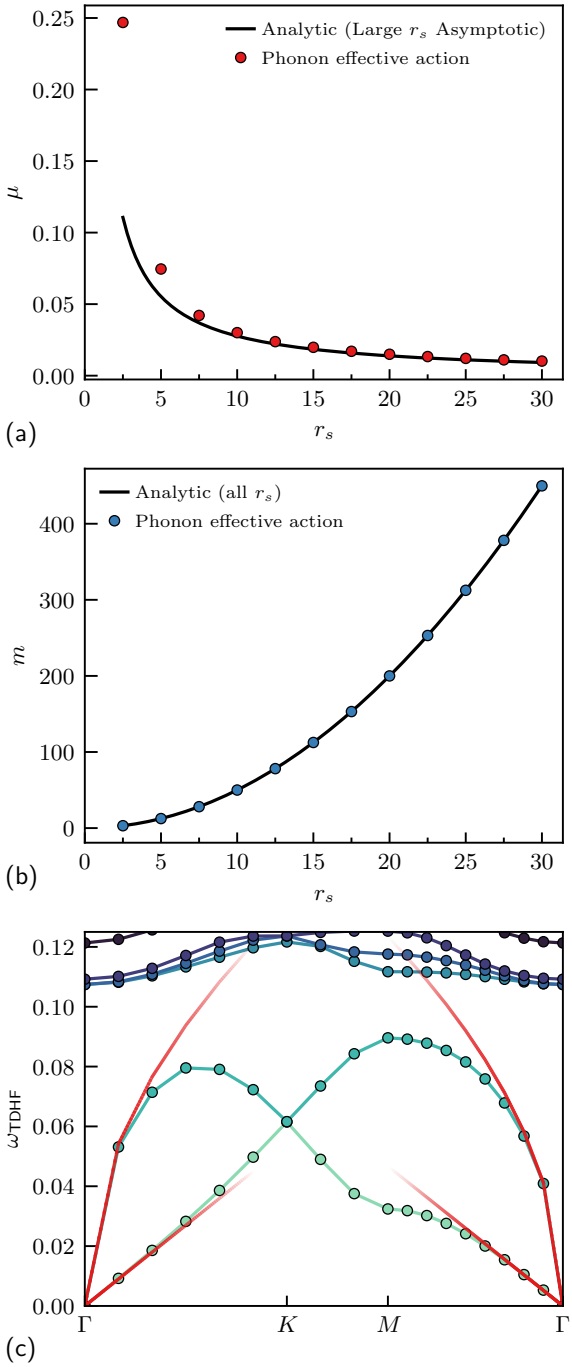


Figure 6.3: Sanity check of the elastic coefficients from linear response theory in the Wigner crystal. (a) Shear stiffness μ from linear response (circles) against the large r_s asymptotic result (lines) from Ref. [386]. The deviation at small r_s is expected. (b) Effective mass m predicted by Galilean invariance (lines) compared to the effective mass we obtain (dots). (c) TDHF spectrum at $r_s = 10$ (dots) compared to the phonon spectrum we extract from the effective action (red lines). All three comparisons (a,b,c) yield excellent agreement. Data in (a,b) are computed from SCHF (18×18) with 13 bands taken. The TDHF in (c) is computed on a 18×18 system with 7 bands taken.

the angular momentum and creates the "halo", after which the remaining electrons dominate the energetics, crystallizing into a triangular Wigner crystal [422].

When $\lambda > 0$, the Hamiltonian lacks Gailean invariance, allowing the mass term to vary. However, inversion symmetry remains, forcing the kineo-elastic term to vanish. We now examine the mass and stiffness of the crystalline states in this regime. We begin by studying triangular unit cells, but will soon find we must broaden our choice of lattice.

Fig. 6.4 shows the linear-response stiffness μ, mass m and the derived speed of sound $\sqrt{\mu/m}$ over the λ-jellium phase diagram. For each we show the phase diagram and line cuts of interest, which we now step through in sequence.

Stiffness is shown in Fig. 6.4(a-c). At asymptotically large r_s, the stiffness generally scales as $1/r_s$; we therefore plot $\mu \cdot r_s$. Fig. 6.4(b) shows linecuts at constant λ, which show the crystal softens considerably at intermediate λ before recovering at large λ to values close to those of the Wigner crystal.

In fact, the stiffness changes drastically upon the first order phase transition [422] from the WC1 to AHC, and becomes negative (Fig. 6.4(c)). The negative shear stiffness(the dashed region in Fig. 6.4a) signals an instability of the triangular AHC, which we analyze in detail in Sec. 6.3.1. The triangular AHC phase becomes stable above $\lambda = 0.75 - 1$, depending on r_s, where the stiffness becomes positive. The stiffness increases monotonically with λ beyond that point, recovering close to the WC value.

This can be understood from the single-particle wavefunctions of the λ-jellium. The electrons have a concentrated Berry curvature in a region of size $1/\lambda$ around the origin. On the other hand, most of the electrons outside the region have spin down, and therefore trivial. When λ becomes large, the trivial electrons dominate the energetics, and they crystalize into a triangular crystal as if they are in the jellium model. This claim is corroborated by the observation that the shear stiffness of the triangular crystal is smallest in the AHC phase when λ is small.

Previously, in Ref. [422], we observed that there is a continuous phase transition between the AHC phase and the halo WC phase. For $r_s = 15$, this phase transition takes place around $\lambda \approx 2.1$. The shear stiffness of the crystalline phases is seemingly continuous throughout the phase transition, again confirming the continuous nature of the phase transition.

In Fig. 6.4(d-f), we show the effective mass. Nonzero λ enhances the effective mass relative to the Galilean invariant value $r_s^2/2$. As the single-particle dispersion remains quadratic at nonzero λ, any change in the effective mass must be entirely driven by the non-Galilean invariance of projected interacting Hamiltonian. Intriguingly,

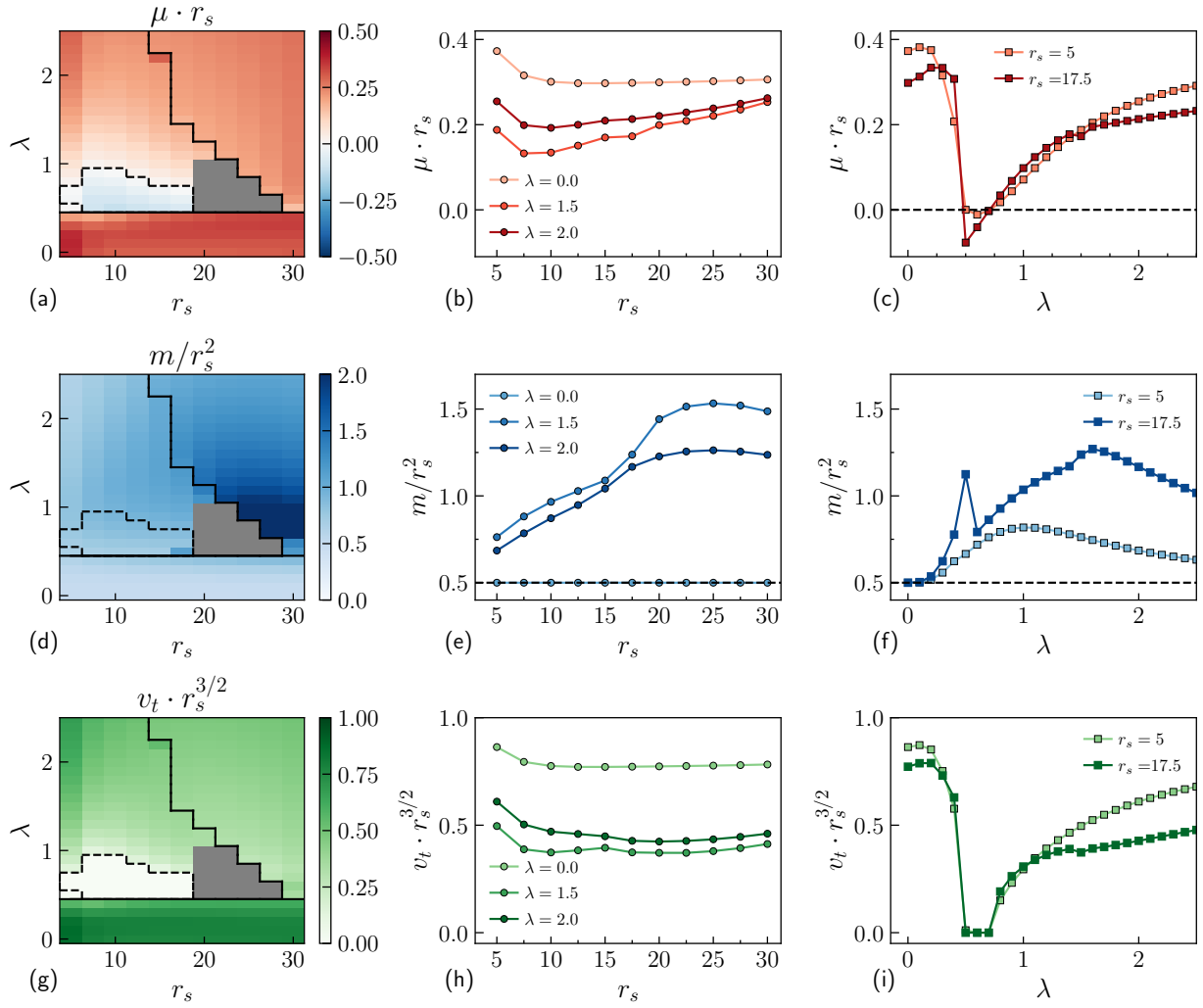


Figure 6.4: Numerical elastic parameters for the low-energy phonon theory of λ-jellium. Panels (a-c) show shear stiffness, panels (d-f) show effective mass, and panels (g-i) show velocity. Quantities are scaled by appropriate powers of r_s to make the large r_s limit order unity (see text). (a) Scaled shear stiffness $\mu \cdot r_s$ as a function of λ and r_s. The gray region suffers from convergence issues, precluding accurate determination of the stiffness. Solid lines mark the phase boundary of different crystals, and the dotted lines surround a region of negative stiffness. (b) Line cuts of scaled shear stiffness against r_s. (c) Line cuts of scaled shear stiffness against λ. Shear stiffness decreases precipitously at the first-order transition between WC and AHC, becoming negative in a small region (see text). (d) The scaled effective mass m/r_s^2 plotted against r_s and λ. (e) Line cut of the scaled effective mass versus r_s, showing interaction effects enhances the mass at nonzero λ. (f) Line cut of the normalized effective mass versus λ. An intermediate value of λ enhances effective mass drastically. (g) Scaled speed of sound $v_t \cdot r_s^{3/2}$ plotted against r_s and λ. (h) Line cut of the scaled speed of sound against r_s. (i) Line cut of the scaled speed of sound against λ. All data are computed from SCHF (18×18) with 13 bands taken, where a triangular unit cell was assumed.

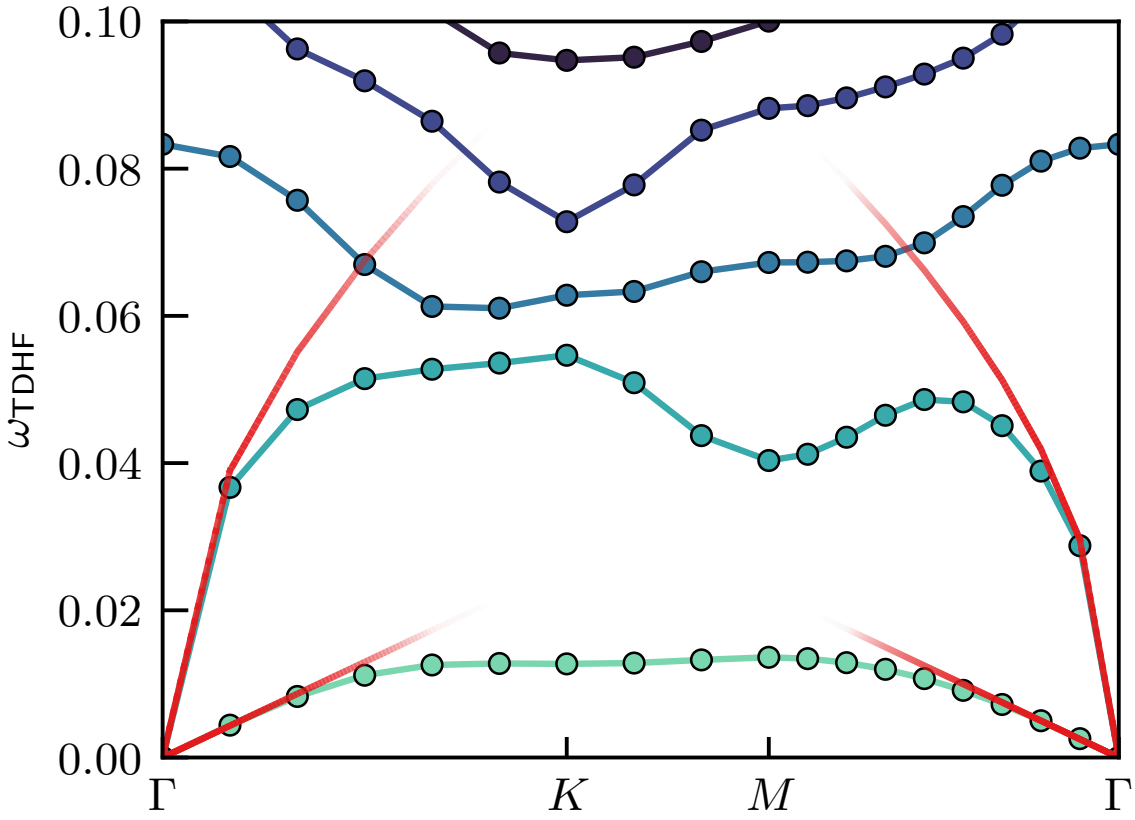


Figure 6.5: Phonon excitation spectrum of the anomalous Hall crystal in $\lambda-$jellium at parameters $(\lambda, r_s) = (1.5, 10)$. Dots are the time-dependent Hartree-Fock spectrum (size 18×18), while red lines come from the effective phonon action with coefficients from linear response. The two methods display excellent quantitative agreement.

the mass increases upon entering the halo WC, reaching up to four times the Galilean value. Understanding the origin of this behavior of quantum geometric quantities is an important future work.

The small stiffness and large mass at large λ conspire to make the phonon modes slow. We show the velocity of the transverse phonon mode, estimated by $\sqrt{\mu/m}$ in Fig. 6.4(g-i). The speed of sound is consistently smaller than the value at $\lambda = 0$ (Fig. 6.4 (h)). This estimate for the transverse velocity, as well as that of the plasmonic dispersion matches TDHF calculations well in small $\boldsymbol{q}$ limit (Fig. 6.5). Beyond acoustic phonons, the optical phonons of the AHC are significantly more dispersive than their WC counterparts. The origin of such behavior is worthy of future investigation.

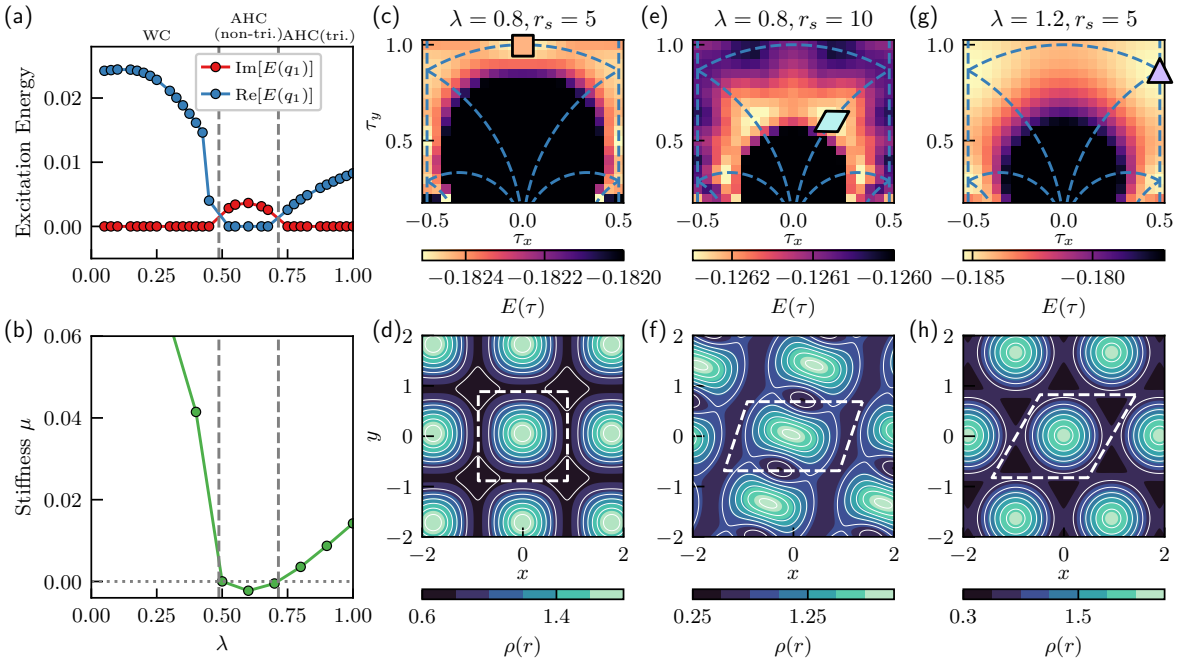


Figure 6.6: The shape of anomalous Hall crystal (AHC) depends on microscopic parameters. (a,b): for $r_s = 5$, the triangular lattice is locally unstable when $\lambda \in [0.5, 0.7]$. TDHF signals an instability for the triangular AHC by a non-zero imaginary part in the TDHF collective mode spectrum at $\boldsymbol{q}_1 = \boldsymbol{G}_1/12$. The shear modulus is also negative within this parameter regime, consistent with TDHF. (c) Schematic phase diagram as a function of λ: At small and large λ, the crystals are triangular WCs and triangular AHCs, respectively. Between these extremes, the AHC can generally take a non-triangular shape. (d): Energy landscape for different lattices parameterized by τ at $\lambda = 0.8, r_s = 5$. The energetically preferred lattice is the square lattice where $\tau = i$. (e): Energy landscape for $\lambda = 0.8, r_s = 10$. The energetically preferred lattice is a rhombic lattice where $\tau = 0.2 + 0.6i$. (f): Energy landscape for $\lambda = 1.2, r_s = 5$. The energetically preferred lattice is the triangular lattice. All data are computed from SCHF (24×24) with 13 bands taken.

AHC Energy Landscape for Different Lattice Choices

We now undertake a comprehensive study of the stability and instability of the triangular unit cell geometry in λ-jellium. As mentioned above, the stiffness of the triangular lattice AHC state is negative in a region near the WC phase boundary, enclosed by dashed lines in Fig. 6.4(a). A negative stiffness measurement implies the self-consistent Hartree-Fock ground state — which is *constrained* to use a triangular unit cell — is not even a local minimum in the energy landscape. As a crosscheck, we perform TDHF calculations at $r_s = 5$ as a function of λ (Fig. 6.6(a,b)). In the same regime where the stability is negative, we find the TDHF spectrum becomes imaginary, which occurs when the Hessian of the Hartree-Fock energy functional over the space of Slater determinants is not positive definite, which again implies the state is not at a local minimum.

This is consistent with our earlier Hartree-Fock work, which found that the square lattice AHC is energetically favored over the triangular lattice AHC in this same region, in stark contrast to the classical Wigner crystal where the square is disfavored [422]. This leads to an obvious question: is the AHC phase itself destabilized, or does it merely choose a different unit cell? To settle this, we now perform a comprehensive analysis of the landscape of unit cell shapes to find its *global* minimum.

As our model is rotationally symmetric, the space of unit cell shapes parametrized by the modular parameter τ of the torus, whose fundamental domain is shown in Fig. 6.1(b). The size is fixed by requiring one electron per unit cell. Fig. 6.6(c-h) shows the τ energy landscape. At $\lambda = 1.2$, Fig. 6.1(g,h), the minimum is at $\tau = e^{i2\pi/3}$, showing the triangular lattice AHC is not just a local minimum but the global minimum there. (One can see the landscape is indeed modular invariant.) At $\lambda = 0.8$ the landscape becomes extremely flat: the relative energy difference between the square lattice and the triangular lattice is around 0.02% for both $r_s = 5$ and $r_s = 10$. By comparison, in the classical limit the triangular Wigner crystal is preferred over the square WC by 0.5% — an order of magnitude larger. It is therefore no surprise that the global minimum shifts significantly with r_s, and is likely sensitive to other details. For $r_s = 5$, the square lattice is indeed the global minimum, Fig. 6.6(c,d). Meanwhile at $r_s = 10$, the energetically preferred lattice is rhombic, Fig. 6.6(e,f), with $\tau = 0.2 + 0.6i$ (note $|1 - \tau| = 1$.) As λ increases, the triangular lattice is increasingly favored, becoming the global minimum by $\lambda \approx 1$. The precise shape of the minimum energy unit cell depends sensitively on physical details; the robust feature of these landscape is their flatness.

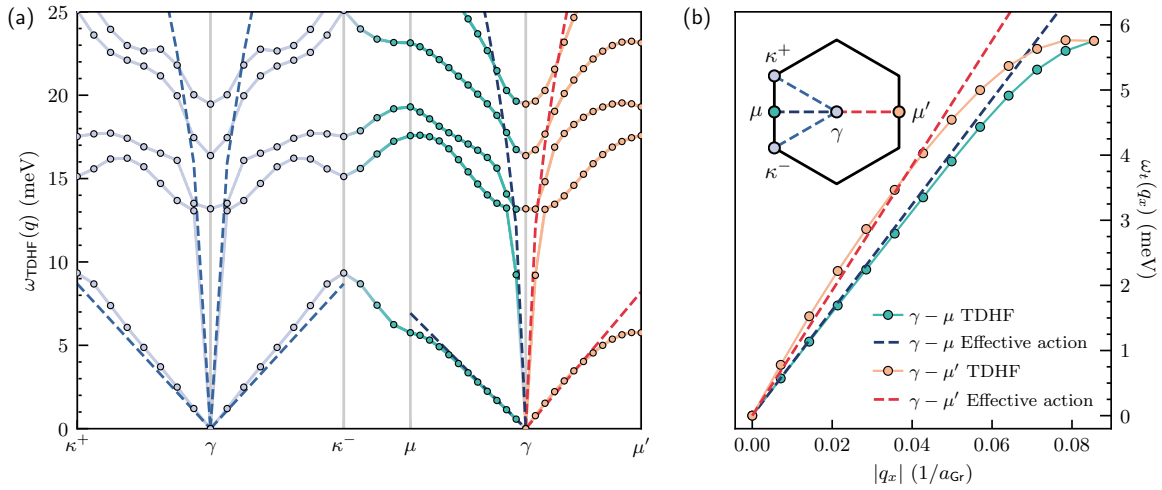


Figure 6.7: Time-dependent Hartree-Fock results for a triangular anomalous Hall crystal in rhombohedral multilayer graphene. Parameters: electronic density $n = 1.06 \times 10^{12}\mathrm{cm}^{-2}$, displacement field $u_D = 40\mathrm{meV}$, relative orientation between the electron lattice and graphene lattice $\phi = \pi/6$, and relative dielectric constant $\epsilon_r = 5$. (a) TDHF spectrum $\omega(\boldsymbol{q})$ along a high-symmetry line cut (dots), computed using a 24×24 torus. Colors represent values of $\cos(3\theta_{\boldsymbol{q}})$ where $\theta_{\boldsymbol{q}}$ is the angle between $\boldsymbol{q}$ and the x axis. Dashed lines correspond to phonon dispersions predicted by effective field theory with parameters extracted numerically: $\mu \approx 1.8\mathrm{meV}, m \approx 0.29 m_e, \ell \approx 1.3 \times 10^{-2}\mathrm{nm}^{-1}$. (b) Due to nonzero kineo-elastic coupling ℓ, the phonon speeds differ by $6\mathrm{km/s}$ along the x axis ($\mu - \gamma - \mu'$ line). All data are computed from SCHF (24×24) with 7 bands taken. For detailed estimates of physical quantities refer to Tab. 6.1.

6.3.2 Rhombohedral Multilayer Graphene

In this section, we turn to rhombohedral multilayer graphene (RMG), an experimental platform for which the anomalous Hall crystal was initially proposed by us and collaborators [320], along with simultaneous works [298, 299]. Recent works have drawn attention to the different possible choices of lattice for AHC phases in RMG, in particular the possible instability of a triangular AHC with one electron per unit cell [301, 380, 384, 423]. Here, we point out that the stability of such an AHC is dependent on microscopic parameters. For instance, the orientation of the emergent lattice is fixed by the trigonal warping of RMG, which breaks continuous rotation symmetry down to C_3 rotation symmetry. Also, and more importantly, we highlight that when a valley polarized crystal forms – i.e. when time reversal and inversion are both broken – the kineo-elastic coupling ℓ_{abc} is generally present. This coupling leads to surprising consequences for the collective modes of the system, such as a giant anisotropy in the phonon speeds along principle axes.

We consider a standard microscopic model of rhombohedral pentalayer graphene with screened Coulomb interactions, detailed in App. F.1. For definiteness, we focus on a low filling $n \approx 1.06 \times 10^{12}\mathrm{cm}^{-2}$ and high displacement field $u_D = 40\mathrm{meV}$, whose flat valence band minimum contains significant Berry curvature. For

comparison, this density would fill one band of a graphene-hBN moiré system with twist angle $0.9°$. On the triangular lattice, the mean field ground state (polarized to a single spin and valley) is an anomalous Hall crystal. Unlike in λ-jellium, its energy depends on the relative orientation between the electronic crystal and the carbon lattice; we find the lowest energy configuration has relative orientation of $\phi = \pi/6$ between its crystal axis and $\boldsymbol{a}_1$ of graphene. * Within TDHF, detailed below, this $\pi/6$ anomalous Hall crystal has a purely real collective mode spectrum across the entire Brillouin zone. We conclude that this mean-field triangular anomalous Hall crystal is locally stable.

Given this local stability, we proceed to examine the triangular AHC phonon spectrum. Fig. 6.7(a) compares the phonon spectrum of the effective field theory, with linear response elastic coefficients from Eq. (6.41), to time-dependent Hartree-Fock numerics. As expected, we find quantitative agreement between the techniques at low energies for both transverse and longitudinal phonons.

Physical Quantity	Numerical Value
Electronic density ρ	$1.06 \times 10^{12}\mathrm{cm}^{-2}$
Displacement field u_D	$40\mathrm{meV}$
Dielectric constant ϵ_r	5
Relative orientation	$30°$
Chern Number $\lvert C\rvert$	1
Charge gap	$15\mathrm{meV}$
Shear stiffness μ	$1.8\mathrm{meV}$
Effective mass m	$0.29m_e$
Kineo-elastic coupling ℓ	$1.3 \times 10^{-2}\mathrm{nm}^{-1}$
Transverse speed v_{+x}	$37\mathrm{km/s}$
Transverse speed v_{-x}	$31\mathrm{km/s}$
Transverse speed v_y	$33\mathrm{km/s}$

Table 6.1: Physical properties of the anomalous Hall crystal in a microscopic model of rhombohedral pentalayer graphene.

Table 6.1 shows the elastic parameters and other physical properties of the stable triangular anomalous Hall crystal — including a non-zero kineo-elastic coupling. Since the AHC is valley polarized, it breaks both inversion

*We note that Ref. [384] has found that the triangular AHC is unstable at a slightly smaller electronic density with $u_D = 36\mathrm{meV}$; this is likely due to a difference in microscopic modeling. Furthermore, the same reference finds that when the triangular lattice is aligned with the graphene lattice, the energy is minimized. We find that in contrast that with our parameters it is most stable when there is a 30-degree relative rotation between them.

and time reversal symmetry. Thus the kineo-elastic terms ℓ_{abc} are symmetry allowed. As discussed in Sec. 6.2.1, the kineo-elastic tensor has only two independent components in the presence of C_3 symmetry:

$$\begin{aligned}\ell_{abc}u_a\partial_b u_c = \ell(\dot{u}_x(u_{xx}-u_{yy}) - \dot{u}_y(2u_{xy})) \\ + \ell'(\dot{u}_y(u_{xx}-u_{yy}) + \dot{u}_x(2u_{xy})).\end{aligned} \tag{6.50}$$

Furthermore, $M_x\mathcal{T}$ symmetry acts within a valley, implying $\ell' = 0$. To measure the single remaining component ℓ, we apply shear strain θ_{xy}, which we find induces a non-zero momentum in the y-direction (fixed by $M_x\mathcal{T}$). Computing the resultant $\langle\hat{p}_y\rangle$ easily yields ℓ via Eq. (6.41).

The presence of kineo-elastic coupling ℓ significantly alters the phonon dispersion. It produces a large anisotropy in the transverse phonon speeds

$$v_\pm = \frac{\sqrt{4m\mu+\ell^2}\pm\ell}{2m} \tag{6.51}$$

along the $\pm x$ directions, respectively. Fig. 6.7(b) shows this clear asymmetry along the the $\mu-\gamma-\mu'$ line: $v_+ - v_- = 6\mathrm{km/s}$, a 20% relative difference. Conversely, there is no asymmetry along the y direction and its C_3 images due to $M_x\mathcal{T}$; the transverse velocity along the $\kappa^+-\gamma-\kappa^-$ line is $v_t = \sqrt{\mu/m} \approx 33\mathrm{km/s}$.

The longitudinal phonons are unaltered by the kineo-elastic coupling. Our Coulomb interactions are screened beyond the gate distance d, so the longitudinal phonons are linear up to $q \approx 1/d$, whereupon they crossover to the characteristic plasmon-like behavior. We take $d = 250$, on the order of the smallest momentum transfer in our 24×24 system, which produces an approximately $\sqrt{q}$ dispersion throughout the Brillouin zone. Furthermore, we observe a series of avoided crossings between the longitudinal phonons and higher bands of collective modes, which are beyond the low-energy effective theory.

6.4 Discussion

In this paper, we analyzed the low energy effective field theory of electronic crystals, with the goal of applying it to the anomalous Hall crystal phase. We derived a method for numerically extracting the coefficients of the low-energy field theory. Two terms in the low-energy field theory are of particular interest: (i) The time reversal breaking Berry phase term β, which can gap out the phonons. We find $\beta = 0$ due to the commuting translation generators. (ii) The kineo-elastic term ℓ_{abc}, which adds a significant anisotropy to the speed of sound, and is only allowed when both time reversal symmetry and inversion symmetry are broken.

We derived on the lowest-order gradient expansion of the phonon effective action. Higher-order gradient terms can reveal other nontrivial features of the crystal. For example, Ref. [397] found that the presence of anomalous velocity modifies the frequency of the phonon modes. We expect other quantum geometric effects to affect higher order terms in the phonon action as well, which we leave for future work.

This paper extracted the parameters of the phonon effective action using Hartree-Fock methods. Such methods can be improved by considering more general schemes to evaluate the ground state energy in the presence of shears and boosts. Two directions are particularly natural: (i) The Hartree-Fock approximation can be viewed as the lowest order conserving approximation to the true effective action, within the 2PI formalism [424] (See also App. F.3). This approximation can be improved by keeping higher order diagrams in the Luttinger-Ward functional. (ii) Viewing Hartree-Fock as a variational method where one searches for the lowest-energy Slater determinant state, one can improve the estimation of the ground state by enlarging the variational manifold with numerical methods such as variational Monte-Carlo methods. Our approach gives us the ability to access dynamical information from ground state energies alone.

The kineo-elastic term ℓ_{abc} we proposed renders the phonons non-reciprocal [425, 426]. For instance, at a point within the AHC phase of RMG, we found that the kineo-elastic term leads to a phonon speed difference up to $6\mathrm{km/s}$ in opposite directions. Previous studies of non-reciprocal phonons have focused on $\mathcal{PT}$-symmetric antiferromagnets [427], the phonon magneto-chiral effect where a magnetic field is applied to a chiral material [428–430], metamaterials [431–433], or surface acoustic waves [434, 435]. In particular, to the best of our knowledge, such a kineo-elastic term was not previously proposed in the literature, although magnon-phonon coupling could induce such an effect [436] [†]; prior discussions of non-reciprocal effects have focused on higher gradient terms [427, 437, 438], which do not affect the speed of sound at $\boldsymbol{q} = 0$. In contrast, the kineo-elastic term dramatically modifies the speed of sound.

Finally, we comment on the implications of this work for rhombohedral graphene systems. The kineo-elastic term will exist in the phonon effective action whenever the electronic crystal is valley polarized. Thus, the anisotropic speed of sound can probe "valley polarization", even within the high-resisitivity phase which is observed at low densities of rhombohedral graphene [21], a putative *Wigner crystal*, whose valley ordering remains an open question. We also find that the triangular anomalous Hall crystal can become unstable for other microscopic

[†]We thank Di Xiao for pointing out to us that in such scenarios the kineo-elastic coupling is several orders of magnitudes smaller due to the magnon-phonon coupling and thus usually ignored.

parameter choices, and indeed Refs. [301, 384] have reported the triangular lattice is destabilized under other parameters. One possibility is that, just as in λ-jellium, the global mean-field ground state at those parameters is also an anomalous Hall crystal — but with a potentially different orientation and shape, such as an oblong rectangle. Exploring the unrestricted landscape of unit cell shapes, orientations, and even expanding the unit cell area is an important direction for future work.

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A

Supplemental Material for Chapter 1

A.1 Holomorphic Boundary conditions and LLL-like wavefunctions

In this section we prove a number of results involving ideal band geometry that we claimed in the main text. First we prove the generality of the boundary conditions (1.10) for a single band with $C > 0$. Then we use a similar argument to Ref. [22], which assumed a gauge-equivalent version of (1.10), to show that all single $C = 1$ bands have an LLL form, in the sense of (1.11).

A.1.1 Classification of holomorphic boundary conditions

We begin by showing that for a single ideal band with Chern number $C > 0$ it is always possible to perform a gauge transformation $u_k \to \lambda_k u_k$ such that

$$\Xi_{\boldsymbol{G}}(k) = \lambda_{k+G}\lambda_k^{-1}\Xi^{(\mathrm{LG})}_{\boldsymbol{G}}(k - k_0) \tag{A.1}$$

where $\boldsymbol{k}_0 \simeq \boldsymbol{k}_0 + \boldsymbol{G}_i/C$ is a vector in the C^2-fold reduced first BZ and

$$\Xi^{(\mathrm{LG})}_{\boldsymbol{G}_1}(k) = \exp(2\pi i C k/G_2), \qquad \Xi^{(\mathrm{LG})}_{\boldsymbol{G}_2}(k) = 1. \tag{A.2}$$

We have added a superscript (LG), relative to the main text, to distinguish the choice (A.2) from other boundary conditions written in this section.

To our knowledge, (A.1) requires the use of some type of cohomology-based classification; we do not know of

an explicit construction. While the Chern number expression

$$2\pi C = f_{G_1}(k+G_2) - f_{G_1}(k) + f_{G_2}(k) - f_{G_2}(k+G_1), \qquad \Xi_{\boldsymbol{G}}(k) = e^{if_{\boldsymbol{G}}(k)} \tag{A.3}$$

suggests a linear form of $f(k)$, such that C is manifestly k-independent as in (A.2), we cannot immediately assume such a linear form of $f_{\boldsymbol{G}}(k)$. Indeed, there are nonlinear functions $f_{\boldsymbol{G}}(k)$ that yield a k-independent C in (A.3). Consider, for example, applying a nonlinear gauge transformation such as $u_k \to \exp(ik^3/G_1^3)u_k$: such a transformation preserves C but takes a linear $f_{\boldsymbol{G}}(k)$ to a nonlinear $f_{\boldsymbol{G}}(k) + 3iGk^2/G_1^3$. It is not clear how to directly show that all non-linearities may be gauged away.

Instead, we rely on the relationship of $\Xi_{\boldsymbol{G}}(k)$, often dubbed a "factor of automorphy" in the mathematics literature, to holomorphic line bundles [439, 440]. In particular, we consider the set of $\Xi_{\boldsymbol{G}}(k)$ that satisfies the cocycle condition

$$\Xi_{\boldsymbol{G}+\boldsymbol{G}'}(k) = \Xi_{\boldsymbol{G}'}(k+G)\Xi_{\boldsymbol{G}}(k) \tag{A.4}$$

and consider two choices of Ξ equivalent if they are related by a gauge transformation (A.1). Then a choice of Ξ corresponds exactly to a holomorphic line bundle over the Brillouin Zone torus; the gauge redundancy (A.1) corresponds exactly to the equivalence of line bundles with different choices of transition functions [439, 440]. The product $\Xi_1\Xi_2$ corresponds to the tensor product of the associated line bundles, and $\Xi \to \Xi^{-1}$ is the mapping to the dual bundle where all transition functions are inverted. The collection of Ξ up to gauge transformations and the set of holomorphic line bundles form isomorphic Abelian groups.

Holomorphic line bundles on the torus, and therefore boundary conditions $\Xi_{\boldsymbol{G}}(k)$ up to gauge transformations, can be classified via the Appell-Humbert theorem [439, 440]. The Appell-Humbert theorem yields a single canonical choice of $\Xi_{\boldsymbol{G}}(k)$ for each distinct line bundle. This canonical choice is usually written in a "symmetric gauge" form rather than our choice (A.2), though they may be related through a suitable gauge transformation.

Instead we will use the Appell-Humbert theorem for $C = 0$, which in practice is how the whole theorem is proved [439, 440]. The theorem states that $\Xi^{(0)}$ has $C = 0$ in (A.3) if and only if it is gauge equivalent to $e^{-i\boldsymbol{G}\cdot\boldsymbol{r}_0}$ where $\boldsymbol{r}_0$ is an element of the "dual torus"[440], i.e. the real-space unit cell. Now consider some $\Xi_{\boldsymbol{G}}(\boldsymbol{r})$ with Chern number C. Then $\Xi^{(0)} = \Xi_{\boldsymbol{G}}(k)/\Xi_{\boldsymbol{G}}^{(\mathrm{LG})}(k)$ has $C = 0$ in (A.3) and therefore is equivalent to

$e^{-i\boldsymbol{G}\cdot\boldsymbol{r}_0}$ for some $\boldsymbol{r}_0$ by Appell-Humbert. Therefore, we may always apply a gauge transformation such that

$$\Xi_{\boldsymbol{G}}(k) = e^{-i\boldsymbol{G}\cdot\boldsymbol{r}_0}\Xi^{(\mathrm{LG})}_{\boldsymbol{G}}(k). \tag{A.5}$$

We now show how to convert a choice of $\boldsymbol{r}_0$ in the real-space unit cell to a choice of $k_0 \equiv k_0 + \boldsymbol{G}_i/C$ for $C \neq 0$. We perform the gauge transformation

$$\Xi_{\boldsymbol{G}}(k) \to \lambda_{\boldsymbol{k}+\boldsymbol{G}}\Xi_{\boldsymbol{G}}(k)\lambda_k^{-1}, \qquad \lambda_k = \exp(i\boldsymbol{G}_2\cdot\boldsymbol{r}_0 k/G_2) \tag{A.6}$$

which has the effect

$$\Xi_{\boldsymbol{G}_1}(k) = e^{-i\boldsymbol{G}_1\cdot\boldsymbol{r}_0}\Xi^{(\mathrm{LG})}_{\boldsymbol{G}_1}(k) \to \Xi^{(\mathrm{LG})}_{\boldsymbol{G}_1}(k-k_0), \qquad \Xi_{\boldsymbol{G}_2}(k) = e^{-i\boldsymbol{G}_2\cdot\boldsymbol{r}_0} \to 1 \tag{A.7}$$

with

$$\boldsymbol{k}_0 = -\frac{A_{\mathrm{BZ}}}{2\pi C}\hat{\mathrm{BZ}}\times\boldsymbol{r}_0. \tag{A.8}$$

Here $A_{\mathrm{BZ}} = \hat{\mathrm{BZ}}\cdot\boldsymbol{G}_1\times\boldsymbol{G}_2$ is the area of the Brillouin Zone. To simplify the first equation of (A.7) we have written $\boldsymbol{r}_0 = r_{1,0}\boldsymbol{a}_1 + \boldsymbol{r}_{2,0}\boldsymbol{a}_2$ and identified $-r_{1,0}\boldsymbol{G}_2 + r_2\boldsymbol{G}_1 = -\frac{A_{\mathrm{BZ}}}{2\pi}\hat{\mathrm{BZ}}\times\boldsymbol{r}_0$ as a map from the unit cell to the first BZ. The further division by C in (A.8) ensures that k_0 is equivalent to $k_0 + G_{1,2}/C$ because $\boldsymbol{r}_0$ is equivalent to $\boldsymbol{r}_0 + \boldsymbol{a}_{1,2}$.

A.1.2 LLL form of $C = 1$ ideal band wavefunctions

We now show how to prove (1.11), copied below for convenience. This argument was previously given in Ref. [22]; we are repeating it here so that our discussion is self-contained.

$$\psi_{\boldsymbol{k}\alpha}(\boldsymbol{r}) = \psi_{\boldsymbol{k}-(\boldsymbol{k}_0-\boldsymbol{k}_0^{\mathrm{LLL}})}(\boldsymbol{r})\mathcal{N}_\alpha(\boldsymbol{r}). \tag{A.9}$$

Here $\psi_{\boldsymbol{k}\alpha}(\boldsymbol{r}) = e^{i\boldsymbol{k}\cdot\boldsymbol{r}}u_k(\boldsymbol{r})$ is the wavefunction for the ideal $C = 1$ band of interest written in the holomorphic gauge corresponding to (A.2) for some k_0. The index α corresponds to a non-positional orbital degree of freedom, for example layer. Similarly, $\psi^{\mathrm{LLL}}_{\boldsymbol{k}}(\boldsymbol{r}) = e^{i\boldsymbol{k}\cdot\boldsymbol{r}}u^{\mathrm{LLL}}_k(\boldsymbol{r})$ is the LLL wavefunction written in holomorphic gauge (A.2) which is constructed explicitly in the next section.

Consider the ratio

$$\frac{\psi_{\boldsymbol{k}\alpha}(\boldsymbol{r})}{\psi^{\mathrm{LLL}}_{\boldsymbol{k}-(\boldsymbol{k}_0-\boldsymbol{k}_0^{\mathrm{LLL}})}(\boldsymbol{r})} = e^{-i(\boldsymbol{k}_0-\boldsymbol{k}_0^{\mathrm{LLL}})\cdot\boldsymbol{r}} \frac{u_{k\alpha}(\boldsymbol{r})}{u^{\mathrm{LLL}}_{k-(k_0-k_0^{\mathrm{LLL}})}(\boldsymbol{r})} \equiv \mathcal{N}_{\alpha k}(\boldsymbol{r}). \tag{A.10}$$

Here, the final equality defines $\mathcal{N}_{\alpha k}(\boldsymbol{r})$, as a holomorphic function which we now show is k-independent. Note that $\mathcal{N}_{\alpha k}(\boldsymbol{r})$ is a periodic under $k \to k + G_i$ because $\psi_{\boldsymbol{k}}$ and $\psi^{\mathrm{LLL}}_{\boldsymbol{k}-(\boldsymbol{k}_0-\boldsymbol{k}_0^{\mathrm{LLL}})}$ have the same boundary conditions.

Furthermore, $\mathcal{N}_{\alpha k}(\boldsymbol{r})$ can have at most one pole: the wavefunction $u^{\mathrm{LLL}}_{\boldsymbol{k}}(\boldsymbol{r})$, for fixed $\boldsymbol{r}$, only has a single zero in the Brillouin Zone. The lack of extra zeros may be seen by through the explicit construction in the next section or proved on general grounds through the expression [22, 23]

$$2\pi i C = \oint_{\partial \mathrm{BZ}} \ln u_k(\boldsymbol{r}) dk = 2\pi i N_z. \tag{A.11}$$

where the integral around the boundary of the Brillouin Zone counts the number of zeros N_z of $u_k(\boldsymbol{r})$ in the first BZ. The relation to the Chern number is recovered by computing the integral using the boundary conditions of u and recovering (A.3) [22, 23].

Periodic holomorphic functions with only one pole per unit cell are constant. To see this, note that Cauchy's integral theorem $\oint_{\mathrm{BZ}} N_{k\alpha}(\boldsymbol{r})dk$ computes the residue of the pole but this integral vanishes since N_k is periodic. Thus N_k is bounded, but bounded holomorphic functions are constant by Liouville's theorem. We have therefore shown that $N_{k\alpha}(\boldsymbol{r})$ is k-independent, such that we recover (A.9).

We emphasize that this argument relies on $C = 1$; higher Chern bands are not related to each other by k-independent factors because their ratio may be an elliptic function with more than one pole per unit cell (with total residue zero). This complexity is difficult to work with, which is one way to view the motivation to decompose higher Chern bands into multiple $C = 1$ bands.

A.2 Magnetic Bloch states of the Lowest Landau level

In this section we review the form of magnetic translations and magnetic Bloch states in symmetric electromagnetic gauge. We then construct the LLL wavefunctions with a suitable momentum-space gauge (1.10) and construct the gauge-dependent magnetic translation form (1.16).

We take our magnetic field to be a constant value of $B > 0$ and choose lattice vectors $\boldsymbol{R}_{1,2}$ that enclose 2π

flux and corresponding reciprocal lattice vectors $\boldsymbol{Q}_{1,2}$ such that $\boldsymbol{Q}_i \cdot \boldsymbol{R}_j = 2\pi\delta_{ij}$:

$$B\hat{\mathrm{BZ}} \cdot \boldsymbol{R}_1 \times \boldsymbol{R}_2 = 2\pi = B^{-1}\hat{\mathrm{BZ}} \cdot \boldsymbol{Q}_1 \times \boldsymbol{Q}_2. \tag{A.12}$$

For our applications to higher Chern bands, the lattice vectors $\boldsymbol{R}$ will describe a $|C|$-fold enlarged real space lattice and the reciprocal lattice vectors $\boldsymbol{Q}$ will describe the folded Brillouin Zone.

The Hamiltonian is

$$H = \frac{|-i\boldsymbol{\nabla} - e\boldsymbol{A}|^2}{2m} = \frac{2}{m}\Pi\overline{\Pi} + \frac{B}{2m}, \quad \overline{\Pi} = (-i\overline{\partial} - \overline{A}) \tag{A.13}$$

where $2\overline{\partial} = \partial_x + i\partial_y$ and $2\overline{A} = A_x + iA_y$. The lowest Landau level is the zero mode space of $\overline{\Pi}$. The zero mode wavefunctions are of the form

$$\psi(\boldsymbol{r}) = f(z)e^{-K(\boldsymbol{r})} \tag{A.14}$$

where $-i\overline{\partial}K = \overline{A}$. We will choose the symmetric gauge vector potential $\boldsymbol{A} = B(-y, x)$ such that $2\overline{A} = B(-y + ix) = -iBz$ and $K = Bz\overline{z}/4$.

The magnetic translation operators are given by

$$T^{\text{mag}}_{\boldsymbol{l}} = e^{-i\zeta_{\boldsymbol{l}}(\boldsymbol{r})}e^{\boldsymbol{l}\cdot\boldsymbol{\nabla}}, \tag{A.15}$$

where

$$\boldsymbol{A}(\boldsymbol{r} + \boldsymbol{l}) - \boldsymbol{A}(\boldsymbol{r}) = \boldsymbol{\nabla}\zeta_{\boldsymbol{l}}(\boldsymbol{r}) \tag{A.16}$$

From this we can derive the commutation relation

$$T^{\text{mag}}_{\boldsymbol{l}_1}T^{\text{mag}}_{\boldsymbol{l}_2} = e^{iB\hat{z}\cdot\boldsymbol{l}_1\times\boldsymbol{l}_2}T^{\text{mag}}_{\boldsymbol{l}_2}T^{\text{mag}}_{\boldsymbol{l}_1} \tag{A.17}$$

such that the magnetic translations commute if $\boldsymbol{l}_i$ are lattice vectors $\boldsymbol{R}$ due to (A.12). Note, however, that there is a subtlety $T^{\text{mag}}_{\boldsymbol{R}_1}T^{\text{mag}}_{\boldsymbol{R}_2} = -T^{\text{mag}}_{\boldsymbol{R}_1+\boldsymbol{R}_2}$ since the triangle spanned by the points $\boldsymbol{R}_1, \boldsymbol{R}_2, \boldsymbol{R}_1 + \boldsymbol{R}_2$ encloses π flux. We therefore take $(T^{\text{mag}}_{\boldsymbol{R}_1})^m(T^{\text{mag}}_{\boldsymbol{R}_2})^n$ to be the Bloch translation associated to the lattice vector $m\boldsymbol{R}_1 + n\boldsymbol{R}_2$, which in general is not the same as $T^{\text{mag}}_{m\boldsymbol{R}_1+n\boldsymbol{R}_2}$.

Consider Bloch wavefunctions $\psi_{\boldsymbol{k}}(\boldsymbol{r})$ such that

$$T^{\text{mag}}_{\boldsymbol{R}}\psi_{\boldsymbol{k}}(\boldsymbol{r}) = e^{i\boldsymbol{k}\cdot\boldsymbol{R}}\psi_{\boldsymbol{k}}(\boldsymbol{r}). \tag{A.18}$$

On general grounds, using (A.17) we have

$$T^{\text{mag}}_{\boldsymbol{R}}T^{\text{mag}}_{\boldsymbol{l}}\psi_{\boldsymbol{k}} = e^{iB\hat{\text{BZ}}\cdot\boldsymbol{R}\times\boldsymbol{l}}T^{\text{mag}}_{\boldsymbol{l}}T^{\text{mag}}_{\boldsymbol{R}}\psi_{\boldsymbol{k}} = e^{iB\hat{\text{BZ}}\cdot\boldsymbol{R}\times\boldsymbol{l}}e^{i\boldsymbol{k}\cdot\boldsymbol{R}}T^{\text{mag}}_{\boldsymbol{l}}\psi_{\boldsymbol{k}} = e^{i(\boldsymbol{k}-B\hat{\text{BZ}}\times\boldsymbol{l})\cdot\boldsymbol{R}}e^{i\boldsymbol{k}\cdot\boldsymbol{R}}T^{\text{mag}}_{\boldsymbol{l}}\psi_{\boldsymbol{k}} \tag{A.19}$$

such that $T^{\text{mag}}_{\boldsymbol{l}}\psi_{\boldsymbol{k}}$ and $\psi_{\boldsymbol{k}-B\hat{\text{BZ}}\times\boldsymbol{l}}$ have the same eigenvalue under translations. We therefore conclude that $T^{\text{mag}}_{\boldsymbol{l}}$ acts as a momentum translation

$$T^{\text{mag}}_{\boldsymbol{l}}\psi_{\boldsymbol{k}} \propto \psi_{\boldsymbol{k}-B\hat{\text{BZ}}\times\boldsymbol{l}}, \tag{A.20}$$

where the proportionality constant is gauge dependent because the relative prefactor between $\psi_{\boldsymbol{k}}$ and $\psi_{\boldsymbol{k}-B\hat{\text{BZ}}\times\boldsymbol{l}}$ is gauge dependent. We now move onto gauge-specific calculations, in both the electromagnetic and momentum-space sense.

In our symmetric gauge, we have the translation operators

$$T^{\text{mag}}_{\boldsymbol{l}} = e^{-\frac{i}{2}B\boldsymbol{l}\times\boldsymbol{r}}e^{\boldsymbol{l}\cdot\boldsymbol{\nabla}}, \qquad T^{\text{mag}}_{\boldsymbol{R}} = \eta_{\boldsymbol{R}}T^{\text{mag}}_{\boldsymbol{R}} \tag{A.21}$$

and the wavefunctions [22, 34]

$$\psi_{\boldsymbol{k}}(\boldsymbol{r}) = \lambda_k e^{\frac{i}{2}\overline{k}'z}\sigma(z + iB^{-1}k')e^{-B|z|^2/4} \tag{A.22}$$

where $k' = k - k_0^{\text{LLL}}$ with $k_0^{\text{LLL}} = -(\boldsymbol{Q}_1 + \boldsymbol{Q}_2)/2$ and $\sigma(z) = \sigma(z|R_1, R_2)$ is the (modified [441]) Weierstrass sigma function which satisfies

$$\sigma(-z) = -\sigma(z), \qquad \sigma(z + R) = \eta_{\boldsymbol{R}}e^{\frac{B}{2}\overline{R}(z+\frac{R}{2})} \tag{A.23}$$

where $\eta_{\boldsymbol{R}} = 1$ if $\boldsymbol{R}/2$ is a lattice vector and -1 otherwise. From (A.23) it is straightforward to verify the Bloch boundary conditions of (A.22). The relations $\hat{\text{BZ}}\cdot\boldsymbol{R}\times\boldsymbol{r} = -\frac{i}{2}(\overline{R}z - R\overline{z})$ and $\boldsymbol{R}\cdot\boldsymbol{r} = \frac{1}{2}(\overline{R}z + R\overline{z})$ for transforming between vector and complex notation are helpful here.

The wavefunction (A.22) is in a holomorphic gauge; this can be straightforwardly verified by computing $u_k(\boldsymbol{r}) = e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}\psi_{\boldsymbol{k}}(\boldsymbol{r}) = e^{-\frac{i}{2}\overline{k}z}e^{-\frac{i}{2}z\overline{k}}\psi_{\boldsymbol{k}}(\boldsymbol{r})$ such that the non-holomorphic $e^{\frac{i}{2}\overline{k}z}$ factor in ψ is cancelled. The holomorphic factor λ_k parameterizes the remaining holomorphic gauge freedom; we now fix λ_k such that the boundary conditions of (A.22) match (1.10), (A.2) for $k_0 = 0$. To do this, we simply engineer λ_k such that $\Xi_{\boldsymbol{Q}_2}(k) = 1$. We obtain

$$1 = \lambda_{k+Q_2}\lambda_k^{-1}e^{i\pi+\frac{1}{2B}\overline{Q}_2\left(k'+\frac{Q_2}{2}\right)} \implies \lambda_k = \exp\left(-i\pi\frac{k'}{Q_2} - \frac{\overline{Q}_2 k'^2}{4BQ_2}\right). \tag{A.24}$$

We now compute, using (A.12)

$$\Xi_{\boldsymbol{Q}_1}(k) = \lambda_{k+Q_1}\lambda_k^{-1}e^{i\pi+\frac{1}{2B}\overline{Q}_1\left(k'+\frac{Q_1}{2}\right)} = e^{-i\pi\frac{Q_1}{Q_2}}\exp\left(\frac{1}{2}B^{-1}(\overline{Q_1}Q_2 - Q_1\overline{Q_2})\left(k'+\frac{Q_1}{2}\right)/Q_2\right) = e^{2\pi ik'/Q_2} \tag{A.25}$$

such that we obtain the desired boundary conditions (A.2) with $k_0 = k_0^{\mathrm{LLL}} = -(Q_1+Q_2)/2$.

We now compute $c_{\boldsymbol{l}}(k)$ where

$$T_{\boldsymbol{l}}^{\mathrm{mag}}\psi_{\boldsymbol{k}} = c_{\boldsymbol{l}}(\boldsymbol{k})\psi_{\boldsymbol{k}-B\hat{\mathrm{BZ}}\times\boldsymbol{l}}. \tag{A.26}$$

Through direct computation we obtain

$$\begin{aligned} c_{\boldsymbol{l}}(\boldsymbol{k}) &= \frac{\lambda_k}{\lambda_{k-iBl}}\exp\left[\frac{i}{2}l\left(\overline{k}' - \frac{iB\overline{l}}{2}\right)\right] \\ &= e^{\pi Bl/Q_2}\exp\left[\frac{-i}{2}\frac{\overline{Q}_2}{Q_2}l\left(k' - \frac{iBl}{2}\right) + \frac{i}{2}l\left(\overline{k}' + \frac{iB\overline{l}}{2}\right)\right] \\ &= e^{i\boldsymbol{k}\cdot\boldsymbol{l}}e^{\frac{i}{2}(\boldsymbol{Q}_1+\boldsymbol{Q}_2)\cdot\boldsymbol{l}}e^{\pi Bl/Q_2}\exp\left[-\frac{i}{2}\left(\frac{\overline{Q}_2}{Q_2}l + \overline{l}\right)\left(k' - \frac{iBl}{2}\right)\right] \\ &= e^{i\boldsymbol{k}_0^{\mathrm{LLL}}}c_{\boldsymbol{l}}^{(0)}(\boldsymbol{k} - \boldsymbol{k}_0^{\mathrm{LLL}}) \end{aligned} \tag{A.27}$$

where

$$c_{\boldsymbol{l}}^{(0)}(\boldsymbol{k}) = e^{i\boldsymbol{k}\cdot\boldsymbol{l}}e^{\frac{i}{2}(\boldsymbol{Q}_1+\boldsymbol{Q}_2)\cdot\boldsymbol{l}}e^{\frac{\pi Bl}{Q_2} - \frac{i}{2}\left(\frac{\overline{Q}_2}{Q_2}l + \overline{l}\right)\left(k - \frac{iBl}{2}\right)} \tag{A.28}$$

as claimed in the main text (1.16).

A.3 Computation details of U_k

In this section we construct and prove the uniqueness of the matrix U_k (A.3) that hybridizes the folded ideal $C = 2$ band into two ideal $C = 1$ bands. We find it easier to compute the wavefunction of the ideal bands after decomposition directly, and then organize the coefficients into the matrix U_k.

We take a general holomorphic combination

$$\phi_{\boldsymbol{k}} = \alpha_0(k)\psi_{\boldsymbol{k}} + \alpha_1(k)\psi_{\boldsymbol{k}+\boldsymbol{G}_1/2} \tag{A.29}$$

and require that

$$\phi_{\boldsymbol{k}+\boldsymbol{G}_1/2} = \xi_{1,g}(k)\phi_{\boldsymbol{k}}, \qquad \phi_{\boldsymbol{k}+\boldsymbol{G}_2} = \phi_{\boldsymbol{k}} \tag{A.30}$$

We will find two distinct solutions α_0, α_1; these two solution sets will comprise the two rows of U_k. Inserting (A.29) into (A.30), we obtain the following conditions for $\alpha_{0,1}$:

$$\begin{aligned} \alpha_0(k + G_1/2) &= \alpha_1(k)\xi_{1,g}(k) \\ \alpha_1(k + G_1/2)\xi_{2,k_0}(k) &= \alpha_0(k)\xi_{1,g}(k) \end{aligned} \tag{A.31}$$

The first equation gives $\alpha_1(k) = \alpha_0(k + G_1/2)\xi_{1,g}^{-1}(k)$; further using the second equation, we have

$$\alpha_0(k + G_1)\xi_{1,g}^{-1}(k + G_1/2)\xi_{2,k_0}(k) = \alpha_0(k)\xi_{1,g}(k), \tag{A.32}$$

which means that

$$\frac{\alpha_0(k + G_1)}{\alpha_0(k)} = \exp(2\pi i(2k_0 - 2g + G_1/2)/G_2) = \exp(\beta). \tag{A.33}$$

for some k-independent β. We also have $\alpha(k + G_2) = \alpha(k)$ from the second equation in (A.30). Defining $\alpha_0(k) = e^{\gamma(k)}$ then yields

$$\gamma(k + G_1) - \gamma(k) = \beta, \qquad \gamma(k + G_2) - \gamma(k) = 2\pi i n. \tag{A.34}$$

Since $\gamma(k)$ has no poles, a contour integral along boundaries of the Brillouin zone yields, via the residue theorem,

$$\beta G_2 - 2\pi i n G_1 = 0. \tag{A.35}$$

We multiply the wavefunction by an overall constant such that $\alpha_0(k=0) = 1$ and $\gamma(0) = 0$. We define $\delta(k) = \gamma(k) - \beta k/G_1$ such that $\delta(k)$ vanishes on reciprocal lattice vectors $aG_1 + bG_2$ for $a, b \in \mathbb{Z}$. Since $\delta(k)$ is a periodic holomorphic function it must be constant, such that $\delta(0) = 0$, $\delta(k) = 0$ and thus $\gamma(k) = 2\pi i n k/G_2$.

From (A.33), we deduce that for each g that satisfies $\exp(2\pi i(2k_0 - 2g + G_1/2 + nG_1)/G_2)$ (A.30). Taking a logarithm, and encoding the associated $2\pi i$ ambiguity with an integer m, we obtain

$$g = k_0 + G_1/4 - nG_1/2 - mG_2/2. \tag{A.36}$$

The shift in $G_1/2$ may be gauged away because $\boldsymbol{G}_1/2$ is now a reciprocal lattice vector (see SI Sec. A.1.1). We can then choose $n = 0$ such that $\alpha_0 = 1$. Since $\boldsymbol{G}_2$ is a reciprocal lattice vector, there are two distinct choices are $m = 0, 1$. We then obtain the two possible boundary conditions

$$\xi_m = \xi_{1,k_0+G_1/4-mG_2/2}(k) = (-1)^m \xi_{1,k_0+G_1/4}(k) \tag{A.37}$$

for which the wavefunctions $\phi_{m\boldsymbol{k}}$ are

$$\phi_{m\boldsymbol{k}} = \psi_{\boldsymbol{k}} + (-1)^m \xi^{-1}_{1,k_0+G_1/4}(k)\psi_{\boldsymbol{k}+\boldsymbol{G}_1/2}. \tag{A.38}$$

We may then read off the matrix U_k

$$U_k = \begin{pmatrix} 1 & \xi^{-1}_{1,k_0+G_1/4}(k) \\ 1 & \xi^{-1}_{1,k_0+G_1/4-G_2/2}(k) \end{pmatrix}. \tag{A.39}$$

A.4 Chern Bands with $C > 2$

For $C > 2$ bands, a similar procedure can be performed. However, in general we can attempt to generate r bands, in which $r < C$. We break translation symmetry in $\boldsymbol{a}_1$ direction by r times:

$$\phi_{\boldsymbol{k}} = \sum_m \alpha_m(k)\psi_{\boldsymbol{k}+m\boldsymbol{G}_1/r} \tag{A.40}$$

and try to "diagonalize" the boundary conditions:

$$\phi_{\boldsymbol{k}+\boldsymbol{G}_1/r} = \xi_{1,g}(k)\phi_{\boldsymbol{k}}, \qquad \phi_{\boldsymbol{k}+\boldsymbol{G}_2} = \phi_{\boldsymbol{k}} \tag{A.41}$$

We find that when r doesn't divide C the two boundary conditions (A.41) are not compatible. When r does divide C we find r solutions to (A.41), leading to an $r \times r$ holomorphic matrix U_k. The resulting bands (A.40) have Chern number C/r. We must solve the following equations similar to (A.31):

$$\begin{aligned}
&\alpha_0(k + G_1/r) = \alpha_1(k)\xi_{C/r,g}(k) \\
&\alpha_1(k + G_1/r) = \alpha_2(k)\xi_{C/r,g}(k) \\
&\cdots \\
&\alpha_{r-1}(k + G_1/r)\xi_{C,k_0}(k) = \alpha_0(k)\xi_{C/r,g}(k)
\end{aligned} \tag{A.42}$$

All coefficients are fundamentally related to α_0. Using techniques analogous to those in the previous section, we find

$$U_{ml} = \alpha_{ml}(k) = e^{-2\pi iml/r} \prod_{n<l} \xi^{-1}_{C/r,k_0+(r-1)G_1/2r}(k + nG_1/r) \tag{A.43}$$

We get r bands $\phi_{m\boldsymbol{k}}$, each with Chern number C/r, with boundary conditions

$$\phi_{m(\boldsymbol{k}+\boldsymbol{G}_1/r)} = \xi_{C/r,k_0+(r-1)G_1/2r}(k)e^{2\pi im/r}\phi_{m\boldsymbol{k}}, \qquad \phi_{m(\boldsymbol{k}+\boldsymbol{G}_2)} = \phi_{m\boldsymbol{k}} \tag{A.44}$$

We now specialize to the case $r = C$ in order to derive the $\mathrm{SU}(C)$ action. We find that $\phi_{m(\boldsymbol{k}+\boldsymbol{G}_2/2-m\boldsymbol{G}_2/C)}$

all have the same boundary conditions. We thus define

$$\tilde{\phi}_{m\boldsymbol{k}} = \phi_{m(\boldsymbol{k}+\boldsymbol{G}_2/2-m\boldsymbol{G}_2/C)} \tag{A.45}$$

Similar to $C = 2$, these states also form a non-unitary representation of $\mathrm{SU}(C)$ [442], with the similar identification

$$\begin{aligned} \mathsf{Z} &= (T^{\text{mag}}_{\boldsymbol{a}_2})^{-1} T_{\boldsymbol{a}_2}, \\ \mathsf{X} &= (T^{\text{mag}}_{\boldsymbol{a}_1})^{-1} T_{\boldsymbol{a}_1}, \ \mathsf{ZX} = e^{2\pi i/C}\mathsf{XZ} \end{aligned} \tag{A.46}$$

Next, we want to understand the ground state degeneracy in this system. In order to obtain the degeneracy of the $\mathrm{SU}(C)$ ferromagnet in a finite system, we need to identify the $\mathrm{SU}(C)$ irreducible representation for the ground states. One definite element of the ground state manifold is the state with full-polarization in a certain $C = 1$ band: $|0\rangle = \mathcal{A}\,|\phi_{0\boldsymbol{k}_1}, \phi_{0\boldsymbol{k}_2}, \ldots\rangle$ in which $\mathcal{A}$ antisymmetrizes particles. Since it is fully symmetrized with respect to the $SU(C)$ index, it is part of the irrep expressed by the following Young diagram:

$$\underbrace{\square\square\cdots\cdots\square}_{N_e} \tag{A.47}$$

whose dimension can be calculated by the Hook's formula:

$$\dim\,[N_e]_C = \frac{(N_e + C - 1)!}{N_e!(C-1)!} \tag{A.48}$$

As the state $|0\rangle$ will generate all other state in the same irrep under the action of $\mathrm{SU}(C)$ symmetry, this corresponds to the groundstate degeneracy of the system. For $C = 2$, $\mathrm{GSD} = N_e + 1$.

As with $C = 2$, because the $\mathrm{SU}(C)$ operators involve momentum translations by $\boldsymbol{G}_i/C$ the full $\mathrm{SU}(C)$ degeneracy only appears in finite size $N_1\boldsymbol{a}_1 \times N_2\boldsymbol{a}_2$ systems when both N_1 and N_2 are divisible by C. If only N_1 or N_2 are divisible by C, but not the other, there are only C ground states, each one obtained by completely filling one of the $\phi_{m\boldsymbol{k}}$ bands.

A.5 General Unit Cell Expansion

In this section we examine the possibility of expanding the unit cell by a factor of r, but not necessarily entirely along the $\boldsymbol{a}_1$ or $\boldsymbol{a}_2$ direction. We will construct r bands with Chern number C/r, though for the $\mathrm{SU}(C)$ action one should follow the previous procedure and expand the unit cell by a factor of C in both directions.

We call our original lattice vectors $\boldsymbol{a}_1, \boldsymbol{a}_2$ and the new lattice vectors $\boldsymbol{a}'_1, \boldsymbol{a}'_2$. The original reciprocal lattice vectors are $\boldsymbol{G}_1, \boldsymbol{G}_2$ and the new ones $\boldsymbol{G}'_1, \boldsymbol{G}'_2$. We write down the coordinates of $\boldsymbol{a}'$ in terms of $\boldsymbol{a}$:

$$\boldsymbol{a}'_i = \sum_j M_{ij} \boldsymbol{a}_j \tag{A.49}$$

where M_{ij} are integer. $\det M$ is the ratio between the volume of the two unit cells: that is, $\det M = r$. We take M to have positive determinant to preserve orientation.

We consider the Smith normal form of the matrix M:

$$M = PRQ, \quad R = \begin{pmatrix} r_1 & 0 \\ 0 & r_2 \end{pmatrix} \tag{A.50}$$

in which $P, Q \in SL(2, \mathbb{Z})$ and r_1 divides r_2. Physically speaking, it tells us that any expansion of unit cell can be decomposed into three steps: redefining the unit cell in the original lattice, expanding the new cell by $r_1 \times r_2$ times and redefining the unit cell again. Lattice redefinitions do not affect the Chern number or holomorphicity of bands ϕ_{mk}, so without loss of generality we may set $P = Q = 1$ by choosing an appropriate basis for the original and final unit cell.

Note that $\det R = r = r_1 r_2$. We will prove that in general such an unit cell expansion for an ideal Chern number C band will generate r ideal bands, each with Chern number C/r.

We have $\boldsymbol{G}'_1 = \boldsymbol{G}_1/r_1, \boldsymbol{G}'_2 = \boldsymbol{G}_2/r_2$. We define the wavefunctions in the folded Brillouin zone as

$$\psi_{lm\mathbf{k}} = \psi_{l\boldsymbol{G}'_1 + m\boldsymbol{G}'_2 + \boldsymbol{k}}, \tag{A.51}$$

which have the boundary conditions

$$\begin{aligned}\left(\Xi_{\boldsymbol{G}_1'}(\boldsymbol{k})\right)_{lm,l'm'} &= \delta_{mm'}\delta_{l,l'+1} \text{ if } 1 \le l' \le r_1,\\ \left(\Xi_{\boldsymbol{G}_1'}(\boldsymbol{k})\right)_{r_1 m,0m'} &= \delta_{mm'}\xi_{C,k_0}(\boldsymbol{k}+m\boldsymbol{G}_2')\end{aligned} \tag{A.52}$$

and

$$\left(\Xi_{\boldsymbol{G}_2'}(\boldsymbol{k})\right)_{lm,l'm'} = \delta_{ll'}\delta_{m,m'+1 \bmod n_2}. \tag{A.53}$$

The "eigenfunctions" after we diagonalize both $\Xi_{\boldsymbol{G}_1'}$ and $\Xi_{\boldsymbol{G}_2'}$ are

$$\varphi_{\alpha\beta\boldsymbol{k}} = \sum_{lm} \exp\left(-2\pi i\left(\frac{\alpha l}{r_1}+\frac{\beta m}{r_2}\right)\right)\prod_{n<l}\xi^{-1}_{C/r_1,k_0+(r_1-1)G_1/2r_1}(k+nG_1')\psi_{lm\boldsymbol{k}} \tag{A.54}$$

for $\alpha \equiv \alpha + r_1$ and $\beta \equiv \beta + r_2$. We therefore obtain $r_1 r_2 = r$ distinct bands in total as expected. The boundary conditions for $\varphi_{\alpha\beta\boldsymbol{k}}$ are

$$\xi_{\boldsymbol{G}_1\alpha\beta}(k) = \exp(2\pi i\frac{\alpha}{r_1})\xi_{C/r_1,k_0+(r_1-1)G_1/2r_1}(k), \tag{A.55}$$

and

$$\xi_{\boldsymbol{G}_2\alpha\beta}(k) = \exp(2\pi i\frac{\beta}{r_2}) \tag{A.56}$$

to convert into the canonical form, we define

$$\phi_{\alpha\beta\boldsymbol{k}} = \varphi_{\alpha\beta\boldsymbol{k}}\exp(-2\pi i\beta k/G_2) \tag{A.57}$$

These ϕ states are holomorphic because they are holomorphic combinations of ideal bands. Furthermore, they have Chern number C/r as may be identified from the boundary condition (A.55) and computed through (1.9).

B

Supplemental Material for Chapter 2

B.1 Continuum Model

In this Appendix we review the continuum model of TMDs [163].

Moiré Geometry. We take monolayer TMD lattice vectors

$$\boldsymbol{R}_{1,2} = a_0 \left(\frac{1}{2}, \mp \frac{\sqrt{3}}{2} \right), \qquad \boldsymbol{G}_{1,2} = \frac{2\pi}{a_0} \left(1, \mp \frac{1}{\sqrt{3}} \right) \tag{B.1}$$

where $a_0 = 3.52$ is the monolayer TMD lattice constant [180].

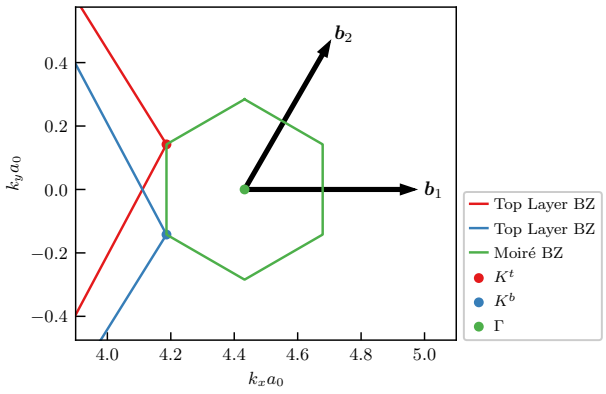


Figure B.1: Geometry of the moiré Brillouin zone.

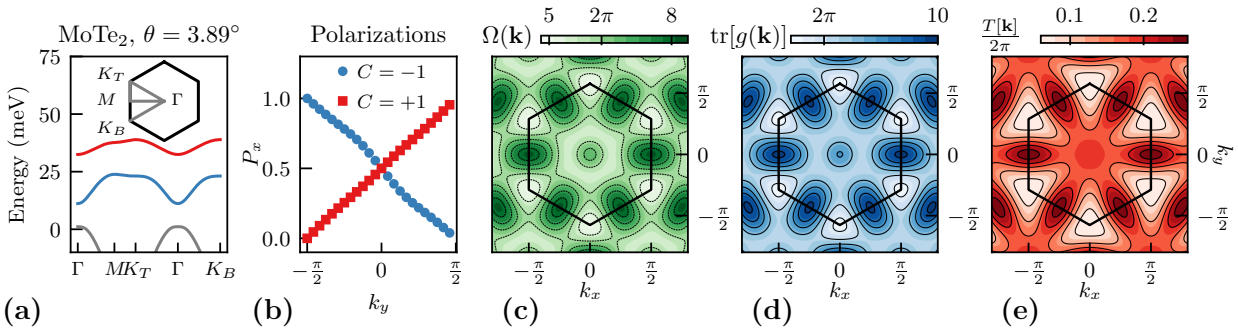


Figure B.2: (a) Band structure of twisted $MoTe_2$ material at parameter $\theta = 3.89°$. The red band and blue bands are the top-most two hole bands. (b) Wilson line of the top-most two hole bands demonstrating they carry Chern number $C = -1$ and $C = +1$, respectively. (c) to (e): the momentum space quantum geometric properties of the top-most band (red), where (c) is the Berry curvature, (d) computes the trace of Fubini-Study metric and (e) depicts the deviation of trace condition in the Brillouin zone.

As usual, the moiré Brillouin zone for the K-valley has moiré lattice vectors

$$\boldsymbol{a}_1 = \frac{a_0}{2\sin\frac{\theta}{2}}\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right), \boldsymbol{a}_2 = \frac{a_0}{2\sin\frac{\theta}{2}}(0,1), \tag{B.2}$$

$$\boldsymbol{b}_1 = \frac{8\pi\sin\frac{\theta}{2}}{\sqrt{3}a_0}(1,0), \boldsymbol{b}_j = R_{(j-1)\pi/3}\boldsymbol{b}_1 \tag{B.3}$$

where R_θ is a rotation matrix. The geometry is shown in Fig. B.1, where one can see that two of the vertices of the hexagonal moiré Brillouin zone match the rotated K-points of the monolayer TMD:

$$\boldsymbol{K}^\ell = R_{\ell\theta/2}\boldsymbol{K}, \quad \boldsymbol{K} = \frac{1}{3}\boldsymbol{G}_1 + \frac{1}{3}\boldsymbol{G}_2 \tag{B.4}$$

We often consider a commmensurate angle $\theta = 3.89°$.

Single-Particle Model. The continuum model for a moiré bilayer TMD [163] consists of four species of electrons, corresponding to two layers in each valley. Since the band gap from the valence band to the conduction band is large, the conduction bands are integrated out. Furthermore, due to spin-orbit coupling, the spin degree of freedom is locked to the valley degree of freedom: valence electrons at the K valley have spin up, and valence electrons at the K' valley have spin down. After these approximations, each valley has two degrees of freedom one for each layer. We focus on the K-valley and obtain the K' valley by time-reversal, described below.

The single-particle Hamiltonian for holes in the K-valley is

$$h_K = \begin{bmatrix} h^b(\boldsymbol{r}) + V/2 & T(\boldsymbol{r}) \\ T^\dagger(\boldsymbol{r}) & h^t(\boldsymbol{r}) - V/2 \end{bmatrix} \tag{B.5}$$

This Hamiltonian consists of three parts: the standard kinetic term, the moiré potentials and tunnelings and the displacement field. The kinetic part is

$$h_{\text{kin}} = \begin{bmatrix} -\frac{(\boldsymbol{p}-\hbar v_F \boldsymbol{K}^b)^2}{2m^*} & 0 \\ 0 & -\frac{(\boldsymbol{p}-\hbar v_F \boldsymbol{K}^t)^2}{2m^*} \end{bmatrix} \tag{B.6}$$

and the moiré potential and tunnelings are

$$h_{\text{moire}} = \begin{bmatrix} \Delta^b(\boldsymbol{r}) & T(\boldsymbol{r}) \\ T^\dagger(\boldsymbol{r}) & \Delta^t(\boldsymbol{r}) \end{bmatrix} \tag{B.7a}$$

$$\Delta^\ell(\boldsymbol{r}) = 2v \sum_{j=1,3,5} \cos(\boldsymbol{b}_j \cdot \boldsymbol{r} + \ell\psi), \tag{B.7b}$$

$$T(\boldsymbol{r}) = w(1 + e^{i\boldsymbol{b}_2\cdot\boldsymbol{r}} + e^{i\boldsymbol{b}_3\cdot\boldsymbol{r}}). \tag{B.7c}$$

Fig. B.2(a) shows the bandstructure of Eq. (B.5) at $V = 0$. As mentioned in the main text, we take parameters from recent first principles calculations of $MoTe_2$ [180]: $(a_0, m^*, V, \psi, \omega) = (3.52\,, 0.6m_e, 20.8\,\text{meV}, -107.7^\circ, -23.8\,\text{meV})$ at $\theta = 3.89^\circ$ (see also [163, 175, 181]). At these parameters, the top valence band for this model has Chern number $C = +1$, and the second-to-top band has Chern number $C = -1$.

Symmetries. Some key symmetries of Eq. (B.5) without displacement field are C_3, C_{2y}, and time-reversal $\mathcal{T}$. We give their action after applying a unitary transform to make their action simpler. Let $\tau = K, K' = 1, -1$ index valleys and define

$$h'_\tau(\boldsymbol{r}) = U_\tau(\boldsymbol{r}) h_\tau(\boldsymbol{r}) U_\tau^\dagger(\boldsymbol{r}), U_\tau(\boldsymbol{r}) = \begin{bmatrix} e^{-i\tau\boldsymbol{K}^b\cdot\boldsymbol{r}} & 0 \\ 0 & e^{-i\tau\boldsymbol{K}^t\cdot\boldsymbol{r}} \end{bmatrix}. \tag{B.8}$$

Then symmetries then act on h' as follows:

$$C_3 h'_\tau(\boldsymbol{r}) C_3^{-1} = h'_\tau(e^{\frac{2\pi i}{3}\ell_y}\boldsymbol{r}), \tag{B.9a}$$

$$C_{2y} h'_\tau(x,y)(C_{2y})^{-1} = \ell_x h'_{-\tau}(-x,y)\ell_x, \tag{B.9b}$$

$$\mathcal{T} h'_\tau(x,y)\mathcal{T}^{-1} = h'_{-\tau}(x,y)^*, \tag{B.9c}$$

where we use layer-space Pauli matrices $\ell_{0,x,y,z}$.

Interactions. Consider creation operators

$$\hat{c}^\dagger_{\boldsymbol{k}\tau b}|0\rangle = |\psi_{\boldsymbol{k}\tau b}\rangle\,, \quad \hat{h}_\tau(\boldsymbol{k})|\psi_{\boldsymbol{k}\tau b}\rangle = \epsilon_{\boldsymbol{k}\tau b}|\psi_{\boldsymbol{k}\tau b}\rangle \tag{B.10}$$

where $\tau \in \{K, K'\}$ and b labels bands. We will often use $\boldsymbol{c}^\dagger_{\boldsymbol{k}}$ as a vector in (band,valley) for concision. As mentioned in the main text we consider an interacting Hamiltonian

$$\hat{H} = -\hat{h} + \frac{1}{2A}\sum_{\boldsymbol{q}} V_{\boldsymbol{q}} :\hat{\rho}_{\boldsymbol{q}}\hat{\rho}_{-\boldsymbol{q}}:, \;\; V_q = \frac{2\pi\tanh(qd)}{\epsilon_r\epsilon_0 q}, \tag{B.11}$$

where

$$\hat{\rho}_{\boldsymbol{q}} = \sum_{\boldsymbol{k}} \hat{\boldsymbol{c}}^\dagger_{\boldsymbol{k}}\Lambda_{\boldsymbol{q}}(\boldsymbol{k})\hat{\boldsymbol{c}}_{\boldsymbol{k}}; \quad \Lambda_{\boldsymbol{q}}(\boldsymbol{k}) = \langle u_{\boldsymbol{k}}|u_{\boldsymbol{k}+\boldsymbol{q}}\rangle \tag{B.12}$$

is the density operator, A is sample area, normal ordering is relative to filling $\nu = 0$, and $\epsilon_r \approx 5-15$ is the dielectric constant. As usual, $V_{\boldsymbol{q}}$ corresponds to double-gate screened Coulomb interactions with gates at distance $d = 100$ to 300 . Here interactions are added directly to the "bare" kinetic term $\hat{h} = \sum_{\boldsymbol{k}}\sum_{\tau\in\{K,K'\}}\hat{c}^\dagger_{\boldsymbol{k}\tau}h_\tau(\boldsymbol{k})\hat{c}_{\boldsymbol{k}\tau}$, where h_K, Eq. (B.5), is determined by, e.g., first principles calculations of the insulating state where the chemical potential is inside the large valence-conduction gap. This is markedly different from interacting gapless systems like twisted bilayer graphene where the single-particle Hamiltonian is quite different from the bare Hamiltonian, and one must apply "Hartree-Fock subtraction" [71, 443] or other renormalization procedures [444] to determine the bare kinetic term. Furthermore, in this situation, integrating out all but the top few valence bands at mean-field level is equivalent to *projecting* to those top few bands; we may form a low-energy effective model of the top valence band by simply restricting the sum over bands in Eqs. (B.11), (B.12).

Self-Consistent Hartree-Fock In the main text and Fig. 1 we make use of standard self-consistent Hartree-

Fock (SCHF) calculations. We briefly detail them here for completeness. Consider a single-particle density matrix $P(\boldsymbol{k})_{ab} = \langle c_{\boldsymbol{k}b} c^\dagger_{\boldsymbol{k}a} \rangle$. Such density matrices are bijective to Slater determinant states. As usual, we define Hartree and Fock Hamiltonians

$$h_H[P](\boldsymbol{k}) = \frac{1}{A} \sum_{\boldsymbol{g}} V_{\boldsymbol{g}} \Lambda_{\boldsymbol{G}}(\boldsymbol{k}) \sum_{\boldsymbol{k}} \mathrm{tr}[P(\boldsymbol{k}) \Lambda^\dagger_{\boldsymbol{G}}(\boldsymbol{k})] \tag{B.13a}$$

$$h_F[P](\boldsymbol{k}) = -\frac{1}{A} \sum_{\boldsymbol{q}} V_{\boldsymbol{q}} \Lambda_{\boldsymbol{q}}(\boldsymbol{k}) P([\boldsymbol{k}+\boldsymbol{q}]) \Lambda_{\boldsymbol{q}}(\boldsymbol{k})^\dagger. \tag{B.13b}$$

Here form factors $\Lambda_{\boldsymbol{q}}(\boldsymbol{k})$ are matrices in (band,valley) space. In terms of these, the energy of a Slater determinant state has energy $E[P] = \langle \hat{H} \rangle_P = \frac{1}{2} \mathrm{Tr}[P(h + h_H[P] + h_F[P])]$ where h is the single-particle Hamiltonian and Tr is the trace over both $\boldsymbol{k}$ and bands. If the self-consistency condition $[P, h_{HF}[P]] = 0$ is satisfied for all $\boldsymbol{k}$, then $E[P]$ is a local minimum. In practice we consider the top four valence bands of both valleys on a k-grid of size 24×24, and use the standard "ODA" algorithm to achieve the self-consistency condition at $\nu = -1$ to numerical precision, imposing valley polarization and C_3 symmetry. Fig. 1 shows the resulting SCHF bandstructure along high-symmetry lines.

B.2 Quantum Geometry

In this section we review momentum-space band geometry [54] and its relationship to the real-space vortexability criterion [23]. We begin by introducing

$$\eta^{\mu\nu}(\boldsymbol{k}) = \sum_a \langle \partial_{k_\mu} u_{\boldsymbol{k}a} | \, Q(\boldsymbol{k}) \, | \partial_{k_\nu} u_{\boldsymbol{k}a} \rangle \,, \tag{B.14}$$

$$Q(\boldsymbol{k}) = 1 - \sum_a |u_{\boldsymbol{k}a}\rangle \langle u_{\boldsymbol{k}a}| \,; \; P = \sum_{\boldsymbol{k},a} |\psi_{\boldsymbol{k}a}\rangle \langle \psi_{\boldsymbol{k}a}| \,, \tag{B.15}$$

where $\eta^{\mu\nu}$ is the so-called quantum geometric tensor. The real and imaginary part of the quantum geometric tensor defines the Fubini-Study metric and Berry curvature, respectively:

$$g^{\mu\nu}(\boldsymbol{k}) = \mathrm{Re}(\eta^{\mu\nu}), \quad \Omega(\boldsymbol{k}) \epsilon^{\mu\nu} = -\frac{1}{2} \mathrm{Im}(\eta^{\mu\nu}). \tag{B.16}$$

The metric $g^{\mu\nu}$ and the Berry curvature two form $\Omega(\boldsymbol{k})\epsilon^{\mu\nu}$ assign the Brillouin zone a Riemann structure and

symplectic structure, respectively. When these two structures are "compatible", holomorphic coordinates can be defined [47–49].

A sufficient compatibility condition is given by $\mathrm{Tr}\, g(\boldsymbol{k}) = \Omega(\boldsymbol{k})$, which we focus on here, though we briefly comment on possible generalizations. Refs. [47–49] discuss the most general momentum-space compatibility condition (the so-called "determinant condition" [44, 54]. Ref. [22] focuses a sub-class related to the trace condition by linear transformations; only this sub-class implies real-space vortexability, and an orthogonal argument based on "forgetting translation symmetry" further distinguishes this class [23]. Vortexability may also be related to a Kähler geometry in real-space, and different choices of vortex functions $z \neq x + iy$ correspond to a wide variety of different real-space Kähler structures, most of which do not correspond to ideal momentum space geometry in the traditional sense [23].

The following conditions are equivalent conditions [22, 23, 47–49]:

1. Trace Condition: $\mathrm{Tr}\, g(\boldsymbol{k}) = \Omega(\boldsymbol{k})$, for all momentum $\boldsymbol{k}$ in the first Brillouin zone. This condition is equivalent to the saturation of the general trace inequality: $\mathrm{Tr}\, g(\boldsymbol{k}) \geq \Omega(\boldsymbol{k})$.

2. Momentum Space Holomorphicity: $Q(\boldsymbol{k}) \left|\overline{\partial}_k u_{\boldsymbol{k}a}\right\rangle = 0$, where $\overline{\partial}_k = \partial_{k_x} + i\partial_{k_y}$, for all momentum $\boldsymbol{k}$ in the first Brillouin zone. This condition is equivalent to the existence of an unnormalized gauge in momentum space for the Bloch wavefunctions $|u_{\boldsymbol{k}a}\rangle$ such that $u_{\boldsymbol{k}a}$ is holomorphic in $k_x + ik_y$.

3. Vortexability: $\hat{z}\psi = P\hat{z}\psi$, where $z = x + iy$, for all states ψ belonging to the bands in the projector P. This condition is equivalent to the one used in text $\hat{z}P = P\hat{z}P$.

In Fig. B.2(c, d, e) we show the quantum geometry of the top-most valence band. Particularly, it almost saturates the trace bound $T = \int d^2\boldsymbol{k}\, (\mathrm{Tr}\, g(\boldsymbol{k}) - \Omega(\boldsymbol{k})) \geq 0$.

The momentum space holomorphicity condition is extremely powerful for constraining the form of flat band wavefunctions, and has been used to prove the generality of the wavefunctions Eq. (3) for single $C = 1$ bands and related results for single bands with $C > 1$ [22, 138, 196].

B.3 Perfect Circular Dichroism

We now outline how to to derive the circular dichroism result Eq. 5. Consider a band (or set of bands) with projector $P = \sum_{\boldsymbol{k},b} |\psi_{\boldsymbol{k}b}\rangle \langle\psi_{\boldsymbol{k}b}|$ and define an orthogonal projector $Q = I - P$. The vortexability condition

$\hat{z}P = P\hat{z}P$ can be rewritten as $Q\hat{z}P = 0$ or, taking the Hermitian conjugate, $P\hat{z}^\dagger Q = 0$. Therefore

$$0 = ||P\hat{z}^\dagger Q|| = \sum_{a\in P, b\in Q} \sum_{\boldsymbol{k}} \left|\langle \psi_{\boldsymbol{k}a}|\hat{z}^\dagger|\psi_{\boldsymbol{k}b}\rangle\right|^2. \tag{B.17}$$

Since each term is non-negative, they must all vanish, implying

$$\langle \psi_{\boldsymbol{k}a}|\hat{z}^\dagger|\psi_{\boldsymbol{k}b}\rangle = 0 \text{ for } a \in P, b \in Q. \tag{B.18}$$

We consider optical response in the velocity gauge, which involve matrix elements

$$v^-_{ab} = \langle \psi_{\boldsymbol{k}a}|\hat{v}^-|\psi_{\boldsymbol{k}b}\rangle \tag{B.19}$$

of the velocity operator $\hat{v}^\mu = -i[\hat{r}^\mu, \hat{H}]$. Since each both $|\psi_{\boldsymbol{k}a}\rangle$ and $|\psi_{\boldsymbol{k}b}\rangle$ are eigenstates, Eq. (B.18) implies

$$v^-_{ab} = -i(\epsilon_{\boldsymbol{k}b} - \epsilon_{\boldsymbol{k}a}) \langle \psi_{\boldsymbol{k}a}|\hat{z}^\dagger|\psi_{\boldsymbol{k}b}\rangle = 0. \tag{B.20}$$

The linear optical response for interband transitions from Q to P under circularly polarized light is given by the Kubo formula

$$\sigma^\pm_{Q\to P}(\omega) = \frac{ie^2}{\hbar} \sum_{\boldsymbol{k}, a\in Q, b\in P} \frac{f_{ab}}{\epsilon_{ab}} \frac{|\langle \psi_{\boldsymbol{k}a}|\hat{v}^\pm|\psi_{\boldsymbol{k}b}\rangle|^2}{\omega - \epsilon_{ab}}. \tag{B.21}$$

If the chemical potential is placed such that some of the states corresponding to Q are filled and P is unfilled, then the right handed optical transitions corresponding to $Q \to P$ will be extinguished: $\sigma^-_{Q\to P}(\omega) = 0$ uniformly, whereas σ^+ is generically non-zero.

Observing this effect — which reflects the quality of the quantum geometry rather than topology — likely requires a large sample of high quality (uniform twist angle, low strain, etc), but does not require low temperature. We note that SCHF typically overestimates the gap at $\nu = -1$, since it neglects, for instance, the effects of disorder.

B.4 Parton Theory of the Zero-Field CFL

In this section we discuss a parton-based approach, first introduced by Refs. [211, 212], that elucidates how a charge neutral composite fermion that feels zero magnetic flux can appear at low energy despite zero magnetic field. We note that the two most prominent composite fermion theories of flux-feeling electrons, those of HLR and Son, rely on beginning with a flux-feeling particle. Indeed, the approach of HLR [141] attaches emergent gauge flux to the electron to cancel magnetic flux, while the particle-vortex duality of Son[142] relates the density of composite fermions to the magnetic field felt by electrons. Naively applied to the zero-field TMD, these approaches would lead to composite fermions that either feel flux or have zero density, respectively, neither of which is not suitable for the phenomenology of the CFL. While other parton-based approaches[213–218] have been developed to discuss composite fermions in Chern bands, to our knowledge none enable *any* Landau level theory to be applied.

We pause to comment that, despite the presence of a zero field Chern band, the TMD electrons feel zero flux. While this is clearly true from the perspective of the kinetic Hamiltonian (B.11), we will see that the zero average flux felt is a property of the low energy Chern band wavefunctions as well. It is easiest to make this notion precise when there is discrete translation symmetry. In magnetic systems, where Hamiltonians depend on derivatives through the covariant derivative, $-i\partial_\mu - a_\mu(\boldsymbol{r})$, translation operators come with position-dependent "twists"

$$T_{\boldsymbol{R}}\psi(\boldsymbol{r}) = e^{-i\varphi(\boldsymbol{r})}\psi(\boldsymbol{r}+\boldsymbol{R}),\ \ \partial_\mu\varphi_{\boldsymbol{R}}(\boldsymbol{r}) = a_\mu(\boldsymbol{r}+\boldsymbol{R}) - a_\mu(\boldsymbol{r}) \tag{B.22}$$

such that the commutator

$$T_{\boldsymbol{R}_1}T_{\boldsymbol{R}_2} = e^{i\Phi}T_{\boldsymbol{R}_2}T_{\boldsymbol{R}_1} \tag{B.23}$$

is related to the flux enclosed

$$\begin{aligned}\Phi &= \varphi_{\boldsymbol{R}_1}(\boldsymbol{r}+\boldsymbol{R}_2) - \varphi_{\boldsymbol{R}_1}(\boldsymbol{r}) + \varphi_{\boldsymbol{R}_2}(\boldsymbol{r}) - \varphi_{\boldsymbol{R}_2}(\boldsymbol{r}+\boldsymbol{R}_1)\\ &= \oint_{\partial\mathrm{UC}} \boldsymbol{a}\cdot d\boldsymbol{r} = \int_{\mathrm{UC}} \boldsymbol{\nabla}\times\boldsymbol{a}\ d^2\boldsymbol{r}.\end{aligned} \tag{B.24}$$

Here the line integration is taken over the boundary of the parallelogram unit cell spanned by the vectors $\boldsymbol{R}_1$ and $\boldsymbol{R}_2$ and the surface integration is over the area of the parallelogram.

Let us suppose that Φ is an integer times 2π, such that $[T_{\boldsymbol{R}_1}, T_{\boldsymbol{R}_2}] = 0$ and we can find Bloch states such that

$T_{\boldsymbol{R}_{1,2}}\phi_{\boldsymbol{k}} = e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\phi_{\boldsymbol{k}}$. Then we can identify the average flux per unit cell as the winding of the phase of the Bloch state around the unit cell:

$$\begin{aligned}
&\oint \boldsymbol{\nabla}\mathrm{Im}\log\psi_{\boldsymbol{k}l}(\boldsymbol{r})\cdot d\boldsymbol{r} \\
&= \varphi_{\boldsymbol{R}_1}(\boldsymbol{r}+\boldsymbol{R}_2) - \varphi_{\boldsymbol{R}_1}(\boldsymbol{r}) + \varphi_{\boldsymbol{R}_2}(\boldsymbol{r}) - \varphi_{\boldsymbol{R}_2}(\boldsymbol{r}+\boldsymbol{R}_1) \\
&= \Phi(\boldsymbol{R}_1,\boldsymbol{R}_2) = \int_{\mathrm{UC}} \boldsymbol{\nabla}\times\boldsymbol{a}\, d^2\boldsymbol{r}
\end{aligned} \tag{B.25}$$

We emphasize that the above result implies that the average flux is meaningful even without reference to a UV Hamiltonian; it measures the phase winding per unit cell of any translationally symmetric wavefunction. While particles in the LLL feel an average flux, particles in the twisted TMD feel zero flux, even those with wavefunctions in the low-energy Chern band.

To remedy this situation, and make LLL theories of CFLs accessible, we will fractionalize the TMD electron on layer l as $\psi_l = \phi\chi_l$, where ϕ carries the electric charge, and χ is a bosonic "layeron", a layer-space analog of a spinon, that satisfies $\sum_l |\chi_l(\boldsymbol{r})|^2 = 1$ for all $\boldsymbol{r}$. Associated to this decomposition, there is a $U(1)$ gauge redundancy and an associated emergent gauge field α_μ for which ϕ has charge 1 and χ has charge -1. We will choose $\langle\boldsymbol{\nabla}\times\boldsymbol{\alpha}\rangle = B_{\mathrm{eff}} = 2\pi/A_{\mathrm{uc}}$, so that ϕ now feels a net flux of one flux quantum per unit cell. We will condense the bosonic layerons by choosing them to all be in the same magnetic-translationally-symmetric state $\zeta_l(\boldsymbol{r})$; while this can be generalized to any superfluid wavefunction, we will see that this simple "Bose Einstein condensate" form will match well with the vortexable limit of the TMD Chern band wavefunctions.

With this parton ansatz, the Gutzwiller projected wavefunctions always have the form

$$\Psi_{\{l_i\}}(\{\boldsymbol{r}_i\}) = \Psi_\phi\Psi_\chi = \Psi_\phi(\boldsymbol{r}_i)\prod_i \zeta_{l_i}(\boldsymbol{r}_i), \tag{B.26}$$

where Ψ_ϕ is a wavefunction of particles that feel one flux quantum per unit cell. We may now apply any preferred LLL theory to the ϕ particles to generate choices for Ψ_ϕ. Relatedly, the low energy effective actions all have the form

$$L = L_{\mathrm{LLL}}(\phi, A_\mu + \alpha_\mu) \tag{B.27}$$

such that α_μ precisely makes up for the lack of a background magnetic field. This addresses conceptual issues with the HLR[141] and Son[142] theories in Chern bands: flux attachment and particle-vortex duality, respec-

tively, should be applied to the parton ϕ, which feels flux, not the electron ψ.

The parton ansatz $\psi_l = \phi\chi_l$ is *exact* in the vortexable limit. To see this, let us recall the wavefunctions of $C = 1$ vortexable bands, reviewed in the main text and copied below for convenience:

$$\psi_{\boldsymbol{k}l}(\boldsymbol{r}) = \phi_{\boldsymbol{k}}(\boldsymbol{r})\zeta_l(\boldsymbol{r}). \tag{B.28}$$

Here, $\phi_{\boldsymbol{k}}(\boldsymbol{r})$ is a wavefunction of a Dirac particle in an inhomogeneous magnetic field, with one flux quanta per unit cell, and $\zeta_l(\boldsymbol{r})$ is a layer-space vector that is $\boldsymbol{k}$-independent. The wavefunctions ϕ and ζ are symmetric under *magnetic* translations with opposite magnetic twists. Thus, implicit in (B.28) is the same gauge redundancy and charge-layer separation reviewed above. Furthermore, the $\boldsymbol{k}$-independence of ζ implies that *all* many-body wavefunctions associated to the band (B.28) have the form (B.26).

From a more physical perspective, the parton ansatz $\psi_l = \phi\chi_l$ may also be motivated by the well-known skyrmionic character of the layer index in the TMD band (general amongst all vortexable bands) and the "effective field" it generates[163, 164]. Indeed the tunneling in the TMD Hamiltonian of interest, $\boldsymbol{\Delta}(\boldsymbol{r}) \cdot \boldsymbol{\sigma}$, has a skyrmionic pattern in the unit cell:

$$\frac{1}{4\pi}\int_{\mathrm{UC}} \boldsymbol{n}_\Delta \cdot (\partial_x \boldsymbol{n}_\Delta \times \partial_y \boldsymbol{n}_\Delta) = -1, \tag{B.29}$$

where $\boldsymbol{n}_\Delta(\boldsymbol{r}) = \boldsymbol{\Delta}(\boldsymbol{r})/|\boldsymbol{\Delta}(\boldsymbol{r})|$ Outside the vortexable limit, this skyrmionic texture can still be thought of as generating an effective magnetic field in a sense that we now describe. The wavefunctions can still be decomposed as

$$\psi_{\boldsymbol{k}}(\boldsymbol{r}) = \phi_{\boldsymbol{k}}(\boldsymbol{r})\chi_{\boldsymbol{k}l}(\boldsymbol{r}) \tag{B.30}$$

where now χ is $\boldsymbol{k}$-depndent. The winding of χ, computed from $\boldsymbol{n}_\chi = \chi^\dagger \boldsymbol{\sigma} \chi$ at each $\boldsymbol{k}$, is expected to be the same, either by continuity from the vortexable limit, or since it is energetically favorable for it to follow the direction of the tunneling matrix, at least roughly. Note, however, that the skyrmion charge is equal to the number of flux quanta of the gauge field $a^\chi_\mu = -i\chi^\dagger\partial_\mu\chi$ through the unit cell:

$$\frac{1}{4\pi}\int_{\mathrm{UC}} \boldsymbol{n}_\chi \cdot (\partial_x \boldsymbol{n}_\chi \times \partial_y \boldsymbol{n}_\chi) = \frac{1}{2\pi}\int_{\mathrm{UC}} \boldsymbol{\nabla} \times \boldsymbol{a}^\chi d^2\boldsymbol{r}. \tag{B.31}$$

Furthermore it is straightforward to show that the net flux of a^χ corresponds to a net flux felt by χ in the sense of

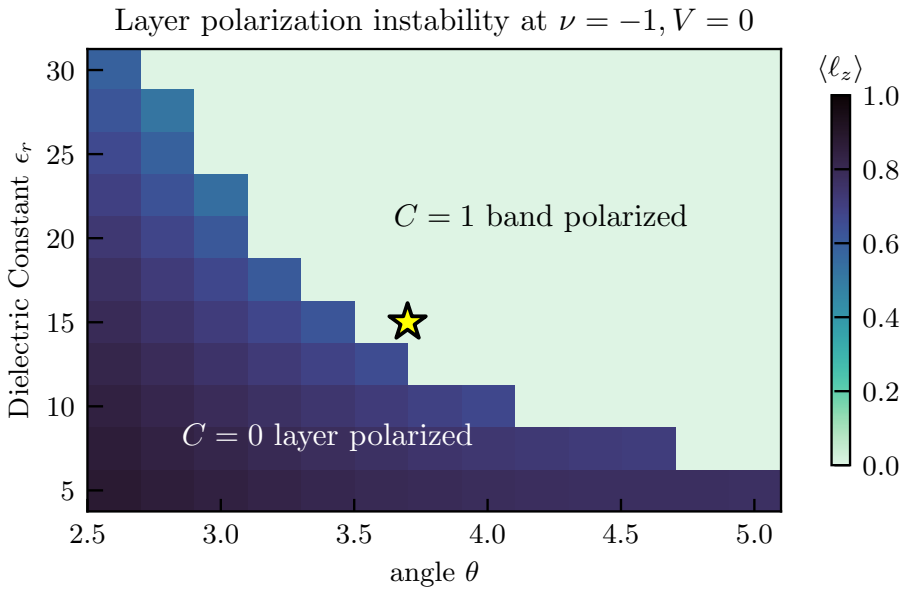


Figure B.3: SCHF phase diagram at $\nu = -1$ in the (θ, ϵ_r) plane, showing an interaction-induced layer polarized phase in the dark region. The band projection approximation is valid at $\nu = -1$ in the light region. The standard parameters used in the main text, $\theta = 3.7°$, $\epsilon_r = 15$, are marked by a yellow star. The SCHF are performed on a 24×24 torus, where we take the top two valence bands for each valley.

(B.25): a nonzero phase winding per unit cell. Indeed, if $\chi_l(\boldsymbol{r}) = e^{-i\varphi(\boldsymbol{r})}\chi_l(\boldsymbol{r} + \boldsymbol{R}_{1,2})$ then we have

$$\begin{aligned} &\oint \boldsymbol{\nabla} \mathrm{Im} \log \chi_l(\boldsymbol{r}) \cdot d\boldsymbol{r} \\ &= \varphi_{\boldsymbol{R}_1}(\boldsymbol{r} + \boldsymbol{R}_2) - \varphi_{\boldsymbol{R}_1}(\boldsymbol{r}) + \varphi_{\boldsymbol{R}_2}(\boldsymbol{r}) - \varphi_{\boldsymbol{R}_2}(\boldsymbol{r} + \boldsymbol{R}_1) \\ &= \int_{\mathrm{UC}} \boldsymbol{\nabla} \times \boldsymbol{a}^{\chi}\, d^2\boldsymbol{r} \end{aligned} \tag{B.32}$$

where we used the boundary conditions of χ to evaluate both integrals, obtaining the middle expression in both cases.

We therefore conclude that the if χ_l has a skyrmion texture, its wavefunctions must have corresponding a winding per unit cell as if χ_l coupled to a gauge field with an average flux of -2π per unit cell. The corresponding parton ϕ must therefore feel $+2\pi$ flux, again leading to the parton ansatz described above.

B.5 Layer Polarization versus Band Projection

In this Appendix, we discuss a subtle issue arising from the interplay of layer polarization versus band projection. To carry out many-body numerics, it is always preferable to work within the smallest possible Hilbert

space, i.e. the Hilbert space of a single flat band. Such an approximation is valid only when the gap to other bands is large, preventing band mixing. For $MoTe_2$, however, there is significant band mixing in a significant range of parameter space. Below we identify the regime where band-projection is permissible. We then discuss how band projected numerics in the 'impermissible' regime can nevertheless be used as a rigorous numerical tool for phase identification via an adiabatic connection.

B.5.1 Layer Polarization Physics

We start at filling $\nu = -1$, where an instability to a layer-polarized (LP) phase competes with the quantum anomalous hall (QAH) phase, even a at displacement field $V = 0$. This instability was previously pointed out in Abouelkomsan *et al.* [174] and Wang *et al.* [228]. In this phase, the top two bands *in the same valley* are strongly mixed, yielding a topologically trivial insulator that remains valley polarized. Fig. B.3 shows the phase diagram at $\nu = -1$ within self-consistent Hartree-Fock (SCHF).

Intuitively, the layer-polarized phase enables electrons to move further apart, thus minimizing the Coulomb energy. On the top $MoTe_2$ layer, the electron densities are concentrated on the MX sites, whereas on the bottom layer, the electron densities are concentrated on the XM sites [164]. In the quantum anomalous Hall phase, the unoccupied band has densities on both layers and both sites, whereas in the layer polarized phase, the unoccupied band has densities only on on layer and one site. If the interaction is strong enough, the layer-unpolarized quantum anomalous hall phase will cost more energy, since the MX-XM distance is shorter than the MX-MX distance. This explains the appearance of the layer-polarized phase at strong interactions (small ϵ_r) in Fig. B.3. At slightly weaker interactions, where the interaction strength is not much larger than the bandgap, one expects the quantum anomalous Hall phase to be more energetically favorable. We note that the phase boundary here depends rather strongly on microscopic parameters. Their values are drawn from DFT calculations that, especially for ψ, may not be representative of experimental values.

B.5.2 Regime of Flat Band Projection

Given this instability to band mixing and layer-polarized physics, when is a band-projected model valid? Were numerical resources not an issue, one could perform calculations in an expanded two-band Hilbert space that included band-mixing exactly. In the present context, however, we expect gapless ground states that require large system sizes to stablize and identify. As a practical matter, this restricts us to either single-band Hilbert spaces or

extremely coarse momentum resolution with larger numbers of bands in exact diagonalization. We are therefore motivated to restrict to the flat band Hilbert space.

We make the assumption that, whenever the $\nu = -1$ state is in the QAH phase, then a band-projected model will faithfully capture the top valence band at partial fillings. This is reasonable since the top valence band is only slightly mixed with the lower bands in SCHF, meaning that the relevant Hilbert space at partial fillings should be that of the top band. Confirming this assumption within large-scale multi-band numerics is an important problem for future work. Hence, to properly identify the CFL in many-body numerics, we have to perform the band projection approximation at $\nu = -1/2$. We note this assumption is standard within the literature on $MoTe_2$ (see e.g. [177, 180, 181, 229]), as well as twisted bilayer graphene where the relevant gaps are somewhat larger. We apply this assumption at parameters $(\theta, \epsilon_r) = (3.7°, 15)$ in the main text, where the system is within the QAH phase at $\nu = -1$. Our expectation is that band projected numerics at these parameters faithfully represent the universal features of the CFL phase that should appear in experiments.

B.5.3 Adiabatic Connection

Even though ground states in the flat-band-projected Hilbert space are not global ground states at small ϵ_r, we can still use these states as a *numerical tool* to understand the phases present. This is of immense practical utility, because the strongly-correlated CFL phase is most easily recognized in the limit of large interactions. To recognize it numerically, we may adopt the following procedure: (1) show a CFL is present in the band-projected model at small ϵ_r, (2) show that the ground state in the band-projected model at a larger ϵ_r is adiabatically connected to the first point, (3) by our assumption above, the band-projected ground is adiabatically connected to the ground state in the unrestricted Hilbert space. We could, of course, have simply started from step (2), but step (1) is much deeper in the CFL. That point is therefore is easier to converge, and numerically cheaper to study in detail.

Below we will carry out numerical studies at $\epsilon_r = 6$ and 8 that are deep in the CFL phase within the band-projected Hilbert space. Within this Hilbert space, they are adiabatically connected to $(\theta, \epsilon_r) = (3.7, 15)$. The fact that they belong to the same phase is confirmed in the many-body spectrum from exact diagonalization (Fig. B.4) and structure factor measurements from DMRG (Fig. B.5), whose interpretation will be discussed below. The parameters $(\theta, \epsilon_r) = (3.7, 15)$, marked with a star in Fig. B.3, is outside the layer-polarized phase and we expect it will have the same ground state physics in either the flat band or two band Hilbert spaces. The

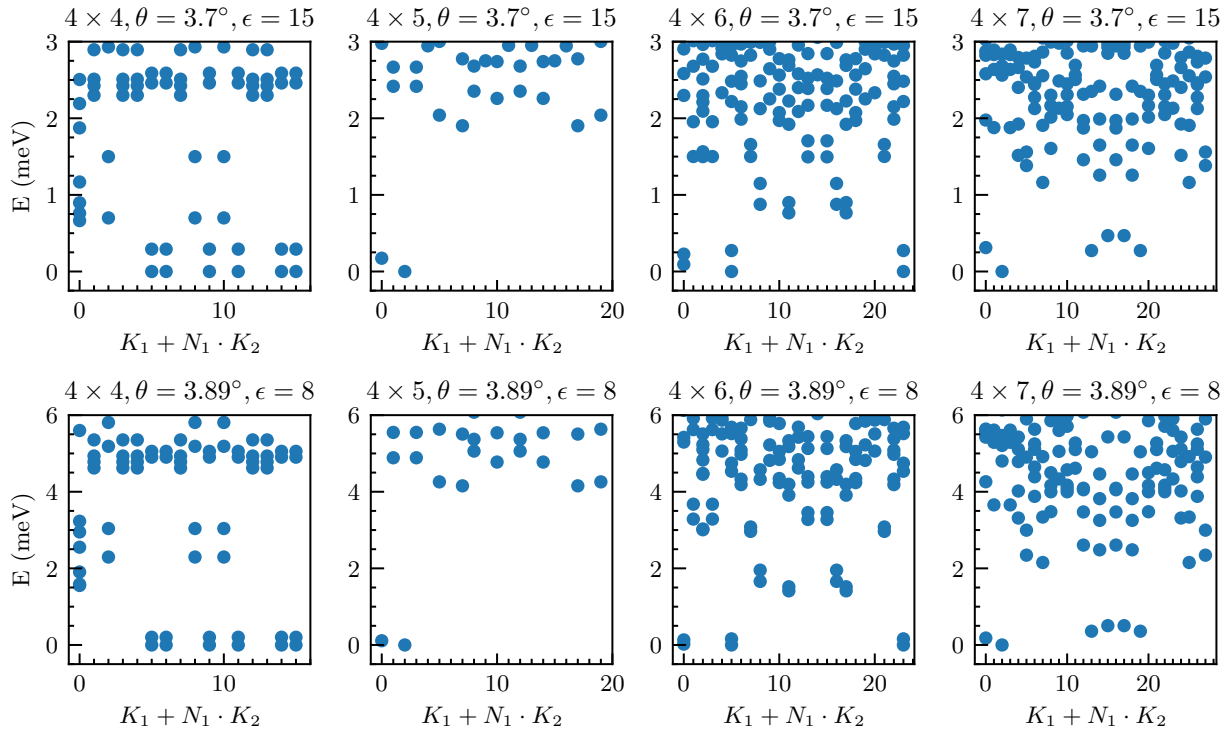


Figure B.4: Comparison of ED spectra at many system sizes within the band projection approximation. The first row uses parameters $(V, \epsilon_r, \theta) = (0, 15, 3.7°)$ and the second row uses parameters $(V, \epsilon_r, \theta) = (0, 8, 3.89°)$. One can construct a one-to-one mapping between the low-energy spectra for these two sets of parameters.

full phase diagram at $\nu = -1/2$ in the expanded many-band Hilbert space will be a topic for future work.

B.6 Exact Diagonalization

We perform exact diagonalization of the *band projected* model Eq. (1) at different system sizes. The validity of the approximation is justified in the previous section. At half filling $\nu = -1/2$, we point out that the CFL phase can be realized in a large portion in the phase diagram (V, ϵ_r, θ) of twisted TMD. In particular, Fig. B.6 lists spectra of moiré TMD at $(V, \epsilon_r, \theta) = (0, 8, 3.89°)$ for various system sizes, and compares them to the spectra of Coulomb interaction in LLL of the same system sizes. The similarities between the spectra of twisted TMD and LLL strongly suggest the interacting low energy states in TMD are CFLs, because Coulomb interaction is known to stabilize the CFL phase in LLL [445]. Moreover, for each system size, the momentum of low energy many-body eigenstates agree with the total momentum of compact composite Fermi seas, again indicating the phase is a CFL [199]. We further confirm the CFL phase by computing the electron occupation $n(\boldsymbol{k})$ in the Brillouin zone and show that it's featureless in the CFL phase but shows a Fermi surface in the FL phase. The CFL phase region

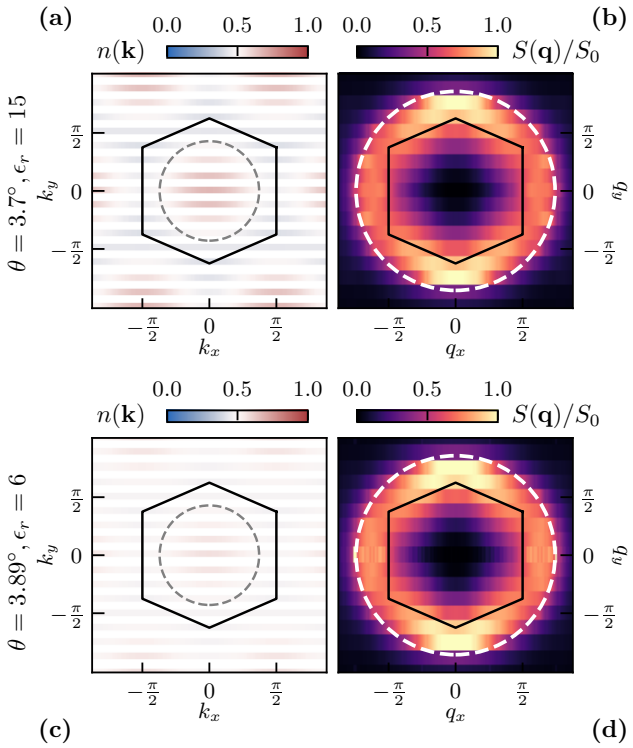


Figure B.5: Comparison of DMRG electron occupations and structure factors at $L_y = 8$ within the band projection approximation. The first row uses parameters $(V, \epsilon_r, \theta) = (0, 15, 3.7°)$ and the second row uses parameters $(V, \epsilon_r, \theta) = (0, 8, 3.89°)$. Both occupations do not contain discontinuities and nor do they change significantly with respect to bond dimension.

is centered around $\theta \approx 3.6°, V = 0$ and small ϵ_r, as expected due to a compromise between the $\nu = -1, 0$ parent-state bandwidths. Finally, we comment on the possibility to see a CFL at $\nu = -3/4$. All the ED results performed on the LLL will be measured in arbitrary units.

B.6.1 Low energy spectrum and the identification of emergent Fermi sea

The similarities between the spectrum of LLL and moiré TMD, shown in Fig. B.6, strongly suggest the ground states of twisted TMD are CFLs. Here we analyze the ED spectrum in more detail and explain why they point to the emergent Fermi sea.

To compare the spectrum between LLL and moiré TMD, we adopt the same real space geometry on these two systems. The vectors that generate the $N_1 \times N_2$ torus are $\boldsymbol{L}_i = N_i \boldsymbol{a}_i$, written in terms of the lattice vectors $\boldsymbol{a}_1$ and $\boldsymbol{a}_2$. We define the corresponding reciprocal lattice vectors as $\boldsymbol{b}_1$ and $\boldsymbol{b}_2$ in which $\boldsymbol{a}_i \cdot \boldsymbol{b}_j = 2\pi\delta_{ij}$. Since we have chosen identical geometries for these two systems, we can also choose $\boldsymbol{a}_{1,2}$ to denote the basis vectors for the magnetic unit cell in LLL as well as the basis vectors for unit cell in moiré TMDs. We start by reviewing translation symmetries and features of the CFL spectrum in LLL; we will later find that similar discussions apply to moiré TMD straightforwardly. We choose the magnetic field such that there is one flux quantum per unit cell, so the number of fluxes $N_\phi = N_1 N_2$.

We first review the magnetic translation boundary conditions in LLL. Without loss of generality, we impose arbitrary twisted boundary conditions on our Hilbert space by threading fluxes ϕ_i through the non-contractible loops of the torus generated by $\boldsymbol{L}_i$. This choice of boundary conditions then defines the single particle Hilbert space, in which every state satisfies the relation [124, 446]

$$T^{\text{mag}}_{\boldsymbol{L}_i} \psi(\boldsymbol{r}) = e^{-i\varphi_{\boldsymbol{L}_i}(\boldsymbol{r})} \psi(\boldsymbol{r} + \boldsymbol{L}_i) = (-1)^{N_\phi} e^{i\phi_i} \psi(\boldsymbol{r}). \tag{B.33}$$

Here, as in the previous section, the magnetic twist $e^{-i\varphi_{\boldsymbol{L}}(\boldsymbol{r})}$ distinguishes magnetic translations from ordinary translations. We emphasize that making a choice of boundary conditions breaks the continuous magnetic translation symmetry of the LLL, and generally applying an arbitrary magnetic translation to each electron in the state will change the boundary condition [124, 446, 447]. We can use this property to relate states with different ϕ_i to each other.

We then define boundary conditions for composite fermions. To analyze composite fermion boundary condi-

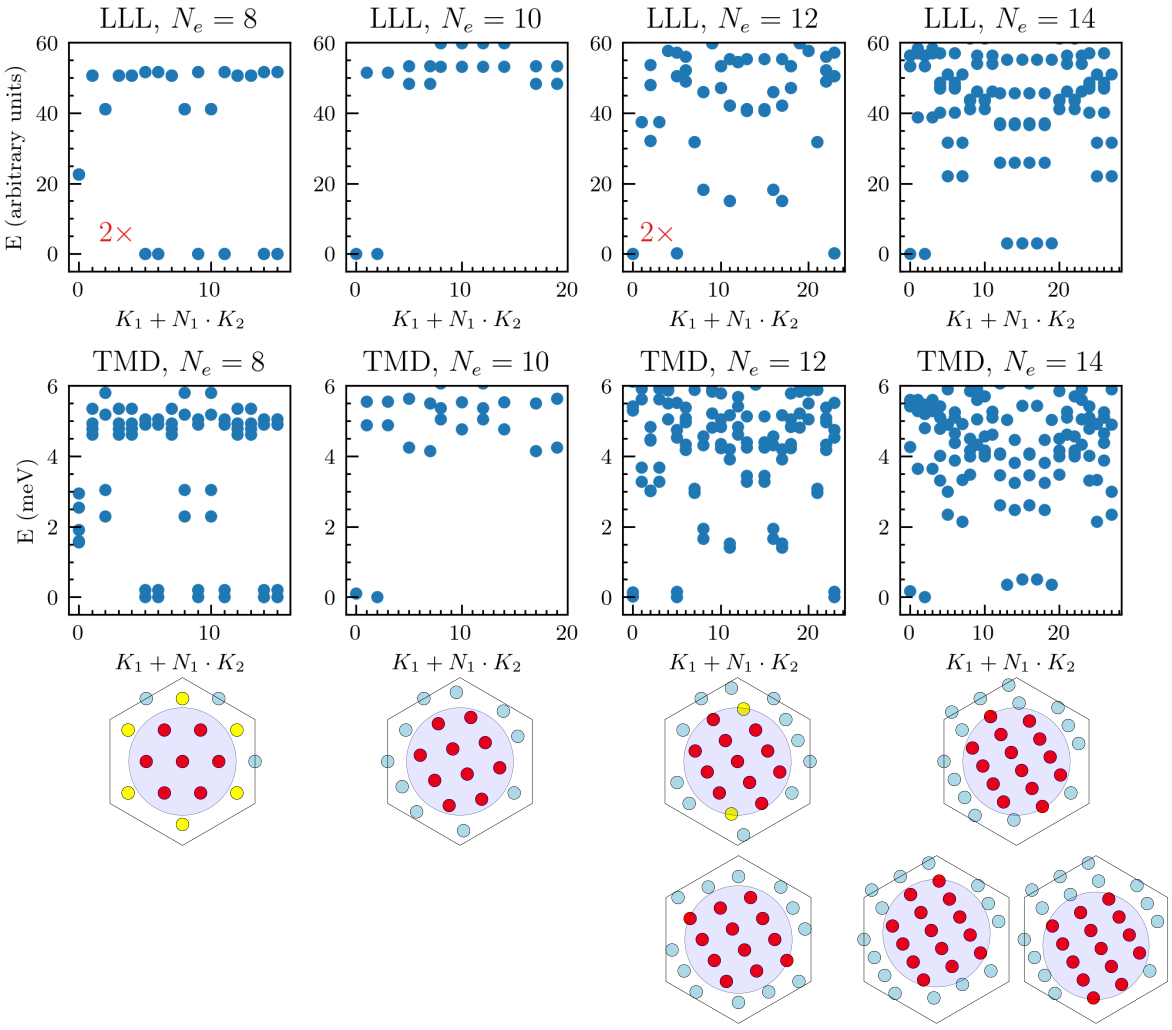


Figure B.6: Comparison between the spectrum of half filled LLL with Coulomb interaction (top row) and that of twisted $MoTe_2$ (middle row) for different system sizes at $(V, \epsilon_r, \theta) = (0, 8, 3.89°)$. The third and fourth row illustrate the composite fermion configuration for the low energy states. We note each point in the spectrum shown in the LLL in $N_e = 8$ and $N_e = 12$ are two fold degenerate, related to the continuous center of mass magnetic translation symmetry; the center of mass partner differs their momentum by $(\delta K_1, \delta K_2) = (2, 0)$ for $N_e = 10$ and $N_e = 14$.

tions we must choose a trial wavefunction to begin with. We choose the widely accepted Read-Rezayi wavefunction, reproduced below [219]:

$$\Psi(\{\boldsymbol{r}_n\}) = \mathcal{P}_{\text{LLL}} \det_{mn} \left(e^{i\tilde{\boldsymbol{k}}_m \cdot \boldsymbol{r}_n} \right) \Psi_L(\{\boldsymbol{r}_n\}) \tag{B.34}$$

This wavefunction can be understood from a parton construction similar to the one in the main text. The first term, $\Psi_f = \det \left(e^{i\tilde{\boldsymbol{k}}_m \cdot \boldsymbol{r}_n} \right)$, describes a composite Fermi sea of fermions that feel no flux. The second term $\Psi_L(\{\boldsymbol{r}_n\})$ describes a bosonic Laughlin state of bosons that do feel the electronic flux. We will see that the composite fermion boundary conditions, manifesting through the quantization of the CF momenta $\tilde{\boldsymbol{k}}_m$, can be completely different than the electron boundary conditions. This is because the CFs boundary condition is set by the *emergent* gauge flux through the torus handles, which is *dynamical*.

To impose the boundary conditions (B.33), we note that individual magnetic translations $T^{\text{mag}}_{j,\boldsymbol{L}_i}$ commute with the projector to the LLL: $[T^{\text{mag}}, \mathcal{P}_{LLL}] = 0$. We then apply the magnetic translation on the jth particle in (B.34). By using the commutativity of magnetic translation and LLL projection we arrive at,

$$T^{\text{mag}}_{j,\boldsymbol{L}_i} \Psi = \mathcal{P}_{\text{LLL}} \left[(t_{j,\boldsymbol{L}_i} \Psi_f) \left(T^{\text{mag}}_{j,\boldsymbol{L}_i} \Psi_L \right) \right], \tag{B.35}$$

where $t_{j,\boldsymbol{L}_i}$ is an ordinary translation, with no magnetic twist. Here we have placed the magnetic twist factor onto the translation of the Laughlin state, anticipating the fact that the Laughlin state can be symmetric under magnetic translations while the zero-flux fermion wavefunction is symmetric under ordinary translations.

We suppose that the bosonic Laughlin wavefunctions satisfy boundary condition $\phi_i^{(b)}$, not necessarily equal to ϕ_i:

$$T^{\text{mag}}_{j,\boldsymbol{L}_i} \Psi_L = (-1)^{N_\phi} e^{i\phi_i^{(b)}} \Psi_L \tag{B.36}$$

Note that Laughlin states have a two-fold degeneracy due to the part of the center of mass translation symmetry. This degeneracy directly translates to the center of mass degeneracy of the composite fermion wavefunction (B.34).

We now impose boundary conditions on the composite fermion part of the wavefunction

$$\begin{aligned} t^{\text{mag}}_{j,\boldsymbol{L}_i}\Psi_f &= \left(\sum_P (-1)^P e^{i\tilde{\boldsymbol{k}}_{P(j)}\cdot\boldsymbol{L}_i} e^{i\tilde{\boldsymbol{k}}_{P(m)}\cdot(\boldsymbol{r}_n)}\right) \\ &= e^{i\phi_i^{(f)}}\mathcal{P}_{\text{LLL}} \det_{mn}\left(e^{i\tilde{\boldsymbol{k}}_m\cdot\boldsymbol{r}_n}\right), \end{aligned} \tag{B.37}$$

which corresponds to $e^{i\boldsymbol{k}_j\cdot\boldsymbol{L}_i} = e^{i\phi_i^{(f)}}$. The quantization of composite fermion momentum is thus

$$\tilde{\boldsymbol{k}} = \frac{n_1 + \frac{\phi_1^{(f)}}{2\pi}}{N_1}\boldsymbol{b}_1 + \frac{n_2 + \frac{\phi_2^{(f)}}{2\pi}}{N_2}\boldsymbol{b}_2. \tag{B.38}$$

The boundary conditions of the two sub-wavefunctions then sum to the electron boundary conditions

$$\phi_i = \phi_i^{(f)} + \phi_i^{(b)}. \tag{B.39}$$

Indeed, the boundary condition for electrons — corresponding to the electromagnetic fluxes through the handles of the torus — are split up between the CFs and the composite bosons that form the Laughlin state. The "relative" degree of freedom, corresponding to increasing $\phi_i^{(f)}$ while decreasing $\phi_i^{(b)}$, such that ϕ_i is held fixed, corresponds to the dynamical flux of the emergent gauge field.

We now compute the center of mass momentum for the state (B.34). To do this, we define the center of mass momentum $\boldsymbol{K}$ for a state as

$$\left(\prod_j T^{\text{mag}}_{j,\boldsymbol{a}_i}\right)\Psi = e^{i\boldsymbol{K}\cdot\boldsymbol{a}_i}\Psi. \tag{B.40}$$

This definition is such that there always exists a bosonic Laughlin state which, when periodic boundary condition $\phi_i^{(b)} = 0$ is imposed, has zero center of mass momentum $K_1 = K_2 = 0$.

Starting with this canonical definition of COM momentum at periodic boundary condition, the COM momentum at arbitrary boundary condition can be obtained from flux insertion:

$$\boldsymbol{K}(\phi) = N_e\left(\frac{\phi_1}{2\pi N_1}\boldsymbol{b}_1 + \frac{\phi_2}{2\pi N_2}\boldsymbol{b}_2\right) + \boldsymbol{K}(\phi = 0). \tag{B.41}$$

Following (B.40), we are ready to compute the center of mass momentum for the state (B.34):

$$\begin{aligned}
&\left(\prod_j T^{\text{mag}}_{j,\boldsymbol{a}_i}\right)\Psi(\{\boldsymbol{r}_n\}) \\
&= e^{i\sum_m \tilde{\boldsymbol{k}}_m\cdot\boldsymbol{a}_i}\mathcal{P}_{\text{LLL}}\det_{mn}\left(e^{i\tilde{\boldsymbol{k}}_m\cdot\boldsymbol{r}_n}\right)\left(\prod_j T^{\text{mag}}_{j,\boldsymbol{a}_i}\right)\Psi_L(\{\boldsymbol{r}_n\}) \\
&= e^{i(\sum_m \tilde{\boldsymbol{k}}_m+\boldsymbol{K}_L(\phi^b))\cdot\boldsymbol{a}_i}\Psi(\{\boldsymbol{r}_n\}),
\end{aligned} \tag{B.42}$$

from which the COM momentum can be read out as

$$\begin{aligned}
\boldsymbol{K}_\Psi &= \sum_m \tilde{\boldsymbol{k}}_m + \boldsymbol{K}_L(\phi^b), \\
&= \sum_m \tilde{\boldsymbol{k}}_m + \frac{N_e\phi_1^{(b)}}{2\pi N_1}\boldsymbol{b}_1 + \frac{N_e\phi_2^{(b)}}{2\pi N_2}\boldsymbol{b}_2,
\end{aligned} \tag{B.43}$$

where we have applied (B.41) for the bosonic Laughlin state in the second line.

From (B.43), (B.38) and (B.39), we notice that the COM momentum of the CFL state $\boldsymbol{K}_\Psi$ is entirely determined by the physical boundary condition $\phi_{1,2}$ and the integer part of composite fermion momenta $\{\tilde{k}_i\}$. This means that the COM momentum is determined by $\phi_{1,2}$ and the relative configuration of the composite fermion momenta. Thus, for the purpose of interpreting CFL many body momentum from ED numerics, the concrete value of $\phi_i^{(f)}$ does not matter.

However, there could be a preferred $\phi_i^{(f)}$ energetically. We assume that $\phi_i^{(f)}$ is self-consistently chosen for each state such that the compact composite Fermi sea is centered at $\boldsymbol{\Gamma}$ to minimize the variational energy, which we conjecture to be captured by a dispersion $E(\tilde{\boldsymbol{k}})$ that grows with $|\tilde{\boldsymbol{k}}|$ for the particle-hole symmetric CFL state in LLL. When we plot the composite Fermi sea in Fig. B.6, we pick particular choices of $\phi_{1,2}^{(f)}$ for each system size such that the composite Fermi sea is centered at $\boldsymbol{\Gamma}$.

We now describe the CFL COM momentum counting for all the low energy states in all system sizes. For all system sizes N_1, N_2 we study here, we can choose the physical boundary condition to be $\phi_i = 0$.

We start with $N_e = 8$ electrons on a 4×4 torus. In this case, $\phi_i^{(b)} = -\phi_i^{(f)} = 0$ already gives a composite fermion Fermi sea centered at $\boldsymbol{\Gamma}$ point. For both LLL and twisted TMD, there are 12 low energy states living in 6 momentum sectors consisting of the ground state manifold. These twelve states, six modulo center of mass degeneracy, correspond to the six possible momenta for the eighth, highest energy, composite fermion shown in

yellow in the third row of Fig. B.6.

We proceed and focus on $N_e = 10$ electrons on a 4×5 torus. We choose $\phi_2^{(b)} = -\phi_2^{(f)} = \pi$ to give a presumably low energy composite fermion configuration centered at $\mathbf{\Gamma}$. From the configuration shown in the third row $\sum \tilde{\boldsymbol{k}}_i = 0, \boldsymbol{K} = N_e \phi_2^{(b)}/2\pi N_2 \boldsymbol{b}_2 = \boldsymbol{b}_2$, which corresponds to the state appearing in sector 0. The state in sector 2 is related to this state by center of mass translation. We numerically study the particle hole symmetry of the CFL in twisted $MoTe_2$ using the 4×5 system by calculating $|\langle \psi_0 | \mathcal{PH} | \psi_2 \rangle|^2 = 0.89$, in which $|\psi_i\rangle$ refers to the ground state in momentum sector i. The action of particle hole is $(\mathcal{PH}) c_{\boldsymbol{k}}^\dagger (\mathcal{PH})^{-1} = c_{\boldsymbol{k}}$, $(\mathcal{PH}) i (\mathcal{PH})^{-1} = -i$. Thus the numerical particle hole breaking per particle is on the order of one percent. In contrast, $|\langle \psi_0 | \mathcal{PH} | \psi_2 \rangle|^2 = 1$ in the LLL, which is known to have perfect particle hole symmetry.

We then study $N_e = 12$ electrons on a 4×6 torus. We argue that two different boundary conditions must be chosen for the low energy states: $\phi_i^{(b)} = -\phi_i^{(f)} = 0$ for the states appearing in energy sectors 5 and 23, shown in the third row, and $\phi_2^{(b)} = -\phi_2^{(f)} = \pi$ which appears in sector 0, shown in the fourth row. We numerically observe these two ED states are close in energy.

We conclude by analyzing $N_e = 14$ electrons on a 4×7 torus. We choose $\phi_2^{(b)} = -\phi_2^{(f)} = \pi$. The symmetric Fermi sea configuration shown in the third row corresponds to the states in sector 0, which is again related to the state in sector 2 by the center of mass magnetic translations. On the other hand, the other low energy states, corresponding to less symmetric composite fermion Fermi seas, are shown in the fourth row.

B.6.2 Explicitly LLL or ideal band projected CFL wavefunction

Here we comment that there exists an explicitly LLL projected CFL wavefunction defined on the torus geometry [199, 447–449], and such a wavefunction can be easily generalized to be an ansatz for the CFL state which is explicitly projected to ideal flatbands. In LLL, such projected wavefunction reads:

$$\begin{aligned} \Psi_{\mathrm{CFL}} &= \det M_{ij} \prod_{k=1}^{2} f(\sum_i z_i - \alpha_k). \\ M_{ij} &= e^{\frac{1}{8}(z_i d_j^* - z_i^* d_j)} \prod_{k \neq i}^{N_e} f(z_i - z_k - d_j + \bar{d}), \\ d_i &= \frac{m L_1 + n L_2}{N_e} + \frac{L^{(f)}}{N_e}; \quad \sum_{k=1}^{2} \alpha_k = L^{(b)}, \end{aligned} \tag{B.44}$$

where d_i is the complex coordinate of the composite fermion dipole $d_i = d_i^x + id_i^y$, and $z = x + iy$ is the complex coordinate of electron. In LLL, the dipole is related to the momentum of composite fermion via dipole-momentum locking $d^a = 2l_B^2 \epsilon^{ab} \tilde{k}_b$. Since the reciprocal lattice is related by a 90° rotation and appropriate rescaling, $\boldsymbol{b}_i \to \epsilon_{ij} \boldsymbol{a}_j$, the dipole is quantized on a $L_{1,2}/N_e = 2L_{1,2}/(N_1 N_2)$ lattice and shifted by an internal flux $L^{(b/f)} = \left[-\phi_2^{(b/f)} L_1 + \phi_1^{(b/f)} L_2\right]/2\pi$. The $\bar{d}$ is a parameter of the wavefunction and can be chosen as center of dipoles $\bar{d} = \sum_i d_i/N$. In the above $\alpha_{1,2}$ labels the COM zeros that determines the COM momentum of the bosonic Laughlin state, which is shifted by the Laughlin state's boundary condition. To obtain a torus wavefunction, we need to use Weierstrass sigma functions $\sigma(z)$ that is quasi-periodic in $L_{1,2}$, and we have defined $f(z) = \sigma(z) \exp(-|z|^2/4N_\phi)$ [441, 447, 450].

The link between the Read-Rezayi wavefunction (B.34) and the projected wavefunction (B.44) can be understood as follows. First, the Jastrow factor from the bosonic Laughlin wavefunction can be pulled into the determinant, so we can rewrite $\Psi_f \Psi_L$ as follows [451],

$$\left(\det_{ij} e^{i\boldsymbol{k}_j \cdot \boldsymbol{r}_i}\right) \prod_{i<j} (z_i - z_j)^2 = \det_{ij} \left[e^{i\boldsymbol{k}_j \cdot \boldsymbol{r}_i} \prod_{k \neq i} (z_i - z_k)\right].$$

Notice the Bloch factor $e^{i\boldsymbol{k}\cdot\boldsymbol{r}} = e^{i\boldsymbol{d}\times\boldsymbol{r}/2} = e^{(\bar{d}z - d\bar{z})/4}$. We then approximate the LLL projection by replacing the anti-holomorphic coordinate $\bar{z}$ with the holomorphic partial derivative $l_B^2 \partial_z$ such that $P_{\mathrm{LLL}} (\Psi_f \Psi_L)$ is approximated by $\det_{ij} \left(e^{\frac{1}{4}\bar{d}_j z_i} \prod_{k \neq i} (z_i - z_k - d_j)\right)$. Finally translating into torus geometry arrives at (B.44). See Ref. [199] for details. Generalizing this wavefunction to $1/2m$ filling is straightforward [449].

In the end we comment that the exact mapping between LLL and ideal band mentioned in the main text, Eq. (3), enables an explicit construction of CFL variational wavefunction without magnetic field: $\Psi_{\mathrm{CFL}}^{\mathrm{band}} = \Psi_{\mathrm{CFL}}(\{z_i\}) \prod_i e^{-K(\boldsymbol{r}_i)} \zeta_{l_i}(\boldsymbol{r}_i)$. We leave a detailed study of this flat band CFL wavefunction for the future.

B.6.3 Phase Diagrams and Phase Identification

In this section, we discuss how to identify different phases from ED. We use multiple indicators, including:

- Momentum of low energy states.

  In the previous section, we have discussed the momentum of low energy states from the CFL phase. They correspond precisely to the total momentum of composite fermions Fermi sea. Moreover, low energy

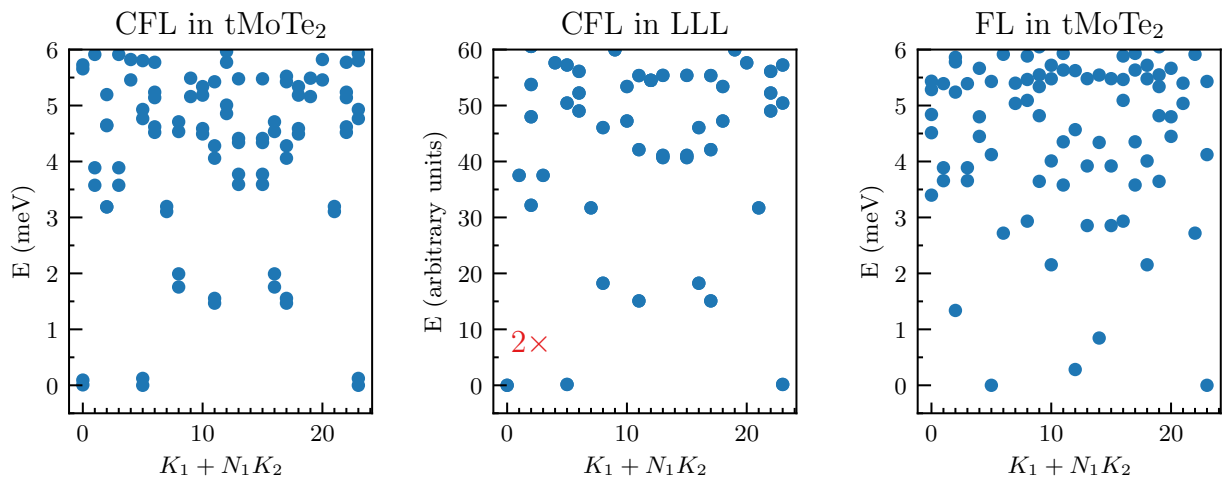


Figure B.7: Representative energy spectrum for the $\nu = -1/2$ CFL phase in twisted $MoTe_2$, the CFL phase in LLL, and the FL phase in twisted $MoTe_2$. All the ED are performed on 4×6 torus with the same geometry. Notably, for CFL phase the two-fold COM degeneracy is exact in LLL, whereas it is weakly broken in twisted $MoTe_2$. When the system transitions into the FL phase, the COM degeneracy is no longer present.

states of CFL are nearly degenerate, corresponding to the exact COM degeneracy in LLL. For instance, for $N_e = 12$ particles, the low energy states of CFL has $K = K_1 + N_1 K_2 = 0, 5, 23$ and each of them are nearly two fold degenerate. Their composite Fermi sea configuration is plotted in Fig. B.6.

Differently, Fermi liquids do not exhibit any signatures of COM degeneracy. See Fig. B.7 for illustrations.

- The occupation number of ground state $n(\boldsymbol{k})$.

  We can also identify the FL phase by computing the ground state occupation of momentum eigenstates in the Brillouin zone. In the CFL phase, the ground state occupations are relatively uniform throughout the Brillouin zone and there are no significant Fermi surface discontinuities in the occupation. On the other hand, in the FL phase, the ground state occupations not only exhibit a discontinuous feature at the putative Fermi surface, but also closely follow the non-interacting Fermi distribution $n_{(\epsilon,\mu)} = \theta(\epsilon - \mu)$ for holes very well, as shown in Fig. B.8(b) and (c). The quasiparticle residue Z, which characterizes the discontinuity at the Fermi surface, is not 1 as the noninteracting FL because it is renormalized by the interactions. We note that this phase identifier is further supported by iDMRG results shown in Fig. 2(a), in which the CFL clearly shows a featureless Brillouin zone in terms of electron occupations, whereas the FL has a clear Fermi surface.

We cross-check the two methods by computing the zero displacement field phase diagram in the (θ, ϵ_r) plane,

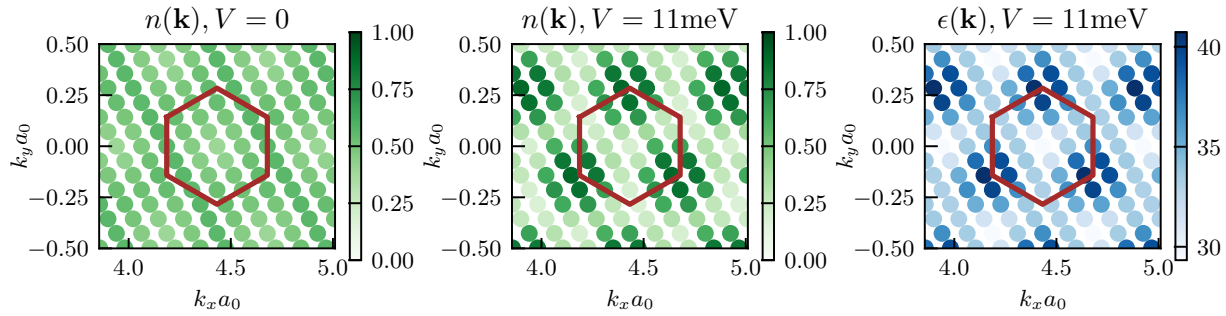


Figure B.8: Occupations of holes in the ED ground state in the CFL phase and the FL phase by tuning displacement field, respectively. All the ED are performed on 4 torus with the same geometry. The FL occupations match excellently to the Fermi distribution $\theta(\epsilon - \mu)$ for holes.

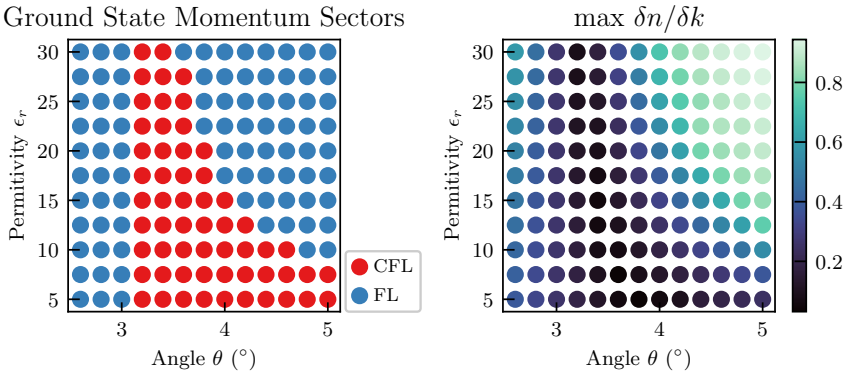


Figure B.9: Band-projected phase diagram of interacting twisted $MoTe_2$ at $\nu = -1/2$ at zero displacement field obtained by two different indicators from ED on a 4×6 torus. The phase diagram of left figure is resolved by the momentum structure of low energy states while that on the right is resolved from the occupation $\langle n(\boldsymbol{k}) \rangle$. See text for details.

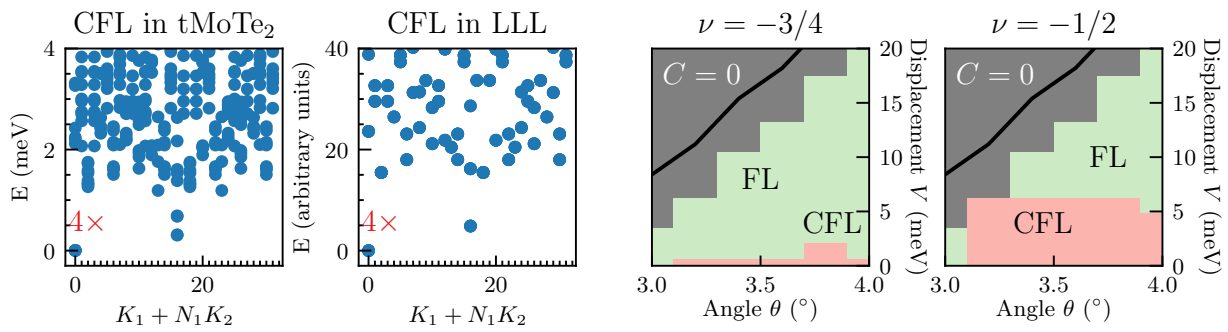


Figure B.10: The first two panels compare the spectrum of twisted $MoTe_2$ and the LLL at $\nu = -3/4$ within the band-projected approximation. The four fold center of mass degeneracy is broken weakly in twisted $MoTe_2$, but is too close to be resolved by eye in the ground state. We then compare the band-projected phase diagrams at $\epsilon_r = 8$ and $\nu = -3/4$ and $\nu = -1/2$. The large extent of the composite Fermi liquid phase is shown in red. The green regions correspond to Fermi liquid states, and the gray regions are near or past the $C = 1 \to 0$ topological transition marked with a black line. The composite Fermi liquid state is much more unstable to displacement field at $\nu = -3/4$ compared to $\nu = -1/2$.

shown in Fig. B.9. In this phase diagram, only the CFL and FL phases appear. The phase diagram computed by the low energy momentum sectors is shown in Fig. B.9(a), in which the CFL phase is shown in red dots and the FL phase is shown in blue dots. The phase diagram computed by the finite differences in occupation is shown in Fig. B.9(b). The dark regions represent a small variation in occupations and thus represent the CFL phase. The bright regions represent the FL phase. The shapes of the phase boundaries agree with each other. The large volume of the CFL state in this band projected phase diagram implies that our result is robust to parameter changes both in twist angle variations and strength of interaction, supporting the possibility that the CFL state can be realized in realistic experiments.

In Fig. 3(b) we identify the phases in the (θ, V) plane using the many-body momentum sectors. We keep $\epsilon_r = 8$ throughout. There is clearly a large portion of the phase diagram exhibiting CFL-like features, represented by the red region centered around 3.6° with a small displacement field. The displacement field drives a phase transition from the CFL into the FL phase, shown in green.

We mark the topological transition line in gray and remark that near the topological phase transition the band gap closes so the band projection approximation in our ED procedure no longer works. We emphasize that this phase diagram is obtained assuming band projection throughout the parameters used in the phase diagram.

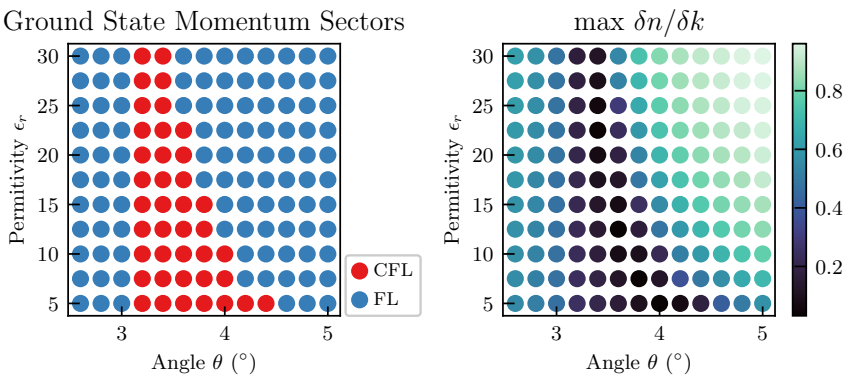


Figure B.11: Band-projected phase diagram of interacting twisted $MoTe_2$ at $\nu = -3/4$ at zero displacement field obtained from two different indicators from ED on a 4×8 torus. The phase diagram of left figure is resolved by the momentum structure of low energy states while that on the right is resolved from the occupation $\langle n(\boldsymbol{k}) \rangle$. See text for details.

B.6.4 $\nu = -3/4$ Composite Fermi Liquid and Associated Phase Diagrams

The possibility of a $\nu = -3/4$ CFL is investigated also using ED on a 4×8 torus. Since such a state is closer to $\nu = -1$, we expect that it generally has a larger spin gap than the possible $\nu = -1/2$ CFL. This means that the valley polarization we assume in the ED procedure is better justified at this filling. However, the layer-polarization instability discussed above will modify this phase diagram in the larger Hilbert space of multiple valence bands.

We compare the spectrum between the $\nu = -3/4$ CFL and the $\nu = 3/4$ CFL from the LLL using dual gate screened Coulomb interactions in Fig. B.10. Each point in the LLL spectrum has a four fold center of mass degeneracy, which is weakly broken in twisted $MoTe_2$. The similarity of the spectrum suggests that a CFL where 8π fluxes are attached to each electron can also be formed at $\nu = -3/4$.

In the last two panels of Fig. B.10 we show the many-body phase diagram at $\nu = -3/4$ and at $\nu = -1/2$ assuming band projection. Since $\nu = -3/4$ is closer to $\nu = -1$ we conjecture that the valley polarization will be more stable to the displacement field. We note that the $\nu = -3/4$ CFL region (red) has significantly shrunk compared to the $\nu = -1/2$ CFL, and most of the phase diagram of the $C = 1$ band is a FL phase (green). We mark the topological transition within the top valence band by a black line, and remark that near the topological phase transition the band gap closes so the band projection approximation in our ED procedure is no longer necessarily (gray region). The layer polarization instability from the second band may occur significantly below this line when a larger Hilbert space is considered.

B.7 Density Matrix Renormalization Group

This Appendix discusses the infinite density matrix renormalization group (iDMRG) numerics used in the main text. We first discuss the computational basis used, then express the structure factor in this basis, and finally give additional details on the phase identification including finite size scaling.

B.7.1 Computational Basis

For DMRG we consider a moiré lattice geometry

$$\boldsymbol{a}_1 = a_M \left(\frac{\sqrt{3}}{2}, \frac{1}{2} \right), \quad \boldsymbol{a}_2 = a_M \left(0, 1 \right) \tag{B.45}$$

$$\boldsymbol{g}_1 = \frac{2\pi}{a_M} \left(\frac{2}{\sqrt{3}}, 0 \right), \quad \boldsymbol{g}_2 = \frac{2\pi}{a_M} \left(-\frac{1}{\sqrt{3}}, 1 \right). \tag{B.46}$$

where $a_M = |\boldsymbol{a}_1|$ is the moiré length. We consider rectangular Brillouin zone $\mathcal{B} = [-G_x/2, G_x/2] \times [-G_y/2 \times G_y/2]$ where $G_x = \frac{4\pi}{\sqrt{3}a}, G_y = \frac{2\pi}{a}$ so that $\boldsymbol{a}_1 = 2\pi \left(\frac{1}{G_x}, \frac{1}{2G_y} \right)$.

We solve the kinetic Hamiltonian Eq. (B.5) as

$$h \left| \phi^{\boldsymbol{k}a} \right\rangle = \epsilon^{\boldsymbol{k}a} \left| \phi^{\boldsymbol{k}a} \right\rangle, \tag{B.47}$$

where $\ell = B, T = -1, +1$ labels layers, and a labels bands. We impose boundary conditions in the y direction corresponding to the momenta discretization

$$k_y[n] = G_y \begin{cases} -\frac{1}{2} + \frac{n + \frac{\Phi_y}{2\pi}}{L_x} & \text{if } L_y \equiv 0 \pmod 2 \\ -\frac{1}{2} + \frac{n + \frac{\Phi_y}{2\pi} + \frac{1}{2}}{L_x} & \text{if } L_y \equiv 1 \pmod 2 \end{cases}. \tag{B.48}$$

We therefore have L_y evenly-spaced"wires" across the Brillouin zone. The value of $L_x \gg 1$ does not affect the physics, provided that it is sufficiently large. If $\Phi_y = 0$ then one wire will go through Γ, giving electrons periodic boundary conditions (PBC); $\Phi_y = 1/2$ corresponds to anti-periodic boundary conditions (APBC).

Following previous DMRG work on moiré systems [71, 206, 452, 453], we take a computational basis of hybrid Wannier functions that are maximally (exponentially) localized along $\boldsymbol{a}_1$ and periodic along $\boldsymbol{a}_2 \propto \hat{y}$.

Let $\hat{P}$ be a projector to the band (or bands) of interest. We transform to the Wannier-Qi states [59]

$$|w^{n,k_y,b}\rangle = \frac{1}{\sqrt{L_x}} \sum_{k_x} |\varphi^{\boldsymbol{k}b}\rangle \, e^{-i\boldsymbol{k}\cdot\boldsymbol{a}_1 n}, |\varphi^{\boldsymbol{k}b}\rangle = |\phi^{\boldsymbol{k}a}\rangle \, U^{ab}(\boldsymbol{k}) \tag{B.49}$$

where $U_{nb}(\boldsymbol{k})$ is a unitary matrix that transforms into a Wannier gauge such that

$$\hat{T}_1 \, |w^{n,k_y,b}\rangle = |w^{n-1,k_y,b}\rangle \tag{B.50a}$$

$$\hat{T}_2 \, |w^{n,k_y,b}\rangle = e^{i2\pi k_y/G_y} \, |w^{n,k_y,b}\rangle \tag{B.50b}$$

$$\hat{P} e^{-i\hat{x}G_x} \hat{P} \, |w^{n,k_y,b}\rangle = e^{i2\pi[n+P_x(k_y,b)]} \, |w^{n,k_y,b}\rangle \tag{B.50c}$$

where the polarizations $P_x(k_y, b) = \frac{G_x}{2\pi} \langle w^{0,k_y,b}|\hat{x}|w^{0,k_y,b}\rangle$ are the centers of the Wannier orbitals in the first unit cell, in accordance with the modern theory of polarization [204]. The polarizations for the top two valence bands of the K-valley are shown in Fig. B.2(b). These have winding $C = \int dk_y \frac{dP_x}{dky} = +1, -1$ for the respectively, matching the Chern number [204]. We now specialize to the case where $\hat{P}$ projects to the top $C = +1$ valence band of the K-valley (though this method applies equally well to both valleys and multiple bands, which will be the subject of future work).

At the many-body level, we consider Fermi creation operators

$$|\varphi^{\boldsymbol{k}}\rangle = \hat{c}^\dagger_{\boldsymbol{k}} \, |0\rangle \, , \quad \hat{c}^\dagger_{\boldsymbol{k}} = \frac{1}{\sqrt{L_x}} \sum_n e^{i\boldsymbol{k}\cdot\boldsymbol{a}_1 n} \hat{d}^\dagger_{n,k_y}, \tag{B.51}$$

which correspond to Bloch states and Wannier-Qi states in the top valence valence band of the K valley, respectively. We then express the Hamiltonian Eq. (B.11) in the $\hat{d}^\dagger_{n,k_y}$ basis, which should be interpreted as the putting the TMD Hamiltonian onto a cylinder geometry that is infinite along the x direction and periodic along the y direction with circumference L_y. The choice of maximally-localized Wannier orbitals along x ensures that hoppings decay as fast as possible in the x-direction. However, the quasi-long range screened Coulomb interactions decay quite slowly along x.

To perform infinite DMRG, one must express the Hamiltonian as a matrix product operator (see e.g. [454]), which is a tensor network composed of size $2 \times 2 \times D \times D$ tensors where 2 is the physical dimension (for fermions) and D is the virtual "bond dimension". One may naively construct an MPO from a Hamiltonian in the $d^\dagger_{n,k_y}$ basis by imposing a finite cutoff $M > 0$ on the interaction range along the cylinder. However, this has

two serious drawbacks: (i) the MPO bond dimension increases as $O(M^2)$, reaching $D \sim 2000$ by $M = 2$ at $L_y = 6$ [206] and (ii) the relative stability of various many-body ground states (particular quantum Hall states) are known to be sharply modified by the interaction distance. To circumvent this issue, some of us developed a technique of "MPO Compression" [205, 206], which creates a faithful representation of the MPO, including its quasi-long range interactions. In practice we ensure that the error of each matrix element of the Hamiltonian is $< 10^{-3}$ meV for up to 200 sites along the 1D chain. This technique has been previously applied to twisted bilayer graphene in the cases of 1, 2, 4, or 8 degrees of freedom per k-point [71, 206, 452, 453]. In this single-DOF case, the resulting bond dimensions are $D \sim 130 - 250$ for $L_y = 5 - 10$ respectively.

B.7.2 Structure Factor

To identify a composite Fermi liquid, we compute the structure factor, i.e. the density-density correlation function. The standard structure factor is defined as

$$S_0(\boldsymbol{q}) = \langle \hat{\rho}_{\boldsymbol{q}} \hat{\rho}_{-\boldsymbol{q}} \rangle \tag{B.52}$$

in terms of the density operator (B.12). However, the dominant features of $S(\boldsymbol{q})$ here — and in any lattice system — are peaks at the Bragg lattice from $\langle \hat{\rho}_{\boldsymbol{q}} \rangle = \sum_{\boldsymbol{G}} \delta_{\boldsymbol{G}\boldsymbol{q}} \langle \hat{\rho}_{\boldsymbol{G}} \rangle$, where $\boldsymbol{G}$ is a reciprocal lattice vector. To identify the non-trivial features associated to a (composite) Fermi surface, we focus on the connected part of the structure factor

$$S(\boldsymbol{q}) = \langle \hat{\rho}_{\boldsymbol{q}} \hat{\rho}_{-\boldsymbol{q}} \rangle - \langle \hat{\rho}_{\boldsymbol{q}} \rangle \langle \hat{\rho}_{-\boldsymbol{q}} \rangle . \tag{B.53}$$

We now resolve this in the computational basis $\hat{d}_{n,k_y}$. A straightforward, albeit lengthy, computatation shows

$$S(\boldsymbol{q}) = \sum_{\substack{n_0 \in \mathbb{Z} \\ |\Delta N| \leq M \\ |\Delta n|, |\Delta n'| \leq N}} \sum_{k_y, k_y'} e^{i\boldsymbol{q}\cdot\boldsymbol{a}_1(\Delta N + \Delta n' - \Delta n)} \tilde{\Lambda}_{\boldsymbol{q}}(\Delta n, k_y), \tilde{\Lambda}_{-\boldsymbol{q}}(\Delta n', k_y') M_{\Delta n, k_y, \Delta n', k_y', \Delta N, q_y} \tag{B.54a}$$

$$\tilde{\Lambda}_{\boldsymbol{q}}(\Delta n, k_y) = \sum_{k_x} e^{-i\boldsymbol{k}\cdot\boldsymbol{a}_1 \Delta n} \Lambda_{\boldsymbol{q}}(k_x, k_y) \tag{B.54b}$$

$$\begin{aligned} M_{\Delta n, k_y, \Delta n', k_y', \Delta N, q_y} = {} & \langle \hat{d}^\dagger_{n_0, k_y} \hat{d}_{n_0 + \Delta n, [k_y + q_y]} \hat{d}^\dagger_{n_0 + \Delta N, k_y'} \hat{d}_{n_0 + \Delta N + \Delta n', [k_y - q_y]} \rangle \\ & - \langle \hat{d}^\dagger_{n_0, k_y} \hat{d}_{n_0 + \Delta n, [k_y + q_y]} \rangle \langle \hat{d}^\dagger_{n_0 + \Delta N, k_y'} \hat{d}_{n_0 + \Delta N + \Delta n', [k_y - q_y]} \rangle . \end{aligned} \tag{B.54c}$$

Here $[k_y + q_y]$ denotes that the quantity lies within the first Brillouin zone, and $M > 0, N > 0$ are finite-size cutoffs. As it is computationally expensive to compute $L_y^3(2M+1)(2N+1)^2$ observables in DMRG, we restrict to $N = 1$ (which is sufficient for small $|q_y| \lesssim G_y$), and take several hundred values of ΔN, q_x, which is sufficient to resolve fine features in Fourier space.

B.7.3 DMRG Phase Identification

This section details the phase identification of the Fermi liquid and composite Fermi liquid in DMRG. We focus on system sizes $L_y = 5, 8, 10$ and bond dimensions $\chi = 1024 - 4096$, which are sufficiently large to determine the phase unambiguously and converge most observables to high precision. To ensure that we only study differences in interacting physics, we study two representative states at $\nu = -1/2$ with (I) strong interactions $\epsilon_r = 6$ and (II) weak interactions $\epsilon_r = 60$ at $\theta = 3.89°$. Following the discussion in App. B.5, we note that this point is adiabatically connected within the projected Hilbert space to the parameters $(\theta, \epsilon_r) = (3.7°, 15)$ studied in the main text.

We begin by examining the occupations $n(\boldsymbol{k}) = \langle \hat{c}_{\boldsymbol{k}}^\dagger \hat{c}_{\boldsymbol{k}} \rangle$ across the Brilouin zone. These are shown in Fig. 2.2(a) of the main text. For $\epsilon_r = 6$, the occupations are almost uniform, while for $\epsilon_r = 60$, the occupations are highly non-uniform with a sharp jump emerging as bond dimension increases. This behavior is characteristic of a Fermi liquid in cylinder DMRG: since the system is quasi-1D, no true Fermi liquid with quasiparticle weight $Z > 0$ can emerge. Instead, the ground state here is converging to (coupled) Luttinger liquids with a non-analyticity at the Fermi surface [455]. The locations of these features for the weakly-interacting state form an almost-circular Fermi surface centered at Γ — precisely what is expected for the weakly interacting Fermi liquid, given that the top band has a minimum at Γ. So far we have established that the weakly interacting state is highly consistent with a Fermi liquid, but the uniform occupations of the strongly-interacting state are consistent with both trivial insulators or more exotic possibilities.

We now examine entanglement scaling and correlation lengths. Fig. B.12(a,d) shows the scaling of entanglement entropy as a function of bond dimension. We see that both the weakly and strongly interacting states are highly entangled, consistent with a gapless state. To determine the character of potential gapless modes, we examine the spectrum of the transfer matrix. For a Fermi liquid, where adding an electron, hole, or particle-hole pair are all gapless excitations, one expects electron-electron, hole-hole, and density-density correlations to all show power-law behavior, with a diverging correlation length. For $\epsilon_r = 60$, we indeed find diverging correlation

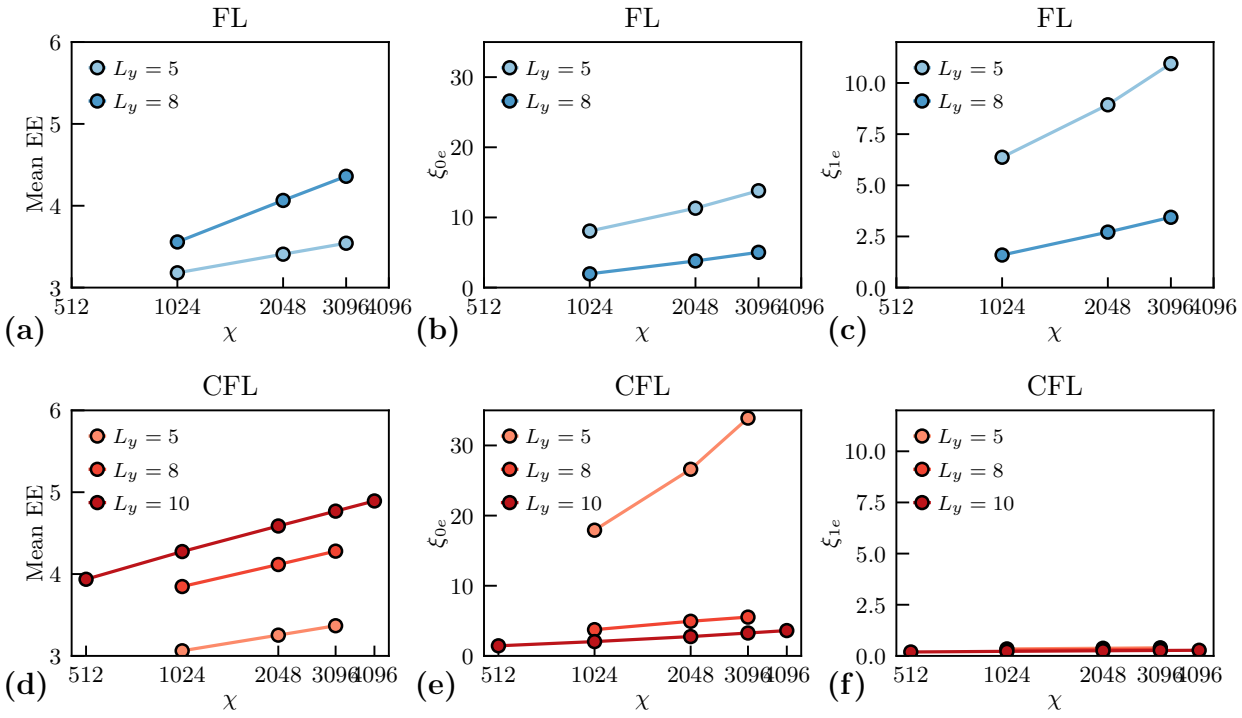


Figure B.12: Entanglement properties as a function of bond dimension χ. (a,d): the mean entanglement entropy measured. (b,e): the correlation length in the $Q_E = 0$ sector. The divergence in correlation length is consistent with the presences of gapless neutral excitations, corresponding to the particle-hole excitations in the Fermi liquid and density fluctuations in the composite Fermi liquid. (c,f): the correlation length in the $Q_E = 1$ sector. The Fermi liquid has a diverging correlation length, reflecting its gapless charged excitations – the electrons. In contrast, the composite Fermi liquid has small correlations in this sector. Parameters: $(\theta, \epsilon_r, V) = (3.89°, 6, 0)$.

lengths for the electric charge $Q_E = +1, -1, 0$ sectors for all k_y, shown in Fig. B.12(b,c). We conclude that the weakly interacting state is indeed a Fermi liquid. For the strongly interacting state, we see a similar increase in $Q_E = 0$ correlations without any accompanying increase in $Q_E = \pm 1$ correlations, shown in Fig. B.12(e,f). This is consistent with a state that has gapless charge neutral excitations, but no gapless electrons and holes.

To understand the character of the neutral degrees of freedom, we turn to the structure factor $S(\boldsymbol{q})$ described above. Fig. B.13 details the structure factor for both the weakly- and strongly-interacting states. As the minimum resolution in q_x is set by the density-density correlation length, we focus on $L_y = 5$ so that we may identify peaks, inflection points, and other sharp features.

We now show that the structure factors $S(\boldsymbol{q})$ for both the weakly and strongly interacting states are consistent with an almost-circular Fermi surface. Explicitly, assume a Fermi surface (not necessarily of electrons) of half the Brillouin zone area, i.e. $k_F = \sqrt{A_{BZ}/2\pi}$. Examining the geometry, we see that four of the five wires intersect a putative Fermi surface. (The assumption that the momenta of the quasiparticles that make up the Fermi surface matches the momenta of the wires holds here, but not in general [184].) On this quasi-1D system, we expect features in $S(\boldsymbol{q})$ whenever particle-hole pairs

$$\hat{c}^\dagger_{\boldsymbol{k}} \hat{c}_{\boldsymbol{k}+\boldsymbol{q}} \subset \hat{\rho}_{\boldsymbol{q}}$$

have $\boldsymbol{k}$ and $\boldsymbol{k}+\boldsymbol{q}$ at intersections between wires and the Fermi surface. We label these intersections

$$\boldsymbol{k}_{A,B,C,D} = \left(-\sqrt{k_F^2 - k_y[n]^2}, k_y[n]\right), n = 1, 2, 3, 4,$$

on the left-hand side of the Fermi surface, and give corresponding labels $\boldsymbol{k}_{A',B',\ldots}$ to the right-hand Fermi surface and higher Brillouin zones (see Fig. B.13).

We identify features in $S(\boldsymbol{q})$ by taking its fractional derivative following [184, 455], defined through the Fourier transformations as follows:

$$\frac{d^\eta}{dq_x^\eta} S(q_x, q_y) = \int dx e^{-iq_x x} |x|^\eta \left(\int dq'_x e^{iq'_x x} S(q'_x, q_y) \right) \tag{B.55}$$

The fractional derivatives magnifies inflections and discontinuities in higher derivatives, making the scattering processes much more visible to the eye. We take various $\eta \in [1/3, 5/2]$ for display.

We observe that the FL structure factors show a much larger *umklapp* scattering than the structure factors of CFL in Fig. B.13. The reason is that the Fermi liquid has a large density modulation in the unit cell, strongly enhancing the amplitude of *umklapp* scatterings, whereas the CFL has a much milder density modulation. We comment that enhancing the density modulation of the CFL may enhance the *umklapp* scattering and introduce deformations of the composite Fermi surface [456].

B.8 More on experimental aspect of Chern band CFL

The CFL is a gapless, compressible phase and in the absence of a quantized response is more challenging to establish as compared to gapped FQHE states. Nevertheless, it has many striking physical responses which are distinct from that of other gapless phases, in particular the Fermi liquid (FL). The experimental signatures include transport properties, tunneling density of states and finite momentum conductivity.

A direct probe of the composite nature of the CFL quasiparticles is revealed in the tunneling density of states (tDOS). The tDOS for a Fermi liquid is finite even at zero bias, a consequence of the low energy electron like quasiparticles. However, physical electrons cannot be injected into a CFL at zero bias, leading to a "pseudogap" density of states. In particular the low energy tDOS of a CFL varies as $A(\omega) \propto e^{-\omega_0/\omega}$ as a function of bias ω [224]. The tDOS can be measured through the tunneling current in two setups. Using scanning tunneling microscope (STM), electrons can be directly injected into the sample. Alternatively, the tunneling current between two CFLs separated by a tunneling barrier can be measured. The contrast in zero bias features have been shown directly in Ref. [143], in which the second setup is employed in the Landau level setting.

Transport measurements give distinct signatures for the CFL. If we denote $\tilde{\rho}$ the (dimensionless) resistivity tensor for the CFL itself, in the half filled Landau level setting, the physical conductivity is given by:

$$\sigma = \frac{e^2}{2h}\left(\epsilon + \frac{\tilde{\rho}}{2}\right), \tag{B.56}$$

where ϵ is the antisymmetric 2×2 tensor. Now, in the absence of disorder and of scattering of the composite fermions, $\tilde{\rho} \to 0$, whch implies the longitudinal conductivity σ_{xx} vanishes. In contrast, in a clean Fermi liquid in a Chern band, the longitudinal conductivity is expected to *grow* as disorder is reduced [225]. One may further seek to measure thermoelectric transport signatures, encapsulated in the tensor $\mathbf{E} = \hat{S}\nabla\mathbf{T}$, where for the CFL in the half filled LL, $\hat{S} = \sigma^{-1}\epsilon\tilde{\rho}\tilde{\alpha}/2$, where $\tilde{\alpha}$ is the thermoelectric conductivity of the composite Fermi surface

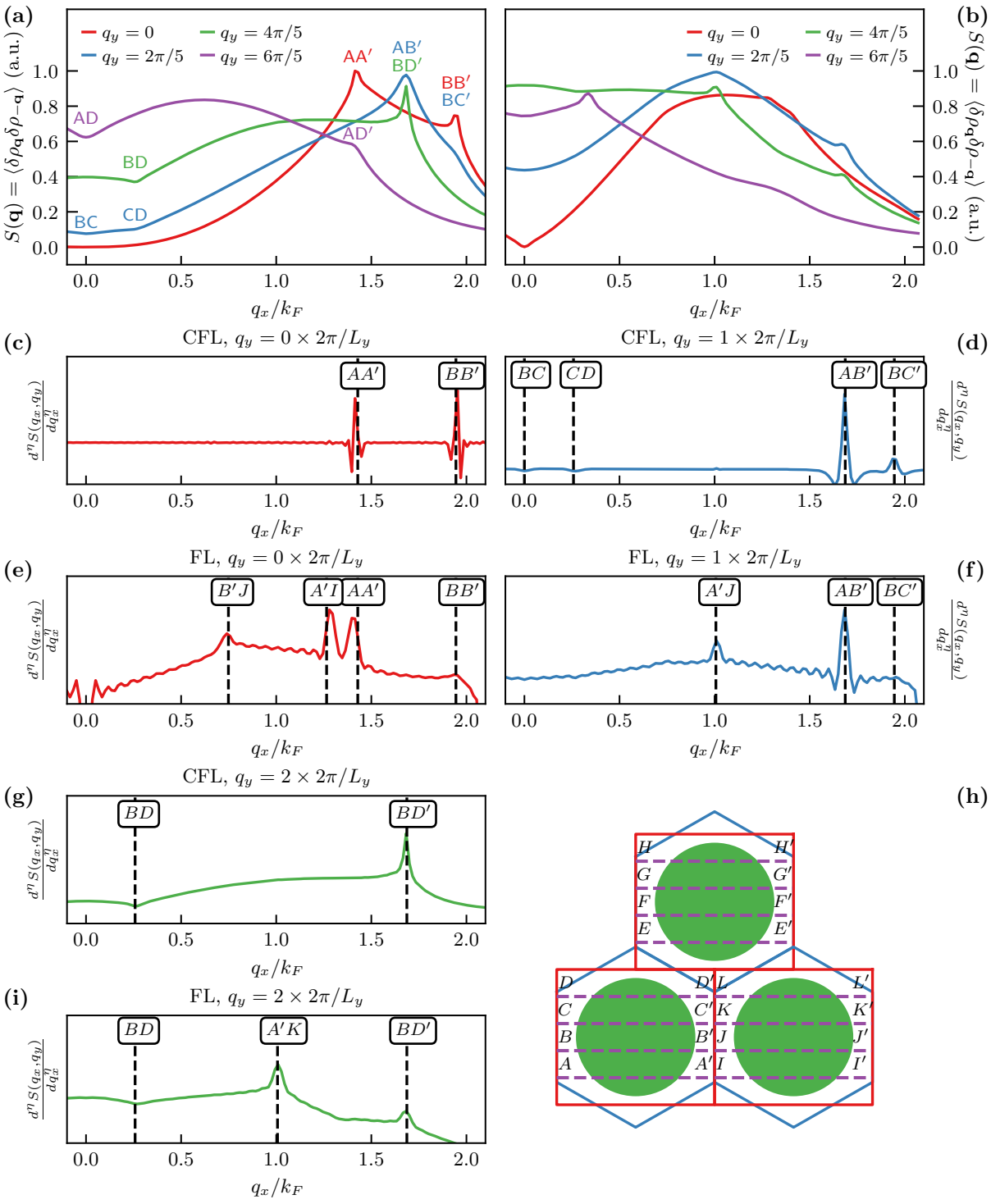


Figure B.13: Quantitative comparison between FL and CFL structure factors with fermi surface scattering momenta. (a) Connected structure factor $S(\boldsymbol{q})$ at constant q_y as a function of q_x for the CFL phase (also shown in main text). (b) Same for the FL at $\epsilon_r = 60$. Fractional derivatives of $S(\boldsymbol{q})$ with respect to q_x to bring out detailed features are shown for $q_y = 0$ (c,e), $q_y = 1$ (d,f), and $q_y = 2$ (g,i). Note that some peaks are only clearly visible for FL or CFL. In particular, *umklapp* peaks between the first and higher Brillouin zones only clearly appear for the FL state. (h) Brillouin zone geometry and labelling scheme for vertical dashed lines. For instance, a line denoted 'BD' corresponds to the q_x position of $\boldsymbol{q} = \boldsymbol{k}_D - \boldsymbol{k}_B$, where $\boldsymbol{k}_B$ and $\boldsymbol{k}_D$ are intersections between the wires and the Fermi surface. Parameters: $(\theta, \epsilon_r, V) = (3.89°, 6 \text{ or } 60, 0)$.

Ref. [226, 227]. Finally, thermal conductivity measurements, although challenging, will serve as a smoking gun signature of the nFL, since the CFL will strongly violate the Wiedemann-Franz law [222].

Away from the CFL filling, the composite fermions experience a effective magnetic field tied to the filling deviation: $\Delta\nu = \nu - 1/2$, leading to Landau levels of the composite Fermi surface. FCIs correspond to fully occupying these Landau levels[139, 141], corresponding to fillings $\nu = -(p+1)/(2p+1)$, and the many-body gap of the FCI is identified with the cyclotron gap of the Landau levels. The observation of FCIs at filling $\nu = -2/3,\ -3/5$ in moiré TMDs [19, 161], corresponding to $p = 1,\ 2$ is consistent with this expectation. On the other hand, an FL would not exhibit such quantum oscillations when the filling is changed. Analogously, imposing a one-dimensional periodic grating is expected to lead to geometric resonances on varying the filling $\Delta\nu$ in the moiré TMD setting, as have been observed for CFLs in Landau levels [141, 144, 145].

A notable prediction of the CFL picture [141], verified in experiments [146] in the Landau level context, relates to the finite wave-vector conductivity $\sigma_{xx}(\boldsymbol{q})$ which can be probed by surface acoustic waves (SAWs). The CFL conductivity is predicted to be *linear* in the wave-vector $|\mathbf{q}|$. This follows directly from the form of the CFL conductivity tensor B.56 combined with the fact that the intrinsic resistivity of the CFL in the clean limit is set by the finite wave-vector of the SAW probe $\tilde{\rho} \propto |\boldsymbol{q}|$.

Implementing the proposals above will require overcoming significant experimental challenges - from performing transport measurements on TMD moiré materials to working with the smaller size of moiré samples compared to GaAs/GaAlAs samples. Nevertheless, given the experimental opportunity to study a zero field non-Fermi liquid which might give rise to of a non-Abelian topological phase, we anticipate there will be strong push to tackle these challenges.

C

Supplementary Material for Chapter 3

C.1 Details of the Microscopic Model

This appendix fully specifies the models discussed in the main text.

C.1.1 Rhombohedral Graphene

Let $a \approx 2.46$ be the lattice constant of graphene and consider real space and reciprocal lattices spanned by

$$\boldsymbol{R}_1 = (a, 0)\,, \qquad \boldsymbol{R}_2 = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \tag{C.1}$$

$$\boldsymbol{G}_1 = \frac{2\pi}{a}\left(1, -\frac{1}{\sqrt{3}}\right), \qquad \boldsymbol{G}_2 = \frac{2\pi}{a}\left(0, \frac{2}{\sqrt{3}}\right). \tag{C.2}$$

For N-layer Rhombohedral graphene [284, 285] (see also: [259, 266, 287, 288, 457] and references therein), consider p_z orbitals of carbon at positions

$$\boldsymbol{r}_{A,\ell} = \left(0, \frac{\ell - 1}{\sqrt{3}}a, c\right), \boldsymbol{r}_{B,\ell} = \left(0, \frac{\ell}{\sqrt{3}}a, c\right) \tag{C.3}$$

for layer $\ell \in [1, N_L]$ with inter-layer spacing c. Let the nearest neighbor vectors be $\boldsymbol{\delta}_n = R_{n2\pi/3}(0, \frac{1}{\sqrt{3}}a)^T$ for $n = 0, 1, 2$ where R_θ is the counterclockwise rotation matrix by angle θ. Below we index sublattice by $\sigma \in \{A, B\} = \{-1, 1\}$. We define standard high-symmetry points $\boldsymbol{K} = \frac{2}{3}\boldsymbol{G}_1 + \frac{1}{3}\boldsymbol{G}_2$ and $\boldsymbol{K}' = \frac{1}{3}\boldsymbol{G}_1 + \frac{2}{3}\boldsymbol{G}_2$.

For each orbital, we have a corresponding second-quantized Fermion operator $\hat{c}^\dagger_{\boldsymbol{R},\sigma,\ell} = \hat{c}^\dagger(\boldsymbol{R} + \boldsymbol{r}_{\sigma,\ell})$. Its Fourier transform is $\hat{c}^\dagger_{\boldsymbol{k}\sigma\ell} = N^{-\frac{1}{2}} \sum_{\boldsymbol{R}} e^{+i\boldsymbol{k}\cdot(\boldsymbol{R}+\boldsymbol{r}_{\sigma,j})}\hat{c}^\dagger(\boldsymbol{R}+\boldsymbol{r}_{\sigma,\ell})$, with normalization so that it obeys canonical

commutation relations $\{\hat{c}_{\boldsymbol{k}\sigma\ell}, \hat{c}_{\boldsymbol{k}\sigma'\ell'}\} = \delta_{\sigma\sigma'}\delta_{\ell\ell'}$. The Hamiltonian for N-layer rhombohedral graphene is $\hat{h}_{RG}^{(N_L)} = \sum_{\boldsymbol{k}\in \mathrm{BZ}} \hat{c}^\dagger_{\boldsymbol{k}\sigma j} \left[h_{RG}^{(N_L)}(\boldsymbol{k})\right]_{\sigma j,\sigma' j'} \hat{c}_{\boldsymbol{k}\sigma' j'}$ where

$$h_{RG}^{(N_L)}(\boldsymbol{k}) = \begin{bmatrix} h_\ell^{(0)} & h^{(1)} & h^{(2)} & & & \\ h^{(1)\dagger} & h_\ell^{(0)} & h^{(1)} & h^{(2)} & & \\ h^{(2)\dagger} & h^{(1)\dagger} & h_\ell^{(0)} & \ddots & \ddots & \\ & h^{(2)\dagger} & \ddots & \ddots & h^{(1)} & h^{(2)} \\ & & \ddots & h^{(1)\dagger} & h_\ell^{(0)} & h^{(1)} \\ & & & h^{(2)\dagger} & h^{(1)\dagger} & h_\ell^{(0)} \end{bmatrix} \tag{C.4}$$

which is a matrix in layer space with entries in sublattice space

$$h_\ell^{(0)}(\boldsymbol{k}) = \begin{pmatrix} u_{A\ell} & -t_0 f_{\boldsymbol{k}} \\ -t_0 \overline{f}_{\boldsymbol{k}} & u_{B\ell} \end{pmatrix} \tag{C.5a}$$

$$h^{(1)}(\boldsymbol{k}) = \begin{pmatrix} t_4 f_{\boldsymbol{k}} & t_3 \overline{f_{\boldsymbol{k}}} \\ t_1 & t_4 f_{\boldsymbol{k}} \end{pmatrix} \tag{C.5b}$$

$$h^{(2)}(\boldsymbol{k}) = \begin{pmatrix} 0 & \frac{t_2}{2} \\ 0 & 0 \end{pmatrix} \tag{C.5c}$$

$$f_{\boldsymbol{k}} = \sum_{i=0}^{2} e^{i\boldsymbol{k}\cdot\boldsymbol{\delta}_i}. \tag{C.5d}$$

We take graphene hopping $t_0 = 3.1\,\mathrm{eV}$ and vertical interlayer hopping $t_1 = 380\,\mathrm{meV}$. Following e.g. [458] we take higher hopping parameters $(t_2, t_3, t_4) = (-21, 290, 141)\,\mathrm{meV}$. Finally, $u_{\sigma\ell}$ are on-site potentials that reflect internal and external displacement fields. Although their value varies with layer [457] to partially screen applied electric fields, for simplicity we consider fields with a uniform potential difference per layer u_D:

$$u_{\sigma,\ell} = u_D \left(\ell + 1 - \frac{N_L - 1}{2}\right). \tag{C.6}$$

Aside from lattice translation, the model enjoys the following symmetries

$$\overline{h(\boldsymbol{k})} = h(-\boldsymbol{k}) \quad \text{(Time reversal)} \tag{C.7a}$$

$$h(\boldsymbol{k}) = h(C_{3z}\boldsymbol{k}) \quad \text{(Three fold rotation)} \tag{C.7b}$$

$$M_{C_{2x}} h(\boldsymbol{k}) M_{C_{2x}}^{-1} = h(C_{2x}\boldsymbol{k}) \quad \text{(Two-fold } x\text{-rotation)} \tag{C.7c}$$

$$h(\boldsymbol{k}) = h(M_x\boldsymbol{k}) \quad \text{(Mirror)} \tag{C.7d}$$

$$V_{\boldsymbol{G}} h(\boldsymbol{k}) V_{\boldsymbol{G}}^\dagger = h(\boldsymbol{k}+\boldsymbol{G}) \quad (\boldsymbol{G}\text{-translation}) \tag{C.7e}$$

where time-reversal acts anti-unitarily, C_{2x} reverses layers as $[M_{C_{2x}}]_{\sigma\ell,\sigma'\ell'} = \delta_{\sigma,-\sigma'}\delta_{\ell,N-(\ell'+1)}$, and other symmetries have trivial orbital part. We note that the last equation is not a symmetry per se, but simply a constraint on the Hamiltonian with $[V_{\boldsymbol{G}}]_{\sigma'\ell',\sigma\ell} = \delta_{\sigma\sigma'}\delta_{\ell\ell'} e^{i\boldsymbol{G}\cdot\boldsymbol{r}_{\sigma,\ell}}$, reflecting the unit cell embedding.

C.1.2 Moiré Rhombohedral Graphene Hamiltonian

The direct lattice of the hBN substrate is is generated by

$$\boldsymbol{R}'_j = M R_\theta \boldsymbol{R}_j; \quad M = (1+\varepsilon) I \tag{C.8}$$

where R_θ is counterclockwise rotation by θ and $\varepsilon = a_{\text{hBN}}/a_{\text{Gr}} \approx 1.018$. We take $\boldsymbol{G}'_j$ such that $\boldsymbol{R}'_j \cdot \boldsymbol{G}'_k = 2\pi\delta_{jk}$ as usual.

Define moiré reciprocal lattice vectors

$$\boldsymbol{g}_j = \boldsymbol{G}_j - \boldsymbol{G}'_j = [I - M^{-1}R_\theta]\boldsymbol{G}_j, \tag{C.9}$$

with corresponding direct vectors $\boldsymbol{a}_j \cdot \boldsymbol{g}_k = 2\pi\delta_{jk}$ whose moiré superlattice period is

$$L_M = |\boldsymbol{a}_j| = a\frac{1+\varepsilon}{\sqrt{\varepsilon^2 + 2(1+\varepsilon)(1-\cos\theta)}} \tag{C.10}$$

and where the lattice scale mismatch rotates the moire-Brillouin zone so that $\boldsymbol{a}_1$ is rotated clockwise from the x axis by an angle

$$\phi = \arctan\left(\frac{-\sin\theta}{1+\varepsilon-\cos\varepsilon}\right). \tag{C.11}$$

As θ changes from 0 to 1, the superlattice scale changes only gradually, but ϕ rotates quickly, as shown in Fig. C.1.

To conveniently express C_3-symmetric quantities, further define $\boldsymbol{g}_3 = -\boldsymbol{g}_1 - \boldsymbol{g}_2$ and likewise for $\boldsymbol{G}_3, \boldsymbol{G}_3'$. We then have high-symmetry points at

$$\begin{aligned} \boldsymbol{\gamma} &= \boldsymbol{K}_{\mathrm{Gr}}, \quad \boldsymbol{\kappa}^+ = \boldsymbol{\gamma} - \frac{1}{3}\boldsymbol{g}_1 + \frac{1}{3}\boldsymbol{g}_2, \\ \boldsymbol{\kappa}^+ &= \boldsymbol{\gamma} - \frac{2}{3}\boldsymbol{g}_1 - \frac{1}{3}\boldsymbol{g}_2, \quad \boldsymbol{\mu} = \boldsymbol{\gamma} - \frac{1}{2}\boldsymbol{g}_1. \end{aligned} \tag{C.12}$$

Crucially, we have taken the center of the moiré Brillouin zone $\boldsymbol{\gamma}$ to be at the $\boldsymbol{K}$-point of graphene.

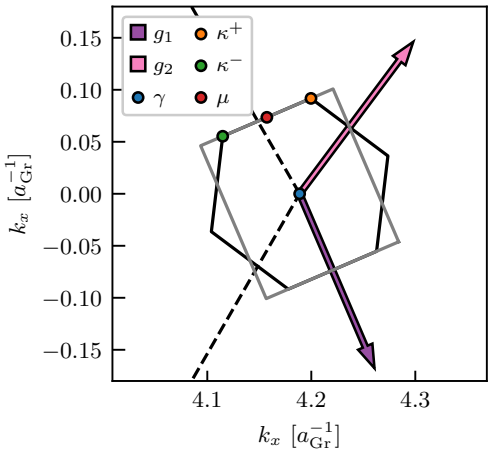


Figure C.1: Geometry of the moiré Brillouin zone at $\theta = 0.77°$. The dashed line is the Brillouin zone of (unrotated) graphene, and the gray rectangle is the Brillouin zone used for DMRG (see text).

We may now express the moiré potential generated by the hBN substrate. We assume that the moiré potential of the hBN substrate only affects the closest layer. We follow [275] (which uses an opposite valley convention), we have

$$V_{\mathrm{hBN}}(\boldsymbol{r}) = V_0\sigma^0 + V_1\boldsymbol{f}(\boldsymbol{r}, \psi)\cdot\boldsymbol{\sigma}, \tag{C.13}$$

where

$$\begin{aligned} \boldsymbol{f}(\boldsymbol{r}, \psi)\cdot\boldsymbol{\sigma} = e^{-i\psi}\Big[& e^{i\boldsymbol{g}_1\cdot\boldsymbol{r}}\begin{pmatrix} 1 & 1 \\ \omega & \omega \end{pmatrix} + e^{i\boldsymbol{g}_2\cdot\boldsymbol{r}}\begin{pmatrix} 1 & \omega^2 \\ \omega^2 & \omega \end{pmatrix} \\ & + e^{-i(\boldsymbol{g}_1+\boldsymbol{g}_2)\cdot\boldsymbol{r}}\begin{pmatrix} 1 & \omega \\ 1 & \omega \end{pmatrix}\Big] + \mathrm{H.c.} \end{aligned} \tag{C.14}$$

Unless otherwise stated we take values $(V_0, V_1, \psi) = (28.9\,\mathrm{meV}, 21\,\mathrm{meV}, -0.29)$ following [275]. The Hamil-

tonian in the K' valley is defined via time-reversal symmetry.

C.1.3 Many-Body Hamiltonian

Let the single-particle eigenstates be given by

$$\hat{h}_{kin} \left|\psi_{\boldsymbol{k}\tau sb}\right\rangle = \epsilon_{\boldsymbol{k}\tau sb} \left|\psi_{\boldsymbol{k}\tau sb}\right\rangle \tag{C.15}$$

where $\tau \in \{K, K'\} = \{-1, 1\}$ labels valley, $s \in \{\uparrow, \downarrow\} = \{-1, 1\}$ labels spin, and b labels bands. Consider creation operators

$$c^\dagger_{\boldsymbol{k}\tau sb} \left|0\right\rangle = \left|\psi_{\boldsymbol{k}\tau sb}\right\rangle, \quad \{c_{\boldsymbol{k}\tau sb}, c_{\boldsymbol{k}'\tau' s' b'}\} = \delta_{\boldsymbol{k}\boldsymbol{k}'}\delta_{\tau\tau'}\delta_{ss'}\delta_{bb'}. \tag{C.16}$$

We consider the interacting model

$$\hat{H} = \hat{h}_{\text{kin}} + \frac{1}{2A}\sum_{\boldsymbol{q}} U_{|\boldsymbol{q}|} : \hat{\rho}_{\boldsymbol{q}}\hat{\rho}_{-\boldsymbol{q}} :, U_q = \frac{2\pi \tanh(qd)}{\epsilon_r \epsilon_0 q} \tag{C.17}$$

where U_q represents Coulomb interactions screened by both top and bottom gates at a distance of $d = 250$.

To make the Hilbert space tractable, we restrict to a limited number of bands N_b above charge neutrality. (Alternatively one could use a plane-wave basis and use some number of shells of Brillouin zones around the first.) More explicitly, we work with the projected density operator,

$$\hat{\rho}_{\boldsymbol{q}} = \sum_{\boldsymbol{k}} \sum_{\alpha,\beta} \sum_{b,b'<N_b} \hat{c}^\dagger_{\boldsymbol{k}\alpha b} \left\langle \psi_{\boldsymbol{k}\alpha b} | e^{-i\boldsymbol{q}\cdot\boldsymbol{r}} | \psi_{\boldsymbol{k}+\boldsymbol{q},\beta,b'} \right\rangle \hat{c}_{\boldsymbol{k}+\boldsymbol{q},\beta,b'} \tag{C.18}$$

where α, β index valley, spin, and b, b' labels band. In some contexts such as twisted bilayer graphene, the operation of restricting to low-energy bands is fraught with difficulties due to double-counting of interaction effects, necessitating procedures such as "Hartree-Fock subtraction" [71, 443] or more sophisticated renormalization-group treatments [444]. Happily, here these issues are ameliorated in the limit where the displacement field opens a gap comparable to or larger than the interaction strength. In that case, the insulating state at charge neutrality is a natural choice of vacuum. The operation of integrating out the "remote" higher-energy conduction bands $N_b + 1, N_b + 2, \ldots$, and valence bands at mean-field level is equivalent to simply *restricting* the band indices in all sums in Eq. (C.17). We note, however, that beyond-mean-field effects like RPA-level screening from

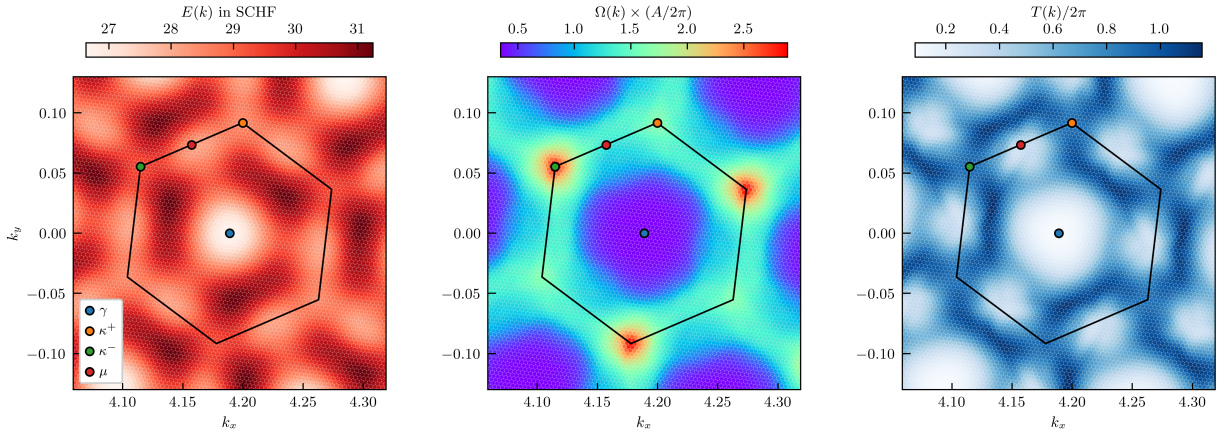


Figure C.2: Details of the SCHF state at $\nu = 1$. (Left) Dispersion. (Middle) Berry curvature. (Right) trace condition violation. Parameters: $(u_D, \theta) = (50\,\text{meV}, 0.77°)$, 48×48 unit cells.

remote bands could substantially influence the system. This is often captured at a phenomenological level by increasing the relative dielectric constant ϵ_r.

C.2 Details of the Hartree-Fock calculation

We use standard self-consistent Hartree-Fock calculations to obtain the SCHF ground state of the electronic system.

C.2.1 Numerical Setup

As usual, we consider single-particle density matrices $P(\boldsymbol{k})_{\alpha\beta} = \langle c^\dagger_{\boldsymbol{k}\beta} c_{\boldsymbol{k}\alpha} \rangle$, where α and β are collective indices for valley, spin, and band. These density matrices are in one-to-one correspondence with Slater determinant states. We define the Hartree and Fock Hamiltonians as

$$h_H[P](\boldsymbol{k}) = \frac{1}{A} \sum_{\boldsymbol{g}} V_{\boldsymbol{g}} \Lambda_{\boldsymbol{g}}(\boldsymbol{k}) \left(\sum_{\boldsymbol{k}} \mathrm{Tr}[P(\boldsymbol{k}) \Lambda_{\boldsymbol{g}}(\boldsymbol{k})^\dagger] \right) \tag{C.19a}$$

$$h_F[P](\boldsymbol{k}) = -\frac{1}{A} \sum_{\boldsymbol{q}} V_{\boldsymbol{q}} \Lambda_{\boldsymbol{q}}(\boldsymbol{k}) P([\boldsymbol{k} + \boldsymbol{q}]) \Lambda_{\boldsymbol{q}}(\boldsymbol{k})^\dagger \tag{C.19b}$$

where $[\Lambda_{\boldsymbol{q}}(\boldsymbol{k})]_{\alpha\beta} = \langle \psi_{\boldsymbol{k}\alpha} | e^{-i\boldsymbol{q}\cdot\boldsymbol{r}} | \psi_{\boldsymbol{k}+\boldsymbol{q}\beta} \rangle$ are form factors, and are treated as matrices whose labels are identical to the single-particle ones. The sum over $\boldsymbol{g}$ runs over reciprocal vectors while $\boldsymbol{q}$ runs over all momentum

transfers. Via Wick's theorem, the energy of the Slater-determinant state is

$$E[P] = \frac{1}{2}\,\mathrm{Tr}[P(2h_{\mathrm{kin}} + h_H[P] + h_F[P])] \tag{C.20}$$

where the trace is over momentum and all other band labels.

To differentiate different ground states obtained in HF, we compute the Chern number and the HF gap. We regard any states with gap $\Delta_{\mathrm{HF}} < 5\,\mathrm{meV}$ as a metal. For insulators, we show their Chern number in the phase diagram. We note that AHC and WLI have identical symmetry and therefore cannot be probed individually within HF by fixing symmetry. Instead, we use 50 random seeds and a few physically motivated seeds to ensure a global minimum is found and the phase boundary between AHC and WLI is better resolved.

C.2.2 Hartree-Fock Convergence

We comment briefly on the stringent criteria we have used to ensure convergence of our SCHF numerics.

We use the optimal-damping algorithm (ODA) to converge to states satisfying the self-consistency condition

$$[P, h_{\mathrm{kin}} + h_H[P] + h_F[P]] = 0 \tag{C.21}$$

to tolerances approaching the square root of machine precision (i.e. machine precision in $E[P]$). We use both rectangular and Monkhorst-Pack grids with 24×24, 30×30, 36×36, or 48×48 unit cells, depending on the sensitivity of the state to finite-size effects. We impose C_3 symmetry explicitly in the Monkhorst-Pack case, but find comparable results in both cases for sufficiently large system sizes. We ensure that the range of momentum transfers $\boldsymbol{q}$ considered is sufficiently large to converge the energy of the state, out to several Brillouin zone distances. To avoid non-global minima, we initialize SCHF in a variety of states for each parameter point, often using a mix of physical ansatzes as well as random initial states. We also assume electrons are spin- and valley-polarized.

We now detail the convergence of our SCHF calculations as a function of the number of k-points and the number of single-particle bands. Fig. C.3(a) shows the ground state energy per electron within SCHF on a system of $N_k \times N_k$ unit cells with a Monkhorst-Pack lattice with the number of bands N_b=7. One can see that the energy is decreasing monotonically with the system size, and 1% relative accuracy is achieved by 24×24 unit cells. Furthermore, we note that the Berry curvature and energetic details of the isolated Chern band are qualitatively stable with system size (with the notable exception of 18×18 which is often qualitatively different, yielding dif-

ferent phase boundaries). We therefore conclude that 24×24 unit cells is sufficient to capture quantitative details of phase boundaries to reasonably good fidelity.

Our SCHF is carried out in a Hilbert space spanned by the projector into the lowest N_b conduction bands. We take $N_b = 7$ bands throughout the main text, a choice we now justify. At an initial qualitative level, we note that the conduction band wavefunctions come from folding the bands of rhombohedral graphene into the moiré Brillouin zone, then mixing them between $\boldsymbol{k}$ and $\boldsymbol{k} + \boldsymbol{g}$ under the moiré potential. We note that the first seven bands are largely supported on the first moiré Brillouin zone together with the first shell of six higher Brillouin zones. Indeed, the first seven bands span a range of kinetic energy much greater than the interaction scale. This suggests the first 7 bands span an adequately large Hilbert space to understand the physics at filling $\nu = 1$. We now investigate this claim quantitatively. Fig. C.3(b) show the SCHF energy as a function of N_b (with values chosen to match centered hexagon numbers, i.e. complete shells of Brillouin zones). One can see that the energy is converged to better than 0.5% relative error by $N_b = 7$. Furthermore, we investigate the participation of the higher bands in the Chern band that appears in SCHF at $\nu = 1$. Explicitly, we compute the participation $P_n = \frac{1}{\mathcal{N}} \left(\sum_{\boldsymbol{k}} | \langle \phi^{\mathrm{HF}}_{\boldsymbol{k},1} | \psi_{\boldsymbol{k},n} \rangle |^2 \right)$ of the single-particle bands in the wavefunction of the isolated Chern band $\phi^{\mathrm{HF}}_{\boldsymbol{k},1}$ at $\nu = 1$ with normalization so that $\sum_n |P_n|^2 = 1$. We find these overlaps decrease precipitously after $n = 3$ and, furthermore, we find the participations are essentially stable after $N_b = 7$. In conclusion, the SCHF ground state of our model is captured to good quantitative accuracy by 24×24 systems with $N_b = 7$.

As a final note, we emphasize that the errors from experimental and modelling uncertainties are likely to be vastly larger than numerical errors in this work.

C.2.3 Metal versus Anomalous Hall Crystal Competition

To examine the competition between the metallic "quarter metal" state and the anomalous Hall crystal shown in Fig. 1(b), we must determine the ground state energy both with and without continuous translation symmetry. To do this, we first compute the SCHF ground state of standalone rhombohedral graphene in a patch near the K point that is much bigger than the moiré Brillouin zone, finding a metallic state with enforced continuous translation symmetry. We then compute another SCHF ground state with a folded Brillouin zone, which gives the possibility of breaking continuous translation symmetry. No hBN moiré potential is added. At small u_D, the energies from both methods agree to extremely high precision and indeed describe the same quarter-metal state. At a critical u_D, however, the translation-breaking ansatz finds a ground state with an interaction induced gap

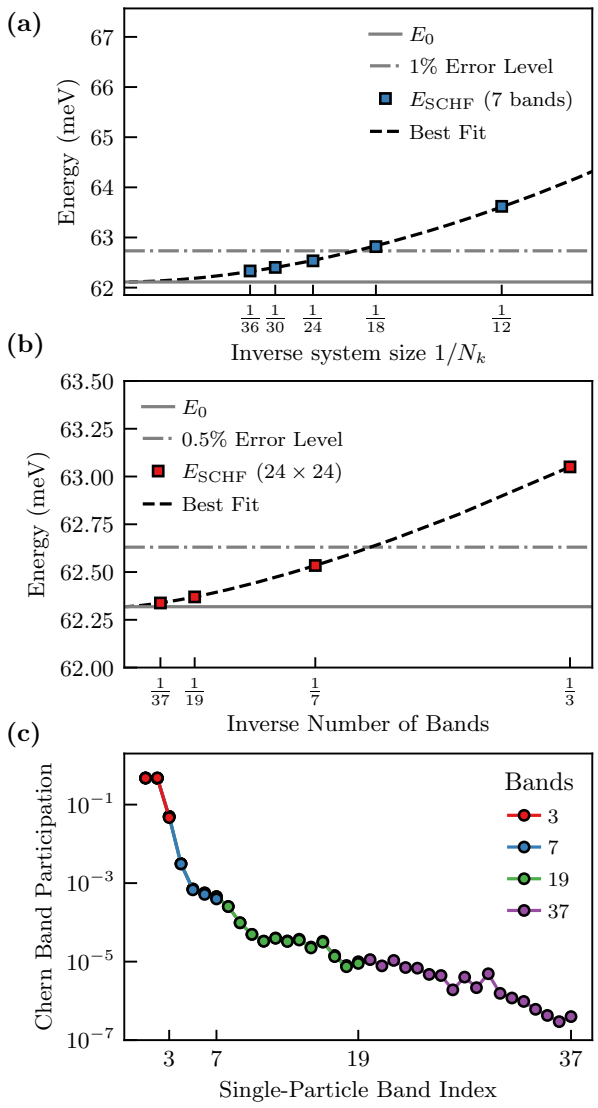


Figure C.3: Convergence details of SCHF computations. (a) SCHF Ground state energy convergence as a function of system size $N_k \times N_k$ with $N_b = 7$ bands. (b) SCHF ground state energy convergence as a function of the number of bands. (c) Participation of the single-particle bands in the isolated Chern band found within SCHF. Note that only the first three bands contribute significantly, and contributions are negligible after $N_b = 7$. Parameters: $(u_D, \theta) = (50\,\mathrm{meV}, 0.77°)$ unless specified.

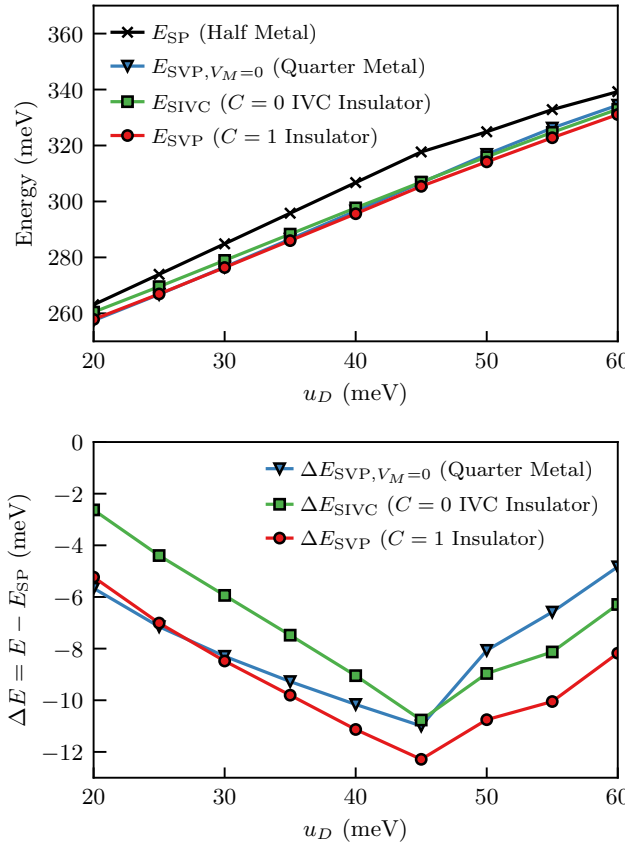


Figure C.4: Energy per electron of states with different flavor ordering patterns. (top panel) The energy per electron of four competing ground state candidate states. (bottom panel) The energy per electron of three competing candidate states relative to the $N_f = 2$ spin polarized state.

that is lower energy than the metal — the anomalous Hall crystal described in the main text.

C.2.4 Isospin symmetry breaking

A universal feature of RMG at high displacement field D and low filling ν is isospin symmetry breaking, regardless of the moiré potential. A plethora of evidence, including quantum oscillation experiments [259, 263, 264, 270] and SCHF numerics [287, 458], support a fully spin and valley polarized state around filling $n \sim 10^{11}\,\mathrm{cm}^{-2}$ (unity moiré filling at twist angle $\theta < 1°$). Therefore, we assume full spin and valley polarization in the SCHF calculation presented in the main text. In this section, we justify this choice by performing SCHF with both spin and valley degrees of freedom in the parameter regime where we have observed AHC.

If we neglect charge orders for the moment, different flavor orders can be classified by the isospin symmetry breaking pattern (Table C.1). We consider states in the following groups: (i) an $N_F = 2$ state, where two out

Group	flavor order	Symmetry		
		$SU(2)_s$	$\tilde{\mathcal{T}}$	$U(1)_v$
(i)	SP	×	✓	✓
	VP	✓	×	✓
	SVL	×	×	✓
(ii)	SVP	×	×	✓
(iii)	SIVC	×	✓	×

Table C.1: Symmetry of various magnetic orders, including the spin $SU(2)_s$ symmetry, spinless time-reversal $\tilde{\mathcal{T}}$ symmetry, and valley $U(1)_v$ symmetry.

of four possible spin-valley flavors are occupied evenly. This includes a spin polarized (SP) state, a valley polarized (VP) state, and a spin-valley locked (SVL) state. These are candidate states for experimentally observed "half metal" states. (ii) a spin-valley polarized (SVP) $N_F = 1$ state, where only one spin-valley species is occupied, and (iii) a spin polarized IVC $N_F = 1$ state (SIVC) that breaks $\mathrm{U}(1)_v$. The latter two groups include the "quarter metal" states seen in experiment.

Since the aforementioned states have different symmetries/quantum numbers Table C.1, we can enforce symmetries in the HF calculation and compare the energy among these states. On top of the magnetic order, we also allow translation symmetry breaking at moiré lattice vectors. Our Hamiltonian does not include any spin-orbit coupling or Hund's term, so the system has an enlarged $\mathrm{SU}(2)_K \times \mathrm{SU}(2)_{K'}$ symmetry. At HF level, candidate ground states within the group (i) are degenerate, whether it be SP, VP, or SVL. We therefore choose an SP state to represent the group (i).

In the top panel of Fig. C.4, we show the energy per electron of $N_f = 2$ SP states, $N_f = 1$ SIVC states, $N_f = 2$ SVP states. For comparison, we also show SVP metallic states that do not break translation symmetry computed with $V_M = 0$.

At $\nu = 1$, we always find the $N_F = 2$ SP state to have the highest energy, so we plot the energy of the other states relative to the SP state in the bottom panel of Fig. C.4. We note that the SP states are always metallic since they correspond to $\nu = 0.5$ per flavor. We find that the isospin symmetry breaking happens at an energy scale at least 5 times bigger than the formation of an insulating state within the SVP phase (see also Fig. 1(b)).

In terms of flavor order, The closest competing state of the SVP AHC is an SIVC AHC. We find that SVP always have lower energy than SIVC, likely due to the winding energy penalty of SIVC hybridizing two Chern

bands with opposite Chern number [193]. The energy splitting, however, is comparable to the energy splitting between the metallic state and the AHC state.

Finally, we make a few comments based on the SCHF data. In RMG, the energy splitting due to different mechanisms form the following hierarchy,

$$\Delta_{\text{Stoner}} \gg \Delta_{\text{AHC}} \approx \Delta_{\text{IVC}} \gg \Delta_{\text{moiré}} \tag{C.22}$$

The system first forms a flavor order due to Stoner ferromagnetism, and then interaction drives the 'quarter metal' to an insulator with finite Chern number, while disfavoring the IVC state by Δ_{IVC}. The moiré potential is not strong enough to alter the energy competition at $\nu = 1$.

C.2.5 Identification of moiré-proximate side from $\nu = 4$ SCHF

To confirm whether our setup of moiré-proximate/distant side corresponds to that in experiments, we identify the experiment observation of correlated insulator at $N_L = 5, \nu = 4$ [21]: At low displacement field $|D| < 0.5\,\text{V/nm}$, corresponding to $u_D \sim 30\,\text{meV}$, an insulator is found at $D < 0$ side up to $|D| = 0.4\,\text{V/nm}$, but not at $D > 0$. We now will show that this agrees with our numerical observation at similar u_D, establishing a matching between the two works.

We perform self-consistent Hartree-Fock (SCHF) with both spin and valley degrees of freedom. We only include the conduction band as in the main text, and restrict ourselves to $u_D \geq 15\,\text{meV}$ where the valence bands are sufficiently separated. At $\nu = 4$, we search for the ground state in the manifold without intervalley coupling, which excludes intervalley coherent (IVC) orders.

We examine the resulting charge gap Δ at different u_D, as shown in Fig. C.5, which shows a clear asymmetry between $u_D > 0$ and $u_D < 0$: a charge gap is only identified for $u_D < 0$ but not for $u_D > 0$, which agrees with the experiment described above. This suggests that the experiment identifies $\nu = 4$ insulator only on the moiré-proximate regime. Although there could be other competing orders unconsidered here, the identification of the gapless phase is expected to be robust even beyond Hartree-Fock approximations.

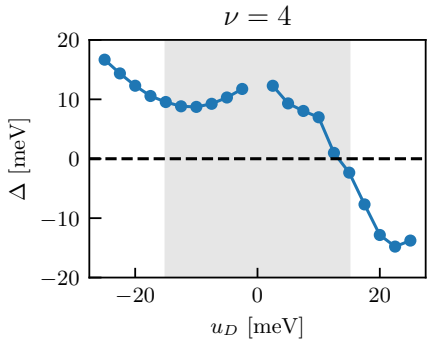


Figure C.5: Charge gap of SCHF ground state at $\nu = 4$ for $N_L = 5$. Model parameters follow Appendix C.2.1, but include both spins and valleys. $|u_D| < 15$ meV results are shaded since they are less reliable in SCHF. Parameters: $\theta = 0.77°$, 30×30 unit cells.

C.2.6 Details of N_L-u_D phase diagram

Fig. C.6 shows the extended N_L-u_D phase diagram in Fig. 1(c) which we extend up to $u_D = 160$ meV. Each N_L shows similar patterns as $N_L = 5$ discussed in Fig. 2(a). First, a broad regime of $C = -1$ AHC phase is stabilized at moderate u_D. The lower critical u_D, referred to as u_D^* in the main text, decreases with N_L, indicating the stronger relevance of AHC in higher-N_L rhombohedral graphenes. Meanwhile, the higher critical u_D is controlled by the development of an indirect band gap between $C = -1$ band and higher bands, which leads to a first-order transition into metal (and occasionally the "reentrance" of $C = -1$ state at $N_L = 4$.) At even higher u_D, other phases such as the $C = 0$ insulator take over, but given the stronger band mixing it is necessary to include more bands in SCHF to accurately resolve the phase competition. We therefore caution against trusting the phase diagram at large u_D.

We note that the charge gap, shown in Figs. C.6 and C.7, has sensitive and non-monotonic dependence on displacement u_D. However, it is not directly proportional to the moiré potential, shown in Fig. 3(a) of the main text. Here we also comment on the observable effects of the charge gap. In addition to the thermal activation behavior expected in transport measurements, the width of the plateau with quantized Hall conductance also depends on the charge gap [459, 460].. If we assume that impurities induce subgap localized states with a constant density of states, the Hall conductance will remain quantized until the doped charge fully fills all subgap states, whose total density is proportional to the charge gap. In Ref. [21], the plateau width of the $\nu = 1$ Chern insulator also behaves non-monotonically with the displacement field, possibly related to this behavior of the charge gap.

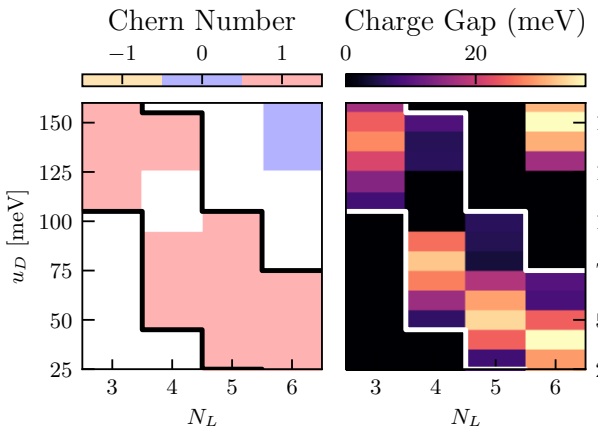


Figure C.6: (a) Chern-number and (b) charge-gap phase diagram at $\nu = 1$ for rhombohedral graphene with $N_L = 3 - 6$ and $u_D = 0 - 160\,\text{meV}$, with fixed $\theta = 0.77°$ and $\epsilon_r = 5$ at each N_L. The boundary of $C = -1$ AHC phase is highlighted. The white region in the Chern-number plot denotes metal with a direct or indirect band gap. Parameters: $\theta = 0.77°$, 36×36 unit cells.

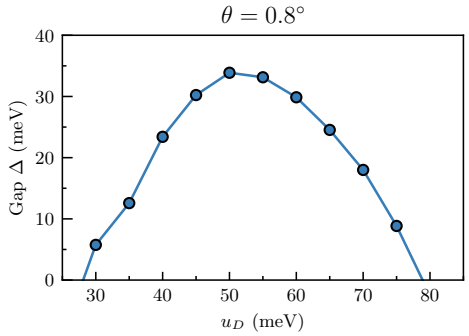


Figure C.7: Dependence of charge gap on u_D at $N_L = 5$, $\theta = 0.8°$ and $\epsilon_r = 5$. We take a 30×30 unit cell. A clear non-monotonic dependence of the charge gap on u_D is observed.

C.2.7 Effect of Graphene-Scale Parameters

This section considers the effect of the graphene-scale parameters on our results. An important phenomenon in graphene and its related systems is the renormalization of the Fermi velocity by Coulomb interactions, which has been observed experimentally [306, 394, 461]. Namely, the effective velocity (i.e. t_0) increases as the energy scale or filling approaches charge neutrality. When discussing phenomena such as those here that depend on both a small moiré scale and a much larger displacement field scale, it is unclear which Fermi velocity to employ. Estimates in the literature for $u_D = 0$ vary between $t_0 \approx 2.6\,\text{eV}$ and $t_0 \approx 3.1\,\text{eV}$, depending on the relevant energy scales, number of layers, and other details [259, 284]. Moreover, this problem is exacerbated by the fact that one must use a "bare" Fermi velocity in models to which interactions are added, but renormalization of the

velocity is a beyond-mean-field effect.

We now show changing the Fermi velocity dramatically shifts the phase boundaries but not the phases present. Fig. C.8 shows phase diagrams at $\nu = 1$ within SCHF for $t_0 = 2.6\,\mathrm{eV}$ and $3.1\,\mathrm{eV}$, respectively. The latter option is used in the main text and throughout the rest of this work. Given that the various ground states candidates are separated on the $O(1)\mathrm{meV}$ scale, this $500\,\mathrm{meV}$ perturbation has a remarkably small effect. Comparing Fig. C.8 (a) and (b), one sees that the smaller t_0 favors $C = 0$ over $C = 1$. This remarkable resilience of the Chern band to these parameter changes suggests a much smaller number of hidden parameters that control the phase diagram. Future work will explicate this phenomenon.

In addition, it turns out that even large differences in the graphene-scale parameters do not strongly affect our results at $\nu = 1$. Fig. C.9 demonstrates the robustness of AHC as the higher hopping terms are varied. Explicitly, we replace hopping terms $t_{2,3,4}$ in Eq. (F.2) with $\lambda t_{2,3,4}$, where $\lambda = 1$ correspond to realistic RMG and $\lambda = 0$ correspond to simplified RMG with only nearest-neighbor hoppings. We then compute its ground state at $\lambda \in (0, 1)$, which all result in AHC phases with $C = -1$. The charge gap of the ground state remains large through the tuning, suggesting the stability of the phase. The bandwidth of the $C = -1$ band gradually increases at lower λ, which indicates its deviation from the optimal condition realizing FQAH.

C.3 Details of the Many-Body Numerics at Fractional Filling

To determine the many-body ground state at fractional filling, we undertake a two-step procedure: (1) solve for the SCHF ground state at filling $\nu = 1$ and then (2) solve the many-body problem restricted to the lowest SCHF band (which is spin and valley-polarized) at fractional filling using either exact-diagonalization or the density-matrix renormalization group numerics.

An explicit assumption of such numerics is spin- and valley-polarization within the whole range $0 \leq \nu \leq 1$. When should this procedure be valid? It is first helpful to review the experimental phenomena. Ref [21] reported Laughlin and Jain sequence states at fillings $\nu = 2/5, 3/7, 4/9, 4/7, 3/5$ as well as $\nu = 2/3$. Crucially, they also reported that the region of large Hall conductance only extends down to $\nu = 0.2 - 0.4$ (depending on u_D), whereupon it transitions to a "speckled" region that may be due to Wigner crystallization. Our assumption that the many-body ground state is within the many-body Hilbert space of the $\nu = 1$ Chern band is clearly violated at this point, rendering our model inapplicable for sufficiently low fillings. It is also helpful to note the simpler case of the lowest Landau level, where the fractional state at $\nu = 2/3$ can be spin unpolarized [131, 291, 292] despite

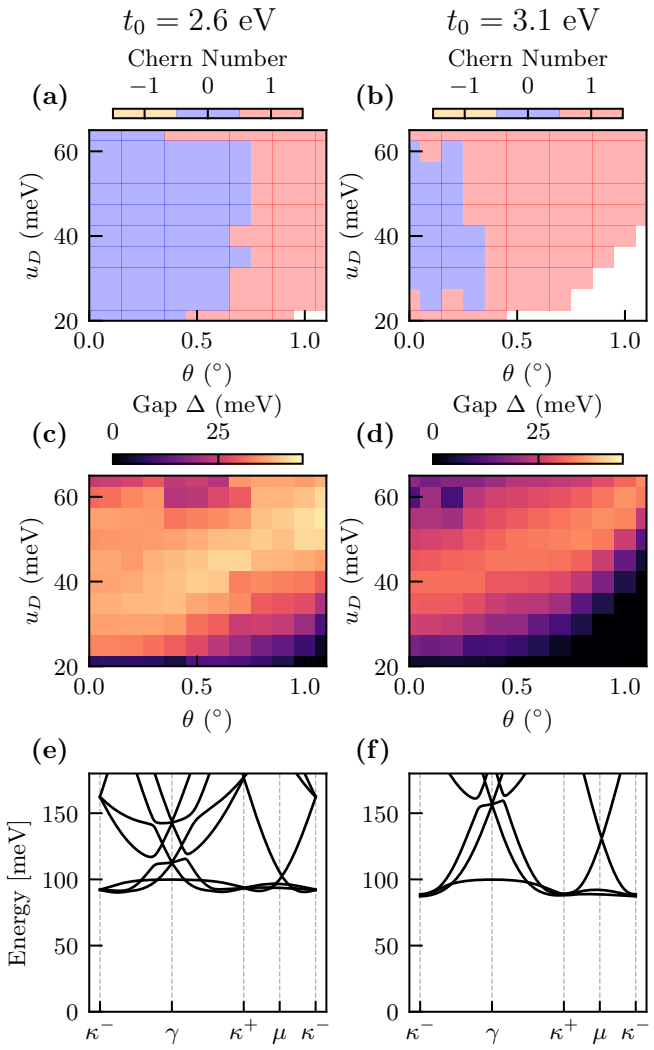


Figure C.8: The effect of large versus small Fermi velocity is minimal. (a,b) SCHF phase diagrams at $\nu = 1$ as a function of twist angle θ and vertical displacement field u_D, showing the difference between large Fermi velocity, corresponding to $t_0 = 3100\,\text{meV}$, versus small Fermi velocity with $t_0 = 2600\,\text{meV}$. Here colors show the Chern number of insulators, and gray data points have gaps of 2 meV or less. (c,d) Charge gaps above $\nu = 1$ within SCHF. (e,f) Single-particle ($\nu = 0$) bandstructures at large and small Fermi velocity. Note that the gap to the second band at γ increases dramatically at large t_0.

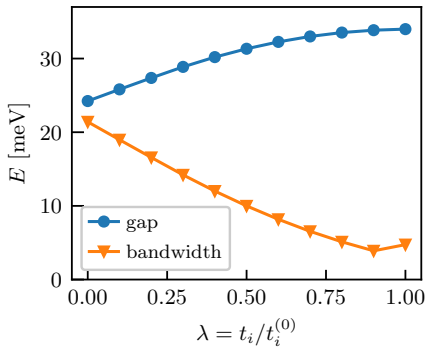


Figure C.9: Charge gap and bandwidth of RMG ground states with modified hopping strength $\lambda t_{2,3,4}$, $\lambda \in (0, 1)$. All ground states are found to be AHC phase with $C = -1$. Parameters: $(u_D, \theta) = (50\,\text{meV}, 0.77°)$, 30×30 unit cells.

a spin-polarized LL being favored at $\nu = 1$.

It is unclear how to perform the corresponding computation without spin polarization, as it is unclear what the 2-band low-energy (i.e. 1 band with spin) Hilbert space should be considered. One *cannot* simply add spin to the $\nu = 1$ Chern band; instead one must restrict to a spinful Hilbert space from the full Hilbert space. Determining such a minimal model may require a four-flavor many-body calculation with many-band and is beyond the scope of the current work. Nevertheless, we expect that spin- and valley-polarized states are at least low-energy candidate ground states through most of the range. We therefore work within the one-band model to examine the presence of Jain sequence states in the simplest scenario.

Explicitly, this is done as follows. Consider the SCHF Hamiltonian $h_{HF}[P]$ at $\nu = 1$, which may be diagonalized as

$$h_{HF}[P]_{\alpha\beta}(\boldsymbol{k})U_{\beta\gamma}(\boldsymbol{k}) = U_{\alpha\gamma}(\boldsymbol{k})E_{\gamma}(\boldsymbol{k}). \tag{C.23}$$

We may then pass to the basis of Hartree-Fock bands

$$|\phi_{\boldsymbol{k}\alpha}\rangle = |\psi_{\boldsymbol{k}\beta}\rangle\, U_{\beta\alpha}(\boldsymbol{k}), \tag{C.24}$$

which constitutes a unitary rotation of our set of N_b low-energy bands. We may then integrate out the $N_b - 1$ bands above the $\nu = 1$ gap at mean-field level — which again is simply an index restriction — to arrive at a one-band Hamiltonian

$$\hat{H} = \sum_{\boldsymbol{k}} t_{\boldsymbol{k}}\hat{n}_{\boldsymbol{k}} + \frac{1}{2A}\sum_{\boldsymbol{q},\boldsymbol{k},\boldsymbol{k}'} U_{\boldsymbol{q}}\lambda_{\boldsymbol{q}}(\boldsymbol{k})\lambda_{-\boldsymbol{q}}(\boldsymbol{k}')\hat{d}^{\dagger}_{\boldsymbol{k}}\hat{d}^{\dagger}_{\boldsymbol{k}'}\hat{d}_{\boldsymbol{k}'-\boldsymbol{q}}\hat{d}_{\boldsymbol{k}+\boldsymbol{q}}, \tag{C.25}$$

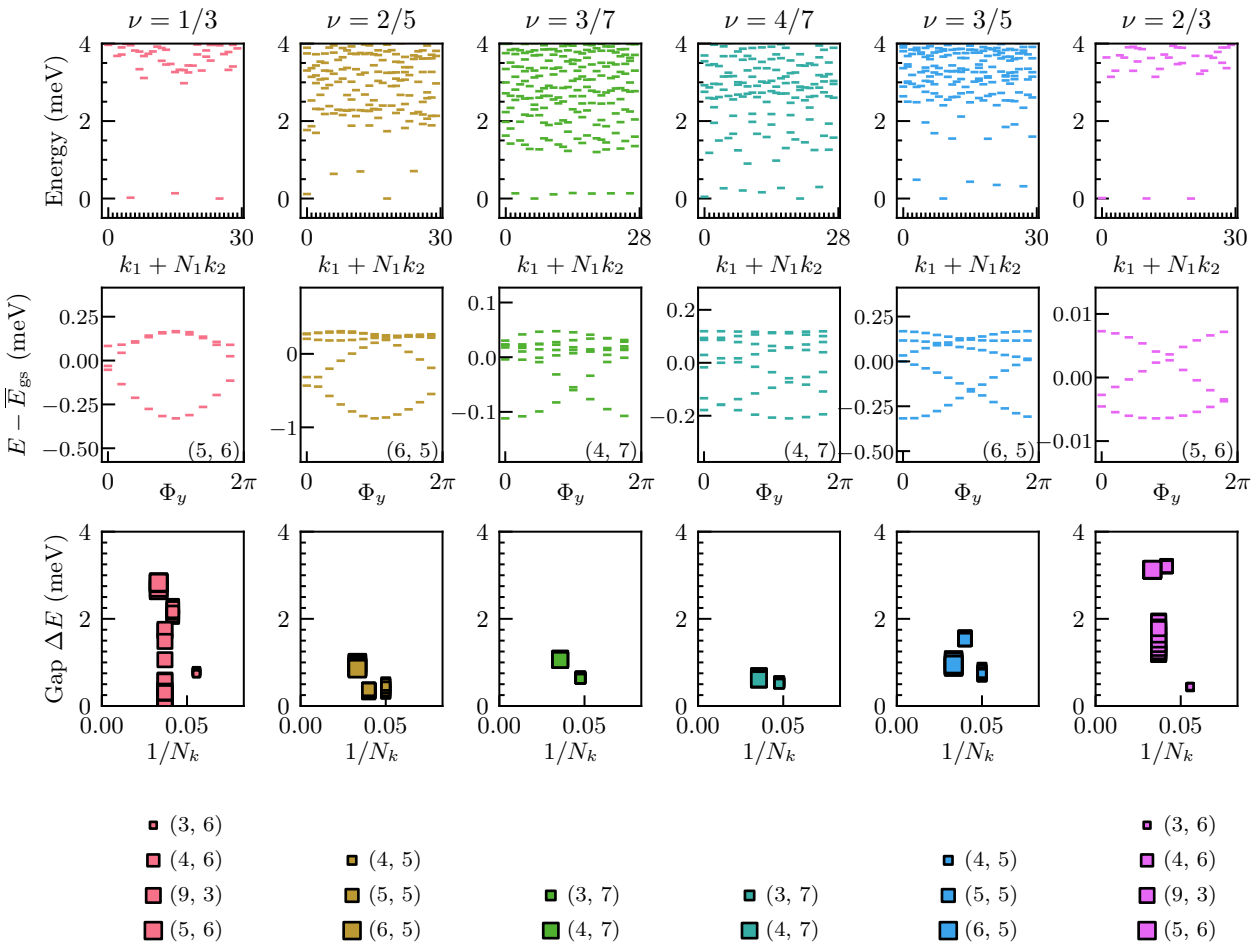


Figure C.10: Exact diagonalization at various fractional fillings over the lowest $\nu = 1$ band of RMG. Top row: spectra at the largest system sizes available at each filling. Second row: spectral flow under flux threading. Third row: system size dependence, showing the gap between the quasidegenerate ground states and the first excited state for all fluxes at each available system size. Details in text. Parameters: $\theta = 0.77°$, $u_D = 50\,\text{meV}$, other parameters match Fig. 2 of the main text

where $\hat{d}_{\boldsymbol{k}}|0\rangle = |\phi_{\boldsymbol{k}0}\rangle$ is a creation operator for the lowest SCHF band, a band with form factors $\lambda_{\boldsymbol{q}}(\boldsymbol{k}) = \langle\phi_{\boldsymbol{k}0}|e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}|\phi_{\boldsymbol{k}+\boldsymbol{q}0}\rangle$, and $\hat{n}_{\boldsymbol{k}} = \hat{d}^\dagger_{\boldsymbol{k}}\hat{d}_{\boldsymbol{k}}$. Note that while we have projected into the Hilbert space of the first SCHF band, the kinetic energy of the band is determined relative to $\hat{H}_{\rm RMG}$ at $\nu = 0$ to avoid double-counting interactions.

C.3.1 Exact Diagonalization at Fractional Filling

To perform exact diagonalization, we compute Eq. (C.25) on a momentum space Monkhorst-Pack grid $\boldsymbol{k}[n_1, n_2]$ of size $N_{k_1} \times N_{k_2}$. We then subsample that grid on points

$$\{\boldsymbol{k}[s_1 m_1 + \phi_1, s_2 m_2] \ : \ 0 \leq m_1 < M_1, 0 \leq m_2 < M_2\}\,, \tag{C.26}$$

to produce a grid of size $M_1 \times M_2$ where the integer ratios s_i obey $s_1 M_1 = N_{k_1}$ and $s_2 M_2 = N_{k_2}$. Here $0 \leq \phi_1 < s_1$ is an offset in the subsampling to enable flux threading. For the main text, we take $N_{k_1} \times N_{k_2} = 36 \times 36$ and $M_1 \times M_2 = 4 \times 6$.

Exact diagonalization was performed at a range of fractional fillings to determine their suitability for fractional quantum Hall states: $\nu = 1/3, 2/5, 3/5, 4/7, 3/5, 2/3$. At each filling we computed spectrum over a range of system sizes and performed flux threading along k_x for the largest system size available, with $N_k = N_{k_1} \times N_{k_2}$ up to 30. The results are shown in Fig. C.10.

The top row of Fig. C.10 shows the low-lying spectrum at all center of mass momenta for each system size. In each case, we observe a quasi-degenerate ground states with degeneracy q at filling $\nu = p/q$, with a gap of 1 meV to 3 meV, depending on filling. This suggests that the ground states here are all fractional Chern insulators of Laughlin and Jain type.

To support this inference, the second row of Fig. C.10 shows the spectral flow under flux threading for each filling. To make the permutation of states clear, we plot energies $E - \overline{E}_{\text{gs}}$, where $\overline{E}_{\text{gs}}(\Phi_y)$ is the average energy of the quasidegerate ground states at each flux Φ_y. For instance $\nu = 2/3$, the spectral flow clearly permutes the ground states on a 5×6 site cluster (see main text on 4×6 sites). Indeed, we see a permutation of the quasidegenerate ground states for each filling for at least some system sizes. Explicitly, we see such a spectral flow for 5×6 at $\nu = 1/3$, but the 9×3 cluster is gapless at the same filling. This flux pumping evidence supports the inference that FCIs of Laughlin and Jain types are ground state competitors at each filling we consider. However, they may not be the true ground state at all fillings in the thermodynamic limits.

Finally, the bottom row of Fig. C.10 shows the system size dependence of the gaps. For each system size, we show the gap between the top quasidegenerate ground state and the next state in the spectrum at the same filling with any momentum. Each box represents that gap at one value of the flux, and the sizes of the boxes increase with system size. We note that the system size dependence is highly non-monotonic and far from the scaling

regime, suggesting that finite size effects dominate the results — as expected for a system with such concentrated Berry curvature pockets and other quantum geometric features. Of particular note is the system sizes 9×3 at $\nu = 1/3$ filling, where we observe gap closings under spectral flow. This suggests a qualitatively different state is stabilize at these oblong aspect ratios, whose energy per electron is slightly less than the competing FCI state (at these small system sizes).

Overall we may conclude that FCIs are low-energy ground state *candidates* at all fillings $\nu = 1/3, 2/5, 3/7, 4/7, 3/5, 2/3$. However, the many-body gaps for these states are small at $\nu = 2/5$, and can vanish entirely at $\nu = 1/3$, depending on the cluster considered. Moreover, the gaps have not stabilized on even the largest sizes we could consider, and we find small changes in parameters such as u_D often favor CDW orders or Fermi liquids. Finally, modelling uncertainty greatly exceeds finite-size effects here. The balance of the evidence supports the following conclusions: filling $1/3$ is inconclusive; $2/5$ and $4/7$ are likely FCIs with small gaps, and $3/7$,$3/5$, $2/3$ have FCI ground states within our model.

C.3.2 DMRG at Fractional Filling

To apply the density-matrix renormalization group, we follow previous work on moiré DMRG systems [71, 206, 452, 453] to place the system onto an infinite cylinder geometry. We start with a rectangular moiré Brillouin zone, shown in Fig. C.1, with side lengths $B_1 = |\boldsymbol{g}_1|$, $B_2 = \frac{\sqrt{3}}{2}|\boldsymbol{g}_2|$. We impose periodic boundary conditions in the "2" direction corresponding to momentum discretization

$$k_2[n] = B_2 \times \begin{cases} -\frac{1}{2} + \frac{n+\frac{\Phi_2}{2\pi}}{N_2} & \text{if } N_2 \equiv 0 \pmod 2 \\ -\frac{1}{2} + \frac{n+\frac{\Phi_2}{2\pi}+\frac{1}{2}}{N_2} & \text{if } N_2 \equiv 1 \pmod 2 \end{cases}, \tag{C.27}$$

relative to an origin at moiré-γ. This gives N_2 evenly spaced "wires" across the Brillouin zone, which intersect the γ point for flux $\Phi_2 = 0$.

We then take a basis of hybrid Wannier functions along each wire, i.e. maximally localized along $\boldsymbol{a}_1$ and periodic along $\boldsymbol{a}_2$:

$$|w_{n,k_2}\rangle = \frac{1}{\sqrt{N_1}} \sum_{\boldsymbol{k}_1} |\varphi_{\boldsymbol{k}}\rangle \, e^{-in\boldsymbol{k}\cdot\boldsymbol{a}_1}; \quad |\varphi_{\boldsymbol{k}}\rangle = |\phi_{\boldsymbol{k}0}\rangle \, e^{i\zeta(\boldsymbol{k})}. \tag{C.28}$$

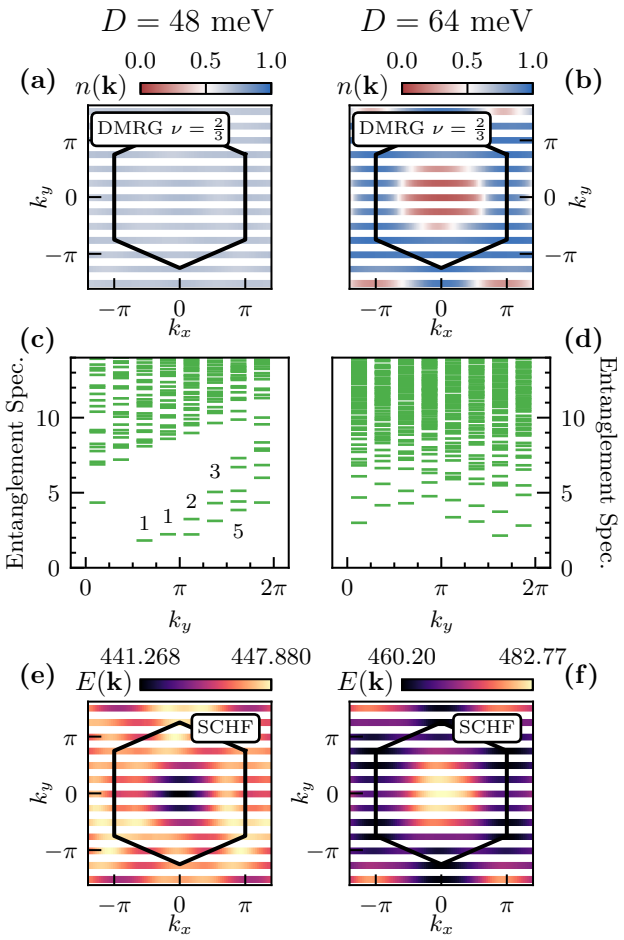


Figure C.11: DMRG results at $u_D = 48$ meV (left) and 64 meV (right). Here (a,b) give occupations over the Brillouin zone, (c,d) give entanglement spectra, and (e,f) show the dispersion of the corresponding lowest SCHF Chern band at $\nu = 1$. See text for details.

Here the gauge choice $\zeta(\boldsymbol{k})$ is chosen such that

$$\hat{T}_1 \left|w^{n,k_2}\right\rangle = \left|w^{n-1,k_2}\right\rangle \tag{C.29a}$$

$$\hat{T}_2 \left|w^{n,k_2}\right\rangle = e^{i2\pi k_2/B_2} \left|w^{n,k_2}\right\rangle \tag{C.29b}$$

$$\hat{P} e^{-i\hat{r}_1 B_1} \hat{P} \left|w^{n,k_2}\right\rangle = e^{i2\pi P_1(k_2)} \left|w^{n,k_2}\right\rangle \tag{C.29c}$$

where the polarizations $P_1(k_2) = \frac{B_1}{2\pi} \langle w^{0,k_2} | \hat{r}_1 | w^{0,k_2} \rangle$ are the centers of the Wannier orbitals in the first unit cell, in accordance with the modern theory of polarization [204]. The Hamiltonian (C.25) can be straightforwardly expressed in terms of fermion operators $\hat{f}^\dagger_{n,k_2} \left|0\right\rangle = \left|w_{n,k_2}\right\rangle$ on an infinite cylinder cylinder geometry. As this is a long-ranged, two-dimensional Hamiltonian, working with it efficiently as a matrix product operator (MPO) requires the technique of "MPO Compression" developed by some of us [205, 206], which creates a faithful low-rank approximation of the Hamiltonian with errors below the 10^{-2} meV level.

We then apply standard DMRG using the open-source tenpy [202] library. We work at cylinder circumference $L_y = 8$. For FCI states, we converge the ground state to norm error below 10^{-13} on bond dimensions up to $\chi = 1536$ — since the state is fully gapped, only moderate bond dimensions are required To diagnose the fractional Chern insulator we use the entanglement spectrum technique first proposed by [289], finding the characteristic degeneracy counting matching partition numbers $1, 1, 2, 3, 5, 7, \ldots$ for each bond. We have also verified this holds for $L_y = 5, 6$.

Fig. C.11 compares the results of DMRG at two difference displacement fields, $u_D = 48\,\text{meV}$ and $u_D = 64\,\text{meV}$. The first is selected as the location where the FCI gap is maximized in ED. The second is in a regime where the $\nu = 1$ state is still a Chern band, but whose bandwidth is now $\sim 20\,\text{meV}$ and trace condition violation is ≈ 1. Both systems are at $\theta = 0.77°$ and all other parameters are identical.

Fig C.11(a,b) shows the occputaion numbers $n(\boldsymbol{k}) = \langle \hat{c}^\dagger_{\boldsymbol{k}} \hat{c}_{\boldsymbol{k}} \rangle$ across the Brillouin zone for both displacement fields. For $u_D = 48\,\text{meV}$, $n(\boldsymbol{k}) = \frac{2}{3} \pm 0.04$ over the whole Brillouin zone, whereas for $u_D = 64\,\text{meV}$ the occcupations $n(\boldsymbol{k})$ vary from 0.08 to 0.95 at $\chi = 1536$. Moreover, the occupations are almost entirely depleted near γ, precisely where the SCHF bandstructure, Fig C.11(f), is peaked. We further check that the charge 0 and charge 1 correlation lengths are increasing with bond dimension, which gives further confirmation of gapless modes in the these channels as expected from a Fermi liquid. The entanglement spectrum Fig C.11(d) does not have a clear low-lying branch separated from the rest of the spectrum, and is clearly not chiral with the expected

CFT degeneracies. The large displacement field system at $u_D = 64\,\mathrm{meV}$ is therefore a Fermi liquid. However, we caution that this Fermi liquid from doping the $\nu = 1$ Chern band may not correspond to the experimentally-observed Fermi liquid at larger displacement fields.

D

Supplemental Material for Chapter 4

D.1 Derivation of Hartree and Fock energies for RMG spinors

We evaluate the Hartree and Fock energies Eqs. (4.20,4.21) explicitly. Recall that the spinor is given by

$$\chi_{n+1}^{(N_L)} \coloneqq \chi^{(N_L)}(\boldsymbol{\kappa}_{n+1}) = \mathcal{N}(1, \eta e^{2\pi i n/6}, \eta^2 e^{4\pi i n/6}, \ldots, \eta^{N_L-1} e^{2\pi i (N_L-1) n/6}), \tag{D.1}$$

where $\mathcal{N} = 1/\sqrt{\sum_{a=0}^{N_L-1} \eta^{2a}}$.

The form factor is defined by $\Lambda_{n,m} = \chi_n^\dagger \chi_m$. Then,

$$\Lambda_{3,1} = \Lambda_{6,4} = \mathcal{N}^2 \sum_{n=0}^{N_L-1} \eta^{2n} e^{-2\pi i n/3}, \tag{D.2}$$

$$\Lambda_{1,2} = \Lambda_{4,5} = \mathcal{N}^2 \sum_{n=0}^{N_L-1} \eta^{2n} e^{2\pi i n/6}. \tag{D.3}$$

Therefore, we get

$$\epsilon_H(C) = \frac{2}{3}\mathrm{Re}[e^{2\pi iC/3}\Lambda_{1,3}\Lambda_{4,6}] \tag{D.4}$$

$$= \frac{2}{3}\mathrm{Re}\left[e^{2\pi iC/3}\left(\mathcal{N}^2\sum_{n=0}^{N_L-1}\eta^{2n}e^{2\pi in/3}\right)^2\right] \tag{D.5}$$

$$= \frac{2}{3}\mathcal{N}^4\sum_{a,b=0}^{N_L-1}\eta^{2(a+b)}\mathrm{Re}[e^{2\pi iC/3}e^{i2(a+b)\phi}] \tag{D.6}$$

$$= \frac{2}{3}\mathcal{N}^4\sum_{a,b=0}^{N_L-1}\eta^{2(a+b)}\cos\left(\frac{2\pi(a+b+C)}{3}\right) \tag{D.7}$$

Similarly, we get for Fock

$$\epsilon_F(C) = -\frac{2}{3}\mathrm{Re}[e^{-2\pi iC/3}\Lambda_{1,2}\Lambda_{4,5}] \tag{D.8}$$

$$= -\frac{2}{3}\mathrm{Re}\left[e^{-2\pi iC/3}\left(\mathcal{N}^2\sum_{n=0}^{N_L-1}\eta^{2n}e^{2\pi in/3}\right)^2\right] \tag{D.9}$$

$$= -\frac{2}{3}\mathcal{N}^4\sum_{a,b=0}^{N_L-1}\eta^{2(a+b)}\mathrm{Re}[e^{-2\pi iC/3}e^{2\pi i(a+b)/3}] \tag{D.10}$$

$$= -\frac{2}{3}\mathcal{N}^4\sum_{a,b=0}^{N_L-1}\eta^{2(a+b)}\cos\left(\frac{2\pi(a+b-2C)}{6}\right) \tag{D.11}$$

D.2 Proof of degeneracy under contact interaction

We will now prove the curious fact that if κ and κ' spinors only have two components (with angular momentum 0 and 1, respectively), then the Chern number 0 and 1 states will remain degenerate under contact interaction.

We now define

$$\chi(\boldsymbol{\kappa}_1) = \begin{pmatrix} a_\kappa \\ b_\kappa \end{pmatrix}, \quad \chi(\boldsymbol{\kappa}_4) = \begin{pmatrix} a_{\kappa'} \\ b_{\kappa'} \end{pmatrix}. \tag{D.12}$$

The C_3 action is given again by $\hat{C}_3 = \mathrm{diag}(1, e^{2\pi i/3})$. We fix the gauge such that $\chi(\boldsymbol{\kappa}_{j+2}) = \hat{C}_3\chi(\boldsymbol{\kappa}_j)$.

Making use of Eqs. (4.15,4.16), we obtain

$$\begin{aligned}\epsilon_H(C) &= \frac{2}{3}\mathrm{Re}\left[e^{2\pi iC/3}\Lambda_{1,3}\Lambda_{4,6}\right] \\ &= \frac{2}{3}\mathrm{Re}\left[e^{2\pi iC/3}(|a_\kappa|^2+e^{2\pi i/3}|b_\kappa|^2)(|a_{\kappa'}|^2+e^{2\pi i/3}|b_{\kappa'}|^2)\right] \\ &= \frac{2}{3}\left(|a_\kappa|^2|a_{\kappa'}|^2\cos(2\pi C/3)+(|a_\kappa|^2|b_{\kappa'}|^2+|b_\kappa|^2|a_{\kappa'}|^2)\cos(2\pi(C+1)/3)+|b_\kappa|^2|b_{\kappa'}|^2\cos(2\pi(C+2)/3)\right), \end{aligned} \tag{D.13}$$

$$\begin{aligned}\epsilon_F(C) &= -\frac{2}{3}\mathrm{Re}\left[e^{-2\pi iC/3}\Lambda_{1,2}\Lambda_{4,5}\right] \\ &= -\frac{2}{3}\mathrm{Re}\left[e^{-2\pi iC/3}(\overline{a_\kappa}a_{\kappa'}+e^{-2\pi i/3}\overline{b_\kappa}b_{\kappa'})(\overline{a_{\kappa'}}a_\kappa+e^{-2\pi i/3}\overline{b_{\kappa'}}b_\kappa)\right] \\ &= -\frac{2}{3}\left(|a_\kappa|^2|a_{\kappa'}|^2\cos(2\pi C/3)+(\overline{a_{\kappa'}}a_\kappa\overline{b_\kappa}b_{\kappa'}+\overline{a_\kappa}a_{\kappa'}\overline{b_{\kappa'}}b_\kappa)\cos(2\pi(C+1)/3)+|b_\kappa|^2|b_{\kappa'}|^2\cos(2\pi(C+2)/3)\right) \end{aligned} \tag{D.14}$$

In contact interactions, we can take $V_{\boldsymbol{g}} = V_\kappa = 1$. Thus

$$E(C) = \epsilon_H(C)+\epsilon_F(C) = \frac{2}{3}\cos\left(\frac{2\pi(C+1)}{3}\right)|\overline{a_\kappa}a_{\kappa'}-\overline{b_\kappa}b_{\kappa'}|^2 \tag{D.15}$$

which means that $E(0) = E(1)$. We note that contact interaction annihilates all intra-orbital interactions, and one can generally prove that the $C = 0$ and $C = 1$ states are still degenerate under any inter-orbital interaction.

D.3 Real space properties of orbitals at κ and κ'

In this appendix, we will explain two elementary characteristics of C_3 symmetric orbitals at κ and κ' points. Recall that Wyckoff positions are points in the unit cell that are invariant under a spatial symmetry action. With the current choice of the lattice, these points are integer multiples of

$$\tilde{\boldsymbol{a}} = \frac{1}{3}(\boldsymbol{a}_1+\boldsymbol{a}_2). \tag{D.16}$$

Below we will establish the following two facts:

1. The orbital-resolved charge densities of C_3 symmetric states at κ and κ' points have peaks and zeros at the Wyckoff points, determined by both the orbital angular momentum ℓ_a and the state angular momentum ℓ_ξ;

2. The action of the translation $T_{\tilde{\boldsymbol{a}}}$ translates $\ell_\kappa \to \ell_\kappa+1$, and $\ell_{\kappa'} \to \ell_{\kappa'}-1$.

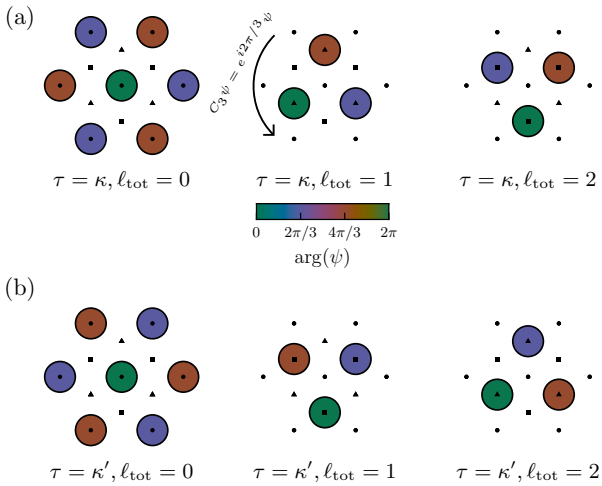


Figure D.1: Real-space plot of high-symmetry point wavefunctions $\phi_{\tau,\ell_{\rm tot}}(\boldsymbol{r})$. The size of the circles corresponds to the amplitudes of the wavefunctions at the Wyckoff points, and the color corresponds to the phase of the wavefunctions. The rows correspond to different high symmetry points (a) κ and (b) κ', and the columns correspond to different angular momenta $\ell_{\rm tot}$. Different wavefunctions at the same high-symmetry point are related by translations.

To understand the first statement, take the $\boldsymbol{\kappa}$ point without loss of generality. Now, consider the state $\psi_{\kappa,\ell_\kappa}(\boldsymbol{r})$: its orbital-resolved charge density is given by

$$\rho^a_\kappa(\boldsymbol{r},\ell_\kappa) = \left|\psi^a_{\kappa,\ell_\kappa}(\boldsymbol{r})\right|^2. \tag{D.17}$$

We now make use of the definition Eq. (4.5). Consider the momentum point $\boldsymbol{\kappa}_1$: Let's say $\chi^a(\boldsymbol{\kappa}_1) = \chi_0$. Then, due to the gauge choice $\chi(\boldsymbol{\kappa}_3) = \hat{C}_3\chi(\boldsymbol{\kappa}_1)$, we have $\chi^a(\boldsymbol{\kappa}_3) = \exp[2\pi i\ell_a/3]\chi_0$, and similarly $\chi^a(\boldsymbol{\kappa}_5) = \exp[4\pi i\ell_a/3]\chi_0$.

We may now compute the angular-momentum resolved charge density:

$$\begin{aligned}
\rho^{\ell_a}_\kappa(\boldsymbol{r},\ell_\kappa) &= \left|\psi^a_{\kappa,\ell_\kappa}(\boldsymbol{r})\right|^2 \\
&= \frac{1}{3}\left|\sum_{j=1,3,5}\exp\left[\frac{2\pi i\ell_\kappa}{3}\frac{j}{2}\right]e^{i\boldsymbol{\kappa}_j\cdot\boldsymbol{r}}\chi(\boldsymbol{\kappa}_j)^a\right|^2 \\
&= \frac{|\chi(\boldsymbol{\kappa}_1)^a|^2}{3}\left|\sum_{j=1,3,5}\exp\left[\frac{2\pi i(\ell_a+\ell_\kappa)}{3}\frac{j}{2}\right]e^{i\boldsymbol{\kappa}_j\cdot\boldsymbol{r}}\right| \\
&= |\chi(\boldsymbol{\kappa}_1)^a|^2\,|\varphi_{\kappa,\ell_\kappa+\ell_a}(\boldsymbol{r})|^2
\end{aligned} \tag{D.18}$$

Since $\boldsymbol{\kappa}_j\cdot\tilde{\boldsymbol{a}} = 2\pi j/3$ for $j\in 1,3,5$ and $-2\pi j/3$ for $j=2,4,6$ (modulo 2π), we may immediately deduce

$$\rho^a_\kappa(m\tilde{\boldsymbol{a}}, \ell_\kappa) = 3\left|\chi(\boldsymbol{\kappa}_1)^a\right|^2 \delta_{m,\ell_a+\ell_\kappa}. \tag{D.19}$$

where the $\delta_{m,n} = 1$ if $3|(m-n)$ and 0 otherwise.

This motivates a definition of

$$\ell_{\text{tot}} = \ell_a + \ell_\kappa. \tag{D.20}$$

Indeed, a similar calculation for the κ' shows that

$$\rho^a_\tau(m\tilde{\boldsymbol{a}}, \ell_\tau) \propto \delta_{m,\tau\ell_{\text{tot}}}. \tag{D.21}$$

where $\tau = \pm$ labels the κ and κ' points.

This establishes the first fact: the peak of the charge density is at a Wyckoff position determined by a signed total angular momentum, $\tau(\ell_a + \ell_\tau)$. Particularly, one can see in Fig. D.1 that although the charge density peaks of $\ell_{\text{tot}} = 1$ at κ point and $\ell_{\text{tot}} = -1$ at κ' point share the same Wyckoff positions, the wavefunctions have different phase structures and thus different angular momenta.

As for the second fact, we show it explicitly:

$$\begin{aligned}
(T_{\tilde{\boldsymbol{a}}}\psi_\kappa)(\boldsymbol{r}, \ell_\kappa) &= \psi_\kappa(\boldsymbol{r} - \tilde{\boldsymbol{a}}, \ell_\kappa) \\
&= \frac{1}{\sqrt{3}} \sum_{j=1,3,5} \exp\left[\frac{2\pi i \ell_\kappa}{3}\frac{j}{2}\right] e^{i\boldsymbol{\kappa}_j\cdot(\boldsymbol{r}-\tilde{\boldsymbol{a}})}\chi(\boldsymbol{\kappa}_j) \\
&\propto \frac{1}{\sqrt{3}} \sum_{j=1,3,5} \exp\left[\frac{2\pi i (\ell_\kappa+1)}{3}\frac{j}{2}\right] e^{i\boldsymbol{\kappa}_j\cdot\boldsymbol{r}}\chi(\boldsymbol{\kappa}_j) \\
&= \psi_\kappa(\boldsymbol{r}, \ell_\kappa + 1).
\end{aligned} \tag{D.22}$$

The κ' point calculation is exactly similar. Both of these statements can be checked graphically in Fig. D.1.

D.4 Justification of the three-patch model

In this appendix, we will motivate the building blocks of the three-patch model from simple physical conditions of the single-particle band structure. We will also discuss how we arrive at the restricted Hilbert space in Eq. (4.5) and below.

Spontaneous translation symmetry breaking can occur in the case that the interaction strength U is much

larger than the bandwidth of the lowest band within the mBZ:

$$|E(\boldsymbol{\gamma}) - E(\boldsymbol{\kappa}_i)| \lesssim U \tag{D.23}$$

In the limit where the lowest band is completely flat, the Fermi surface can be arbitrarily deformed without being penalized by the kinetic energy. In this way, we can create low-energy Fermi surfaces with perfect nesting conditions, which are in general unstable to the creation of insulating states by the exchange interaction. This thus corresponds to condition (i) stated at the beginning of Sec. 4.2. Once we take the dispersion into account, a gap equation must be solved to investigate the existence of such instabilities, which is performed under various approximations in Sec. 4.3.6.

The "weak coupling mean field" assumption has two subparts. It firstly states that the state is weakly coupled. This can also be implemented by an energetic constraint: we need a large gap between the $\boldsymbol{\gamma}$ point and its higher harmonics, which means that there will be a large direct gap around the $\boldsymbol{\gamma}$ point in the folded band structure. In the case where

$$E(\boldsymbol{\gamma} + \boldsymbol{g}_i) - E(\boldsymbol{\gamma}) > U, \tag{D.24}$$

interactions cannot hybridize the $\boldsymbol{\gamma}$ point wavefunction with other plane waves, and thus the state will be weakly coupled. It also states that the state can be described as a Slater determinant state. While this mean-field approximation is uncontrolled, as long as we consider a variational space of Hartree-Fock states, this assumption is always satisfied.

The three-patch assumption is harder to justify *a priori* from the knowledge of the single-particle Hamiltonian $\hat{h}$ alone. Physically, the Fock interaction is strong between nearby momentum given the small momentum exchange, and thus the single particle wavefunctions can be plausibly inferred to be similar. However, realistically, the best we can do is an *a posteriori* justification by inspecting the nature of the SCHF ground state, which we have performed carefully for the ground states of the RMG Hamiltonian.

In the remainder of this section, we will explain in detail how to physically obtain the reduced Hilbert space Eq. (4.5) from the assumptions above.

We only consider Slater determinant states. After we fold the Brillouin zone, the states are then labeled by

crystalline momentum $\boldsymbol{k}$:

$$|\Psi\rangle = \prod_{\boldsymbol{k}} \prod_{j=1}^{n_{\boldsymbol{k}}} d^\dagger_{\boldsymbol{k},j} |0\rangle \,, \tag{D.25}$$

where $n_{\boldsymbol{k}}$ is the occupation number at mBZ momentum $\boldsymbol{k}$, and $d^\dagger_{\boldsymbol{k},j}$ is the creation operator with mBZ momentum $\boldsymbol{k}$. Since we assumed the state to be an insulator, we fill the low-energy band of the mean-field Hamiltonian in the mBZ such that $n_{\boldsymbol{k}} = 1$ for all crystalline momentum $\boldsymbol{k}$.

The weak-coupling assumption then allows us to reduce to a low-energy Hilbert space with the lowest energy plane waves. Explicitly, we can restrict to a Slater determinant states generated by low energy plane waves:

$$\hat{d}^\dagger_{\boldsymbol{k}n} = \sum_{\boldsymbol{g}} \psi^{\boldsymbol{g}}_{\boldsymbol{k}n} \hat{c}^\dagger_{\boldsymbol{k}+\boldsymbol{g}}. \tag{D.26}$$

Here the sum over $\boldsymbol{g}$ is only non-zero when $E(\boldsymbol{k}+\boldsymbol{g}) - E(\boldsymbol{k})$ is small relative to the interaction strength.

Finally, the three-patch enables a further reduction to an analytically tractable model. Since the wavefunction is assumed to be homogeneous within each patch, we pick one representative momentum to describe the whole patch. The most natural choice is the set of high symmetry points $\xi = \{\gamma, \kappa, \kappa'\}$. Schematically, this reduction in Hilbert space can be written as:

$$|\Psi\rangle = \prod_{\boldsymbol{k}\in \mathrm{mBZ}} d^\dagger_{\boldsymbol{k}} |0\rangle \xrightarrow[\text{transformation}]{\text{three-patch}} \prod_{\substack{\boldsymbol{k}\in \\ \{\gamma,\kappa,\kappa'\}}} d^\dagger_{\boldsymbol{k}} |0\rangle = |\Psi_{3\mathrm{p}}\rangle \tag{D.27}$$

The high symmetry points $\{\gamma, \kappa, \kappa'\}$ have one, three, and three low-energy states respectively. We will assume that electronic states from the higher shells have a gap to these low-energy states that is much larger compared to the interaction energy. Following (D.26), and C_3 symmetry, the associated creation operators are $\hat{d}^\dagger_\gamma, \hat{d}^\dagger_{\kappa,\ell_\kappa}$, and $\hat{d}^\dagger_{\kappa',\ell_{\kappa'}}$ with C_3 angular momentum ℓ_ξ. Their single-particle wavefunctions $\psi() = \langle|\hat{d}^\dagger|0\rangle$ are thus given by Eq. (4.5), which we reproduce here

$$\begin{aligned}
\psi_{\gamma,\ell_\gamma=0} &= \chi(\gamma) \\
\psi_{\kappa,\ell_\kappa}(\boldsymbol{r}) &= \frac{1}{\sqrt{3}} \sum_{j=1,3,5} \exp\left[\frac{2\pi i \ell_\kappa}{3}\frac{j}{2}\right] e^{i\boldsymbol{\kappa}_j\cdot\boldsymbol{r}} \chi(\boldsymbol{\kappa}_j), \\
\psi_{\kappa',\ell_{\kappa'}}(\boldsymbol{r}) &= \frac{1}{\sqrt{3}} \sum_{j=2,4,6} \exp\left[\frac{2\pi i \ell_{\kappa'}}{3}\frac{j}{2}\right] e^{i\boldsymbol{\kappa}_j\cdot\boldsymbol{r}} \chi(\boldsymbol{\kappa}_j).
\end{aligned} \tag{D.28}$$

By the assumption that we have an insulating state with $\nu = 1$, we have one electron per representative state in the patch. Together with the Slater determinant assumption, we arrive at variational states

$$\Psi_{\ell_\kappa,\ell_{\kappa'}}(\boldsymbol{r}_1,\boldsymbol{r}_2,\boldsymbol{r}_3) = \mathcal{A}[\psi_\gamma(\boldsymbol{r}_1),\psi_{\kappa,\ell_\kappa}(\boldsymbol{r}_2),\psi_{\kappa',\ell_{\kappa'}}(\boldsymbol{r}_3)]. \tag{D.29}$$

This establishes the three-patch model.

D.5 Energetic competition of the three-patch states

We now use the charge density pattern above to infer some features of the Hartree and Fock energies. The Hartree energy between ψ_κ and $\psi_{\kappa'}$ corresponds to the interaction energy of classical charge configurations. It now becomes a simple visual exercise: we can visually figure out which charge densities are on top of each other by looking at Fig. D.2. It is easy to see that the $C = 1$ state has the least density overlap between $\rho_\kappa, \rho_{\kappa'}$. This is confirmed in the Hartree energy plot in Fig. D.2.

At small η, the Fock term can be similarly graphically evaluated. This is because Fock exchange energy is controlled by the alignment of pseudospins, and at small η most weight is on the top layer. We conclude the $C = 0$ state has the lowest Fock energy due to the large overlap between the top layer charge densities, as confirmed in the Fock energy plot in Fig. D.2. We note, however, that this visual exercise misses other contributions such as intercomponent interactions, and is qualitatively incorrect at large η, as evidenced by the fact that $C = 1$ state can have lower Fock energy.

We note an interesting implication of the density patterns. Consider a potential such that $V(\boldsymbol{r}) = V_0 \sum_i \cos(\boldsymbol{g}_i \cdot \boldsymbol{r})$, where $V_0 < 0$. This potential has a honeycomb shape, where the electrons are favored to form a triangular lattice at the superlattice points. If we apply this potential to the first component of the spinor, the $C = 0$ state, which has a triangular charge density in the first component, gains more energy than $C = \pm 1$ states. On the other hand, if the potential is applied to the second component, the $C = 1$ state is favored over the other states, since the charge density of the second component forms a triangular lattice. This consideration implies that the phase boundary can be shifted by such external potentials.

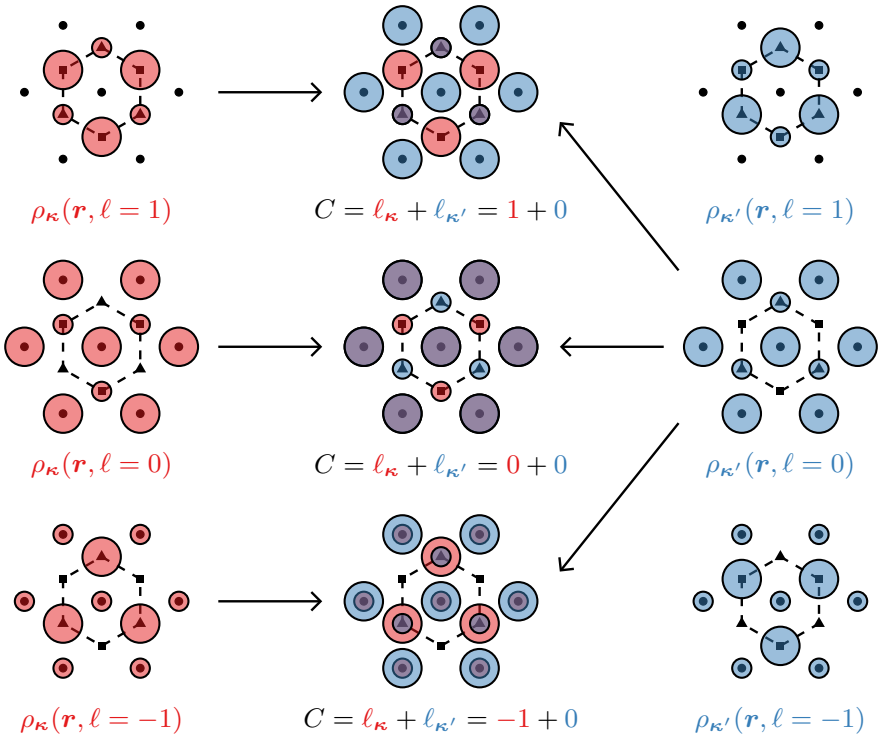


Figure D.2: Reproduction of Fig. 4.4. Charge densities of competing candidate states of the three-patch model. (left): the charge densities at the κ point. The size of the circles corresponds to different amplitudes of charge densities, which come from different components of the spinor. Different angular momentum states labeled by ℓ are related to each other by translation. (right): the charge densities in the κ' point. (center): the total charge densities of states with different Chern numbers.

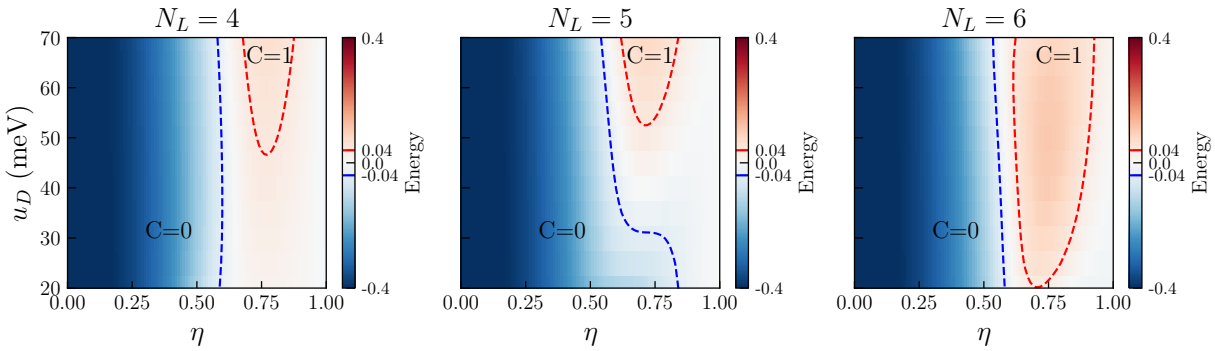


Figure D.3: The energy difference between $C = 0$ and $C = 1$ states using the realistic spinors with the three-patch model, as described in the App. D.6. The blue and red dashed lines correspond to the contour lines with energy -0.1 and 0.1 respectively.

D.6 The three-patch model using realistic spinors

In the main text, we used the sublattice projected spinor χ_B (Eq. (4.35)) to compute the energetic competition between different candidate states. In this section, we consider the three-patch model with the full spinor generated by the single particle Hamiltonian Eq. (4.26), which takes the presence of the other sublattice A into account as well. We note a similar computation was done in Ref. [365].

In Fig. D.3, we show the Coulomb energy difference between the $C = 1$ and $C = 0$ states, computed using the three-patch model. The input spinors for the three-patch model were taken to be the eigenstates of the single particle Hamiltonian of RMG at different values of N_L. As in Fig. 4.8, the spinors are taken to be at the corners of the BZ, whose size is parametrized by $\eta = v_F|\boldsymbol{\kappa}_i(n) - \boldsymbol{K}|/t_1$.

As shown in Fig. 4.8, smaller η stabilizes the $C = 0$ state consistently for different values of N_L. At higher η, the $C = 1$ state becomes more favorable, consistent with the computation using χ_B. In particular, at large u_D, the boundary between $C = 0$ and $C = 1$ is in quantitative agreement with that computed from χ_B, since the large displacement field suppresses mixing between the two zero modes. On the other hand, at small u_D, there is a significant deviation from the prediction from χ_B, especially at $N_L = 5$.

We note, however, that the small u_D region is part of a large "gray area" where the energy difference between the two competing states is small. In this region, we should not take the phase diagram from the three-patch model at face value, as second-order effects neglected in our modelling likely determine the ground state in this regime.

D.7 Dependence of the phase diagram on gate distance

In this Appendix, we examine the effect of gate distance on the phase diagram of hombohedral pentalayer graphene. This allows us to trace the evolution of the phase diagram from the Coulomb limit to the contact limit as the gate distance decreases discussed in Fig. 4.3 of the main text.

Recall that the double-gated Coulomb interaction is given by $V_{\boldsymbol{q}} = 2\pi \tanh |\boldsymbol{q}|d/(\epsilon_r\epsilon_0|\boldsymbol{q}|)$, where d is the distance to the metallic gates. In the long-distance limit, $V_{\boldsymbol{q}\to 0} = 2\pi d/(\epsilon_r\epsilon_0)$. In the following calculation, we use fixed d/ϵ_r so that $V_{\boldsymbol{q}\to 0}$ remains constant.

In Fig. D.4, we show the phase diagram of rhombohedral pentalayer graphene at various gate distances. Towards the contact limit (small gate distances), the $C = 1$ state relative to the $C = 0$ state — consistent with the

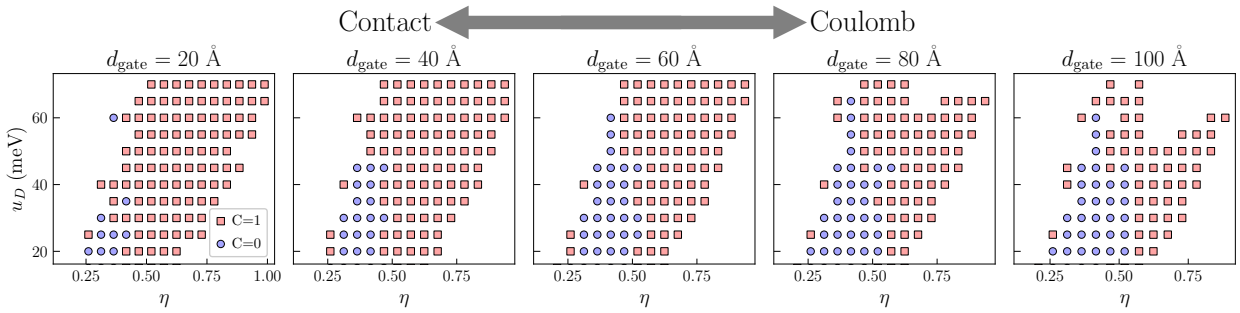


Figure D.4: The phase diagram of rhombohedral pentalayer graphene at various gate distances, computed using SCHF. The interaction strength used is $V_{\boldsymbol{q}\to 0} = 73.55\text{eV}\cdot\text{nm}$, or $\epsilon_r = d/d_0$ where $d_0 = 2\text{nm}$. Missing datapoints correspond to metallic states.

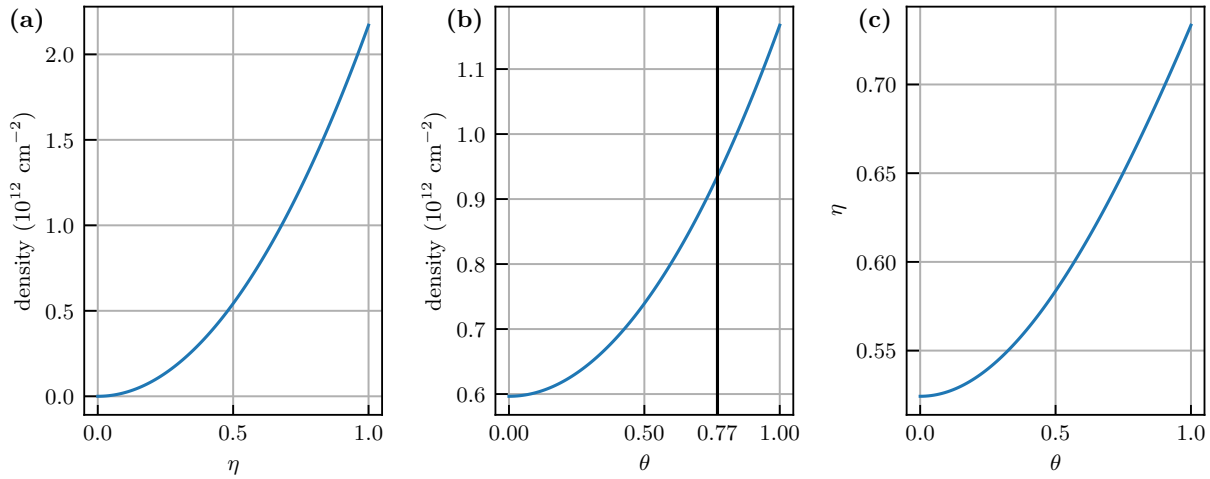


Figure D.5: Conversions between the dimensionless parameter η and experimentally measurable quantities. (a) Electron density versus η. (b) Electron density versus the moiré twist angle θ. (c) η versus θ.

prediction of the three-patch model.

To facilitate comparison between these results and experiments, Fig. D.5 shows the correspondence between the dimensionless parameters $\eta = \frac{v_F|\boldsymbol{\kappa}_i - \boldsymbol{K}|}{t_1}$ and experimental parameters of charge density and twist angle. For instance, $\theta = 0.77^\circ$ corresponds to a charge density $n \approx 0.93 \times 10^{12}\,\text{cm}^{-2}$, which in turns corresponds to $\eta \approx 0.66$.

E

Supplemental Material for Chapter 5

E.1 Modeling Details

This appendix describes modeling details of the two-dimensional topological electron gas.

E.1.1 Geometry

We consider electrons in the infinite plane subject to the Hamiltonian in Eq. (5.5) with a compensating background charge. To make the computation tractable, we impose a periodic boundary condition, and consider a Hilbert space defined by a finite torus M with area $A = N\rho$, where N is the number of particles. When imposing discrete translation symmetry, we choose M to be compatible with the choice of the Bravais lattice.

Triangular lattice

Consider a direct lattice $\mathbb{L}$ with lattice vectors

$$\boldsymbol{a}_1 = L\left(\tfrac{1}{2}, \quad -\tfrac{\sqrt{3}}{2}\right), \quad \boldsymbol{a}_2 = L\left(\tfrac{1}{2}, \quad \tfrac{\sqrt{3}}{2}\right). \tag{E.1}$$

Here $L = \sqrt{\frac{2\pi}{\sqrt{3}}}a$ ensures the density $\rho = \frac{1}{\pi a^2}$ matches the unit cell size $A_{\text{uc}} = L^2\frac{\sqrt{3}}{2}$. The reciprocal lattice is $\mathbb{L}^*$ with reciprocal basis vectors

$$\boldsymbol{g}_1 = \frac{2\pi}{L}\left(1, \quad -\tfrac{1}{\sqrt{3}}\right), \quad \boldsymbol{g}_2 = \frac{2\pi}{L}\left(1, \quad \tfrac{1}{\sqrt{3}}\right). \tag{E.2}$$

The full torus is a parallelogram with $N = N_1 \times N_2$ unit cells and total area $A = NA_{\text{uc}}$ with $A_{\text{uc}} = \pi a^2$.

Square lattice

The direct lattice is given by

$$\boldsymbol{a}_1 = L\begin{pmatrix}1, & 0\end{pmatrix}, \quad \boldsymbol{a}_2 = L\begin{pmatrix}0, & 1\end{pmatrix}. \tag{E.3}$$

Here we take $L = \sqrt{\pi}a$. The reciprocal lattice is $\mathbb{L}^*$ with reciprocal basis vectors

$$\boldsymbol{g}_1 = \frac{2\pi}{L}\begin{pmatrix}1, & 0\end{pmatrix}, \quad \boldsymbol{g}_2 = \frac{2\pi}{L}\begin{pmatrix}0, & 1\end{pmatrix}. \tag{E.4}$$

The full torus is a parallelogram with $N = N_1 \times N_2$ unit cells and total area $A = NA_{\text{uc}}$ with $A_{\text{uc}} = \pi a^2$.

Brillouin Zone & Momentum Grid

We consider a parallelogram Brillouin zone

$$\text{BZ} = \{k_1\boldsymbol{g}_1 + k_2\boldsymbol{g}_2 \ : \ -\pi \leq k_{1,2} < \pi\}. \tag{E.5}$$

We denote elements of the Brillouin zone by $\boldsymbol{k} \in \text{BZ}$, and unrestricted momenta by $\boldsymbol{q}$. Such momenta can always be split as

$$\boldsymbol{q} = \{\boldsymbol{q}\} + [\boldsymbol{q}] = \boldsymbol{k} + \boldsymbol{g}, \tag{E.6}$$

where $\boldsymbol{k}$ is in the first Brillouin zone and $\boldsymbol{g} = n_1\boldsymbol{g}_1 + n_2\boldsymbol{g}_2$ is a reciprocal lattice vector.

We take a Monkhorst-Pack discretization of the Brillouin zone

$$\boldsymbol{k}_{n_1,n_2} = \frac{n_1 + \Phi_1/2\pi}{N_1}\boldsymbol{g}_1 + \frac{n_2 + \Phi_2/2\pi}{N_2}\boldsymbol{g}_2, \tag{E.7}$$

where $0 \leq \Phi_{1,2} \leq 2\pi$ are the two fluxes through the torus and

$$n_i = \begin{cases} -\frac{N_i}{2}, -\frac{N_i}{2}+1, \ldots, \frac{N_i}{2}-1 & \text{for } N_i \text{ even} \\ -\frac{N_i}{2}+\frac{1}{2}, -\frac{N_i}{2}+\frac{3}{2}, \ldots, \frac{N_i}{2}-\frac{1}{2} & \text{for } N_i \text{ odd.} \end{cases} \tag{E.8}$$

This choice ensures that the lattice goes through Γ for $\Phi_{1,2} = 0$.

Momentum Shell Structure

To describe continuous functions within the unit cell (such as wavefunctions), we consider a "shell structure" of reciprocal momenta

$$S = \{\boldsymbol{g} = m_1 \boldsymbol{g}_1 + m_2 \boldsymbol{g}_2 \ : \ |\boldsymbol{g}| < N_s \max(|\boldsymbol{g}_1|, |\boldsymbol{g}_2|)\} . \tag{E.9}$$

In practice we usually take $N_s = 5 - 7$, giving $|S| \approx 100 - 200$.

E.1.2 First-Quantized Wavefunctions

We now set out basis conventions for first-quantized wavefunctions.

Microscopic Basis

Consider a microscopic basis $|\boldsymbol{r}, I\rangle$ where $I = 1, 2$ is a spinor index with corresponding basis vectors $\check{e}_I$. Wavefunctions are then written as

$$\psi^I(\boldsymbol{r}) = \langle \boldsymbol{r}, I | \psi \rangle . \tag{E.10}$$

Spinor indices are usually left implicit for concision.

Bloch Theorem Conventions

Suppose $\hat{h}$ has discrete translation symmetry: $[\hat{h}, \hat{T}_{\boldsymbol{R}}] = 0$ where $\hat{T}_{\boldsymbol{R}} f(\boldsymbol{r}) = f(\boldsymbol{r} + \boldsymbol{R})$ is translation by a direct lattice vector. We then consider Bloch wavefunctions

$$\begin{aligned}
\hat{h} \, |\phi_{\boldsymbol{k}n}\rangle &= \varepsilon_{\boldsymbol{k}n} \, |\phi_{\boldsymbol{k}n}\rangle \\
\hat{T}_{\boldsymbol{R}} \, |\psi_{\boldsymbol{k}n}\rangle &= e^{i\boldsymbol{k}\cdot\boldsymbol{r}} \, |\phi_{\boldsymbol{k}n}\rangle \\
\phi^I_{\boldsymbol{k}n}(\boldsymbol{r}) &= \langle \boldsymbol{r}, I | \psi_{\boldsymbol{k}n} \rangle .
\end{aligned}$$

Here n labels bands. The periodic part of the Bloch waves are defined as

$$u^I_{\boldsymbol{k}n}(\boldsymbol{r}) = N^{\frac{1}{2}} e^{-i\boldsymbol{k}\cdot\boldsymbol{r}} \phi^I_{\boldsymbol{k}n}(\boldsymbol{r}), \tag{E.11}$$

normalized on the unit cell. These are eigenstates of $\hat{h}[\boldsymbol{k}] = e^{-i\boldsymbol{k}\cdot\hat{\boldsymbol{r}}}\hat{h}e^{i\boldsymbol{k}\cdot\hat{\boldsymbol{r}}}$. The Bloch states comprise a unitary transform:

$$\langle \phi_{\boldsymbol{k}n} | \phi_{\boldsymbol{k}'n'} \rangle = \sum_I \int_M d^2\boldsymbol{r}\ \overline{\phi^I_{\boldsymbol{k}n}(\boldsymbol{r})} \phi^I_{\boldsymbol{k}'n'}(\boldsymbol{r}) = \delta_{\boldsymbol{k}\boldsymbol{k}'}\delta_{nn'}, \tag{E.12}$$

$$\langle \boldsymbol{r}, I | \boldsymbol{r}', I' \rangle = \sum_n \sum_{\boldsymbol{k}\in \mathrm{BZ}} \overline{\phi^I_{\boldsymbol{k}n}(\boldsymbol{r})} \phi^{I'}_{\boldsymbol{k}n}(\boldsymbol{r}') = \delta^{II'}\delta(\boldsymbol{r}-\boldsymbol{r}'). \tag{E.13}$$

Note that the $\boldsymbol{k}$ sum is discrete due to the finite area A.

We adopt a periodic gauge so that

$$|\psi_{\boldsymbol{k}+\boldsymbol{g},n}\rangle = \hat{V}_{\boldsymbol{g}} |\psi_{\boldsymbol{k},n}\rangle, \quad \hat{V}_{\boldsymbol{g}} = e^{-i\boldsymbol{g}\cdot\hat{\boldsymbol{r}}}. \tag{E.14}$$

Though this operator is unitary in principle, in practice there is a parametrically small violation $\hat{V}_{\boldsymbol{g}}\hat{V}^\dagger_{\boldsymbol{g}} = I + O(N_s)$.

Computational Basis

The computational basis is in momentum space, with unbounded momentum $\boldsymbol{q}$ partitioned as $\boldsymbol{k}+\boldsymbol{g}$ as above. Explicitly, we use

$$\langle \boldsymbol{r}, I | \boldsymbol{k}, \boldsymbol{g}, I' \rangle = \frac{1}{\sqrt{A}} e^{i(\boldsymbol{k}+\boldsymbol{g})\cdot\boldsymbol{r}} \delta_{II'} \tag{E.15}$$

where $\boldsymbol{g} = m_1\boldsymbol{g}_1 + m_2\boldsymbol{g}_2$ and $\boldsymbol{k} \in \mathrm{BZ}$. The discretization of $\boldsymbol{k}$ is given above, as is the cutoff on $\boldsymbol{g}$.

We use the following Fourier conventions:

$$\phi^I_{\boldsymbol{k}n}(\boldsymbol{r}) = \sum_{\boldsymbol{k}',\boldsymbol{g},I'} \langle \boldsymbol{r}, I|\boldsymbol{k}', \boldsymbol{g}, I'\rangle \langle \boldsymbol{k}', \boldsymbol{g}, I'|\psi_{\boldsymbol{k},n}\rangle \tag{E.16}$$

$$= \frac{1}{\sqrt{A}} \sum_{\boldsymbol{g}} e^{i(\boldsymbol{k}+\boldsymbol{g})\cdot\boldsymbol{r}} \phi^{\boldsymbol{g}I}_{\boldsymbol{k}n}, \tag{E.17}$$

$$\phi^{\boldsymbol{g},I}_{\boldsymbol{k},n} = \langle \boldsymbol{k}, \boldsymbol{g}, I|\psi_{\boldsymbol{k},n}\rangle = \sum_I \int_M d^2\boldsymbol{r}\ \frac{e^{-i(\boldsymbol{k}+\boldsymbol{g})\cdot\boldsymbol{r}}}{\sqrt{A}} \phi^I_{\boldsymbol{k},n}(\boldsymbol{r}). \tag{E.18}$$

The computational eigenvectors are thus normalized as

$$\sum_{\boldsymbol{g},I} \overline{\phi^{\boldsymbol{g},I}_{\boldsymbol{k},n}} \phi^{\boldsymbol{g},I}_{\boldsymbol{k}',n'} = \delta_{nn'}. \tag{E.19}$$

E.1.3 Interactions

We consider Coulomb interactions between electrons

$$V(\boldsymbol{r}) = \frac{2}{r_s\,|\boldsymbol{r}|} = \frac{1}{A} \sum_{\boldsymbol{q}} V(\boldsymbol{q}) e^{i\boldsymbol{q}\cdot\boldsymbol{r}}, \tag{E.20}$$

$$V(\boldsymbol{q}) = \frac{4\pi}{r_s q}. \tag{E.21}$$

This Hamiltonian suffers from IR divergence due to the long-range Coulomb interaction, and thus needs to be accompanied by a neutralizing background charge. We review the procedure for correcting for neutralizing background charge in App. E.2. The resulting interaction on the torus takes the form

$$H_{\text{int}} = \frac{1}{2A} \sum_{\boldsymbol{q}\neq 0} V(\boldsymbol{q}) : \hat{\rho}_{\boldsymbol{q}} \hat{\rho}_{-\boldsymbol{q}} : + \Delta\epsilon, \tag{E.22}$$

where $: \cdots :$ is the normal ordering operator with respect to the vacuum. We remove $\boldsymbol{q} = 0$ from the sum due to cancellation with the background charge, and $\Delta\epsilon$ encodes the energy correction from self-images. Formally, it is written as

$$\Delta\epsilon = \frac{1}{N}\left(-\frac{\rho}{2}V_{\boldsymbol{q}=0} + \frac{1}{2}N\sum_{\boldsymbol{R}\neq 0} V(\boldsymbol{R})\right), \tag{E.23}$$

where $\boldsymbol{R}$ is the set of vectors that define the periodic boundary condition of the torus M. Although both of the terms are formally divergent, the divergences cancel each other. In practice, we need to perform an Ewald summation to compute this summation. Including the correction $\Delta\epsilon$ handles the dominant contribution to finite-size energy correction of the system (see Fig. E.1 below from the last Appendix).

E.1.4 Second quantization and band projection

We now perform second quantization and band projection of the Hamiltonian. Let us denote by $c^\dagger_{\boldsymbol{k}}$ the creation operator for the lower band. The kinetic part of the Hamiltonian is simply given by

$$H_{\text{kin}} = \sum_{\boldsymbol{k}} \frac{|\boldsymbol{k}|^2}{r_s^2} c^\dagger_{\boldsymbol{k}} c_{\boldsymbol{k}} \tag{E.24}$$

The non-constant part of the interaction is given by

$$\frac{1}{2A}\sum_{\boldsymbol{q}\neq 0} V(\boldsymbol{q}) : \hat{\rho}_{\boldsymbol{q}}\hat{\rho}_{-\boldsymbol{q}} : \tag{E.25}$$

We can write the band projected interaction as

$$\tilde{\rho}_{\boldsymbol{q}} = \sum_{\boldsymbol{k}} \Lambda_{\boldsymbol{q}}(\boldsymbol{k}) c^\dagger_{\boldsymbol{k}} c_{\boldsymbol{k}+\boldsymbol{q}} \tag{E.26}$$

where $\Lambda_{\boldsymbol{q}}(\boldsymbol{k}) = \langle u_{\boldsymbol{k}} | u_{\boldsymbol{k}+\boldsymbol{q}}\rangle$. The projected interaction is therefore written as

$$H_{\text{int}} = \frac{1}{2A}\sum_{\boldsymbol{q}\neq 0}\sum_{\boldsymbol{k},\boldsymbol{k}'} V(\boldsymbol{q})\Lambda_{\boldsymbol{q}}(\boldsymbol{k})\Lambda_{-\boldsymbol{q}}(\boldsymbol{k}') : c^\dagger_{\boldsymbol{k}} c_{\boldsymbol{k}+\boldsymbol{q}} c^\dagger_{\boldsymbol{k}'} c_{\boldsymbol{k}'-\boldsymbol{q}} : . \tag{E.27}$$

E.2 Review of Ewald summation and Madelung energy

We now review the method of Ewald summation and computation of the Madelung energy. Some useful references are found in Ref. [462, 463].

We consider a system of electrons in the infinite plane. We denote their interaction by $V(\boldsymbol{r})$. The interaction energy is given by a formal sum

$$E_{\text{int}} = \sum_{i<j}^{\infty} V(\boldsymbol{r}_i - \boldsymbol{r}_j). \tag{E.28}$$

Our goal is to evaluate the energy density of this system by imposing periodic boundary condition. We consider a parallelogram $P = \boldsymbol{R}_1 \wedge \boldsymbol{R}_2$ spanned by $\boldsymbol{R}_1$ and $\boldsymbol{R}_2$. This defines the lattice $\Lambda = \{m\boldsymbol{R}_1 + n\boldsymbol{R}_2 | m, n \in \mathbb{Z}\}$. It is convenient to write $\boldsymbol{R}_j = L_j \boldsymbol{r}_j$ with $L_i \in \mathbb{Z}$. We take $|\hat{z} \cdot \boldsymbol{r}_1 \times \boldsymbol{r}_2| = 1/\rho$, where ρ is the electron density. The parallelogram therefore contains $N = L_1 L_2$ electrons.

We can now truncate the formal sum to define energy density. To do so, let us rewrite the formal sum as follows:

$$\begin{aligned}
E_{\text{int}} &= \sum_{i<j}^{\infty} V(\boldsymbol{r}_i - \boldsymbol{r}_j) && \text{(E.29)}\\
&= \frac{1}{2}\sum_{i\neq j}^{\infty} V(\boldsymbol{r}_i - \boldsymbol{r}_j) && \text{(E.30)}\\
&= \frac{1}{2}\sum_{i\neq j}^{N} \sum_{\boldsymbol{R}_a, \boldsymbol{R}_b \in \Lambda} V(\boldsymbol{r}_i - \boldsymbol{R}_a - \boldsymbol{r}_j - \boldsymbol{R}_b) + \frac{1}{2}\sum_{i}^{N} \sum_{\boldsymbol{R}_a \neq \boldsymbol{R}_b} V(\boldsymbol{R}_a - \boldsymbol{R}_b) && \text{(E.31)}\\
&= \sum_{\boldsymbol{R}_a \in \Lambda} \frac{1}{2}\sum_{i\neq j}^{N} \sum_{\boldsymbol{R}_b \in \Lambda} V(\boldsymbol{r}_i - \boldsymbol{r}_j - \boldsymbol{R}_b|) + \sum_{\boldsymbol{R}_a} \frac{1}{2}\sum_{i}^{N} \sum_{\boldsymbol{R}_b \neq 0} V(\boldsymbol{R}_b) && \text{(E.32)}\\
&= \sum_{\boldsymbol{R}_a} \left(\sum_{i<j}^{N} \sum_{\boldsymbol{R}_b \in \Lambda} V(\boldsymbol{r}_i - \boldsymbol{r}_j - \boldsymbol{R}_b|) + \frac{1}{2}\sum_{i}^{N} \sum_{\boldsymbol{R}_b \neq 0} V(\boldsymbol{R}_b). \right) && \text{(E.33)}
\end{aligned}$$

In going from the second line to the third line, we introduced the sum over positions within the unit cell i.e. $\sum_i^{\infty} f(\boldsymbol{r}_i)$ is the sum over all particle positions, and $\sum_i^N f(\boldsymbol{r}_i)$ is a sum over particle positions within the unit cell. We can split the sum to the part $i \neq j$, in which case there is no restriction on values of $\boldsymbol{R}_a$ and $\boldsymbol{R}_b$, and $i = j$, in which case we require $\boldsymbol{R}_a \neq \boldsymbol{R}_b$. The energy par particle is then given by

$$\epsilon_{\text{int}} = \frac{1}{N}\left(\sum_{i<j}^{N}\sum_{\boldsymbol{R}\in\Lambda} V(\boldsymbol{r}_i - \boldsymbol{r}_j - \boldsymbol{R}) + \frac{1}{2}\sum_{i}^{N}\sum_{\boldsymbol{R}\neq\boldsymbol{0}} V(\boldsymbol{R})\right), \tag{E.34}$$

where $\boldsymbol{r}_i$'s label the location of N electrons within the parallelogram.

If $V(\boldsymbol{r})$ is the Coulomb interaction, this sum is infrared divergent. We can regularize this divergence by adding a neutralizing background:

$$\epsilon_{\text{bkg}} = \frac{1}{N}\left(-\sum_{i}^{N}\int \rho_b(\boldsymbol{r})V(\boldsymbol{r})d^2r + \frac{1}{2}\int_P d^2\boldsymbol{r}\int_{\mathbb{R}^2} d^2\boldsymbol{r}'\rho_b(\boldsymbol{r})\rho_b(\boldsymbol{r}')V(\boldsymbol{r}-\boldsymbol{r}')\right) \tag{E.35}$$

$$= \frac{1}{N}\left(-N\rho\int V(\boldsymbol{r})d^2\boldsymbol{r} + \frac{1}{2}\frac{N}{\rho}\rho^2\int d^2\boldsymbol{r}V(\boldsymbol{r})\right) \tag{E.36}$$

$$= -\rho\int V(\boldsymbol{r})d^2\boldsymbol{r} + \frac{1}{2}\int d^2\boldsymbol{r}V(\boldsymbol{r}) = -\frac{\rho}{2}\int d^2\boldsymbol{r}V(\boldsymbol{r}) \tag{E.37}$$

where we have chosen the neutralizing background to be uniform and exactly cancel the charges of the electrons: $\rho_b(\boldsymbol{r}) = \rho$. In the sum, the first term is the electron-background interaction, and the second term is the background-background interaction.

We now wish to evaluate this expression in momentum space. To do so, we first assume $V(r)$ is properly regularized such that $V(\boldsymbol{k}=0) = \int d^2\boldsymbol{r}V(\boldsymbol{r})$ is finite. The precise regularization scheme does not matter. The interaction term can then be rearranged as

$$\epsilon_{\text{int}} = \frac{1}{N}\left(\sum_{i<j}^{N}\sum_{\boldsymbol{R}\in\Lambda} V(\boldsymbol{r}_i - \boldsymbol{r}_j - \boldsymbol{R}) + \frac{1}{2}\sum_{i}^{N}\sum_{\boldsymbol{R}\neq 0} V(\boldsymbol{R})\right) \tag{E.38}$$

$$= \frac{1}{N}\left(\frac{1}{|P|}\sum_{i<j}^{N}\sum_{\boldsymbol{G}\in\Lambda^*}\tilde{V}(\boldsymbol{G})e^{i\boldsymbol{G}\cdot(\boldsymbol{r}_i-\boldsymbol{r}_j)} + \frac{1}{2}\sum_{i}^{N}\sum_{\boldsymbol{R}\neq 0} V(\boldsymbol{R})\right) \tag{E.39}$$

$$= \frac{1}{N}\left(\frac{(N-1)\rho}{2}\tilde{V}(0) + \frac{\rho}{N}\sum_{i<j}^{N}\sum_{\boldsymbol{G}\neq\boldsymbol{0}}\tilde{V}(\boldsymbol{G})e^{i\boldsymbol{G}\cdot(\boldsymbol{r}_i-\boldsymbol{r}_j)} + \frac{1}{2}\sum_{i}^{N}\sum_{\boldsymbol{R}\neq 0} V(\boldsymbol{R})\right) \tag{E.40}$$

where we used Poisson summation: $\sum_{\boldsymbol{R}\in\Lambda} f(\boldsymbol{R}) = \frac{1}{|P|}\sum_{\boldsymbol{G}\in\Lambda^*}\tilde{f}(\boldsymbol{G}), |P| = N/\rho$. From now on, we will switch to the notation where $\tilde{f}$ is the Fourier transform of f. The neutralizing background, on the other hand, gives

$$\epsilon_{\text{bkg}} = -\frac{\rho}{2}\int d^2\boldsymbol{r} V(\boldsymbol{r}) = -\frac{1}{2}\rho\tilde{V}(0). \tag{E.41}$$

Combining these together, we get

$$\epsilon_{\text{int}} + \epsilon_{\text{bkg}} = \frac{1}{N}\left(-\frac{\rho}{2}\tilde{V}(0) + \frac{\rho}{N}\sum_{i<j}^{N}\sum_{\boldsymbol{G}\neq\boldsymbol{0}}\tilde{V}(\boldsymbol{G})e^{i\boldsymbol{G}\cdot(\boldsymbol{r}_i-\boldsymbol{r}_j)} + \frac{1}{2}\sum_{i}^{N}\sum_{\boldsymbol{R}\neq 0}V(\boldsymbol{R})\right) \tag{E.42}$$

The first term and the last term (Madelung term) do not scale with the number of particles N, so they can be dropped to give us

$$\epsilon_{\text{int}} + \epsilon_{\text{bkg}} = \frac{\rho}{N^2}\sum_{i<j}^{N}\sum_{\boldsymbol{G}<\boldsymbol{0}}\tilde{V}(\boldsymbol{G})e^{i\boldsymbol{G}\cdot(\boldsymbol{r}_i-\boldsymbol{r}_j)} + O\left(\frac{1}{N}\right). \tag{E.43}$$

We now treat the $O(1/N)$ term carefully to tame the finite-size effect. The correction term is

$$\Delta\epsilon = \frac{1}{N}\left(-\frac{\rho}{2}\tilde{V}(0) + \frac{1}{2}N\sum_{\boldsymbol{R}\neq 0}V(\boldsymbol{R})\right). \tag{E.44}$$

We will now use the Ewald summation trick and split the interaction as $V(r) = V^{\ell}(r) + V^{s}(r)$. We demand $V^{\ell}(0) < \infty$ (UV convergent, IR divergent) and $\tilde{V}^{s}(k=0) < \infty$ (UV divergent, IR convergent). We can now rewrite the sum as

$$\Delta\epsilon = \frac{1}{N}\left(-\frac{\rho}{2}\tilde{V}^{\ell}(0) + \frac{1}{2}N\sum_{\boldsymbol{R}\neq 0}V^{\ell}(\boldsymbol{R}) - \frac{\rho}{2}\tilde{V}^{s}(0) + \frac{1}{2}N\sum_{\boldsymbol{R}\neq 0}V^{s}(\boldsymbol{R})\right) \tag{E.45}$$

$$= \frac{1}{N}\left(-\frac{\rho}{2}\tilde{V}^{\ell}(0) + \frac{1}{2}N\sum_{\boldsymbol{R}}V^{\ell}(\boldsymbol{R}) - \frac{N}{2}V^{\ell}(0) - \frac{\rho}{2}\tilde{V}^{s}(0) + \frac{1}{2}N\sum_{\boldsymbol{R}\neq 0}V^{s}(\boldsymbol{R})\right) \tag{E.46}$$

$$= \frac{1}{N}\left(-\frac{\rho}{2}\tilde{V}^{\ell}(0) + \frac{\rho}{2}\sum_{\boldsymbol{G}}\tilde{V}^{\ell}(\boldsymbol{G}) - \frac{N}{2}V^{\ell}(0) - \frac{\rho}{2}\tilde{V}^{s}(0) + \frac{1}{2}N\sum_{\boldsymbol{R}\neq 0}V^{s}(\boldsymbol{R})\right) \tag{E.47}$$

$$= \frac{1}{N}\left(\frac{\rho}{2}\sum_{\boldsymbol{G}\neq 0}\tilde{V}^{\ell}(\boldsymbol{G}) - \frac{N}{2}V^{\ell}(0) - \frac{\rho}{2}\tilde{V}^{s}(0) + \frac{1}{2}N\sum_{\boldsymbol{R}\neq 0}V^{s}(\boldsymbol{R})\right) \tag{E.48}$$

All terms in this expression are manifestly finite, and we can evaluate them directly.

E.2.1 Error function for the Ewald trick

We now make a choice for decomposition $V(\boldsymbol{r}) = V(|\boldsymbol{r}|) = V^\ell(|\boldsymbol{r}|) + V^s(|\boldsymbol{r}|)$. The standard method is to take the following:

$$V^s(r) = \frac{\mathrm{erfc}(\alpha r)}{r}, \quad V^\ell(r) = \frac{\mathrm{erf}(\alpha r)}{r}, \tag{E.49}$$

where $\mathrm{erf}(r) = \frac{2}{\sqrt{\pi}} \int_0^r e^{-t^2} dt$, $\mathrm{erfc}(r) = 1 - \mathrm{erf}(r)$. There is a simple physical picture for this choice. $V^\ell(r)$ is the potential coming from a gaussian cloud of electron with charge density $e^{-\alpha r^2}$, as can be checked from the Gauss law.

E.2.2 3D Ewald

Even though we are interested in 2D, it is convenient to start from evaluating the 3D integral first. The 3D Fourier transform of $V^\ell(r)$ is

$$\tilde{V}^\ell(k) = 4\pi \frac{e^{-\frac{k^2}{4\alpha^2}}}{k^2} \tag{E.50}$$

We check this by performing Fourier transform:

$$\begin{aligned}
\mathcal{F}[\tilde{V}^\ell](r) &= 4\pi \int \frac{d^3k}{(2\pi)^3} \frac{e^{-\frac{k^2}{4\alpha^2}}}{k^2} e^{ik\cdot r} && \text{(E.51)} \\
&= \frac{1}{\pi} \int_0^\infty dk e^{-\frac{k^2}{4\alpha^2}} \int \sin\theta d\theta e^{ik|r|\cos\theta} && \text{(E.52)} \\
&= 4\pi \frac{1}{\pi} \int_0^\infty dk \frac{1}{ik|r|} [e^{ik|r|} - e^{-ik|r|}] e^{-\frac{k^2}{4\alpha^2}} && \text{(E.53)} \\
&= \frac{1}{\pi|r|} \int_{-\infty}^\infty dk \frac{1}{ik} e^{ik|r|} e^{-\frac{k^2}{4\alpha^2}} && \text{(E.54)}
\end{aligned}$$

Let $I(r) = \int_{-\infty}^\infty dk \frac{1}{ik} e^{ik|r|} e^{-\frac{k^2}{4\alpha}}$. Then

$$\frac{dI(r)}{dr} = \int_{-\infty}^\infty dk e^{ik|r|} e^{-\frac{k^2}{4\alpha^2}} = 2\alpha\sqrt{\pi} e^{-\alpha^2|r|^2} \implies I(r) = 2\sqrt{\pi} \int_0^{\alpha r} e^{-t^2} dt = \pi \mathrm{erf}(\alpha r) \tag{E.55}$$

So we have $\mathcal{F}[\tilde{V}^\ell](r) = I(r)/\pi r = V^\ell(r)$. *

E.2.3 2D Ewald

We will now perform 2D Ewald transform. We start from the 3D Fourier transform of the long range potential.

$$\tilde{V}^\ell(k) = 4\pi \frac{e^{-\frac{k^2}{4\alpha^2}}}{k^2} \tag{E.56}$$

We can now integrate this in the z direction to get 2D Fourier component:

$$\tilde{V}^\ell_{2D}(k) = e^{-\frac{|k|^2}{4\alpha^2}} \int_{-\infty}^{\infty} \frac{dk_z}{2\pi} e^{-\frac{k_z^2}{4\alpha^2}} \frac{4\pi}{k^2 + k_z^2} \tag{E.57}$$

We use the following standard integral equality:

$$\int_0^\infty dk \frac{e^{-k^2}}{k^2 + t^2} = \frac{\pi \mathrm{erfc}(t)}{2t} e^{t^2} \tag{E.58}$$

Using this, we get

$$\tilde{V}^\ell_{2D}(k) = \frac{2\pi \mathrm{erfc}(\frac{k}{2\alpha})}{k}. \tag{E.59}$$

Rather interestingly, the error function shows up in real space instead of momentum space in this case. We can also evaluate the integral of the short-range part of the interaction:

$$\int d^2r \frac{\mathrm{erfc}(\alpha r)}{r} = 2\pi \int_0^\infty dr\, \mathrm{erfc}(\alpha r) = \frac{2\sqrt{\pi}}{\alpha} \tag{E.60}$$

The spatial limit of the interaction is easy to see. As $\mathrm{erf}(r) = 2r/\sqrt{\pi} + O(r^3)$, we have

$$V^\ell(0) = \frac{2\alpha}{\sqrt{\pi}}. \tag{E.61}$$

*We could have gotten this result by noticing the charge distribution corresponding to this potential is nothing but a Gaussian

Combining these results together, the energy correction using Ewald summation is given by

$$\Delta\epsilon_C = \frac{1}{N}\left(\frac{\rho}{2}\sum_{\boldsymbol{G}\neq 0}\frac{2\pi\mathrm{erfc}(\frac{k}{2\alpha})}{k} - \frac{N}{2}\frac{2\alpha}{\sqrt{\pi}} - \frac{\rho}{2}\frac{2\sqrt{\pi}}{\alpha} + \frac{1}{2}N\sum_{\boldsymbol{R}\neq 0}\frac{\mathrm{erfc}(\alpha R)}{R}\right). \tag{E.62}$$

For 2D electron gas in particular, in which the interaction is given by $\frac{2}{r_s}$, we have

$$\Delta\epsilon_{2DEG} = \frac{2}{Nr_s}\left(\frac{\rho}{2}\sum_{\boldsymbol{G}\neq 0}\frac{2\pi\mathrm{erfc}(\frac{k}{2\alpha})}{k} - \frac{N}{2}\frac{2\alpha}{\sqrt{\pi}} - \frac{\rho}{2}\frac{2\sqrt{\pi}}{\alpha} + \frac{1}{2}N\sum_{\boldsymbol{R}\neq 0}\frac{\mathrm{erfc}(\alpha R)}{R}\right), \tag{E.63}$$

where the $\boldsymbol{r}_j$ must be chosen to have area π. These summations can then be performed easily by choosing an appropriate α and truncating the summation at finite $|\boldsymbol{G}|$ and $|\boldsymbol{R}|$, which only leaves an exponentially small error.

E.3 Hartree-Fock procedure

We use standard self-consistent Hartree-Fock calculations to obtain the SCHF ground state of the electronic system.

E.3.1 Continuous translation symmetry

In the presence of continuous symmetry, the mean-field states are fully characterized by $n(k)$, and its energies are given by Eq. (5.9) (5.11), which reproduce for convenience below:

$$E_{\text{kin}} = \sum_{\boldsymbol{k}} n(\boldsymbol{k})\frac{|\boldsymbol{k}|^2}{r_s^2}, \tag{E.64}$$

$$E_{\text{int}} = -\frac{1}{2A}\sum_{\boldsymbol{k},\boldsymbol{k}'}\tilde{V}(\boldsymbol{k},\boldsymbol{k}')n(\boldsymbol{k})n(\boldsymbol{k}'). \tag{E.65}$$

This simple formula allows us to evaluate the energy of Fermi liquid candidate states efficiently. As our Fermi liquid ansatz, we take the following $n(k)$:

$$n(k) = \begin{cases} 0 & k < k_{\min} \\ 1 & k_{\min} \leq k \leq k_{\max} \\ 0 & k > k_{\max} \end{cases}. \tag{E.66}$$

Here $k_{\max}^2 - k_{\min}^2 = 4/a^2$ to ensure the same electron density. Setting $k_{\min} = 0$ recovers the circular Fermi surface. By minimizing the sum of kinetic and interaction energy with respect to $k_{\min}$, we can evaluate the energy of the Fermi liquid. This reduces to evaluating numerical integrals, which allows us to achieve the thermodynamic limit at a small energy cost.

E.3.2 Discrete translation symmetry

We now consider performing the self-consistent Hartree Fock while only imposing discrete translation symmetry. As usual, we consider single-particle density matrices $P(\boldsymbol{k})_{\beta\alpha} = \langle c^\dagger_{\boldsymbol{k}\beta} c_{\boldsymbol{k}\alpha} \rangle$, where α and β are collective indices for valley, spin, and band. These density matrices are in one-to-one correspondence with Slater determinant states. We define the Hartree and Fock Hamiltonians as

$$h_H[P](\boldsymbol{k}) = \frac{1}{A} \sum_{\boldsymbol{g}} V(\boldsymbol{g}) \Lambda_{\boldsymbol{g}}(\boldsymbol{k}) \left(\sum_{\boldsymbol{k}} \mathrm{Tr}[P(\boldsymbol{k}) \Lambda_{\boldsymbol{g}}(\boldsymbol{k})^\dagger] \right) \tag{E.67a}$$

$$h_F[P](\boldsymbol{k}) = -\frac{1}{A} \sum_{\boldsymbol{q}} V(\boldsymbol{q}) \Lambda_{\boldsymbol{q}}(\boldsymbol{k}) P([\boldsymbol{k} + \boldsymbol{q}]) \Lambda_{\boldsymbol{q}}(\boldsymbol{k})^\dagger \tag{E.67b}$$

where $[\Lambda_{\boldsymbol{q}}(\boldsymbol{k})]_{\alpha\beta} = \langle \psi_{\boldsymbol{k}\alpha} | e^{-i\boldsymbol{q}\cdot\boldsymbol{r}} | \psi_{\boldsymbol{k}+\boldsymbol{q}\beta} \rangle$ are form factors, and are treated as matrices whose labels are identical to the single-particle ones. The sum over $\boldsymbol{g}$ runs over reciprocal vectors while $\boldsymbol{q}$ runs over all momentum transfers. Via Wick's theorem, the energy of the Slater-determinant state is

$$E[P] = \frac{1}{2} \mathrm{Tr}[P(2h_{\text{kin}} + h_H[P] + h_F[P])] \tag{E.68}$$

where the trace is over momentum and all other band labels.

We use the optimal-damping algorithm (ODA) to converge to states satisfying the self-consistency condition

$$[P, h_{\text{kin}} + h_H[P] + h_F[P]] = 0 \tag{E.69}$$

to tolerances approaching the square root of machine precision (i.e. machine precision in $E[P]$). We use Monkhorst-Pack grids with 12×12 up to 36×36 unit cells, and ensure that the range of momentum transfers $\boldsymbol{q}$ considered is sufficiently large to converge the energy of the state, out to several Brillouin zone sizes. The choices of system sizes N_k are always multiples of 6 to ensure that the high-symmetry points, such as the M and K points when the lattice is triangular, are properly sampled. We further truncate the single-particle Hilbert spaces to $N_b = 7, 13, 19, 37$ bands, until energy convergence is reached. To avoid non-global minima, we initialize SCHF in a variety of states for each parameter point using a mix of physical ansätze as well as random initial states.

E.4 Finite-size scaling of the energy

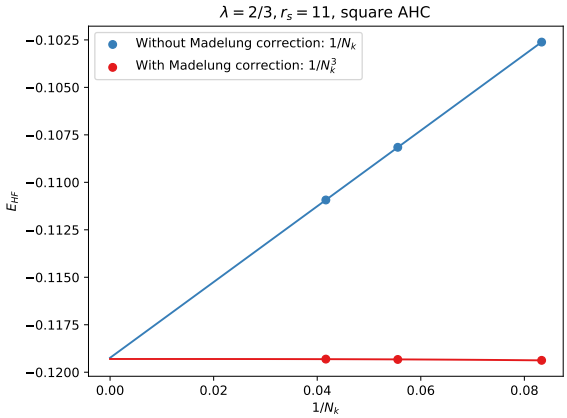
In this section, we detail how we perform finite-size scaling.

As discussed in App. E.3.1, our energy evaluation is already in the thermodynamic limit, and therefore does not require finite-size scaling.

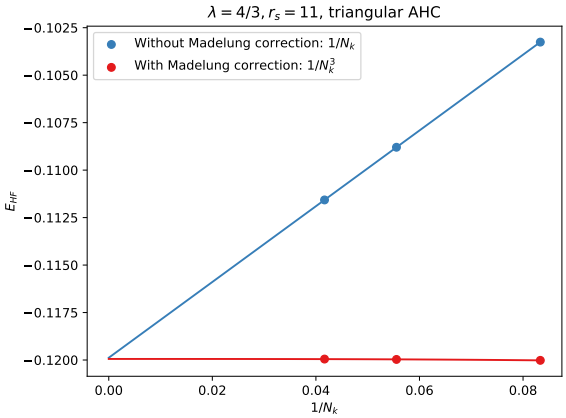
As for the Hartree-Fock ground states found for discrete translation symmetry, we take into account the finite-size corrections detailed in App. E.2:

$$E_{\text{corrected}}[P] = E[P] + \frac{2}{r_s}\Delta\epsilon_C, \tag{E.70}$$

where $\Delta\epsilon_C$ is given by Eq. (E.62). This improves the convergence of energies greatly, from a $1/L$ asymptotic to a $1/L^3$ asymptotic [385]. In Fig. E.1a,E.1b, we show the result of such finite-size scaling. The inclusion of the Madelung energy correction indeed reduces the finite-size effect considerably.



(a) Finite-size scaling for ground state at $\lambda = 2/3, r_s = 11$.



(b) Finite-size scaling for ground state at $\lambda = 4/3, r_s = 11$.

Figure E.1: Finite-size scaling plots for the square and triangular AHC states, with and without the Madelung correction. The Madelung correction significantly reduces the finite-size effect.

F

Supplemental Material for Chapter 6

F.1 Microscopic Hamiltonian of rhombohedral multilayer graphene

This Appendix reviews the microscopic Hamiltonian of rhombohedral multilayer graphene. We follow the conventions from Ref. [320]. Here, we provide the minimal summary necessary for reproducing the Hamiltonian.

The direct lattice for graphene is given by

$$\boldsymbol{R}_1 = a\,(1,0)\,, \qquad \boldsymbol{R}_2 = a\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \tag{F.1}$$

where we take the lattice constant to be $a \approx 0.246\text{nm}$.

The Hamiltonian for rhombohedral multilayer graphene [284, 285] (see also: [259, 266, 287, 288, 457] and references therein) in Fourier space is a $2\ell \times 2\ell$ dimensional matrix with the block structure

$$h_{RG}^{(N_L)}(\boldsymbol{k}) = \begin{bmatrix} h_\ell^{(0)} & h^{(1)} & h^{(2)} & & & \\ h^{(1)\dagger} & h_\ell^{(0)} & h^{(1)} & h^{(2)} & & \\ h^{(2)\dagger} & h^{(1)\dagger} & h_\ell^{(0)} & \ddots & \ddots & \\ & h^{(2)\dagger} & \ddots & \ddots & h^{(1)} & h^{(2)} \\ & & \ddots & h^{(1)\dagger} & h_\ell^{(0)} & h^{(1)} \\ & & & h^{(2)\dagger} & h^{(1)\dagger} & h_\ell^{(0)} \end{bmatrix} \tag{F.2}$$

whose subblocks are parameterized as

$$h_\ell^{(0)}(\boldsymbol{k}) = \begin{pmatrix} u_{A\ell} & -t_0 f_{\boldsymbol{k}} \\ -t_0 \overline{f}_{\boldsymbol{k}} & u_{B\ell} \end{pmatrix}, \quad h^{(1)}(\boldsymbol{k}) = \begin{pmatrix} t_4 f_{\boldsymbol{k}} & t_3 \overline{f_{\boldsymbol{k}}} \\ t_1 & t_4 f_{\boldsymbol{k}} \end{pmatrix}, \qquad h^{(2)}(\boldsymbol{k}) = \begin{pmatrix} 0 & \frac{t_2}{2} \\ 0 & 0 \end{pmatrix}, \quad f_{\boldsymbol{k}} = \sum_{i=0}^{2} e^{i\boldsymbol{k}\cdot\boldsymbol{\delta}_i}, \tag{F.3a}$$

where we defined the nearest neighbor vectors to be $\boldsymbol{\delta}_n = R_{n2\pi/3}(0, \frac{1}{\sqrt{3}}a)^T$ for $n = 0, 1, 2$ where R_θ is the counterclockwise rotation matrix by angle θ.

For microscopic parameters, we take parameters extracted from Ref. [458]:

$$\begin{aligned} t_0 &= 3100\text{meV}, \\ t_1 &= 380\text{meV}, \\ t_2 &= -21\text{meV}, \\ t_3 &= 290\text{meV}, \\ t_4 &= 141\text{meV}. \end{aligned} \tag{F.4}$$

The displacement field is implemented as a uniform gradient of potential energy throughout the graphene layers:

$$u_{\sigma,\ell} = u_D \left(\ell + 1 - \frac{N_L - 1}{2} \right). \tag{F.5}$$

The full many-body Hamiltonian is given by

$$\hat{H} = \hat{h}_{RG}^{(N_L=5)} + \hat{V} \tag{F.6}$$

where $\hat{V}$ encodes standard double-gate screened Coulomb interactions

$$V(q) = \frac{2\pi \tanh qd}{\epsilon_r \epsilon_0 q}, \tag{F.7}$$

where $\epsilon_r = 5$ is the relative dielectric constant and $d = 250$ is the gate distance. We use the "charge neutrality" subtraction scheme [320].

Our self-consistent Hartree-Fock calculations are projected into seven bands with system sizes up to 24×24.

We consider a single valley and spin only. When performing time-dependent Hartree-Fock, the mean-field correlation matrix P was converged so that the commutator error obeys $[H_{HF}[P], P] = 0$ with accuracy close to machine precision. This avoids spurious imaginary eigenvalues in the collective mode spectrum. The TDHF eigenvalue problem is solved using standard Arnoldi methods *without* shift-inversion. We ensure good convergence in the case of degenerate eigenvalues, such as the exact zero modes at γ.

F.2 TDHF

This Appendix derives the time-dependent Hartree-Fock formulation of particle-hole excitations [464]. We follow the notation and presentation of Appendix A of [465], with some use of [466]. The main goal is to derive formulas that are well-suited to numerical calculations using band-projected Hartree-Fock.

F.2.1 The Idea of TDHF

Time-dependent Hartree-Fock is a special case of the time-dependent variational principle, a lens through which the procedure becomes simple. Given a Hamiltonian, we can compute the energy $E[P]$ of a Slater determinant P. To second order, the energy of nearby Slater determinants is (using a schematic notation)

$$E[P + dP] = E[P] + J_P \cdot dP + \frac{1}{2} dP \cdot \mathcal{H}_P \cdot dP, \tag{F.8}$$

where J is a Jacobian and $\mathcal{H}$ is a Hessian. Self-consistent Hartree-Fock minimizes the energy over the variational manifold of Slater determinants, producing a local minimum $E[P_0] = E_0$. As a local minimum, the Jacobian vanishes: $J_{P_0} = 0$, while the Hessian $\mathcal{H}_{P_0}$ controls the potential well. The low-frequency dynamics of any small perturbation around the minima,

$$P(t) = P_0 + \delta P(t), \tag{F.9}$$

are therefore controlled by the smallest eigenvalues of the Hessian at the minima. Our goal is to compute them. Instead of using the positive semi-definite Hessian directly, it is more convenient in practice to use a slightly different operator with a particle-hole-like structure that is called $\mathcal{L}$ below.

F.2.2 Setup

Consider a Hamiltonian

$$\hat{H} = \hat{h} + \hat{V} = \sum_k \hat{c}^\dagger_{\boldsymbol{k},a} \left[h(\boldsymbol{k})\right]_{ab} \hat{c}_{\boldsymbol{k},b} + \frac{1}{2A} \sum_{\boldsymbol{q}} V_q \hat{\rho}_{\boldsymbol{q}} \hat{\rho}_{-\boldsymbol{q}} \tag{F.10}$$

with $\hat{\rho}_{\boldsymbol{q}} = \sum_k \hat{c}^\dagger_{\boldsymbol{k},a} \left[\Lambda_{\boldsymbol{q}}(\boldsymbol{k})\right]_{ab} \hat{c}_{\boldsymbol{k}+\boldsymbol{q},b}$ as normal. Let

$$\widehat{\phi}_{\boldsymbol{q}} = \sum_{\boldsymbol{k},a,b} \hat{c}^\dagger_{\boldsymbol{k}} \left[\phi_{\boldsymbol{q}}(\boldsymbol{k})\right]_{ab} \hat{c}_{\boldsymbol{k}+\boldsymbol{q}} \tag{F.11}$$

be a neutral collective mode where ϕ_q (no hat) denotes its corresponding matrix. Its true equation of motion is

$$i\partial_t \hat{\phi}_{\boldsymbol{q}} = [\hat{H}, \hat{\phi}_{\boldsymbol{q}}] = \mathcal{L}\hat{\phi}_{\boldsymbol{q}}, \tag{F.12}$$

where $\mathcal{L} = [\hat{H}, \cdot]$ is the Liouvillian. We study the collective mode within TDHF over a particular mean field ground state with correlation matrix P, which is equivalent to the Bethe-Saltpeter collective mode at RPA level [467].

The TDHF collective modes are eigenvectors [465]

$$i\partial_t \hat{\phi}_q = \mathcal{L}^{\mathsf{MF}}_{\boldsymbol{q}} \hat{\phi}_{\boldsymbol{q}} = \omega_{\boldsymbol{q}} \hat{\phi}_{\boldsymbol{q}} \tag{F.13}$$

where $\mathcal{L}^{\mathsf{MF}}_{\boldsymbol{q}}$ is the "mean-field Liouvillian" at momentum $\boldsymbol{q}$. Since we wish to remain within the space of mean-field operators, we define

$$\mathcal{L}^{\mathsf{MF}}_{\boldsymbol{q}} = \mathcal{P}^{\mathsf{MF}} \mathcal{L}, \tag{F.14}$$

where P_{MF} projects an operator to mean-field level (with respect to a correlation matrix P) via

$$\hat{O}^{\mathsf{MF}} = \mathcal{P}^{\mathsf{MF}} \hat{O} = \sum_k \hat{c}^\dagger_{\boldsymbol{k},a} \left[O(\boldsymbol{k}, \boldsymbol{k}')\right]_{ab} \hat{c}_{\boldsymbol{k}',b} \tag{F.15}$$

where

$$\left[O(\boldsymbol{k}, \boldsymbol{k}')\right]_{ab} = \frac{\partial}{\partial P(\boldsymbol{k}, \boldsymbol{k}')_{ab}} \left\langle \hat{O} \right\rangle_P . \tag{F.16}$$

For instance,

$$\hat{H}^{\mathsf{MF}} = \mathcal{P}^{\mathsf{MF}}\hat{H} = \hat{h}^{\mathsf{HF}}[P] = \hat{h} + \hat{h}_H[P] + \hat{h}_F[P], \tag{F.17}$$

where these are these normal Hartree and Fock Hamiltonians.

We focus on particle-hole excitations only, i.e. those that that create or annihilates a PH excitation, i.e.

$$\langle \Psi_{\mathsf{MF}} | \hat{\phi}_{\boldsymbol{q}} | \Psi_{\mathsf{MF}} \rangle = 0. \tag{F.18}$$

This is true whenever

$$\hat{\phi}_{\boldsymbol{q}} = \hat{P}^{\perp}\hat{\phi}_{\boldsymbol{q}}\hat{P} \text{ or } \hat{\phi}_{\boldsymbol{q}} = \hat{P}^{\perp}\hat{\phi}_{\boldsymbol{q}}\hat{P} \tag{F.19}$$

where $\hat{P}$ is the projector onto the mean-field ground state and $\hat{P}^{\perp} = \hat{I} = \hat{P}$. Such states are in the image of the PH super-projector

$$\mathcal{P}^{\mathsf{PH}}\hat{\phi}_{\boldsymbol{q}} = \sum_{\boldsymbol{k},a,b} \hat{c}^{\dagger}_{\boldsymbol{k},a} \left[P^{\perp}(\boldsymbol{k})\phi_q(\boldsymbol{k})P(\boldsymbol{k}+\boldsymbol{q}) + P(\boldsymbol{k})\phi_{\boldsymbol{q}}(\boldsymbol{k})P^{\perp}(\boldsymbol{k}+\boldsymbol{q})\right]_{ab} \hat{c}_{\boldsymbol{k}+\boldsymbol{q},b} \tag{F.20}$$

where $P^{\perp} = I - P$. These two terms create and destroy particle-hole excitations above the ground state, respectively.

We therefore wish to solve the (non-Hermitian) eigenvalue problem

$$\mathcal{L}^{\mathsf{PH}}_{\boldsymbol{q}}\hat{\phi}_{\boldsymbol{q}} = \omega_{\boldsymbol{q}}\hat{\phi}_{\boldsymbol{q}} \text{ where } \mathcal{L}^{\mathsf{PH}} = \mathcal{P}^{\mathsf{PH}}\mathcal{L}^{\mathsf{MF}}\mathcal{P}^{\mathsf{PH}}, \tag{F.21}$$

which is an operator entirely within the space of mean-field particle-hole excitations over P.

F.2.3 Boosted Hartree-Fock Hamiltonians

The TDHF equation is best expressed in terms of two simple building blocks: the "boosted commutator" and the "boosted Hartree-Fock Hamiltonians".

Suppose $\hat{A} = \sum_{\boldsymbol{k}} \hat{c}^{\dagger}_{\boldsymbol{k}} A(\boldsymbol{k}) \hat{c}_{\boldsymbol{k}}$ and $\hat{B}_{\boldsymbol{q}} = \sum_{\boldsymbol{k}} \hat{c}^{\dagger}_{\boldsymbol{k}} B_{\boldsymbol{q}}(\boldsymbol{k}) \hat{c}_{\boldsymbol{k}+\boldsymbol{q}}$. Here and below band indices will be implicit. Then

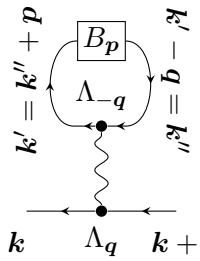
$$[\hat{A}, \hat{B}_{\boldsymbol{q}}] = \sum_{\boldsymbol{k}} \hat{c}^{\dagger}_{\boldsymbol{k}} \left[A, B_{\boldsymbol{q}}\right]_{\boldsymbol{q}}(\boldsymbol{k}) \hat{c}_{\boldsymbol{k}+\boldsymbol{q}} \tag{F.22}$$

where the boosted commutator is

$$[A, B_{\boldsymbol{q}}]_{\boldsymbol{q}}(\boldsymbol{k}) := A(\boldsymbol{k})B_{\boldsymbol{q}}(\boldsymbol{k}) - B_{\boldsymbol{q}}(\boldsymbol{k})A(\boldsymbol{k}+\boldsymbol{q}). \tag{F.23}$$

Now we consider the Hartree-Fock equations but with boosted correlation matrices, which have the form of $\hat{B}_{\boldsymbol{p}}$. The boosted Hartree Hamiltonian is

$$\hat{h}_H[B_{\boldsymbol{p}}] = 2 \times \tag{F.24}$$



$$= \frac{1}{A}\sum_{\boldsymbol{q}}\sum_{\boldsymbol{k},\boldsymbol{k}',\boldsymbol{k}''} V_{\boldsymbol{q}}\Lambda_{\boldsymbol{q}}(\boldsymbol{k})\,\mathrm{tr}[\Lambda_{-\boldsymbol{q}}(\boldsymbol{k}')B_{\boldsymbol{p}}(\boldsymbol{k}'')] \tag{F.25}$$

$$\times\, \delta_{\boldsymbol{k}',[\boldsymbol{k}''+\boldsymbol{p}]}\delta_{\boldsymbol{k}'',[\boldsymbol{k}'-\boldsymbol{q}]}\hat{c}^\dagger_{\boldsymbol{k}}c_{[\boldsymbol{k}+\boldsymbol{q}]}. \tag{F.26}$$

The total momentum around the loop must be zero $(\mathrm{mod}\ \boldsymbol{G})$:

$$\delta_{\boldsymbol{k}',[\boldsymbol{k}''+\boldsymbol{p}]}\delta_{\boldsymbol{k}'',[\boldsymbol{k}'-\boldsymbol{q}]} = \delta_{\boldsymbol{k}',[\boldsymbol{k}''+\boldsymbol{p}]}\sum_{\boldsymbol{G}}\delta_{\boldsymbol{q},\boldsymbol{p}+\boldsymbol{G}}, \tag{F.27}$$

so

$$\hat{h}_H[B_{\boldsymbol{p}}] = \frac{1}{A}\sum_{\boldsymbol{G}}\sum_{\boldsymbol{k},\boldsymbol{k}''} V_{\boldsymbol{p}+\boldsymbol{G}}\Lambda_{\boldsymbol{p}+\boldsymbol{G}}(\boldsymbol{k}) \tag{F.28}$$

$$\times\, \mathrm{tr}[\Lambda_{-\boldsymbol{p}-\boldsymbol{G}}(\boldsymbol{k}''+\boldsymbol{p})B_{\boldsymbol{p}}(\boldsymbol{k}'')]\hat{c}^\dagger_{\boldsymbol{k}}c_{[\boldsymbol{k}+\boldsymbol{p}+\boldsymbol{G}]}. \tag{F.29}$$

Finally using $\Lambda_{\boldsymbol{q}}(\boldsymbol{k}) = \Lambda_{-\boldsymbol{q}}(\boldsymbol{k}+\boldsymbol{q})^\dagger$ we get

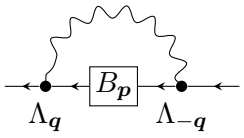
$$\hat{h}_H[B_{\boldsymbol{p}}] = \sum_{\boldsymbol{k}}\hat{c}^\dagger_{\boldsymbol{k}}h_H[B_{\boldsymbol{p}}](\boldsymbol{k})c_{[\boldsymbol{k}+\boldsymbol{p}]} \tag{F.30}$$

where

$$h_H[B_{\boldsymbol{p}}](\boldsymbol{k}) = \frac{1}{A}\sum_{\boldsymbol{G}} V_{\boldsymbol{p}+\boldsymbol{G}}\Lambda_{\boldsymbol{p}+\boldsymbol{G}}(\boldsymbol{k})\sum_{\boldsymbol{k}'}\mathrm{tr}[\Lambda_{\boldsymbol{p}+\boldsymbol{G}}(\boldsymbol{k}')^\dagger B_{\boldsymbol{p}}(\boldsymbol{k}')]. \tag{F.31}$$

Similarly, the Boosted Fock Hamiltonian is

$$\hat{h}_F[B_{\boldsymbol{p}}] = \tag{F.32}$$



$$= \frac{1}{A} \sum_{\boldsymbol{q},\boldsymbol{k},\boldsymbol{k}',\boldsymbol{k}''} V_{\boldsymbol{q}} \Lambda_{\boldsymbol{q}}(\boldsymbol{k}) B_{\boldsymbol{p}}(\boldsymbol{k}') \Lambda_{-\boldsymbol{q}}(\boldsymbol{k}'') \tag{F.33}$$

$$\times \delta_{[\boldsymbol{k}+\boldsymbol{q}],\boldsymbol{k}'} \delta_{[\boldsymbol{k}'+\boldsymbol{p}],\boldsymbol{k}''} \hat{c}^\dagger_{\boldsymbol{k}} \hat{c}_{\boldsymbol{k}''-\boldsymbol{q}}. \tag{F.34}$$

We can eliminate $\boldsymbol{k}' = [\boldsymbol{k}+\boldsymbol{q}]$ and $\boldsymbol{k}'' = [\boldsymbol{k}'+\boldsymbol{p}] = [\boldsymbol{k}+\boldsymbol{q}+\boldsymbol{p}]$ to get

$$\hat{h}_F[B_{\boldsymbol{p}}] = \frac{1}{A} \sum_{\boldsymbol{q},\boldsymbol{k},\boldsymbol{k}',\boldsymbol{k}''} V_{\boldsymbol{q}} \Lambda_{\boldsymbol{q}}(\boldsymbol{k}) B_{\boldsymbol{p}}([\boldsymbol{k}+\boldsymbol{q}]) \Lambda_{-\boldsymbol{q}}(\boldsymbol{k}+\boldsymbol{p}+\boldsymbol{q}) \hat{c}^\dagger_{\boldsymbol{k}} \hat{c}_{\boldsymbol{k}+\boldsymbol{p}+\boldsymbol{q}-\boldsymbol{q}}. \tag{F.35}$$

or

$$\hat{h}_F[B_{\boldsymbol{p}}] = \sum_{\boldsymbol{k}} \hat{c}^\dagger_{\boldsymbol{k}} h_F[B_{\boldsymbol{p}}](\boldsymbol{k}) c_{[\boldsymbol{k}+\boldsymbol{p}]} \text{ where } h_F[B_{\boldsymbol{p}}](\boldsymbol{k}) = \frac{1}{A} \sum_{\boldsymbol{q}} V_{\boldsymbol{q}} \Lambda_{\boldsymbol{q}}(\boldsymbol{k}) B_{\boldsymbol{p}}([\boldsymbol{k}+\boldsymbol{q}]) \Lambda_{\boldsymbol{q}}(\boldsymbol{k}+\boldsymbol{p})^\dagger. \tag{F.36}$$

In summary, we have found the boosted Hartree-Fock Hamiltonian

$$\hat{h}_{HF}[B_{\boldsymbol{p}}] = \sum_{\boldsymbol{k}} \hat{c}^\dagger_{\boldsymbol{k}} h_{HF}[B_{\boldsymbol{p}}](\boldsymbol{k}) c_{[\boldsymbol{k}+\boldsymbol{p}]}, \tag{F.37}$$

$$h_{HF}[B_{\boldsymbol{p}}] = h_H[B_{\boldsymbol{p}}] + h_F[B_{\boldsymbol{p}}], \tag{F.38}$$

$$h_H[B_{\boldsymbol{p}}](\boldsymbol{k}) = \frac{1}{A} \sum_{\boldsymbol{G}} V_{\boldsymbol{p}+\boldsymbol{G}} \Lambda_{\boldsymbol{p}+\boldsymbol{G}}(\boldsymbol{k}) \sum_{\boldsymbol{k}'} \mathrm{tr}[\Lambda_{\boldsymbol{p}+\boldsymbol{G}}(\boldsymbol{k}')^\dagger B_{\boldsymbol{p}}(\boldsymbol{k}')], \tag{F.39}$$

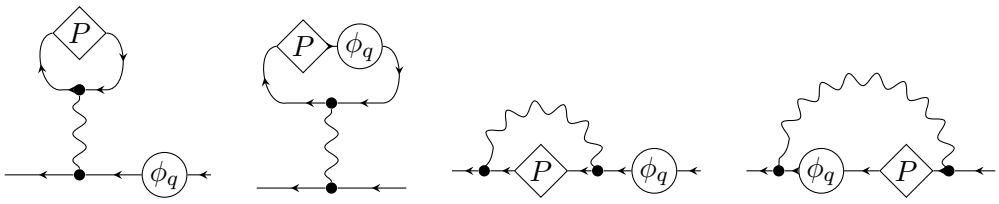
$$h_F[B_{\boldsymbol{p}}](\boldsymbol{k}) = \frac{1}{A} \sum_{\boldsymbol{q}} V_{\boldsymbol{q}} \Lambda_{\boldsymbol{q}}(\boldsymbol{k}) B_{\boldsymbol{p}}([\boldsymbol{k}+\boldsymbol{q}]) \Lambda_{\boldsymbol{q}}(\boldsymbol{k}+\boldsymbol{p})^\dagger. \tag{F.40}$$

F.2.4 Derivation of TDHF

We now reduce the TDHF operator $\mathcal{L}^{\mathsf{MF}}$ to an operator on matrices $\phi_{\boldsymbol{q}}$. We therefore evaluate

$$\mathcal{L}^{\mathsf{MF}}_{\boldsymbol{q}} \hat{\phi}_q = \mathcal{P}^{\mathsf{MF}}[\hat{h}, \hat{\phi}_{\boldsymbol{q}}] + \mathcal{P}^{\mathsf{MF}}[\hat{V}, \hat{\phi}_{\boldsymbol{q}}]. \tag{F.41}$$

We interpret Eq. (F.16) diagramatically: write all Wick contractions diagrams of $\hat{O}$ by P with exactly two dangling legs (so that inserting one more P would give $\langle \hat{O} \rangle_P$). The projector is trivial for the first term, yielding $[\hat{h}, \hat{\phi}_{\boldsymbol{q}}]$. For the second term there are four non-trivial diagrams for $\mathcal{P}^{\mathsf{MF}} \hat{V} \hat{\phi}_{\boldsymbol{q}}$:



The first two are Hartree-like, while the second two are Fock-like (and come with a minus sign). Since $\hat{\phi}_{\boldsymbol{q}}$ acts before $\hat{V}$, there cannot be a P separating $\phi_{\boldsymbol{q}}$ from the interaction vertex. Comparing to the terms above, these diagrams gives

$$\hat{h}_H[P]\hat{\phi}_{\boldsymbol{q}} + \hat{h}_H[\phi_{\boldsymbol{q}} P] + \hat{h}_F[P]\hat{\phi}_{\boldsymbol{q}} + \hat{h}_F[\phi_{\boldsymbol{q}} P]. \tag{F.42}$$

Similarly, $-\mathcal{P}^{\mathsf{MF}} \hat{\phi}_{\boldsymbol{q}} \hat{V}$ gives

$$-\hat{\phi}_{\boldsymbol{q}} \hat{h}_H[P] - \hat{h}_H[P\phi_{\boldsymbol{q}}] - \hat{\phi}_{\boldsymbol{q}} \hat{h}_F[P] - \hat{h}_F[P\phi_{\boldsymbol{q}}]. \tag{F.43}$$

Combining these

$$\mathcal{P}^{\mathsf{MF}}[\hat{V}, \hat{\phi}_{\boldsymbol{q}}] = \big[\hat{h}_H[P] + \hat{h}_F[P], \hat{\phi}_{\boldsymbol{q}}\big] + \hat{h}_H\big[[\phi_{\boldsymbol{q}}, P]\big] + \hat{h}_F\big[[\phi_{\boldsymbol{q}}, P]\big] = \big[\hat{h}_H[P] + \hat{h}_F[P], \hat{\phi}_{\boldsymbol{q}}\big] - \hat{h}_{HF}\big[[P, \phi_{\boldsymbol{q}}]_{\boldsymbol{q}}\big] \tag{F.44}$$

The kinetic term $\mathcal{P}^{\mathsf{MF}}[\hat{h}, \hat{\phi}_{\boldsymbol{q}}] = [\hat{h}, \hat{\phi}_{\boldsymbol{q}}]$ combines with the first term here to give the full (non-boosted) Hartree-Fock Hamiltonian. Altogether then the action of $\mathcal{L}^{\mathsf{MF}}_{\boldsymbol{q}}$ on matrices $\phi_{\boldsymbol{q}}$ is

$$\mathcal{L}^{\mathsf{MF}}_{\boldsymbol{q}} \phi_q = \big[h_{HF}[P], \phi_{\boldsymbol{q}}\big]_{\boldsymbol{q}} - h_{HF}\big[[P, \phi_{\boldsymbol{q}}]_{\boldsymbol{q}}\big]. \tag{F.45}$$

The first term is the boosted commutator of the un-boosted HF Hamiltonian, and the second term is (minus) the boosted HF Hamiltonian of the boosted commutator.

F.3 2PI Effective Action

While we have focused on extracting the phonon action via Hartree-Fock in the main text, it is worth noting that our procedure is more general. In particular, any approximate scheme for computing the effective action $\Gamma[\boldsymbol{\theta}]$ will do. The existence of such an effective action guarantees that computing for example the kineo-elastic coefficient by first shearing the unit cell, and then boosting, or boosting and the shearing will give the same result. Let us briefly discuss how how (Time dependent)Hartree-Fock can be considered such a scheme. In particular by considering the effective action of the Hamiltonian $H[\boldsymbol{\theta}]$ as a function of the *Greens function*, within the 2PI formalism [424, 468]

$$\Gamma[G, \boldsymbol{\theta}] = \operatorname{tr} \log G^{-1} - \operatorname{tr}[G G_0^{-1}[\boldsymbol{\theta}] - 1] - \Phi[G, \boldsymbol{\theta}] \tag{F.46}$$

where, $G_0^{-1}[\boldsymbol{\theta}] = \partial_t - h[\boldsymbol{\theta}]$ is the free propagator and Φ is the Luttinger-Ward functional, which perturbatively is the sum of all 2 particle irreducible diagrams of the gauge twisted interaction. The effective action in terms of $\boldsymbol{\theta}$ is obtained by finding the saddle point G of Γ. In particular note that the equation of motion for G is precisely the usual Dyson equations

$$0 = \frac{\delta \Gamma}{\delta G} = G^{-1} - G_0^{-1} + \Sigma[G] \tag{F.47}$$

where $\Sigma[G] = \frac{\delta \Phi[G]}{\delta G}$ is the self energy functional. Hartree-Fock emerges as the lowest order approximation to Φ in terms of interaction, which in this case is

$$\Phi^{\mathrm{HF}}[G, \boldsymbol{\theta}] = \frac{1}{2} \int d\boldsymbol{r}_1 d\boldsymbol{r}_2 \, V(\boldsymbol{r}_1 + \boldsymbol{\theta}(\boldsymbol{r}_1, t) - \boldsymbol{r}_2 - \boldsymbol{\theta}(\boldsymbol{r}_2, t)) G_{\alpha\alpha}(\boldsymbol{r}_1, \boldsymbol{r}_1) G_{\beta\beta}(\boldsymbol{r}_2, \boldsymbol{r}_2) \tag{F.48}$$

$$- \frac{1}{2} \int d\boldsymbol{r}_1 d\boldsymbol{r}_2 \, V(\boldsymbol{r}_1 + \boldsymbol{\theta}(\boldsymbol{r}_1, t) - \boldsymbol{r}_2 - \boldsymbol{\theta}(\boldsymbol{r}_2, t)) G_{\alpha\beta}(\boldsymbol{r}_1, \boldsymbol{r}_2) G_{\beta\alpha}(\boldsymbol{r}_2, \boldsymbol{r}_1) \tag{F.49}$$

The equations of motion $\frac{\delta}{\delta G(x,y)} \log Z^{\mathrm{HF}} = 0$ are solved precisely by the solutions to the self-consistent Hartree-Fock problem, with both stationary and time dependent solutions corresponding to the conventional Hartree-Fock approach, and general time dependent Hartree-Fock. Making the connection to the variational formulation can be done easily considering multiplying equation (F.47) with G, and taking the commutator:

$$[\partial_t, G] = [h + \Sigma^{\mathrm{HF}}[G], G] \tag{F.50}$$

At the Hartree-Fock mean field level, Σ becomes local in time, and we recover the most general time dependent Hartree Fock equations. By considering stationary solutions $[\partial_t, G] = 0$, we recover the usual stationary Hartree-Fock equations $[h+\Sigma^{\mathrm{HF}}[G], G] = 0$. By contrast the linearized TDHF we perform consist of taking a stationary solution $\bar{G}$, and considering small fluctuations $G = \bar{G} + \phi$. The linearized equations of motion for ϕ are

$$[\partial_t, \phi] = \left[h + \Sigma^{\mathrm{HF}}[\bar{G}], \phi\right] + \left[\frac{\delta\Sigma[\bar{G}]}{\delta G}\phi, \bar{G}\right] = \mathcal{L}[\phi] \tag{F.51}$$

which is precisely equation (F.45). Suppose $U = e^{\alpha A}$ generates a continuous symmetry that commutes with the Hamiltonian, then $\phi_A = [A, \bar{G}]$ is an exact zero mode of equation (F.51), which is a manifestation of the Goldstone theorem [400]. Keeping higher order terms in Φ is a natural extension of HF.

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[198] See Supplemental Material [url] for a detailed description of the continuum model, a brief introduction to quantum geometry, a derivation of vortexability induced perfect circular dichroism, a parton theory for the zero field composite Fermi liquid in Chern bands, a discussion of interaction driven layer polarization physics, details and discussions on exact diagonalization numerics, details and discussions on density matrix renormalization group numerics, and more experimental signatures, which includes Refs. [445–456, 469, 473–476].

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